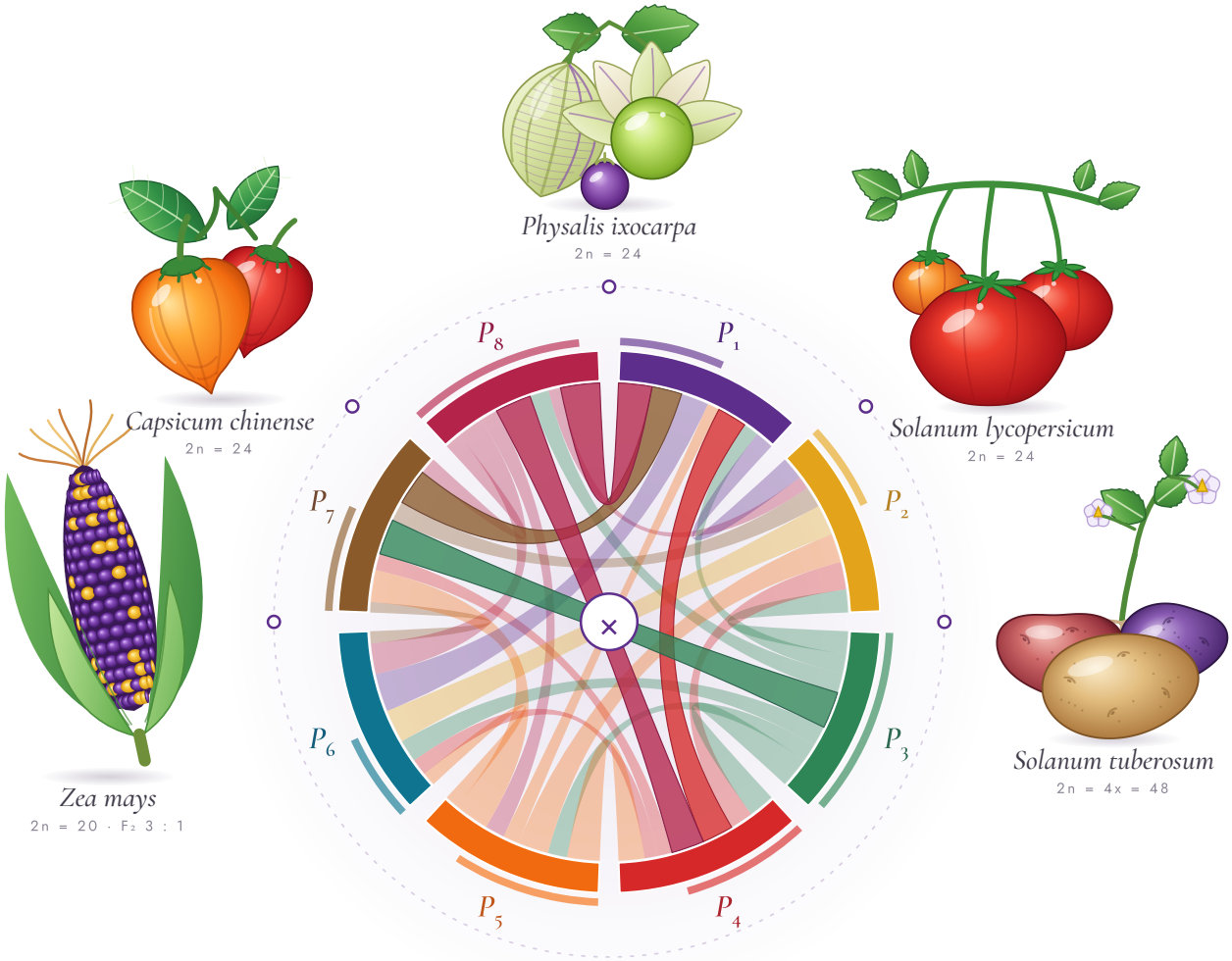
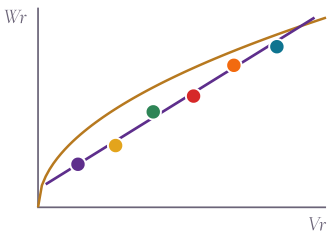
BreedingPro

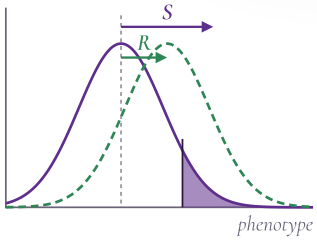
User's Manual
Manual de usuario

PLANT AND ANIMAL BREEDING · MEJORAMIENTO GENÉTICO

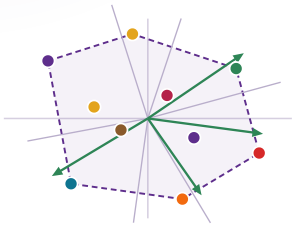




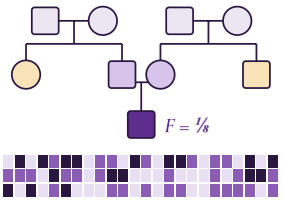
HAYMAN · Wr – Vr



SELECTION · R = h²S



G × E · GGE BIPLOT



BLUP · PEDIGREE AND MARKERS

Luis Ángel Barrera-Guzmán

ENGLISH EDITION · VERSION 1.0.1 · VERSIÓN 1.0.1 · 2026

BreedingPro

User's Manual · Plant and animal breeding

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PROGRAM	BreedingPro, version 1.0.1 (September 2026)
MANUAL	First English edition, 2026, translated from the first Spanish edition. Describes version 1.0.1 of the program.
PROGRAM LICENSE	GNU General Public License, version 3 or later (GPL-3.0-or-later).
MANUAL RIGHTS	© 2026 Luis Ángel Barrera-Guzmán. All rights reserved.
INTERFACE LANGUAGE	Spanish or English, with the ES · EN switch on the top bar. The app opens in Spanish; in this manual the names of buttons and menus are written as they appear on screen in English, for example Build the report.

HOW TO CITE THE PROGRAM

Barrera-Guzmán, L.Á. (2026). *BreedingPro: a browser-based platform for diallel crosses, mating designs, genetic parameters and prediction in plant and animal breeding* (Version 1.0.1) [Computer software]. https://doi.org/10.5281/zenodo.23005938

HOW TO CITE THIS MANUAL

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That digital object identifier (DOI) is the concept DOI: it always leads to the most recent version. Each archived version also has its own (version 1.0.0, 10.5281/zenodo.23005939). The citation of the program also appears at the end of every Block 12 report and in the How to cite section of the app's home page.

> **PRIVACY**
>
> BreedingPro runs entirely inside your browser. The files you open are read on your computer and **are never sent to any server**. You can analyze unpublished trials, data from a thesis in progress or materials protected by agreements with no risk of them leaving your machine.

The illustrations, diagrams and screenshots in this manual are original: they were generated with the program itself or with code written for the manual. The example data sets are described in Section I.7, each with its source; the simulated ones were generated with fixed seeds, and their true values are known.

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How to read this manual

The manual follows the same order as the app: twelve numbered blocks that appear, from top to bottom, in the sidebar on the left. Each chapter explains one block and carries its number (Chapter 3 is Block 3, and its sections, figures and tables are numbered 3.1, 3.2…). It begins with the theory you cannot do without, describes what the block calculates and how to read it, and ends with step-by-step examples using the data that already come inside the program. The introduction comes before the blocks and is numbered separately: I.1, I.2…

A color for each block

The block's color appears on the opening page of its chapter, on the section numbers, on the tab at the edge of the page and in the table of contents, and it is the same as in the strip on the cover. That way you can tell at a glance which part of the app you are in.

Block	Name in the app	What it does
1	Home	Home page, two labs with simulated genetics, theory in plain language, references and how to cite
2	Plan	Crossing plan, seed and pollinations, trial layout, field book and power before sowing
3	Data	File loading, the role of each column, recognized design, data quality and field analysis
4	Griffing	General and specific combining ability with Griffing's four methods
5	Hayman–Jinks	Hayman's partition, genetic components, Wr–Vr graph, Gardner–Eberhart and heterosis
6	Mating designs	North Carolina I, II and III, triple test cross, line × tester and partial diallels
7	Generations	Scaling tests, generation mean models, variances, heterosis and effective factors
8	Selection	Heritability, correlations and paths, selection indices and gain from recurrent selection
9	GxE	Genotype × environment interaction: regressions, stability statistics, AMMI and GGE biplot
10	BLUP	Pedigree, animal and sire models, breeding values, REML and crossbreeding parameters
11	Genomics	Markers, genomic relationships, GBLUP, cross-validation and hybrid prediction
12	Report	Report with written methods and references, PDF, HTML and ZIP package of results

Types of box

NOTE

Clarifications and practical tips that save time.

IMPORTANT

What can change your conclusions if overlooked.

THEORY IN BRIEF

The idea behind an analysis, without unnecessary formulas.

EXAMPLE

A case worked step by step with the data included in the program.

DECISION RULE

If the result falls in this case…	**then** this is the recommended reading.

These are guides for interpretation, not laws: what you know about your species and your program always comes first.

900 dpi

KEY FACT

A figure or a fact worth remembering; for example, the maximum resolution at which figures are exported.

Writing conventions

Written like this	Means
Fit the model	A button, checkbox or menu of the app, with its text as it appears on screen in English.
Edit figure › Style & typography	A path: open the first item and then the second.
`Yield`, `.xlsx`	A column name, file name or value typed as it stands.
σ^2_A, h^2, $\hat{g}_i$, $\hat{s}_{ij}$	Parameters and estimators; their meaning is explained the first time they appear and gathered in the glossary.
GCA, SCA	The abbreviations of the literature in English, the same ones the app uses in English: general and specific combining ability.
● ◐ —	Available · partly available or with conditions · does not apply.

In

INTRODUCTION · BEFORE THE BLOCKS

Introduction to BreedingPro

What the program is, which questions of a breeding program it answers, which data it accepts and how to go through it from beginning to end. It includes the five ideas of quantitative genetics on which the rest of the manual rests.

I.1 What is BreedingPro?

I.2 Questions it answers

I.3 Five key ideas of breeding

I.4 Data it accepts

I.5 What you need and how to open it

I.6 A tour of the interface

I.7 The examples included

I.8 Recommended workflow

I.9 P values, standard errors and intervals

In

I.1 What is BreedingPro?

BreedingPro is a platform for analyzing the experiments of a breeding program **from the cross to the decision**: it helps you plan the crosses and the trial, recognizes the design of the table that comes back from the field, estimates combining ability, gene action, heritability and stability, predicts breeding values from a pedigree or from markers, and delivers a report with figures, written methods and references.

Everything happens inside the web browser. There is nothing to install, no programming is needed and the data never leave the computer. The same workflow serves plants and animals: the diallels and mating designs of plant breeding, and the animal model and crossbreeding of animal breeding.

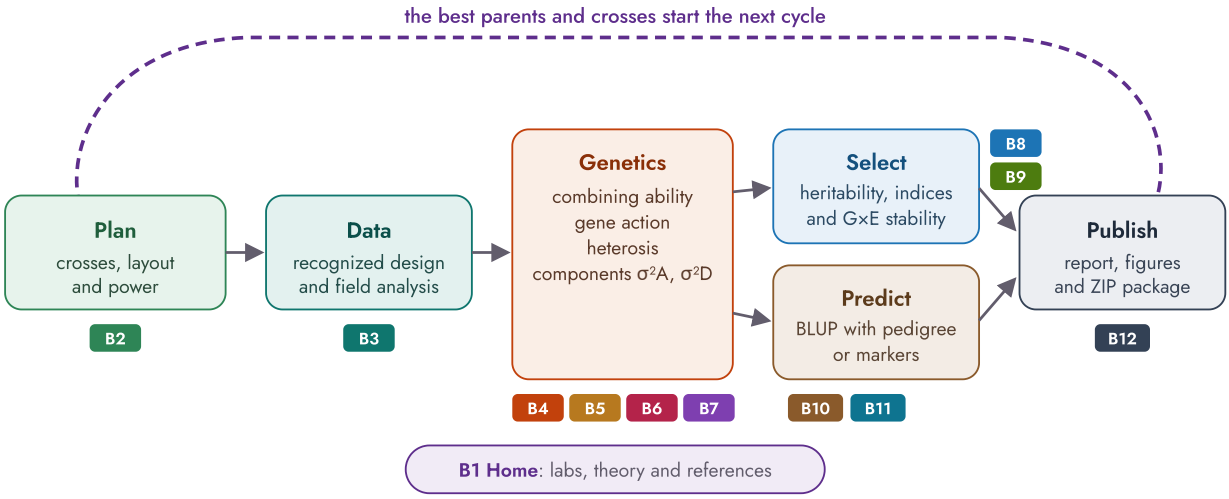


Figure I.1. A breeding cycle traveled through the blocks of BreedingPro. The genetics blocks (4 to 7) are chosen according to the mating design; the selection blocks (8 and 9) and the prediction blocks (10 and 11), according to the question. Block 1 accompanies the whole route with labs and theory, and Block 12 gathers the results of all of them.

What it does, in one table

Table I.1. The twelve blocks and what each one calculates.

Block	Calculates or delivers
1	Diallel lab (parents with actual genes, plots with error, and Griffing and Hayman analyses) and selection lab (observed response against R = h^2S); theory and 63 references.
2	Crossing plan for 11 designs, plants per parent, seed and pollinations, schedule of cycles; randomized layout in complete blocks, α-lattice, completely randomized or augmented blocks; field book; standard errors, detectable differences and power of the F tests.
3	Reading of spreadsheets and text, the role of each column, recognition of the mating design and of the field design, quality and outliers, analysis of variance or REML by environment and combined, adjusted means and heritability of the trial.

Table I.1. The twelve blocks and what each one calculates. *(continued)*

Block	Calculates or delivers
4	Griffing's Methods 1 to 4, Models I and II, GCA, SCA and reciprocal effects (maternal and non-maternal) with exact standard errors, σ^2A, σ^2D, Baker's ratio, best parents and crosses, several environments.
5	Hayman's partition (a, b_1, b_2, b_3, c, d) or Morley Jones's, components D, H_1, H_2, F, h^2 and E, Wr–Vr graph and dominance order, Gardner–Eberhart Analyses II and III, heterosis of each cross and identities between methodologies.
6	North Carolina I, II and III, triple test cross, line × tester and partial diallels, with expected mean squares computed for the actual design, effects, components and epistasis.
7	Scaling tests A, B, C and D, joint scaling test and mean models with epistasis, components D, H, F and E of the variances, heterosis, inbreeding depression and number of effective factors.
8	Variances, heritability with an exact interval, coefficients of variation and expected gain; genetic correlations and path analysis; eight selection indices; gain per cycle and per year of seven recurrent selection methods.
9	Regression on the environmental index, ecovalence, stability variance, superiority, reliability, ranks, yield–stability, AMMI with WAAS, GGE biplot in seven views and interaction variance by REML.
10	Pedigree with selfing, inbreeding and relationship; animal or sire model with permanent environment and maternal effect; breeding values with their reliability; BLUP of the genotypes of the trial; Dickerson's crossbreeding parameters.
11	Marker quality control, genomic relationship matrix, structure, GBLUP and marker effects, cross-validation, response per year and prediction of untested single crosses, three-way and double crosses.
12	Report with the results of each block, methods written with the settings used, and references; PDF, HTML and ZIP package with tables, results workbooks, figures up to 900 dpi and data.

Who is it for?

For **plant and animal breeders** who need to analyze their trials with classical and modern methods without depending on a different program for each design; for **graduate students** preparing a thesis who want to understand what they calculate; and for **teachers** of quantitative genetics, who will find support for their classes in the labs of Block 1 and in the simulated examples, whose true values are known.

I.2 Questions it answers

A breeding program asks itself concrete questions. Table I.2 shows which ones BreedingPro answers and in which block.

Table I.2. Frequent questions of a breeding program and where they are answered.

Question	Block	What it delivers
How many crosses, plants, replicates and locations do I need to detect the differences I care about?	2	Crossing plan, seed, layout and power
Do my data have errors? What design does the table have?	3	Recognized design, quality and outliers
Which parents combine best? Which is the best cross?	4 6	GCA, SCA and their significance

In

Table I.2. Frequent questions of a breeding program and where they are answered. *(continued)*

Question	Block	What it delivers
Is inheritance additive or dominant? Is there epistasis or are there maternal effects?	5 6 7	Components, degree of dominance and scaling tests
How much heterosis is there and where does it come from?	5 7	Average, variety and specific heterosis
How much will I gain if I select? With which method and which index?	8	Heritability, gain and indices
Which variety is stable, and which one wins in each environment?	9	Stability, AMMI and GGE
Which animal or line has the best breeding value, even without records of its own?	10 11	BLUP with pedigree or with markers
Which breed cross or crossbreeding system pays off?	10	Dickerson's parameters and predictions
Which untested hybrids will be the best?	11	Prediction of single, three-way and double crosses
How do I write it up in the thesis or the paper?	12	Report with methods and references

I.3 Five key ideas of breeding

Everything BreedingPro calculates rests on five ideas of quantitative genetics. They are worth keeping in mind; each chapter takes them up again in more detail.

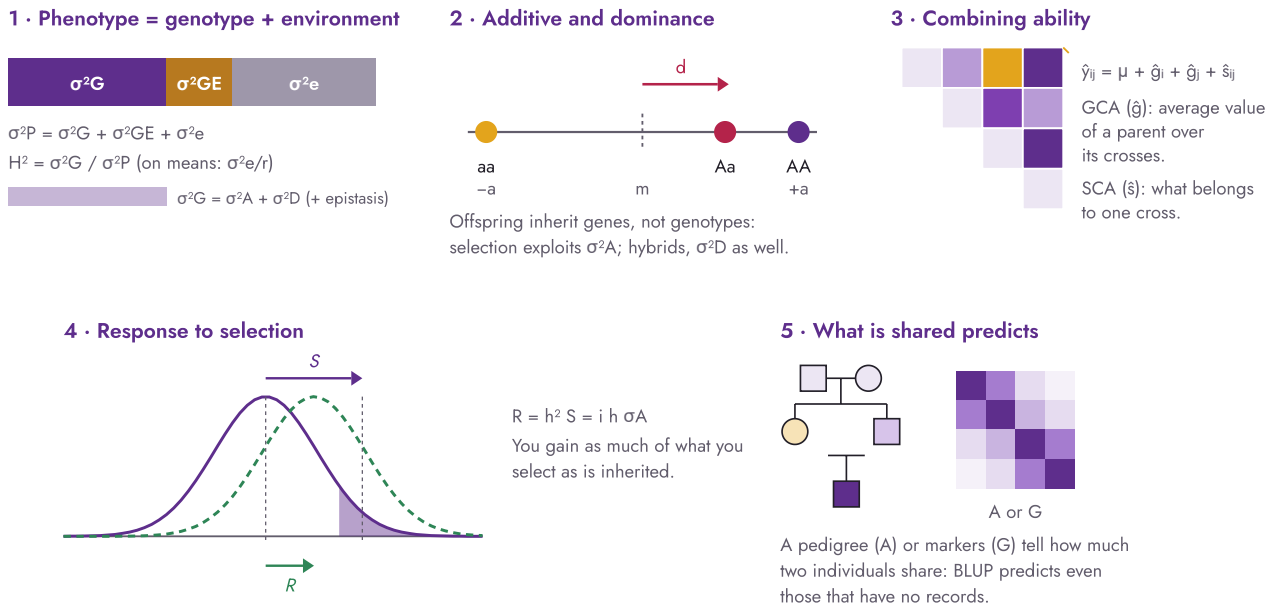


Figure I.2. The five ideas of quantitative genetics on which the program rests.

1. **Phenotype = genotype + environment.** What is measured in a plot mixes the genetic value of the material, the effect of the environment, their interaction and the error. The field designs (Block 3) and the models of the rest of the blocks separate those parts. Heritability, $H^2 = \sigma^2G/\sigma^2P$, tells what fraction of the observed variation is genetic.

2. **Additive and dominance.** At one locus, the heterozygote may sit halfway between the homozygotes (additive action) or be shifted toward one of them (dominance, d). Offspring inherit genes, not genotypes; that is why selection mainly exploits the additive variance σ^2A, while hybrids also exploit the dominance variance σ^2D.

3. **Combining ability.** In a diallel, the performance of a cross is split into the general combining ability of its two parents (GCA, tied to σ^2A) and the specific combining ability of the cross (SCA, tied to σ^2D and to epistasis). It is the basis of Blocks 4 to 6.

4. **Response to selection.** If the best individuals are selected with a differential S, the next generation advances $R = h^2S$. With the selection intensity i, the accuracy h and the additive standard deviation σ_A, the same idea gives the gain per cycle and per year of Blocks 8 and 11.

5. **What is shared predicts.** Two relatives resemble each other in proportion to the genes they share. The pedigree (matrix A) or the markers (matrix G) measure that relationship, and BLUP uses it to predict the breeding value of every individual, including those with no records of their own (Blocks 10 and 11).

I.4 Data it accepts

Most analyses start from a **table of plots** that is loaded in Block 3: one row per plot (or per plant) and one column per item. The app recognizes the design from the names and the contents of the columns, in Spanish or in English, and shows you what it assumed before analyzing.

Table I.3. Designs that Block 3 recognizes in a table of plots.

Mating design	Columns that identify it	Analyzed in
Diallel (Methods 1 to 4)	Female and male of each cross; parents with female = male; reciprocals if both orientations appear	4 5 8
Partial diallel	Each parent crossed with s of the others	6
North Carolina I and II	Males and females (nested or crossed) and sets	6 8
North Carolina III and triple test cross	F_2 plants or lines crossed with P_1, P_2 (and F_1)	6
Line × tester	Lines, testers and, if they were evaluated, the parents	6 8
Generation means	A generation column: P_1, P_2, F_1, F_2, BC_1, BC_2…	7
Genotype trial	Entries (varieties, lines, clones, hybrids) without parents	8 9 10

The field design is recognized in the same way: completely randomized, randomized complete blocks, α-lattice and incomplete blocks, augmented blocks with replicated checks, rows and columns, or means without replicates. With a location or year column the table is multi-environment, and with a plant column the plot means and the sampling error are analyzed separately.

Other blocks take tables of their own: **Block 10**, records with individual, sire and dam (and a separate pedigree if needed); **Block 11**, a marker matrix (as dosages 0/1/2, –1/0/1, presence or absence, or letters) with phenotypes per individual or with a table of hybrids.

FILE FORMATS

- Spreadsheets `.xlsx`, `.xls`, `.xlsm` and `.ods` (you choose the sheet).
- Delimited text `.csv`, `.tsv` and `.txt`, with comma, semicolon or tab.
- Pasting a block copied from a spreadsheet into the Data sheet.
- The Block 2 field book, with or without simulated values.

MISSING DATA

- Empty cells and codes such as `NA`, `.`, `-`, `?` or `faltante` are read as missing.
- Lost crosses and unequal replicates are analyzed without filling in: the standard errors are adjusted.
- The decimal separator may be a point or a comma; the app detects it.

> **IMPORTANT**
>
> The role of each column decides the analysis. A column of parents read as a trait, or a replicate read as a location, changes the results. Block 3 assigns the roles automatically, but it **always shows you what it assumed** in Data › Role of every column and lets you correct it before analyzing.

1.5 What you need and how to open it

REQUIREMENTS

- A desktop or laptop computer with a **modern, up-to-date web browser**.
- A screen at least 1,280 pixels wide; the app adapts to smaller screens and to phones, but tables and figures are best used on a large one.
- 4 GB of memory is enough for almost everything; genomics with thousands of individuals benefits from 8 GB.
- **No** internet connection, installation or administrator rights are needed.

CONTENTS OF THE FOLDER

`index.html`	The app. Open it with a double click.
`Open BreedingPro.bat`	Optional start-up with a local server.
`js/`, `css/`	The program's code, with the examples included.
`vendor/`	The spreadsheet reader and its license.
`tests/`	The program's tests: open them with a double click, and they should all come out green.
`manual/`	This manual.

Opening BreedingPro in three steps

1. **Copy the whole `BreedingPro` folder** to your computer, for example to Documents. Do not separate `index.html` from the rest of the folders.
2. **Double-click `index.html`**. It opens in your default browser, on the Block 1 home page.
3. **Click** Start with my data → to go to Block 3 and load a file or an example, or browse the home page, the labs and the theory.

> **NOTE · OPENING WITH A LOCAL SERVER**
>
> With a double click everything works. If your institution blocks pages opened as local files, use `Open BreedingPro.bat`: it starts a small server on your own computer (port 9200) and opens the same app in the browser. Nothing leaves your machine.

> **NOTE · WHAT IS REMEMBERED AND WHAT IS NOT**
>
> The app remembers in your browser the language, the light or dark theme, the style of the figures and the Block 3 data sheet. **Loaded data and results are not saved**: when you reload the page you have to load the file again. Download the results of each block (Results (.xlsx)) or the Block 12 package before closing.

1.6 A tour of the interface

The top bar and the two side panels accompany every block: on the left, the list of the twelve blocks; on the right, the index of the block you are in. In the middle, each block takes up a single page that is read from top to bottom.

The page of a block

All the analysis blocks share the same layout, so you only need to learn it once:

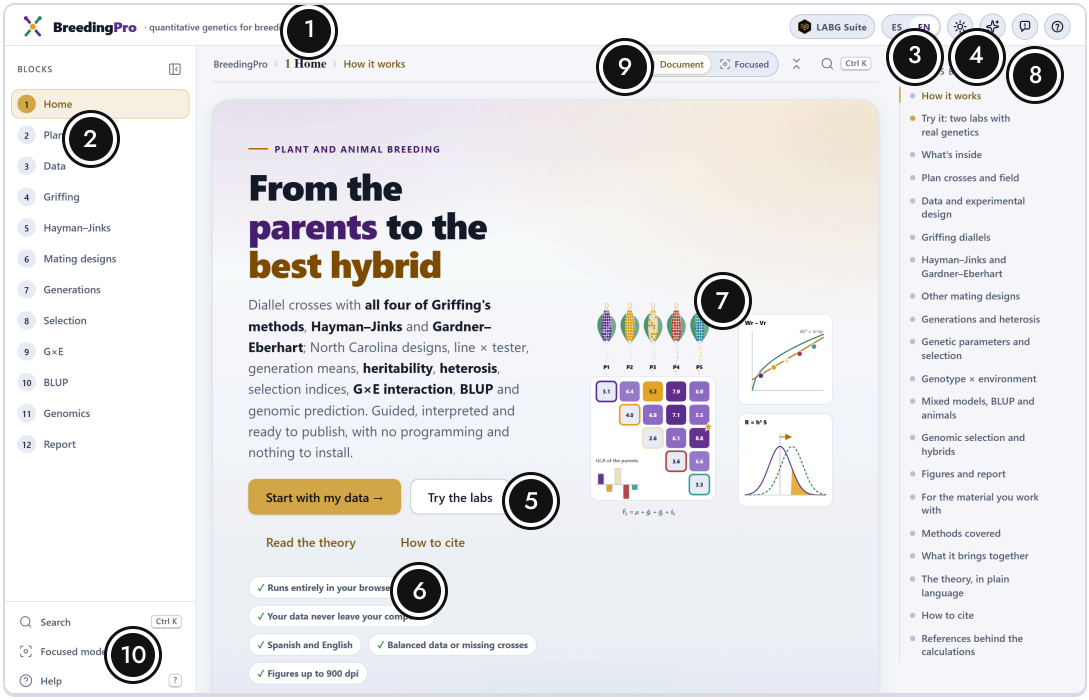


Figure I.3. The app's home page (Block 1) with the top bar and the two side panels.

1. Name of the program: one click goes back to the home page; `LABG Suite` leads to the suite's portal.
2. The list of the twelve blocks. The active block is highlighted; any of them opens with a click.
3. `ES` · `EN`: changes the language of the whole app, including the figures and the reports.
4. Light or dark theme; then animations, report a problem and keyboard shortcuts.
5. Start with your data, try the labs, read the theory or see how to cite.
6. What the app guarantees: nothing to install, no data sent, two languages, missing crosses and figures up to 900 dpi.
7. Live illustration: a diallel with its GCAs and SCAs, a Wr–Vr graph and the response to selection.
8. `In this block`: the sections of the block, one click away.
9. Where you are, the `Document` and `Focused` modes, and the search.
10. `Search`, `Focused mode` and `Help`.

Part	What it contains
Numbered cards	Each block is divided into cards (`1 · Data and model`, `2 · Analysis of variance of the diallel`…). The first ones ask for the settings and the following ones show the results. In `Focused` mode only one section is shown at a time.
Colored messages	Purple informs, green confirms, amber warns and red stops: for example, a design that does not allow the analysis or a variance that was estimated as negative.
Summary boxes	The main numbers in large type, with a line of context underneath.
Tables	With the P values and their asterisks. Long tables show the first rows and download them all.

Part	What it contains
Figures	Editable and exportable (Figure I.5).
Final card and bar	The last card gathers assumptions, formulas and references, and the Results (.xlsx) button with all the tables of the block; the bar at the bottom goes back to the previous block or continues to the next one.

2 · Analysis of variance of the diallel

Partition of the entries into GCA, SCA and reciprocal effects (and these into maternal and non-maternal).

Source of variation	df	SS	MS	F · Model I	p	Denominator	F · Model II
Entries	**63**	**26,230.9682**	**416.3646**	—	—	—	—
General combining ability (GCA)	7	3,804.6427	543.5204	18.89	< 0.0001 ***	Experimental error	0.69 ns
Specific combining ability (SCA)	28	22,060.7885	787.8853	27.38	< 0.0001 ***	Experimental error	27.38 ***
Reciprocal effects	28	365.5369	13.0549	0.45	0.9923 ns	Experimental error	0.45 ns
Maternal	7	77.6843	11.0978	0.39	0.9100 ns	Experimental error	0.39 ns
Non-maternal	21	287.8526	13.7073	0.48	0.9758 ns	Experimental error	0.48 ns
Experimental error	189	5,438.9907	28.7777	—	—	—	—

Sums of squares on the entry-mean scale; the error is MS_{error}/r = 28.7777 with 189 df.

Figure I.4. A results card of Block 4 with the 8 × 8 diallel of the textbook example: the analysis of variance of the diallel with the F tests of Model I (against the error) and of Model II (with the appropriate denominator), the P value with its significance mark, and a note on the scale of the sums of squares.

The figure editor

Every figure has a folded editor underneath. Every change redraws the figure instantly, and the style you choose (theme, font, sizes) is remembered for the other figures of all the blocks. The ✎ button above the figure opens a fuller editor of titles, axes, series, legend, texts and annotations, and the icon at the right of the card opens it in the figure studio, with the screen split to edit and export.

Language and theme

The EN button switches the whole app to English: the texts, the tables, the titles and axes of the figures, and the Block 12 report. That way you can prepare the figures of a paper in English without analyzing again. The dark theme rests your eyes in long sessions; figures keep their own white background for export.

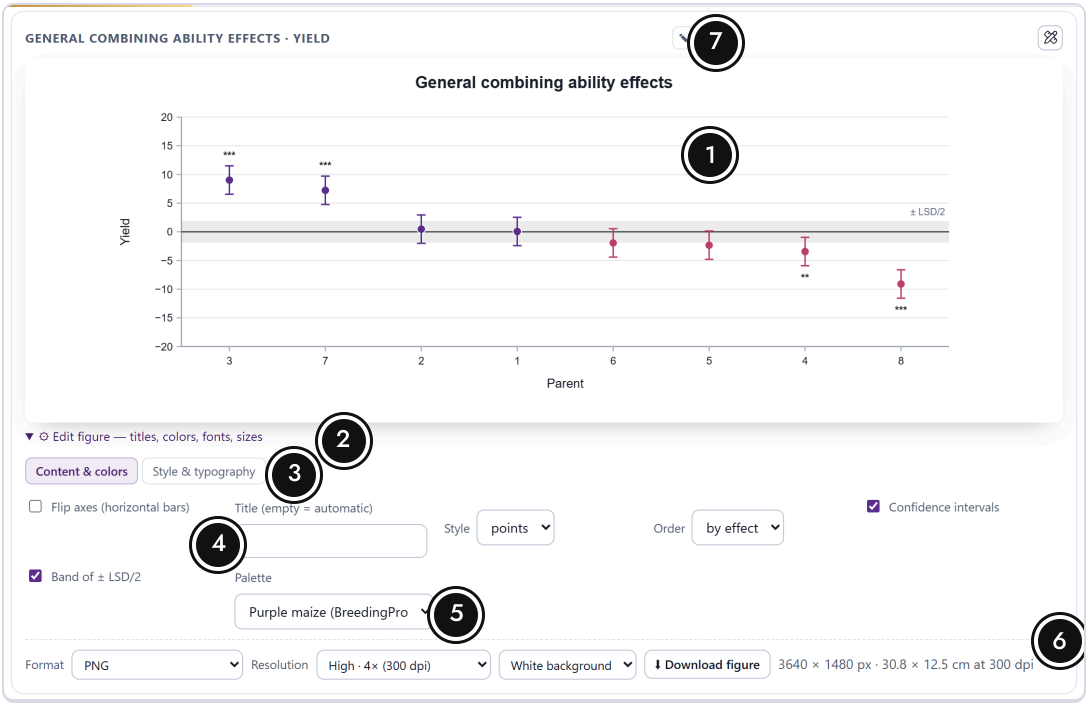


Figure I.5. The figure of the GCA effects of Block 4 with its editor open.

1. The figure, with significance asterisks and confidence intervals.
2. Edit figure — titles, colors, fonts, sizes opens and closes the editor.
3. Two tabs: Content & colors and Style & typography.
4. Options of the figure itself: flip the axes, style, order, bands and intervals.
5. Color palette; the second tab changes theme, font and sizes.
6. Format (PNG, TIFF, SVG, JPG or WEBP), resolution up to 900 dpi, background and download, with the size in pixels and centimeters.
7. ✎ opens the full editor of the figure; the icon at the right opens the figure studio.

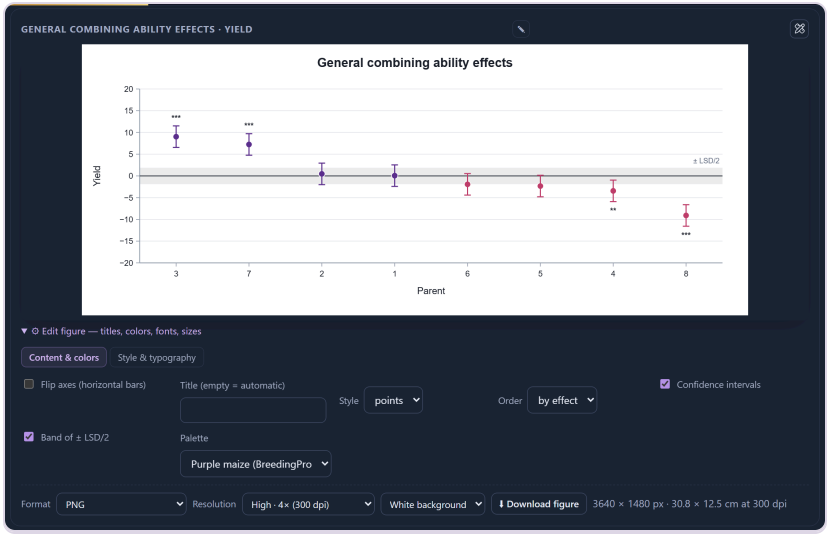


Figure I.6. The same figure as in Figure I.5 with the app in the dark theme.

1.7 The examples included

The program comes with 21 data sets in Block 3 and others in Blocks 2, 10 and 11. They are embedded in the app, so they load without a connection. The **published** ones reproduce tables from papers and books; the **simulated** ones were generated with fixed seeds and come with their true values, so you can check how well each method estimates.

Table 1.4. Examples in Block 3, by family. **P** = published, **S** = simulated.

Family	Example in the app		Blocks
Diallels	Maize: full 4 × 4 diallel	P	4 5
	Nicotiana rustica: Hayman's 8 × 8 diallel	P	4 5
	Full 8 × 8 diallel (textbook)	P	4 5 8
	Diallel with parents in two environments	P	4 5
	Maize: means of 36 single crosses	P	4
	Maize intervarietal crosses	P	5
Partial	Maize: partial diallel of 8 lines	P	6
North Carolina	North Carolina I	P	6
	Potato: North Carolina II	P	6 8
	North Carolina III	P	6
	Simulated: triple test cross	S	6
Line × tester	Line × tester with parents	P	6
Genotypes and traits	Maize: 13 hybrids and 15 traits	P	8
Multi-environment	Maize: 13 hybrids in 4 environments	P	8 9
	Oat: 10 genotypes in 14 environments	P	9
	Potato: 28 clones in 6 localities	P	9
	Means of 17 genotypes in 12 environments	P	9
	Simulated: inbred lines with a pedigree	S	9 10
	Simulated: diallel in an α-lattice, 3 locations	S	3 4
Generations	*Nicotiana rustica*: six generations	P	7
	Simulated: generation means, plant by plant	S	7

In addition, **Block 2** generates the plan of any of its 11 designs with the parents you type; **Block 10** comes with five published examples of animal and sire models, a simulated sheep flock, the pedigree of the simulated lines and five crossbreeding examples; and **Block 11**, two published sets (wheat lines with DArT markers and dent × flint maize hybrids) and two simulated ones. The full sources are in the references appendix and on the card of each example.

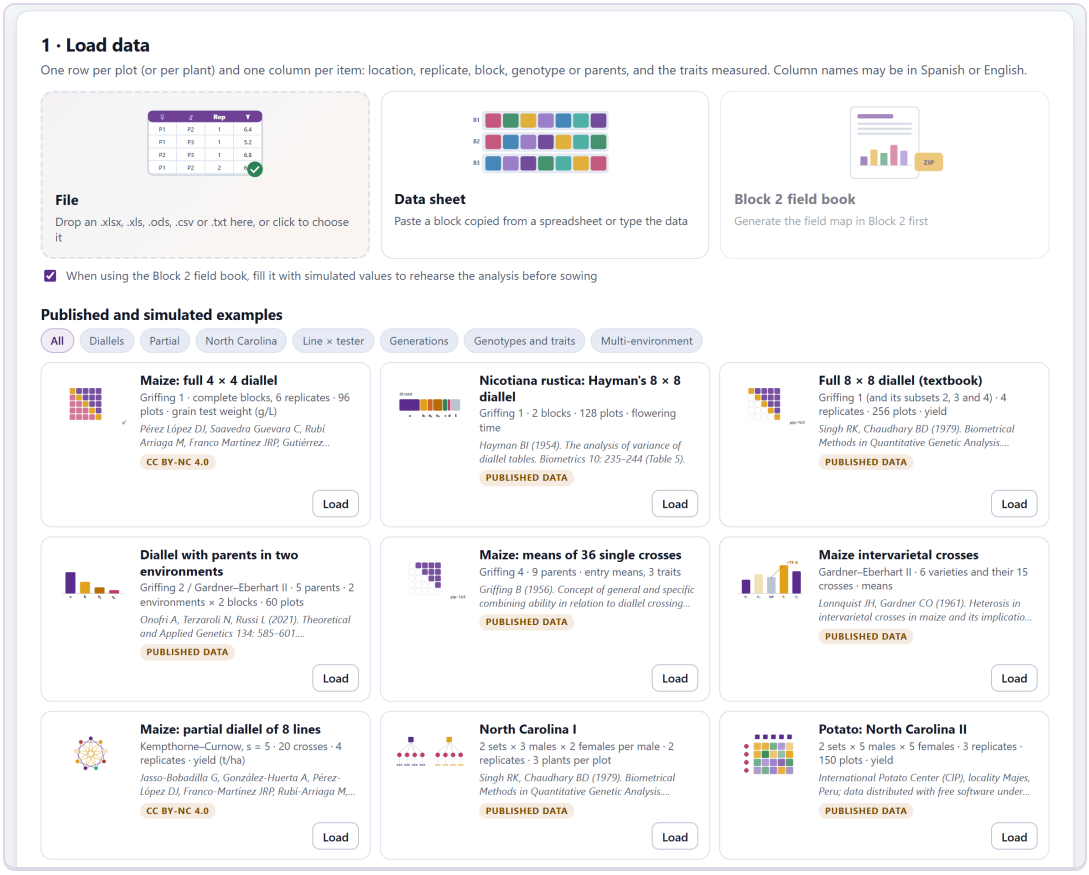


Figure I.7. The `1 · Load data` card of Block 3: file, data sheet or Block 2 field book, and underneath the examples, filtered by family. Each example shows its design, its source and its license.

YOUR FIRST ANALYSIS IN TWO MINUTES

1. Open the app and click `Start with my data →`.
2. Under `Published and simulated examples`, click `Load` on `Full 8 × 8 diallel (textbook)`. The app recognizes a diallel of Griffing's Method 1 with 8 parents, 64 entries and 4 replicates in randomized complete blocks.
3. Review the recognized design and the quality of the data, and click `Griffing` (4) in the list of blocks.
4. On the `2 · Analysis of variance of the diallel` card you will see that GCA ($F = 18.89$) and SCA ($F = 27.38$) are significant ($P < 0.0001$) and the reciprocal effects are not ($P = 0.99$), as in Figure I.4. Under `3 · Effects`, parents 3 and 7 have the largest GCAs (9.03 and 7.24), which can also be seen in Figure I.5.

I.8 Recommended workflow

The blocks are ordered as a breeding program is carried out, but not all of them apply to every data set. Each kind of experiment travels its own route, like subway lines that stop only at some stations (Figure I.8). Block 3 is the common station: that is where the design is recognized and the quality is checked before any genetic analysis.

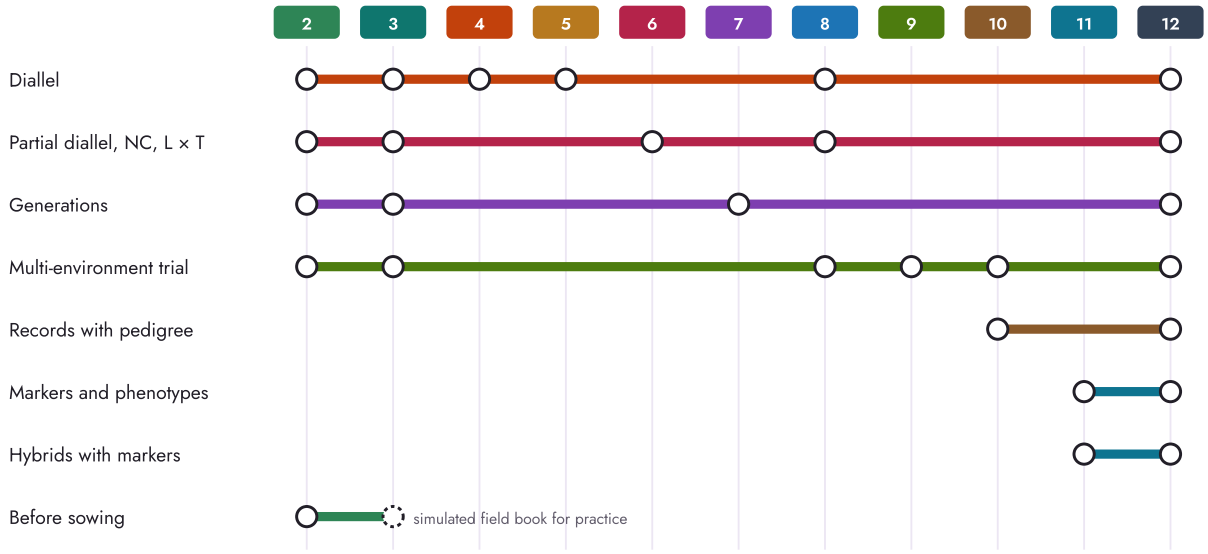


Figure I.8. Analysis routes by kind of experiment. Each circle is a station where that experiment has something to calculate. A diallel or a mating design in several environments also goes through Block 9; records with a pedigree and markers are loaded in their own blocks and do not need Block 3; a trial with a pedigree or with markers joins its route to that of Blocks 10 and 11 to predict in a single stage.

DECISION RULE · WHICH BLOCK TO USE

If the table has the female and male of a group of parents crossed among themselves…	**then** Griffing (4), Hayman–Jinks (5) and parameters (8).
If males and females are different groups (NC, L × T) or the diallel is partial…	**then** mating designs (6) and parameters (8).
If there is a generation column (P_1, P_2, F_1, F_2…)…	**then** generations (7).
If they are genotypes in several locations or years…	**then** parameters (8), stability (9) and, with a pedigree, BLUP (10).
If you have a pedigree or markers…	**then** BLUP (10) or genomics (11).

Block 3 makes this choice for you when it recognizes the design, and states it on the 3 · Recognized design card.

I.9 P values, standard errors and intervals

BreedingPro's tables accompany each estimate with its **standard error** (SE) and each test with its **P value**, written with four decimals (or "< 0.0001") and with the significance mark of the agronomic literature:

Mark	P value	Reading
***	$P < 0.001$	Very strong evidence
**	$0.001 \le P < 0.01$	Strong evidence
*	$0.01 \le P < 0.05$	Moderate evidence
ns	$P \ge 0.05$	Not significant at 5%

- **A P value does not measure the importance of an effect.** With many replicates, a small effect turns out significant; with few, a large one may not. Always look at the magnitude and its standard error as well.
- **Confidence intervals** are given at 95% unless the block says otherwise. Heritability uses the exact interval of Knapp, Stroup and Ross, which is not symmetric.
- **Variance components** may occasionally be estimated as negative; the app warns about it and interprets them as zero. Those it estimates by restricted maximum likelihood (REML) stay on the boundary, at zero, and are reported as such, without a standard error.
- **The least significant differences** (LSD) that accompany the effects allow pairs to be compared; for many comparisons at once it is advisable to be stricter with the significance level.

> **! IMPORTANT · FIXED OR RANDOM MODEL**
>
> The same table gives different tests depending on the model. If the parents or the environments are the only ones of interest, they are **fixed** and are tested against the error; if they are a sample from a population, they are **random**, their variances are estimated and some tests use another denominator. Blocks 4 and 9 show both readings and say which one fits each case.

B1

CHAPTER ONE · BLOCK 1 · HOME

Home, labs and theory

The app's home page is not just a welcome page. It is the map of the whole program, it holds two labs where quantitative genetics can be seen in motion (a diallel whose genes you choose and a population selected cycle after cycle), and it gathers the theory and the references behind every calculation.

1.1 The app's home page

1.2 The diallel lab

1.3 Five exercises with the diallel lab

1.4 The selection lab

1.5 Six exercises with the selection lab

1.6 The theory, in plain language

1.7 How to cite and references

1.1 The app's home page

When you open `index.html`, Block 1 appears. It reads from top to bottom, like a brochure: first what the program does and how to go through it, then two labs to experiment with, the map of the eleven analysis blocks, the methods, the theory and, at the end, how to cite it. The In this block index on the right lists its sections and takes you to any of them with a click.

B1

Table 1.1. Sections of the home page, in the order in which they appear.

Section in the app	What it contains	What it is for
Introduction four buttons and five guarantees	Start with my data →, Try the labs, Read the theory and How to cite; an illustration with a diallel, its GCAs and SCAs, a Wr–Vr graph and the response to selection.	Going straight to Block 3 or to any of the sections below.
How it works	The seven steps of a breeding program (Figure 1.1).	Placing each block within the breeding cycle.
Try it: two labs with real genetics	The diallel lab and the selection lab (Sections 1.2 to 1.5).	Understanding, by playing, what a Wr–Vr graph, GCA or the breeder's equation say.
What's inside	One card per analysis block, from 2 to 12, with its illustration and what it calculates.	Jumping to a block with a click (or with Enter from the keyboard).
For the material you work with	Six kinds of material and what is worth analyzing with each (Figure 1.2).	Knowing which analyses make sense with your parents.
Methods covered	Gallery of 32 methods in seven families, with a filter (Table 1.4).	Seeing at a glance everything that can be calculated.
What it brings together	Ten capabilities of the program in a single workflow.	Justifying its use in a project or a thesis.
The theory, in plain language	Fifteen expandable topics with formulas, practical rules and references (Section 1.6).	Reviewing the theory before interpreting or defending results.
How to cite and References behind the calculations	The citation of the program ready to copy, and 63 references in seven families (Section 1.7).	Citing correctly in your thesis or paper.

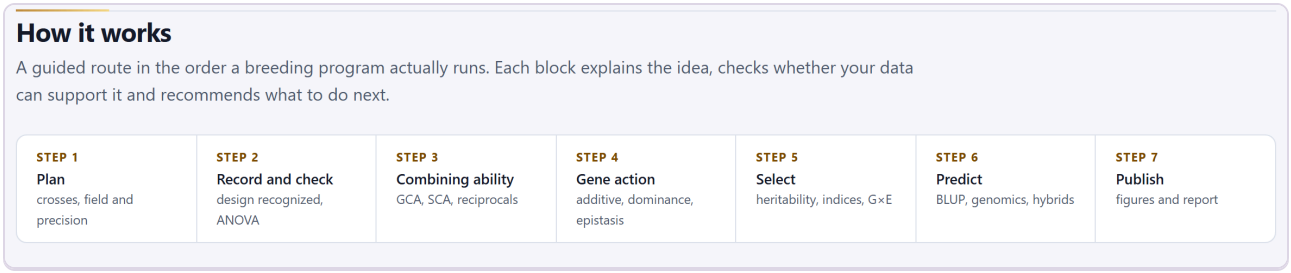


Figure 1.1. The How it works section: seven steps in the order in which a breeding program is actually carried out, from the crossing plan to the paper.

Table 1.2. The seven steps of the home page and the blocks that solve them.

Step	In the app	What is done	Blocks
1	Plan	Crossing plan, seed and pollinations, trial layout and power	2
2	Record and check	Data loading, recognized design, quality and field analysis	3
3	Combining ability	Effects and variances of GCA, SCA and reciprocals	4 6

Table 1.2. The seven steps of the home page and the blocks that solve them. *(continued)*

Step	In the app	What is done	Blocks
4	Gene action	Genetic components, degree of dominance, epistasis and heterosis	5 6 7
5	Select	Heritability, expected response, indices and stability	8 9
6	Predict	Breeding values with a pedigree or markers, and untested hybrids	10 11
7	Publish	Report, figures and results package	12

What's inside

Eleven cards, one per analysis block, show its number, an illustration, the family of methods it belongs to and a paragraph with what it calculates. A click anywhere on the card opens the block; it is the same information as Table I.1 of the introduction, told from the app.

For the material you work with

The kind of material decides which analyses make sense: a Hayman–Jinks diallel requires homozygous lines, Gardner–Eberhart heterosis was designed for varieties, and the animal model needs a pedigree. The strip in Figure 1.2 sums it up in six cards, and Table 1.3 adds the blocks that correspond to each one.

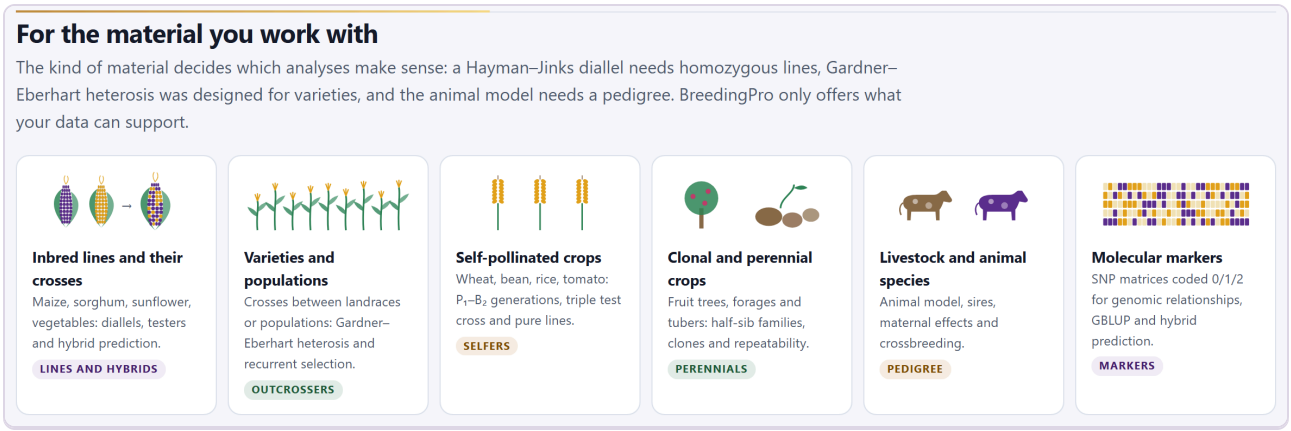


Figure 1.2. The For the material you work with section. Each card carries a tag underneath with the kind of material.

Table 1.3. Material, the analyses that suit it and the blocks where they are done.

Material	Analyses that make sense	Blocks
Inbred lines and their crosses	Griffing and Hayman–Jinks diallels, line × tester, hybrid prediction	4 5 6 11
Varieties and populations	Gardner and Eberhart heterosis, North Carolina designs, recurrent selection	5 6 8
Self-pollinated crops	Generations P_1 to BC_2, triple test cross, pure lines in trials	6 7 8
Clonal and perennial crops	Half-sib families, clones in several environments, repeatability	8 9 10

Table 1.3. Material, the analyses that suit it and the blocks where they are done. *(continued)*

Material	Analyses that make sense	Blocks
Livestock and animal species	Animal model, sires, maternal effects and crossbreeding	10
Molecular markers	Genomic relationships, GBLUP and hybrid prediction	11

Methods covered

The gallery shows 32 methods, each with an illustration of its own. The buttons above filter it by family and say how many methods each family has. Table 1.4 gathers them with the block where they are calculated.

Table 1.4. The 32 methods of the gallery, by family.

Family	Methods	Blocks
Diallels · 9	Griffing Methods 1, 2, 3 and 4; reciprocal effects; Wr–Vr graph; Hayman's ANOVA; Gardner and Eberhart; partial diallel	4 5 6
Mating · 5	North Carolina I and II; North Carolina III and triple test cross; line × tester; epistasis test	6
Generations · 3	Generation means; generation variances; heterosis	7
Selection · 4	Heritability; response to selection; selection indices; correlations and paths	8
G×E · 4	Stability regression; ecovalence and superiority; AMMI; GGE biplot	9
Mixed models · 3	REML and BLUP; animal model; crossbreeding	10
Genomics · 4	GBLUP; untested hybrids; double and three-way crosses; crosses between two groups	11

What it brings together

Ten boxes sum up what the program brings together in a single workflow: from the crossing plan to the paper; every mating design; lost crosses and unbalanced data; assumptions put to the test; interpretation in words; plants, animals and markers; publication-ready figures; a reproducible report; Spanish and English; and data that never leave your computer.

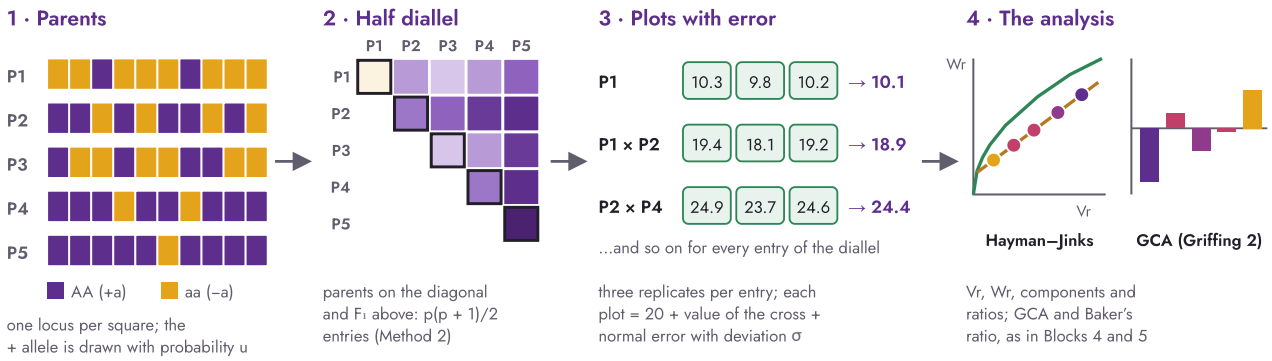
> **NOTE · THE FOUR BUTTONS OF THE INTRODUCTION**
>
> Start with my data → opens Block 3. Try the labs, Read the theory and How to cite scroll down to their section of the same page; Read the theory also opens the first topic. The name BreedingPro on the top bar goes back to this home page from any block.

1.2 The diallel lab

Faced with a Wr–Vr graph, every breeding student wonders which genes would produce those points. The diallel lab travels the road backward: you decide how the genes act, and the app creates parents with those genes, crosses them, "sows" the plots with experimental error and analyzes the table with the same formulas it will apply to your data in Blocks 4 and 5. Since you know the truth, you can see when the analysis recovers it and when it goes wrong.

THEORY IN BRIEF · HAYMAN'S ADDITIVE–DOMINANCE MODEL

Each locus has two alleles. The homozygote AA adds $+a$ to the trait, aa subtracts a and the heterozygote Aa contributes d; in the lab $a = 1$, so the Dominance d/a control gives the degree of dominance directly: 0 is additive action, between 0 and 1 partial dominance, 1 complete and above 1 overdominance (Hayman, 1954b). The parents are homozygous: at each locus they carry the allele that increases the trait with probability u and the one that decreases it with probability $v = 1 - u$. The value of a cross is the sum over its loci, and each plot adds 20 units and a normal error.



Value of a cross = Σ over its loci: +a if AA, –a if aa, d if Aa (d positive or negative depending on the locus)
Epistasis d × d: loci are grouped in pairs (1–2, 3–4...); if the cross is heterozygous at both, the value of the control is added

Figure 1.3. What the lab does every time you move a control. The parents are drawn with actual genes, crossed in a half diallel with parents, each entry is "sown" in three plots with error, and the table of means is analyzed as if it were real data. The plot values are illustrative; the means are those of the partial dominance scenario.

The seven controls in Table 1.5 are below the graphs. Each one changes one piece of the genetic model of Figure 1.3, and the app creates, sows and analyzes the diallel again as soon as you release it.

Table 1.5. Controls of the diallel lab.

Control	Range	Initial	What it represents
Parents	4 to 12	7	Number of parents p; the diallel has $p(p + 1)/2$ entries.
Loci	1 to 30	12	Genes that may differ between parents.
Dominance d/a	0 to 2	0.50	Degree of dominance at each locus.

Table 1.5. Controls of the diallel lab. *(continued)*

Control	Range	Initial	What it represents
Toward high values	0 to 100%	100%	Fraction of loci at which the dominant allele is the one that increases the trait; at the rest, the one that decreases it is dominant.
Frequency u	0.10 to 0.90	0.50	Probability that a parent carries, at each locus, the allele that increases the trait.
Epistasis dxd	−3 to +3	0	Dominance × dominance interaction between pairs of loci: negative, duplicate; positive, complementary.
Plot error σ	0 to 3	0.60	Standard deviation of the error of each of the three plots per entry.

What you see on screen

Table 1.6. The six summary boxes and their values when the model holds.

Box	What it is	Without epistasis and with little error
Slope b ± SE	Regression of Wr on Vr; underneath, the t of the test b = 1	Close to 1; if it moves away, there is epistasis or another broken assumption
$(H_1/D)^{½}$	Mean degree of dominance; underneath, the d/a of the model	Close to the d/a you chose
$H_2/4H_1$	Proportion of alleles that increase and that decrease the trait	¼ when $u = v = ½$; smaller if the frequencies differ
KD/KR	Dominant against recessive alleles in the parents	1 if they are balanced; larger with more dominant ones
Baker's ratio	$2\sigma^2_{GCA}/(2\sigma^2_{GCA} + \sigma^2_{SCA})$, from Griffing's Method 2	1 with additive action; falls toward 0 as dominance weighs more
Mean heterosis	Mean of the F_1s against the mean of the parents, as a percentage	0 without dominance; grows with d/a if dominance goes in a single direction

Without dominance, $H_2/4H_1$ and KD/KR make no sense, and their boxes show a dash with the caption no dominance. The components D, H_1, H_2, F and h^2 that feed the ratios are calculated as in Block 5 and are explained in the chapter on that block.

> **THEORY IN BRIEF · THE ERROR IN A HALF DIALLEL**
>
> Hayman's classical error corrections assume a full diallel in which each cross is the average of itself and its reciprocal, with half the error. In a half diallel each cross is observed only once, and the corrections change: H_1 is corrected with $-(5p - 4)E/p$ instead of $-(3p - 2)E/p$; H_2 with $-4E + 4(p - 1)E/p^2$ instead of $-2E$, and h^2 with $-4(p^2 - 1)E/p^3$ instead of $-4(p - 1)E/p^2$. D and F do not change. In 20,000 diallels simulated during the validation of the program, the classical corrections inflated H_1 by about 1.7E and H_2 by 1.5E; those of the half diallel had no bias. The lab and Block 5 use the latter.

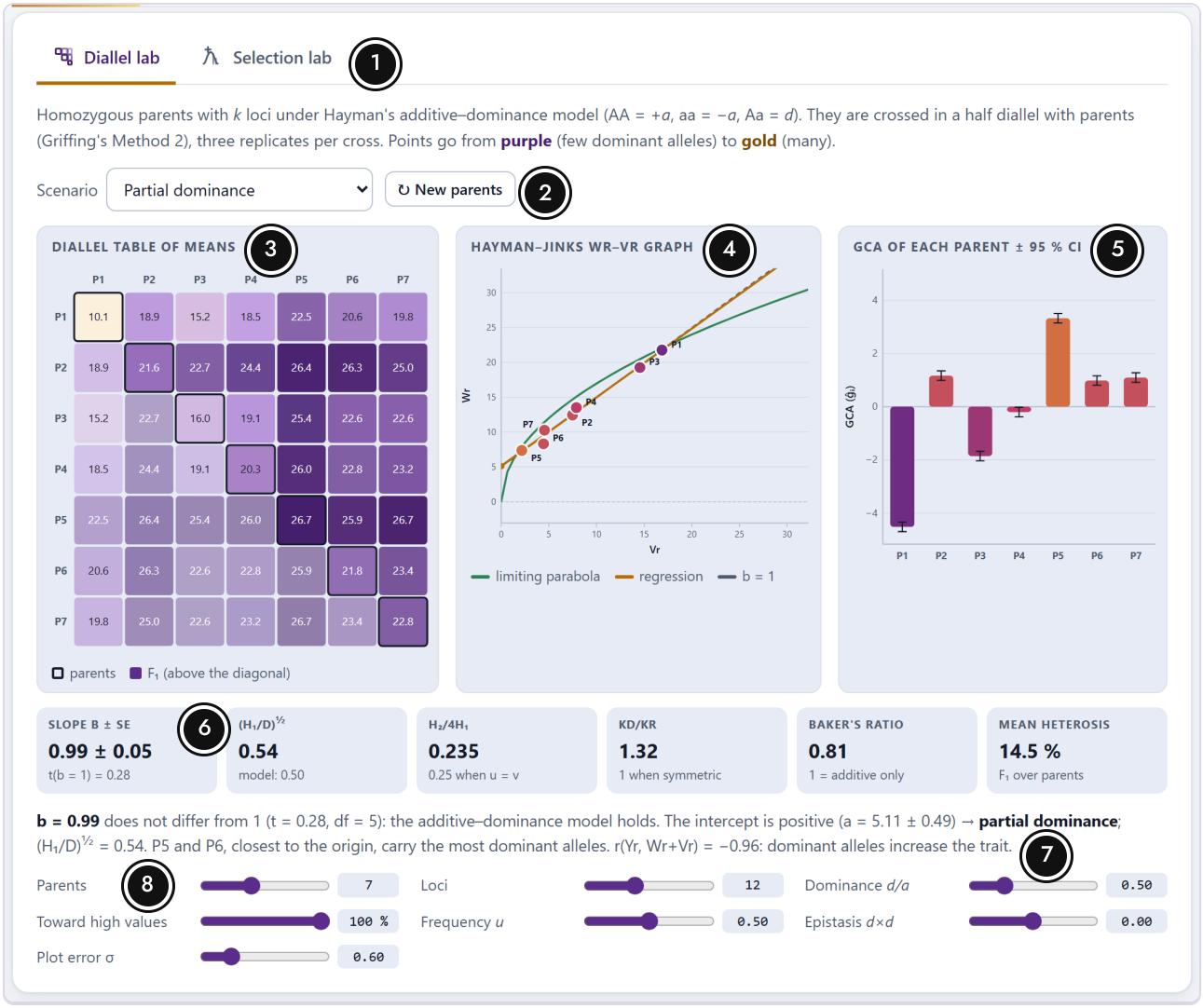


Figure 1.4. The diallel lab with the initial scenario, Partial dominance: seven parents, twelve loci and d/a = 0.50.

1. The two tabs: diallel lab and selection lab.
2. Scenario, with eight ready-made configurations (Table 1.7), and ↻ New parents, which draws a new group with the same settings.
3. Diallel table of means: parents on the diagonal, with a black border; F_1s above and their fainter copy below. Hovering the mouse shows each mean.
4. Wr–Vr graph: limiting parabola (green), regression line (amber) and line of slope 1 through the midpoint (gray). Each parent goes from purple (few dominant alleles) to gold (many).
5. GCA of each parent with its 95% confidence interval (Griffing's Method 2, Model I).
6. Six summary boxes (Table 1.6).
7. The interpretation in words.
8. The seven controls of the genetic model (Table 1.5).

NOTE · THE SAME NUMBERS EVERY TIME

Each scenario starts with a fixed seed, which is why the values in this chapter are reproduced exactly. Moving a control keeps the seed: with the same parents and the same error, only what you moved changes. ↻ New parents changes the seed, and choosing a scenario again brings back the original one.

DECISION RULE · HOW THE APP WORDS THE INTERPRETATION

If b does not differ from 1 (|t| smaller than the critical value at 5% with p – 2 degrees of freedom)…	**then** the additive–dominance model holds.
If b departs from 1…	**then** there is non-allelic interaction or another broken assumption, and the components are biased.
If $(H_1/D)^{½} < 0.25$…	**then** gene action is practically additive.
If the intercept of the line is positive and significant…	**then** partial dominance.
If it is negative and significant…	**then** overdominance.
If it does not differ from zero and $(H_1/D)^{½}$ is between 0.8 and 1.2…	**then** complete dominance. Outside that interval, the app warns that the intercept test has little power (Exercise 4).
If r(Yr, Wr + Vr) < –0.3 · > 0.3…	**then** the dominant · recessive alleles increase the trait; between both values the direction is unclear.

The app also names the two parents closest to the origin: they are the ones that carry the most dominant alleles.

1.3 Five exercises with the diallel lab

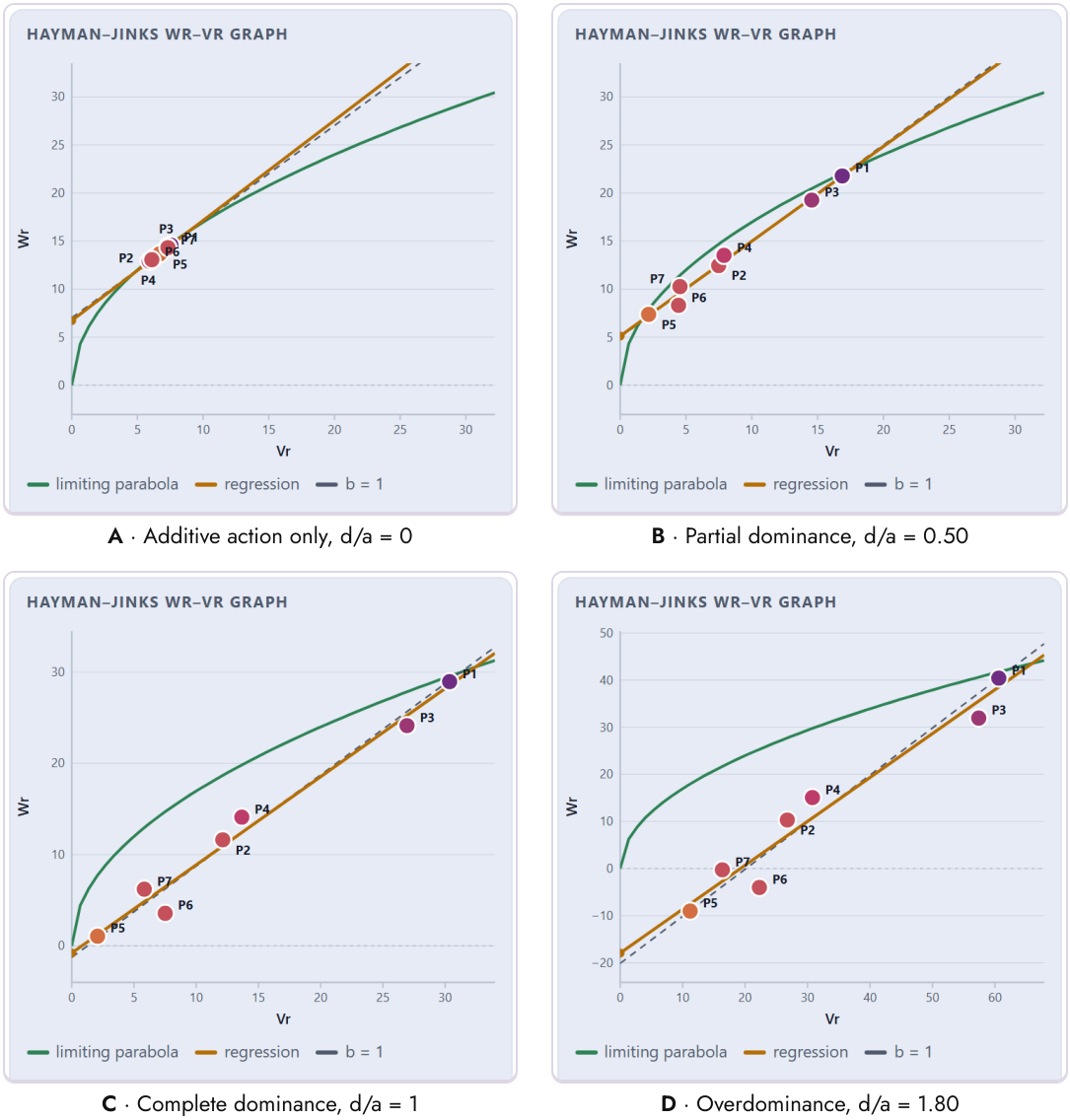
The eight scenarios of the menu are meant to be worked through in order. Table 1.7 says what each one changes with respect to the initial one and in which exercise it is used.

Table 1.7. Scenarios of the diallel lab. Settings not listed keep their initial values (7 parents, 12 loci, d/a = 0.50, 100% toward high values, u = 0.50, no epistasis, σ = 0.60).

Scenario	What changes	Exercise
Partial dominance	Nothing: it is the initial scenario	1
Complete dominance	d/a = 1	1
Overdominance	d/a = 1.80	1
Additive action only	d/a = 0	1
Dominance in both directions	8 parents, 16 loci, d/a = 0.80, 50% toward high values	2
Duplicate epistasis	8 parents, d/a = 0.80, epistasis –3, σ = 0.30	3
Complementary epistasis	8 parents, d/a = 0.80, epistasis +3, σ = 0.30 (another seed)	3
Large experimental error	σ = 2.50	4

Exercise 1 · Four degrees of dominance

Choose **Additive action only** and then **Partial dominance**, **Complete dominance** and **Overdominance**. The four scenarios use the same parents and the same error; only d/a changes. Watch how the line moves down in the Wr–Vr graph (Figure 1.5) and write down the boxes (Table 1.8).



A · Additive action only, d/a = 0

B · Partial dominance, d/a = 0.50

C · Complete dominance, d/a = 1

D · Overdominance, d/a = 1.80

Figure 1.5. The Wr–Vr graph of the same seven parents with four degrees of dominance. The line cuts the Wr axis lower and lower: above the origin (A, B), almost at the origin (C) and below it (D). Note that the scales of the axes change from one panel to another.

Table 1.8. What the lab shows in Exercise 1.

Scenario	d/a	$(H_1/D)^{½}$	Intercept a ± SE	Baker	Mean heterosis
Additive action only	0	0.11	6.73 ± 0.16	0.99	–1.1%
Partial dominance	0.50	0.54	5.11 ± 0.49	0.81	14.5%

Table 1.8. What the lab shows in Exercise 1. *(continued)*

Scenario	d/a	$(H_1/D)^{½}$	Intercept a ± SE	Baker	Mean heterosis
Complete dominance	1.00	1.08	−0.79 ± 1.14	0.45	30.1%
Overdominance	1.80	1.95	−17.96 ± 3.49	0.07	55%

Three lessons come out of this exercise. **The line moves down as dominance grows**, and in all four cases the slope does not differ from 1 (from 0.93 to 1.04), because there is no epistasis. **$(H_1/D)^{½}$ recovers the d/a of the model** with a small error that comes from having only seven parents; the 0.11 of additive action comes from the experimental error. **Baker's ratio falls from 0.99 to 0.07** and mean heterosis rises from −1.1 to 55%: with additive action the value of a cross is predicted from the GCA of its parents, and with overdominance the crosses have to be tested.

Notice also the order of the parents along the line. P5, which carries the dominant allele at 9 of the 12 loci, always stays close to the origin, and P1, with only one, at the far end. With additive action the app omits that reading, because without dominance there are no dominant alleles to rank.

Exercise 2 · Dominance that cancels out

The Dominance in both directions scenario has d/a = 0.80 at every locus, but at half of them the allele that increases the trait is dominant and at the other half the one that decreases it. The lab shows $(H_1/D)^{½}$ = 0.60, that is, clear dominance at each locus, and yet the **mean heterosis is −2.1%**: the dominance deviations cancel out when the loci are added up. This is the situation in which a diallel has a lot of SCA and almost no average heterosis.

> **! IMPORTANT · THE DIRECTION OF DOMINANCE**
>
> In this scenario the app reads r(Yr, Wr + Vr) = 0.77 and concludes that the recessive alleles increase the trait. That correlation reflects the direction that prevails in *these eight* parents, not that of the model, which goes both ways. When dominance is bidirectional, the most reliable signal is the combination of a clear degree of dominance with a mean heterosis close to zero.

Exercise 3 · When there is epistasis

The Duplicate epistasis and Complementary epistasis scenarios add a dominance × dominance interaction of −3 and +3 to the partial dominance model (d/a = 0.80). In both, the slope departs from 1: b = 0.68 ± 0.08 (t = 3.88 with 6 degrees of freedom) in the duplicate one and b = 0.66 ± 0.07 (t = 4.61) in the complementary one, and the app warns that the components are biased. The bias is large: in the complementary one, $(H_1/D)^{½}$ is 2.51, which would be read as overdominance, although the model has partial dominance (Figure 1.6 A).

The b = 1 test, however, does not always detect epistasis. Click ↻ New parents several times with duplicate epistasis and you will see samples in which the slope does not differ from 1; the app then adds: *"Beware: epistasis is present here"*. Table 1.9 measures how often this happens.

A · Complementary epistasis (Exercise 3) **B** · Large experimental error (Exercise 4)

Figure 1.6. Two ways for the Wr–Vr graph to mislead. **A**: with epistasis, the points leave the line of slope 1 (gray) and the regression (amber) cuts the Wr axis below the origin, as if there were overdominance. **B**: with a lot of error, the points scatter and the intercept loses significance.

Table 1.9. Percentage of samples in which the b = 1 test is rejected at 5%, in 2,000 diallels simulated with the lab (12 loci, d/a = 0.80, u = 0.50, σ = 0.30).

Model	6 par.	8 par.	10 par.	12 par.
Duplicate epistasis (–3)	27%	41%	51%	60%
Complementary epistasis (+3)	38%	60%	71%	79%
No epistasis	5%	6%	10%	11%

41%

KEY FACT

With eight parents, the b = 1 test detects duplicate epistasis in only 41 of every 100 samples. **A slope equal to 1 does not prove that there is no epistasis.** To test it directly there are the triple test cross (Block 6) and the scaling tests (Block 7).

The last row of Table 1.9 has a lesson of its own: without epistasis, the test is wrong in 5 to 11% of the samples, somewhat more than the nominal 5% when there are many parents for few loci, because the genes are then no longer distributed at random among the parents, which is another of Hayman's assumptions.

Exercise 4 · Large experimental error

The Large experimental error scenario repeats the partial dominance one with σ = 2.50 instead of 0.60. The slope stays at 1 (1.00 ± 0.16), but its standard error triples, and the intercept, 4.22 ± 1.78, no longer differs from zero (t = 2.37, smaller than the critical value of 2.57 with 5 degrees of freedom). The app does not conclude "complete dominance": it warns that $(H_1/D)^{1/2}$ = 0.61 points to partial dominance and that, with this error, the intercept test has little power (Figure 1.6 B).

The error affects everything else: $H_2/4H_1$ falls to 0.185 and KD/KR rises to 2.22, when the model has 0.25 and 1, and the standard error of each GCA goes from 0.09 to 0.38. Before sowing a real diallel, Block 2 calculates how many replicates are needed to detect the differences you care about.

Exercise 5 · Unequal frequencies

Choose Complete dominance and move Frequency u to 0.20 and then to 0.80. With $u = 0.20$ the alleles that increase the trait, which here are the dominant ones, are scarce: $H_2/4H_1$ falls to 0.185 (theory gives $uv = 0.16$) and KD/KR to 0.36 (theory, 0.25), and $(H_1/D)^{1/2} = 0.99$ stays at complete dominance. With $u = 0.80$, KD/KR rises to 2.38, but $(H_1/D)^{1/2}$ reaches 1.51 and the intercept, −2.27 ± 0.87, becomes significant: the app reads overdominance where there is only complete dominance.

> **THEORY IN BRIEF · APPARENT OVERDOMINANCE**
>
> With 80% favorable alleles, seven parents differ at few loci and the genes are no longer distributed at random among them. That association inflates H_1 relative to D and produces an apparent overdominance, a phenomenon well known in real diallels. That is why the degree of dominance is best read together with the number of parents and, when possible, confirmed with North Carolina design III (Block 6) or with the generation variances (Block 7).

1.4 The selection lab

The second lab puts the breeder's equation, $R = h^2S$, to the test. A population with actual genes is selected cycle after cycle; in each cycle the app calculates what the equation predicts with the additive variance of that moment and compares it with what the population really gains. This shows why the prediction works for one or two cycles and not for twenty.

> **THEORY IN BRIEF · WHAT IT SIMULATES**
>
> A diploid population of N plants with k loci of equal effect (AA = +1, Aa = d, aa = −1), free recombination and no mutation; the environmental error is set at the start to give the requested heritability. In each cycle the best plants are selected on phenotype, and their N offspring come from random pairs among them ($c = 1$) or from a selected mother with any father ($c = ½$, open pollination). The predicted response is $R = c \cdot h^2 \cdot S$, with $h^2 = \sigma^2_A/\sigma^2_P$ and $\sigma^2_A = \Sigma\, 2pq\alpha^2$, where $\alpha = a + d(q - p)$ is the average effect of an allele substitution (Falconer and Mackay, 1996).

The seven controls in Table 1.10 are below the graphs. Any change creates a new population and takes it back to cycle 0.

Table 1.10. Controls of the selection lab.

Control	Range	Initial	What it represents
Plants N	40 to 1,000	400	Size of the population in each cycle.
Loci	1 to 100	40	Genes of equal effect.
Initial h^2	0.05 to 0.95	0.40	Initial heritability (broad sense; without dominance it is the narrow-sense one).
Selected	1 to 50%	10%	Selected fraction; it defines the intensity.
Initial frequency	0.05 to 0.95	0.30	Initial frequency of the favorable allele at each locus.
Dominance d/a	0 to 1	0	Degree of dominance of the favorable allele.
Parental control	c = 1 or ½	1	Both parents selected, or only the females with open pollination.

What you see on screen

Table 1.11. The eight boxes of the selection lab.

Box	What it is
Cycle	Selection cycles completed, up to 25.
Intensity i	Observed S/σ_P; underneath, the theoretical one for the selected fraction (1.75 with 10%).
Differential S	Mean of the selected plants minus the mean of the population.
R predicted	$c \cdot h^2 \cdot S$: what the equation expects for the next cycle.
R realized	How much the mean advanced in this cycle; underneath, what had been predicted in the previous one.
h^2	σ^2_A/σ^2_P of the current cycle; underneath, that of cycle 0.
Total gain	Cumulative advance, in phenotypic deviations of cycle 0 (σ).
Fixed loci	Loci at which one of the alleles was lost.

NOTE · WHY THE H^2 IN THE BOX IS NOT EXACTLY THE ONE REQUESTED

The control sets the environmental error from the genotypic variance of the population drawn, whereas the box calculates σ^2_A from the allele frequencies. With 400 plants the difference is minimal (0.41 against 0.40); with 60, sampling weighs more (0.29). With dominance the box shows only the additive part: 0.35 when 0.40 was requested.

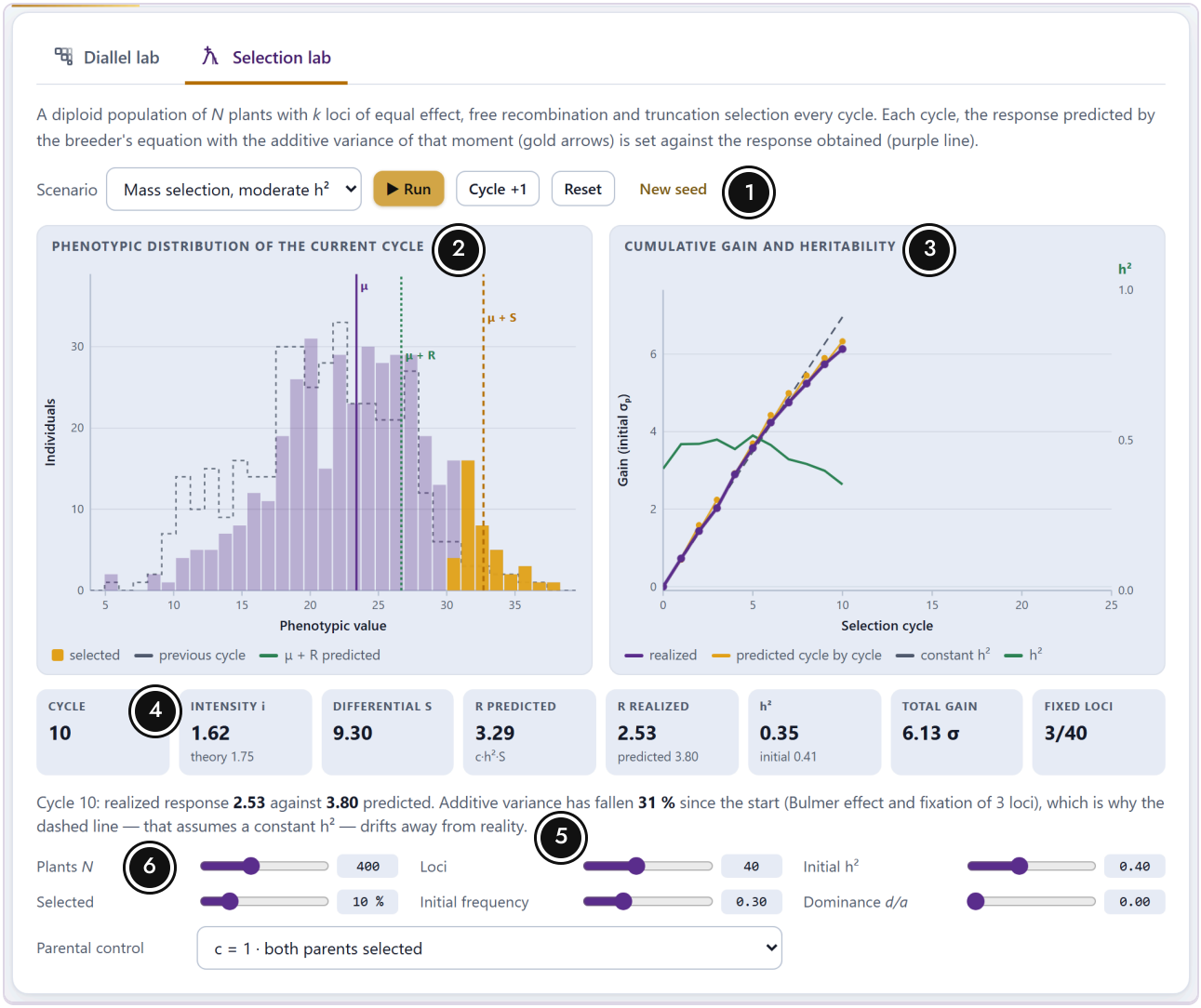


Figure 1.7. The selection lab with the Mass selection, moderate h² scenario after ten cycles.

1. Scenario, with six configurations (Table 1.12), and the buttons: ▶ Run advances one cycle every fraction of a second and turns into ❚❚ Pause; Cycle +1 advances one; Reset goes back to cycle 0; New seed draws another population.
2. Phenotypic distribution of the cycle: selected plants in gold, the previous cycle as a gray outline, and the lines μ, μ + S and μ + R.
3. Cumulative gain in initial phenotypic deviations: realized (purple), predicted cycle by cycle (gold), the one that assumes a constant h^2 (gray), and h^2 on the right axis (green).
4. Eight boxes (Table 1.11).
5. The interpretation of the cycle.
6. The seven controls (Table 1.10).

1.5 Six exercises with the selection lab

Each scenario starts with the same seed, so the six populations start from comparable genotypes. For each one, choose the scenario and click ▶ Run up to cycle 25, or Cycle +1 to go step by step. Table 1.12 summarizes what happens.

Table 1.12. The six scenarios after 5, 10 and 25 cycles. The gain is in phenotypic deviations of cycle 0; the last column is the additive variance lost by cycle 25.

Scenario	What changes	Initial h^2	5	10	25 cycles	Fixed loci	σ^2_A lost
Mass selection, moderate h^2	N 400, 40 loci, h^2 0.40, 10%, p 0.30	0.41	3.58	6.13	8.63	37 / 40	99%
Low heritability	h^2 0.10	0.10	0.69	1.47	3.28	11 / 40	53%
Few genes of large effect	4 loci, h^2 0.50, p 0.20	0.41	3.58	3.62	3.69	4 / 4	100%
No pollen control (c = ½)	c = ½	0.41	1.70	3.37	7.25	1 / 40	56%
Small population (drift)	N 60	0.29	2.53	4.05	5.49	40 / 40	100%
Complete dominance	d/a 1	0.35	2.24	3.24	3.76	5 / 40	95%

Exercise 1 · The breeder's equation and its limits

With Mass selection, moderate h^2, cycle 0 selects the best 10%: the observed intensity is 1.72 (the theoretical one, 1.75), S = 11.03 and h^2 = 0.41, so the equation predicts R = 4.47. In cycle 1 the population advances 4.67: **the prediction works**. In the following cycles the realized response stays somewhat below the predicted one (4.55 against 5.52 in cycle 2), because selection creates negative associations between loci that reduce the additive variance: this is the Bulmer effect (Bulmer, 1971).

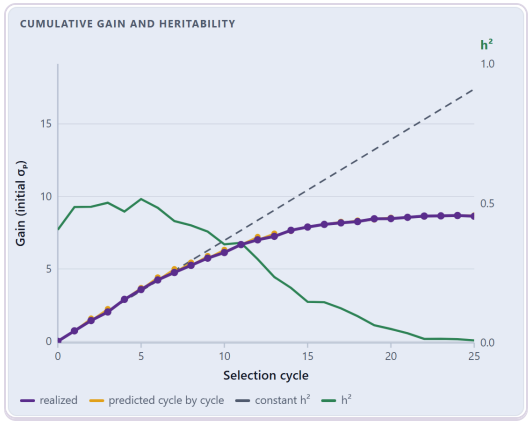
At first h^2 rises, from 0.41 to 0.52 in cycle 5, because the frequency of the favorable allele moves from 0.30 toward 0.50, where $2pq$ is at its maximum. Then it falls: 0.35 in cycle 10, 0.15 in cycle 15 and 0.01 in cycle 25, with 37 of the 40 loci fixed. The cumulative gain levels off at 8.63 σ, while the line that assumes a constant h^2 reaches 17.4 σ (Figure 1.8 A).

Exercise 2 · Low heritability

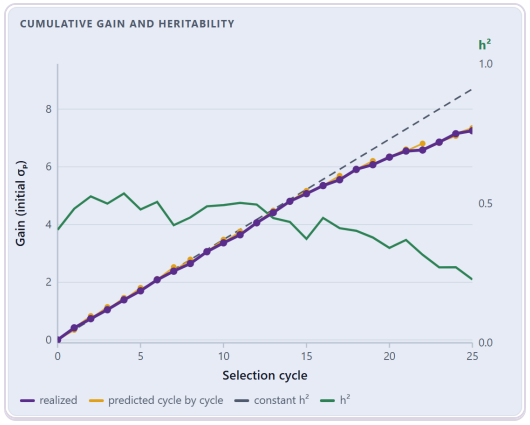
With h^2 = 0.10 the equation predicts little (R = 2.15) and the response of each cycle is erratic: 2.93 in cycle 1, 0.95 in cycle 2 and 0.97 in cycle 3, against predictions of 2.15, 2.45 and 2.59. In 25 cycles the population gains only 3.28 σ, when the constant-h^2 line promised 4.2 σ. That is why, with traits of low heritability, selection is done on family or replicate means, whose heritability is higher (Block 8).

Exercise 3 · Few genes of large effect

With four loci and an initial frequency of 0.20 the response is fast at first (1.12 σ in the first cycle), but by cycle 10 all four loci are fixed, h^2 is 0 and the population stops at 3.6 σ: this is the **selection limit**. With 40 loci the limit is farther away and is reached more slowly (Figure 1.9 A).



A · Mass selection, $c = 1$



B · No pollen control, $c = ½$

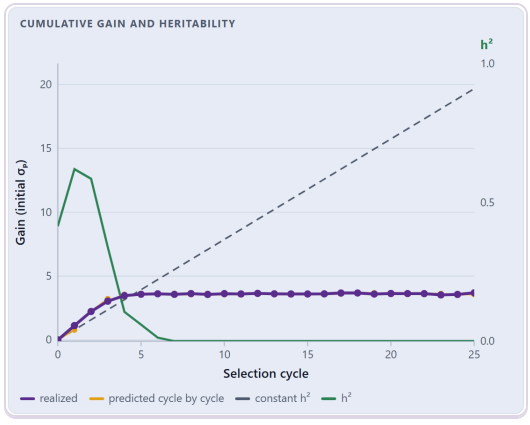
Figure 1.8. Cumulative gain over 25 cycles. With both parents selected (A) the population advances twice as fast, but the additive variance is exhausted around cycle 20 and the gain levels off; without pollen control (B) it advances at half the rate and keeps variance. The vertical scales differ.

Exercise 4 · No pollen control

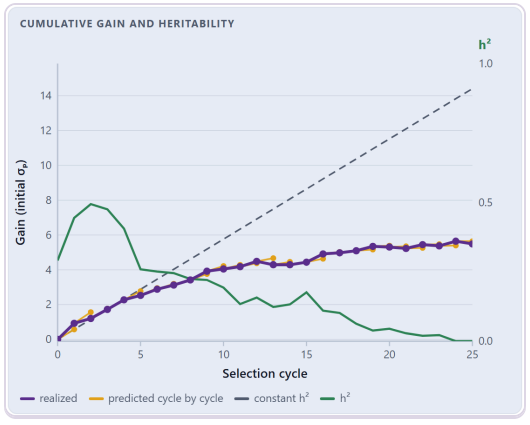
With $c = ½$ only the females are selected and the pollen comes from the whole population: the predicted R falls to 2.24, half that of mass selection, and after 10 cycles the gain is 3.37 σ against 6.13 σ. The counterpart is that the variance is exhausted more slowly: by cycle 25, 56% of σ^2_A has been lost (against 99%) and h^2 is still 0.23 (Figure 1.8 B). Controlling the pollen doubles the gain per cycle: that is the role of c in the breeder's equation.

Exercise 5 · Small population

With 60 plants, selecting 10% leaves six parents per cycle, and drift fixes loci at random: 7 by cycle 5, 24 by cycle 10 and all 40 by cycle 25. Of them, 7 were fixed for the unfavorable allele, and the gain stops at 5.49 σ against 8.63 σ with 400 plants (Figure 1.9 B). A small effective size spends the variability the program will need in future cycles.



A · Few genes of large effect



B · Small population, N = 60

Figure 1.9. Two ways of exhausting the response. With four genes (A), selection fixes them in ten cycles and the gain reaches its limit; with a small population (B), drift fixes the loci at random, some for the unfavorable allele, and the limit is lower.

Exercise 6 · Complete dominance

With d/a = 1 the gain slows down early, 3.24 σ after 10 cycles and 3.76 σ after 25, although only 5 loci were fixed: the additive variance fell by 95%. When the favorable allele is dominant and becomes common, the unfavorable recessive that remains hides in the heterozygotes, where selection on phenotype does not see it; in the formula, $\alpha = a + d(q - p)$ approaches zero. The app explains this in its interpretation, and it is the reason for using progeny tests or selfing in the recurrent selection methods of Block 8.

FOR THE CLASSROOM · A 50-MINUTE SESSION

1. **10 min.** Diallels, Exercise 1: predict where the line will cut before changing the scenario.
2. **10 min.** Exercise 3: click ↻ New parents ten times and compare the detection rate with Table 1.9.
3. **15 min.** Selection, Exercises 1 and 4: the gain over 10 cycles with and without pollen control.
4. **15 min.** Exercises 3 and 5: what a selection limit is and how to protect variability.

1.6 The theory, in plain language

The theory section gathers what is needed to run the analyses and defend them in a seminar. There are fifteen folded topics; a click on the title opens each one. Nine carry an illustration, the formulas appear in a fixed-width font and each topic names the papers they come from. The tag to the right of the title says which block it belongs to.

Table 1.13. The fifteen theory topics and the chapter of this manual that develops them.

#	Topic in the app	Tag	Expanded in
1	What a breeding program estimates	start here	Section I.3
2	General and specific combining ability	Sprague & Tatum 1942	4
3	Griffing's four methods and two models	Block 4	4
4	Reciprocal, maternal and non-maternal effects	Methods 1 and 3	4
5	Hayman and Jinks: how the genes act	Block 5	5
6	Gardner and Eberhart: heterosis between varieties	Block 5	5
7	The methodologies compared: assumptions and similarities	which to use	4 5
8	North Carolina, line × tester and triple test cross	Block 6	6
9	Generation means, epistasis and heterosis	Block 7	7
10	Heritability and response to selection	Block 8 · lab	8
11	Selection indices: several traits at once	Block 8	8
12	Genotype × environment interaction and stability	Block 9	9

Table 1.13. The fifteen theory topics and the chapter of this manual that develops them. *(continued)*

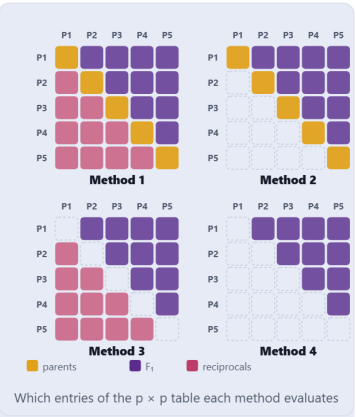
#	Topic in the app	Tag	Expanded in
13	Mixed models, BLUP and the animal model	Block 10	10
14	Genomic selection and hybrid prediction	Block 11	11
15	Which analysis answers which question	decision guide	Section I.8

▸ **General and specific combining ability** SPRAGUE & TATUM 1942

▾ **Griffing's four methods and two models** BLOCK 4

With p parents there are p^2 combinations: the p parents, $p(p-1)/2$ F_1 crosses and as many reciprocals. Griffing (1956b) defined four **methods** according to which part of that table is grown:

Method	Material	Entries	Estimates
1	parents, F_1 and reciprocals	p^2	GCA, SCA, reciprocal
2	parents and F_1	$p(p+1)/2$	GCA, SCA
3	F_1 and reciprocals	$p(p-1)$	GCA, SCA, reciprocal
4	F_1 only	$p(p-1)/2$	GCA, SCA



Which entries of the p × p table each method evaluates

Each method is analyzed under two **models**. In **Model I (fixed)** the parents were deliberately chosen and conclusions are restricted to them: what matters are the effects $\hat{g}_i$, $\hat{s}_{ij}$ and $\hat{r}_{ij}$, their standard errors and the differences between them. In **Model II (random)** the parents stand for a population, and what matters are the variances σ^2GCA and σ^2SCA and, from them, σ^2A, σ^2D and heritability. That makes eight different analyses.

Which one to choose
Including the parents (Methods 1 and 2) biases SCA when they are highly inbred lines, because the parent–F_1 contrast mixes heterosis into specific ability; Methods 3 and 4 estimate it better. Reciprocals (Methods 1 and 3) are only worth growing if you expect maternal or cytoplasmic effects.

When crosses or plots are lost, the closed formulas no longer hold. BreedingPro then fits the same model by least squares or REML (Möhring, Melchinger and Piepho 2011), which reproduce the classical tables exactly when the data are complete.

Figure 1.10. The Griffing's four methods and two models topic open: the table of the four methods, the illustration of the entries each one evaluates, the Which one to choose box and the note on lost crosses.

Topic 7 deserves special attention. A table compares side by side Griffing's Model I and Model II, Hayman–Jinks and Gardner–Eberhart: the question each one answers, which parents it assumes, which Griffing methods it allows, for which species it holds, how it treats reciprocal effects, heterosis and epistasis, what it estimates and the exact equivalences between them (Table 1.14). Block 5 applies those equivalences to your data.

! IMPORTANT · A COMMON MISTAKE

Interpreting σ^2_{GCA} and σ^2_{SCA} as σ^2_A and σ^2_D when the parents were chosen on purpose (Model I). Genetic variances only make sense if the parents are a random sample of a defined population. The app reminds you of this at the end of this topic and in Block 4.

Table 1.14. The comparison of the methodologies in topic 7, as it appears in the app. ● yes; ◐ partly or with conditions; ○ no, or assumed absent.

	Griffing, Model I (fixed)	Griffing, Model II (random)	Hayman–Jinks	Gardner–Eberhart II and III
Question it answers	Which parents and crosses are best in this set?	How much additive and dominance variance is there in the population the parents came from?	How do the genes act: degree and direction of dominance, number of gene groups?	How much heterosis is there, and of which kind, between varieties or populations?
Parents	Chosen on purpose; conclusions apply to them only	Random sample from a reference population	Homozygous (inbred) lines	Varieties or populations in equilibrium; also lines
Applicable Griffing methods	1, 2, 3 and 4	1, 2, 3 and 4; for the ancestral population, preferably 3 or 4	Needs the parents: 1 or 2	Half diallel with parents (as in Method 2)
Species	Selfers, outcrossers and polyploids	Selfers, outcrossers and polyploids	◐ Diploids with inbred parents only	Selfers, outcrossers and polyploids
Reciprocal effects	● Estimated, with maternal and non-maternal split (Methods 1 and 3)	● Reciprocal, maternal and non-maternal variance (Methods 1 and 3)	○ Assumed absent; averaged	◐ Only if added to the model
Heterosis	◐ Only as the parents-versus-crosses contrast	◐ Not estimated directly	◐ Direction and degree of dominance (h^2, b_1)	● Average, per variety and specific
Epistasis and linkage	◐ Not separated: absorbed into GCA and SCA	◐ Bias σ^2_A and σ^2_D if present	○ Assumed absent; tested with b = 1 and t^2	◐ Absorbed into specific heterosis
Estimates	$\hat{g}_i$, $\hat{s}_{ij}$, $\hat{r}_{ij}$ ($\hat{m}_i$, $\hat{n}_{ij}$) and their standard errors	σ^2_{GCA}, σ^2_{SCA}, $\sigma^2_R \rightarrow \sigma^2_A$, σ^2_D, h^2, Baker's ratio	D, H_1, H_2, F, h^2, E, $(H_1/D)^{½}$, $H_2/4H_1$, KD/KR, dominance order	μ_v, v_j, $\overline{h}$, h_j, $s_{jj'}$ (II); GCA and SCA of the crosses (III)
Exact equivalences	Method 1: GCA, SCA and reciprocals = Hayman's items a, b and c + d; Method 2: GCA and SCA = varieties and total heterosis of Analysis II	With inbred parents: $\sigma^2_{GCA} = ¼(D + H_1 - H_2 - F)$ and $\sigma^2_{SCA} = ¼H_2$	Half-diallel b_1, b_2 and b_3 = Gardner–Eberhart $\overline{h}$, h_j and $s_{jj'}$; c = maternal, d = remaining reciprocal	$\hat{g}_j$ (III) = $½\hat{v}_j + \hat{h}_j$ (II) = Method 4 $\hat{g}_j$; $s_{jj'}$ = Method 4 SCA

1.7 How to cite and references

At the end of the home page is the citation of the program, with the installed version, ready to copy. Copy the citation copies it in the language of the interface, and the button confirms with ✓ Copied. The citation carries the program's digital object identifier (DOI), 10.5281/zenodo.23005938, which always leads to the most recent version.

Below are the **63 references** of the methods the program calculates, with authors, year, title and journal. The same family buttons as in the methods gallery filter the list (Table 1.15). Each statistic is cited again where it is used: in the last card of its block and in the Block 12 report.

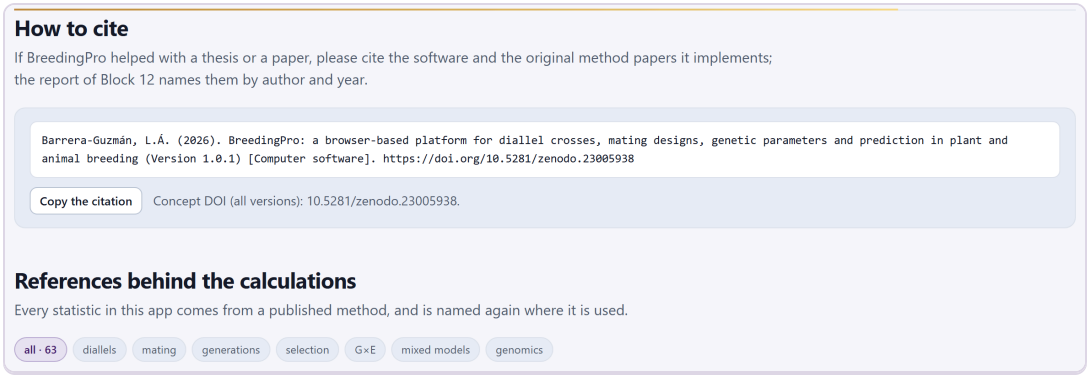


Figure 1.11. The How to cite section and the beginning of References behind the calculations, with the buttons that filter the 63 references by family.

Table 1.15. The references of the home page, by family.

Family	Ref.	Some examples
Diallels	19	Sprague and Tatum (1942); Griffing (1956a, 1956b); Hayman (1954a, 1954b); Jinks (1954); Gardner and Eberhart (1966); Möhring, Melchinger and Piepho (2011)
Mating	4	Comstock and Robinson (1948, 1952); Kempthorne (1957); Kearsey and Jinks (1968)
Generations	6	Mather (1949); Cavalli (1952); Hayman (1958); Jinks and Jones (1958); Lande (1981)
Selection	16	Smith (1936); Hazel (1943); Falconer and Mackay (1996); Holland et al. (2003); Bulmer (1971)
G×E	7	Finlay and Wilkinson (1963); Eberhart and Russell (1966); Shukla (1972); Gauch (1988); Yan et al. (2000)
Mixed models	6	Henderson (1975, 1976); Patterson and Thompson (1971); Meuwissen and Luo (1992); Dickerson (1969)
Genomics	5	Jenkins (1934); Bernardo (1994); Meuwissen, Hayes and Goddard (2001); VanRaden (2008); Technow et al. (2012)

! IMPORTANT · CITE THE METHODS TOO

If BreedingPro helped you with a thesis or a paper, cite the program and also the original papers of the methods you used. You do not have to look for them in the full list: the Block 12 report writes the methods section with the settings you used and includes in its references only the works it cites.

B2

CHAPTER TWO · BLOCK 2 · PLAN

Planning crosses, trial and power

Before the first pollination is made, it pays to know how many crosses must be made, how much seed will be needed, how the trial will be laid out in the field and whether the experiment will have the power needed to answer the question. Block 2 settles all four and delivers the field book that Block 3 will read later.

2.1 What this block does

2.2 The mating design

2.3 Seed and pollinations

2.4 The field trial

2.5 The field book

2.6 Precision and power before sowing

2.7 Other designs, with their numbers

2.8 From the field book to Block 3

2.1 What this block does

Block 2 is worked from top to bottom in four cards. Each one takes what the previous one defined, and everything is recomputed at once: if you change the number of parents, the crosses, the seed, the field map and the power change with it.

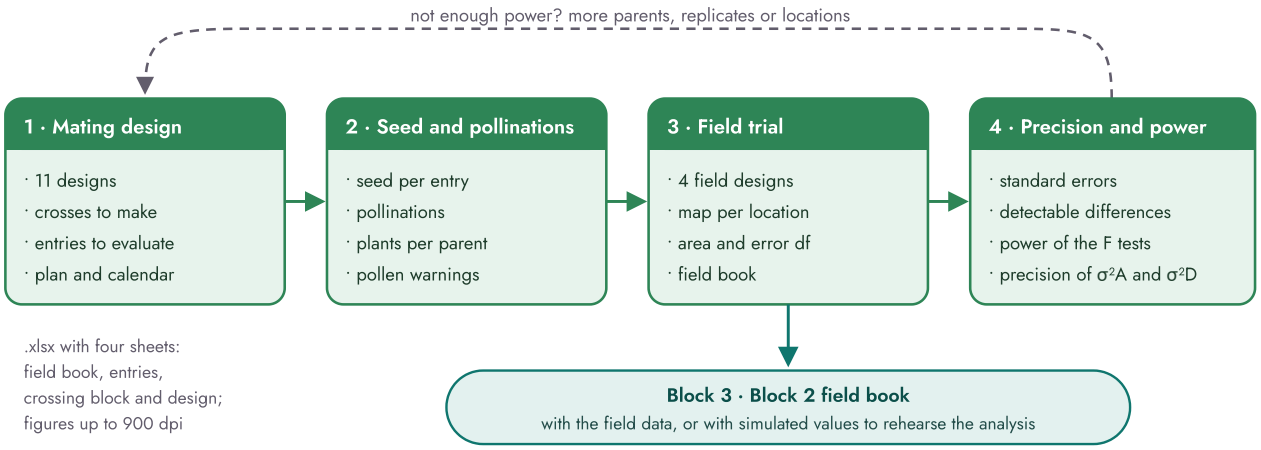


Figure 2.1. The four cards of Block 2. The field book of card 3 is downloaded or goes straight to Block 3; if card 4 shows low power, you go back to card 1 or 2 and adjust the size of the experiment.

The same example is used throughout the chapter: **a diallel of eight maize lines**, L1 to L8, with Griffing's Method 4 (F_1 crosses only), two commercial checks, three replicates and two locations, in an α-lattice. Every number quoted comes from the app with those settings and with its default random seed, 2026.

2.2 The mating design

The card 1 · Mating design opens with eleven tiles, one per design, each with its own illustration. Choosing one brings up the inputs that design needs: the list of parents, the number of males and females, the two parents of a cross or the generations. Below, four tiles summarize the plan: parents or individuals, crosses to make, entries to evaluate and growing seasons up to the evaluation.

Table 2.1. The eleven designs. *p*: parents; *c*: sets; *m* and *f*: males and females per set; *n*: F_2 plants or lines; *l* and *t*: lines and testers.

Design	What is crossed	Entries	Blocks
Griffing 1	Every pair in both directions, plus the parents	p^2	4 5
Griffing 2	Every pair once, plus the parents	$p(p+1)/2$	4 5
Griffing 3	Every pair in both directions, without the parents	$p(p-1)$	4
Griffing 4	Every pair once, without the parents	$p(p-1)/2$	4
Partial diallel	Each parent with *s* of the others, in a circulant scheme	$ps/2$	6
North Carolina I	Each male with *f* females of its own, different from those of the other males	$c \cdot m \cdot f$	6
North Carolina II	Every male with every female of its set	$c \cdot m \cdot f$	6
North Carolina III	Each F_2 plant as male with P_1 and with P_2	$2n$	6
Triple test cross	Each F_2 plant or line as male with P_1, P_2 and F_1	$3n$	6

Table 2.1. The eleven designs. *p*: parents; *c*: sets; *m* and *f*: males and females per set; *n*: F_2 plants or lines; *l* and *t*: lines and testers. *(continued)*

Design	What is crossed	Entries	Blocks
Line × tester	Each line (female) with each tester (male)	*l*·*t*	6
Generations	P_1, P_2, F_1, F_2, $BC_1 = F_1 \times P_1$ and $BC_2 = F_1 \times P_2$; optionally the reciprocal F_1, the F_3 and the selfed backcrosses	one per generation	7

The partial diallel, the North Carolina III and the triple test cross can add the parents (and the F_1) as entries; the line × tester adds the lines and testers by default; and any design accepts **checks**, which are typed separated by commas and are evaluated as one more entry. Parent names are free; if the list is left empty, the app uses generic names (P1, P2… or L1, L2…) and the number you give. When you type names, that number adjusts itself to the count typed and is locked.

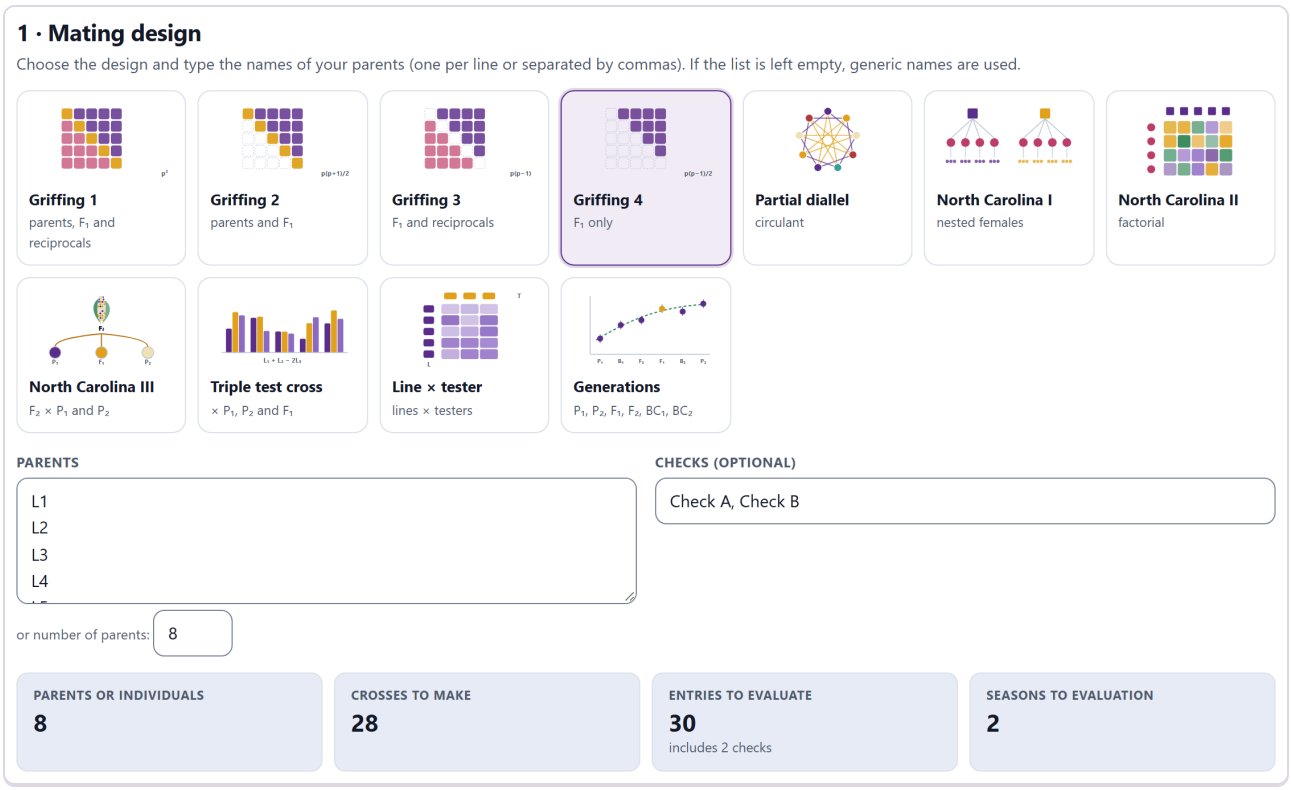


Figure 2.2. Card 1 with the example: Griffing 4, eight lines and two checks. The result is 28 crosses and 30 entries, which are evaluated in the second growing season.

NOTE · HOW CROSSES ARE WRITTEN

Every cross is written **female × male**. In diallels, the direct F_1 crosses go above the diagonal (the female is the row) and the reciprocals below, as in Griffing (1956b). In the North Carolina III and the triple test cross, the testers P_1, P_2 and F_1 are the females and each F_2 plant supplies the pollen. In the line × tester, the lines are the females. Repeated names are ignored with a warning, because the following blocks identify each entry by its name.

Rules of some designs

- **Methods 3 and 4** need at least four parents to estimate SCA; the other diallels, at least three.
- **Partial diallel** (Kempthorne and Curnow, 1961): parent i is crossed with $i + k$, ..., $i + k + s - 1$ (around the circle), with $k = (p + 1 - s)/2$. Each parent then enters exactly s crosses, but only if p and s have opposite parity and $s < p - 1$; the menu Crosses per parent (s) offers only those values. With 10 parents they are $s = 3$, 5 or 7 (15, 25 or 35 crosses, instead of 45).
- **North Carolina I and II** are organized in sets, each with new males and females; they are evaluated as separate experiments. In the II, the checkbox Males and females are plants from a population means they are non-inbred plants: when the design is chosen with it ticked, card 4 takes F = 0 (as in the I, the III and the triple test cross), and card 2 checks whether each plant is enough for all its crosses.
- **North Carolina III and triple test cross**: 20 or more F_2 plants are recommended (Kearsey, 1980). The checkbox I already have F_2 seed (or the lines) removes from the calendar the seasons needed to produce them.
- **Line × tester** with a single tester is a test cross: the app warns that neither SCA nor the GCA of the testers can be estimated that way.
- **Generations**: without P_1, P_2, F_1 or F_2 there are no scaling tests, and without both backcrosses only the additive–dominance model with the C test is fitted; the app points this out. The backcrosses carry the codes BC1 and BC2; Block 3 also recognizes the Spanish RC1 and RC2, and Block 7 shows them as BC_1 and BC_2.

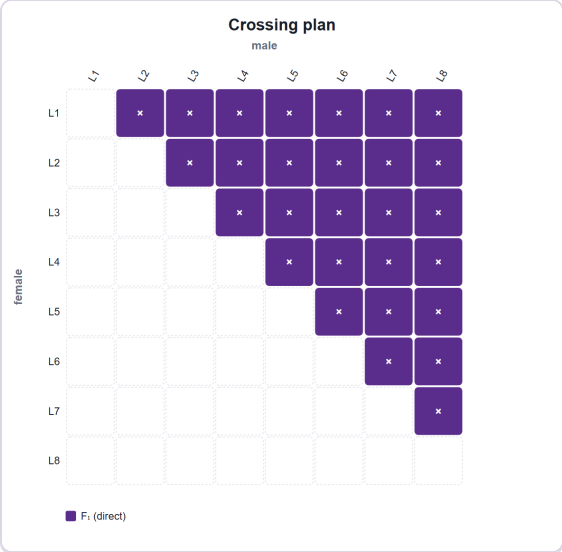
The crossing plan and the calendar

Below the tiles there are two editable figures. The **crossing plan** is the female × male table with each cross colored by type (parent, direct F_1, reciprocal, full-sib family, backcross to each tester); for generations it is replaced by a diagram of the origin of each generation with its plants. The **calendar by growing season** spreads the work over growing seasons, from the crossing block to the evaluation.

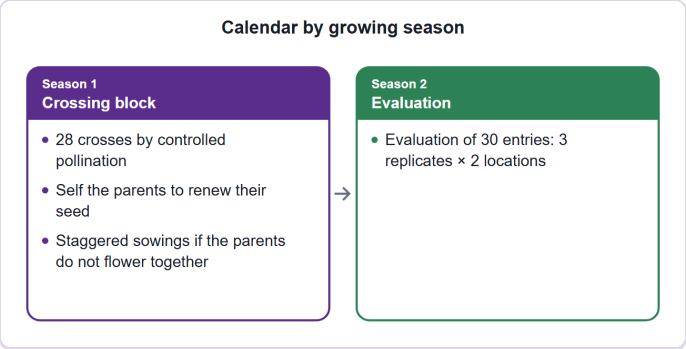
Table 2.2. Growing seasons up to the evaluation, by design.

Design	Seasons	Total
Diallels, partial and line × tester	Crossing block (with selfing of the parents and staggered sowings if they do not flower together) · evaluation	2
North Carolina I and II	Crossing block, with every plant labeled · evaluation, one experiment per set	2
North Carolina III and triple test cross	Produce the F_1 · produce the F_2 · test crosses, each male with all its testers on the same day · evaluation	4
Generations	Produce the F_1 · segregating generations, remaking the F_1 and increasing P_1 and P_2 · (third generation, if requested) · joint evaluation of all of them	3 or 4

The full list of crosses and entries is folded at the end of the card, in List of crosses and entries, with the number, code, female, male and type of each one, and it is downloaded with ⬇ CSV.



A · Crossing plan



B · Calendar by growing season

Figure 2.3. The two figures of card 1 with the example. In Method 4 only the crosses above the diagonal are made; the checks do not appear in the plan because they are not crossed.

How many plants per generation

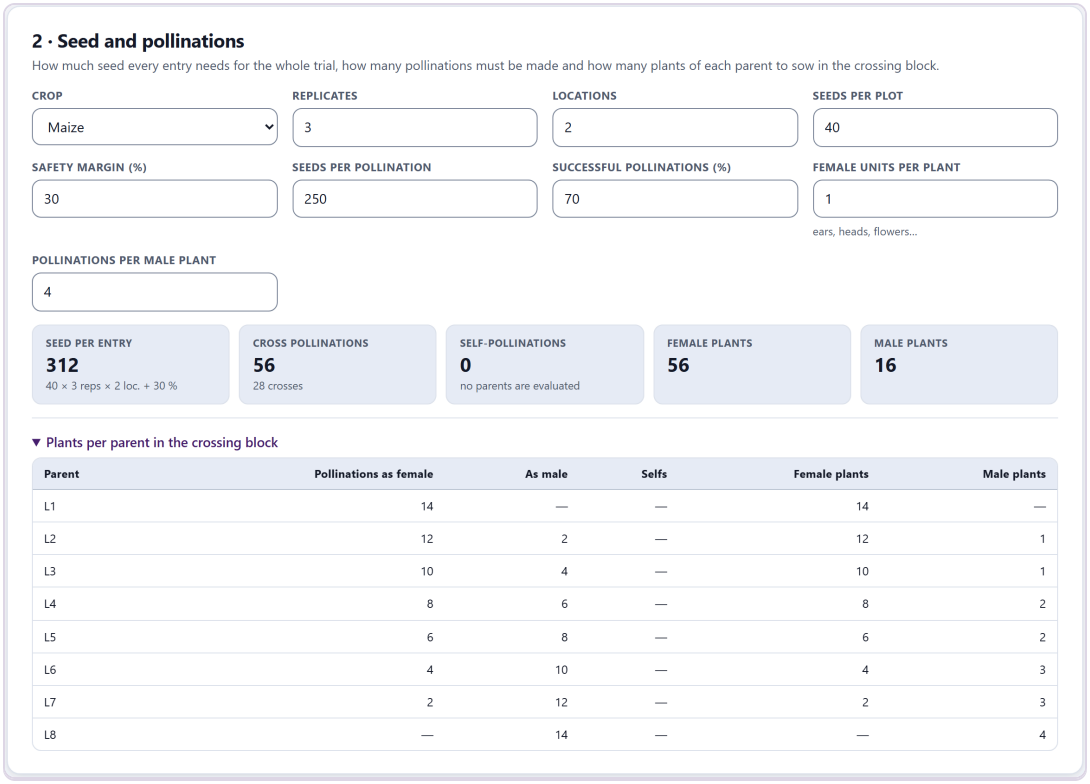
In the generations design each generation is evaluated with a number of plants. The segregating generations vary more than the parents and the F_1, so with the same number of plants their means are more or less precise according to their variance. The button **Share out to equalize precision** distributes a total of plants in proportion to the expected variance of each generation, so that all means have the same standard error (Kearsey, 1980).

$V(P_1) = V(P_2) = V(F_1) = E \cdot V(F_2) = \frac{1}{2}D + \frac{1}{4}H + E \cdot V(BC_1) = V(BC_2) = \frac{1}{4}D + \frac{1}{4}H + E$
with D and H derived from the heritability in the F_2 and from the ratio $(H/D)^{1/2}$ you give

With the default values (600 plants, $H^2 = 0.5$ in the F_2 and $(H/D)^{1/2} = 0.8$) the allocation is 73 plants of each parent and of the F_1, 146 of F_2 and 118 of each backcross, and the standard errors of the six means fall between 0.079 and 0.080. With the initial allocation of 30 and 100 plants they ranged from 0.086 to 0.124 (Section 2.7).

2.3 Seed and pollinations

The card **2 · Seed and pollinations** answers three questions about the crossing block: how much seed each entry needs for the whole trial, how many pollinations must be made and how many plants of each parent must be sown. The menu **Crop** fills the fields with guide values (Table 2.4) that you can correct with those of your own program.



2 · Seed and pollinations

How much seed every entry needs for the whole trial, how many pollinations must be made and how many plants of each parent to sow in the crossing block.

CROP	REPLICATES	LOCATIONS	SEEDS PER PLOT
Maize	3	2	40

SAFETY MARGIN (%)	SEEDS PER POLLINATION	SUCCESSFUL POLLINATIONS (%)	FEMALE UNITS PER PLANT
30	250	70	1

ears, heads, flowers...

POLLINATIONS PER MALE PLANT: 4

SEED PER ENTRY	CROSS POLLINATIONS	SELF-POLLINATIONS	FEMALE PLANTS	MALE PLANTS
312	56	0	56	16
40 × 3 reps × 2 loc. + 30 %	28 crosses	no parents are evaluated		

▼ Plants per parent in the crossing block

Parent	Pollinations as female	As male	Selfs	Female plants	Male plants
L1	14	—	—	14	—
L2	12	2	—	12	1
L3	10	4	—	10	1
L4	8	6	—	8	2
L5	6	8	—	6	2
L6	4	10	—	4	3
L7	2	12	—	2	3
L8	—	14	—	—	4

Figure 2.4. Card 2 with the example and the table Plants per parent in the crossing block opened.

THEORY IN BRIEF · THE ARITHMETIC OF THE CARD

seed per entry = seeds per plot × replicates × locations × (1 + margin)
pollinations per entry = seed ÷ (seeds per pollination × fraction of successful pollinations)
female plants of a parent = (pollinations as female + self-pollinations) ÷ female units per plant
male plants of a parent = pollinations as male ÷ pollinations per male plant

Everything is rounded up. In the generations design the seed comes from the plants of each generation times the number of locations, with the same margin. The self-pollinations are those that provide the seed of the parents that are evaluated; in Methods 3 and 4 there are none, because the parents do not enter the trial.

In the example, each entry needs 40 × 3 × 2 × 1.3 = 312 seeds; with 250 seeds per pollination and 70% success, each pollination yields 175, so two pollinations are made per cross: 56 in all for the 28 crosses. With one ear per plant and four pollinations per male plant, the crossing block needs 56 female plants and 16 male plants.

IMPORTANT · WHO IS FEMALE AND WHO IS MALE

In Methods 2 and 4 the app writes each cross with the parent that comes first in the list as the female. In the example, L1 is the female in its seven crosses (14 female plants and no male ones) and L8 is the male in all of them (4 male plants). Since these methods assume there are no reciprocal effects, you can reverse some crosses in the crossing block to share out the work; keep the entry code as it is in the field book.

Table 2.3. Inputs of card 2 and their initial values (those of maize).

Field	Initial	What it is
Replicates · Locations	3 · 2	Those of the whole block: cards 3 and 4 use them too
Seeds per plot	40	Seeds sown in each plot
Safety margin (%)	30	Extra seed, for resowing and losses
Seeds per pollination	250	Seed produced by one successful pollination
Successful pollinations (%)	70	Fraction of pollinations that set seed
Female units per plant	1	Ears, heads or flowers that can be pollinated on one plant
Pollinations per male plant	4	Pollinations the pollen of one plant can make

Table 2.4. Guide values of the menu Crop. Other (own values) leaves the fields as they are.

Crop	Seeds per pollination	Female units per plant	Pollinations per male plant	Seeds per plot
Maize	250	1	4	40
Sorghum	400	1	3	80
Wheat	30	3	5	300
Rice	20	3	6	200
Common bean	4	10	10	40
Tomato	60	8	10	20
Pepper	40	8	10	20
Squash	200	2	3	10
Sunflower	300	1	3	30
Cotton	25	10	10	40

When each parent is a single plant

In the North Carolina I and III, in the triple test cross and in the North Carolina II with plants from a population, each parent is a plant that cannot be multiplied. The card compares what is asked of it with what it can give and warns when it is not enough. In the North Carolina II with its initial values (three sets of 4 × 4), each male must supply pollen for 8 pollinations, more than the 4 assumed, and each female needs 8 pollinations with a single ear. The app suggests reducing the seed per plot, staggering the sowing of the females, using several tassels or flowers per plant or, in the II, using as females lines or clones that can be multiplied.

2.4 The field trial

The card 3 · Field trial design randomizes the entries at each location and draws the field map. The randomization is **reproducible**: it starts from the Random seed, which is written in the field book, and each location receives its own randomization derived from it. The button ↻ draws another seed.

Table 2.5. The four field designs. *v*: entries; *r*: replicates; *s*: blocks per replicate; *b*: blocks and *c*: checks of the augmented design.

Design	How it is built	Error degrees of freedom
Randomized complete blocks	Each replicate is a block with all the entries	$(r-1)(v-1)$
Alpha lattice (incomplete blocks)	Each replicate is divided into *s* blocks of *k* plots (Patterson and Williams, 1976)	$vr-1-(v-1)-(r-1)-r(s-1)$
Completely randomized	All the plots are randomized together	$v(r-1)$
Augmented (repeated checks)	The checks in every block; the new entries only once (Federer, 1956)	$(b-1)(c-1)$

DECISION RULE · WHICH FIELD DESIGN TO USE

If there are few entries (up to 15 or 20) and the land is uniform over the size of a replicate…	**then** randomized complete blocks.
If there are many entries and a full replicate no longer fits on homogeneous soil…	**then** α-lattice, with blocks of a size close to $\sqrt{v}$.
If there are many new entries and seed for a single plot of each…	**then** augmented, with at least two checks and 12 or more error degrees of freedom.
If the plots are pots or homogeneous units (greenhouse, laboratory)…	**then** completely randomized.

The app requires at least two replicates, except in the augmented design, and warns when the error is left with fewer than 12 degrees of freedom.

The α-lattice

When the lattice is chosen, the app proposes a block size *k* close to $\sqrt{v}$; with 30 entries it proposes 5, that is, six blocks per replicate. If *v* is not a multiple of *k*, the blocks come in two sizes that differ by one plot.

THEORY IN BRIEF · HOW IT IS BUILT AND WHAT THE EFFICIENCY FACTOR IS

When the number of blocks per replicate *s* is prime and $r \le s$ and $k \le s$, a cyclic array gives an $\alpha(0,1)$ design, in which no pair of entries shares a block more than once: with 25 entries in blocks of 5, or 49 in blocks of 7, it reaches the highest efficiency possible. In the other cases, the app generates candidates from random generating arrays and with an exchange algorithm that spreads the concurrences among pairs, picks the one with the highest **efficiency factor E** and, up to 100 entries, polishes it with exchanges that raise E directly.

E compares the variance of the differences between entries with what a complete-block design with the same error would have: E = 0.785 means that the information lost by using incomplete blocks amounts to 21.5%. Next to E, the app shows its **upper bound** for resolvable designs and the **maximum concurrence**, the largest number of times a pair of entries shares a block.

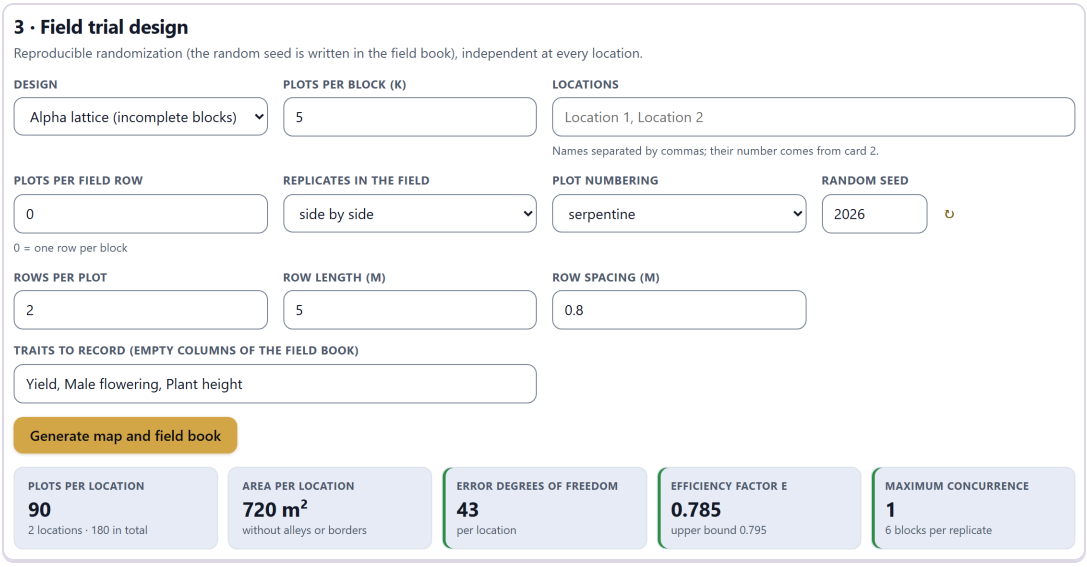


Figure 2.5. Card 3 with the α-lattice of the example: 90 plots per location, 720 m², 43 error degrees of freedom and E = 0.785, with upper bound 0.795; no pair of entries shares a block more than once.

Table 2.6. The α-lattice of the example (30 entries, three replicates) with different block sizes.

k	Blocks per replicate	E	Upper bound	Maximum concurrence	Error df
3	10	0.591	0.682	1	31
5	6	0.785	0.795	1	43
6	5	0.823	0.829	2	46
10	3	0.905	0.906	3	52
30	1 (complete blocks)	1	—	—	58

IMPORTANT · A HIGHER E IS NOT A BETTER DESIGN

E rises with block size and reaches 1 with complete blocks, because it compares with a design that has the *same* error. But the purpose of small blocks is a smaller error, because each block occupies a more uniform patch of land. Choose *k* by the size of the homogeneous land and use E to compare designs with the same *k*. Nor is the upper bound always reached: with blocks of three plots, several seeds give E between 0.591 and 0.592, far from 0.682, because no design attains it.

6 of 7

KEY FACT

In the validation of the program, the α-lattices generated matched or exceeded the efficiency of those from an independent generator with the same number of entries, replicates and block size in 6 of 7 cases; in the seventh they fell 0.03% short. E is computed exactly, from the eigenvalues of the information matrix.

How it is laid out on the land

Table 2.7. Options of card 3 that do not change the statistical design.

Field	What it does
Plots per field row	0 puts each block in one row; any other number splits the blocks into rows of that length.
Replicates in the field	one below the other or side by side, with an alley one plot wide between them.
Plot numbering	serpentine (each row in the opposite direction to the previous one, the way the field is walked), row by row or replicate × 100 + plot (101, 102… or ×1000 if a replicate has 100 or more plots).
Locations	Their names, separated by commas; their number comes from card 2.
Rows per plot, Row length (m), Row spacing (m)	They compute the area: plots × rows × length × spacing, without alleys or borders. In the example, 90 × 2 × 5 × 0.8 = 720 m² per location.
Traits to record	Names separated by commas; each one will be an empty column of the field book.

The field map is drawn for one location at a time; if there are several, a menu switches between them. The figure editor colors the plots by entry type or by replicate and shows or hides the entry codes and the plot numbers. When the plan is small, the map is redrawn by itself whenever the plan changes; if it is large (more than 2,500 plots, or an α-lattice with more than 200 entries), the app asks you to press Generate map and field book.

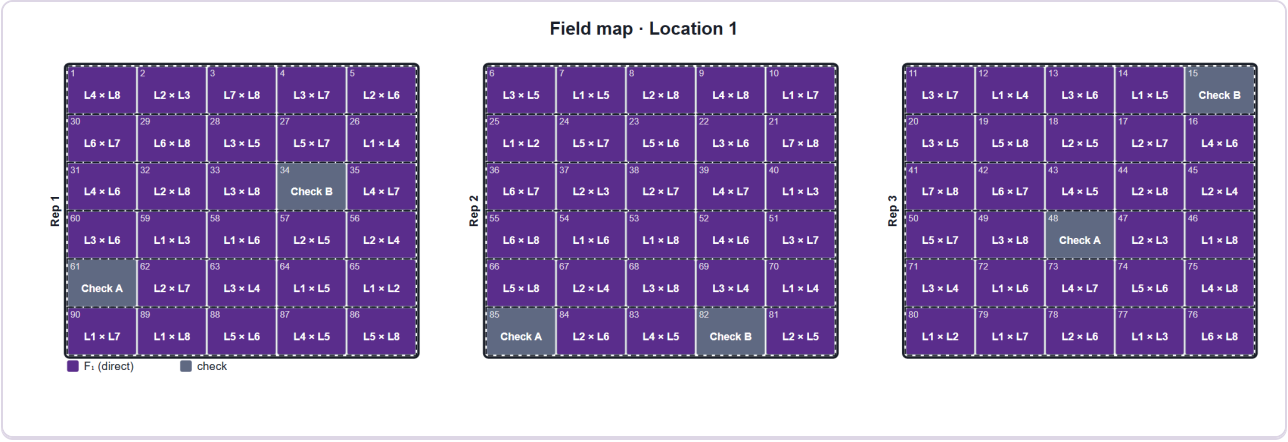


Figure 2.6. Field map of Location 1 with the replicates side by side. Each row of a replicate is an incomplete block of five plots; the serpentine numbering runs across the three replicates row by row. The checks are in gray.

2.5 The field book

Below the map is the **field book**: one row per plot for all locations, ready to be printed or to take the data. It has eleven fixed columns and one more, empty, for each trait typed in the card. The view shows the first 60 rows; the downloads include them all.

Table 2.8. The first rows of the field book of the example (180 plots in all). The last three columns are left empty for the data.

Location	Plot	Replicate	Block	Row	Column	Entry	Genotype	Female	Male	Type	Yield	Male flowering	Plant height
Location 1	1	1	1	1	1	22	L4 × L8	L4	L8	F_1 (direct)			
Location 1	2	1	1	1	2	8	L2 × L3	L2	L3	F_1 (direct)			
Location 1	3	1	1	1	3	28	L7 × L8	L7	L8	F_1 (direct)			
Location 1	4	1	1	1	4	17	L3 × L7	L3	L7	F_1 (direct)			
Location 1	5	1	1	1	5	11	L2 × L6	L2	L6	F_1 (direct)			
Location 1	6	2	1	1	7	15	L3 × L5	L3	L5	F_1 (direct)			
…													

Plot 6 already belongs to replicate 2 and lies in column 7: with the replicates side by side, the first field row crosses the three replicates, and column 6 is the alley between them.

The downloads

- ⬇ Workbook (.xlsx) saves four sheets: **Field book**; **Entries**, with the seed and pollinations of each one; **Crossing block**, with the plants of each parent; and **Design**, with the mating and field designs, the replicates, the locations, the random seed, the error degrees of freedom, the efficiency factor and its bound (in the α-lattice) and the date.
- ⬇ CSV saves the field book only, as UTF-8 text, which spreadsheets open with the accents right.

With the checkbox Include the line that describes the design ticked, the first line of both files declares the plan. In the example it reads:

```
#BreedingPro fieldbook, mating=griffing, method=4, field=alpha, reps=3, k=5, locations=2, entries=30, seed=2026, F=1
```

Block 3 reads from that line the mating design, the method and the field design, so it does not have to guess them; the rest stays as a record. Enter the data in the empty columns without changing the names of the fixed columns.

2.6 Precision and power before sowing

The last card answers the question that is too often asked too late: with this size of experiment, will I be able to detect what matters to me? It is enough to assume a mean, the coefficient of variation of the error and the size of the genetic variances; the app computes everything from the expected mean squares of the balanced analysis, before a single datum exists.

Table 2.9. The assumptions of card 4 and their initial values.

Field	Initial	What it is
Trait mean	8	In the units of the trait; for example, t/ha of grain
Error CV (%)	12	Coefficient of variation of the plot error that the trait usually has

Table 2.9. The assumptions of card 4 and their initial values. *(continued)*

Field	Initial	What it is
Additive CV, σA/mean (%)	10	Size of the expected additive variation, relative to the mean
Dominance CV, σD/mean (%)	8	The same for dominance
Parental inbreeding (F)	1	1 for lines; 0.875, 0.75 or 0.5 for S_3, S_2 or S_1; 0 for non-inbred plants
α · Target power	0.05 · 80 %	Significance level and power sought
Power curve against	parents	Number of parents (or families) or number of replicates

THEORY IN BRIEF · FROM THE ASSUMPTIONS TO THE VARIANCES

$\sigma^2_e = (CV_e \times \text{mean})^2$ · $\sigma^2_A = (CV_A \times \text{mean})^2$ · $\sigma^2_D = (CV_D \times \text{mean})^2$

$\sigma^2_{GCA} = ¼(1 + F)\,\sigma^2_A$ · $\sigma^2_{SCA} = ¼(1 + F)^2\,\sigma^2_D$ · error of an entry mean: $\sigma^2_e/(r \cdot L)$

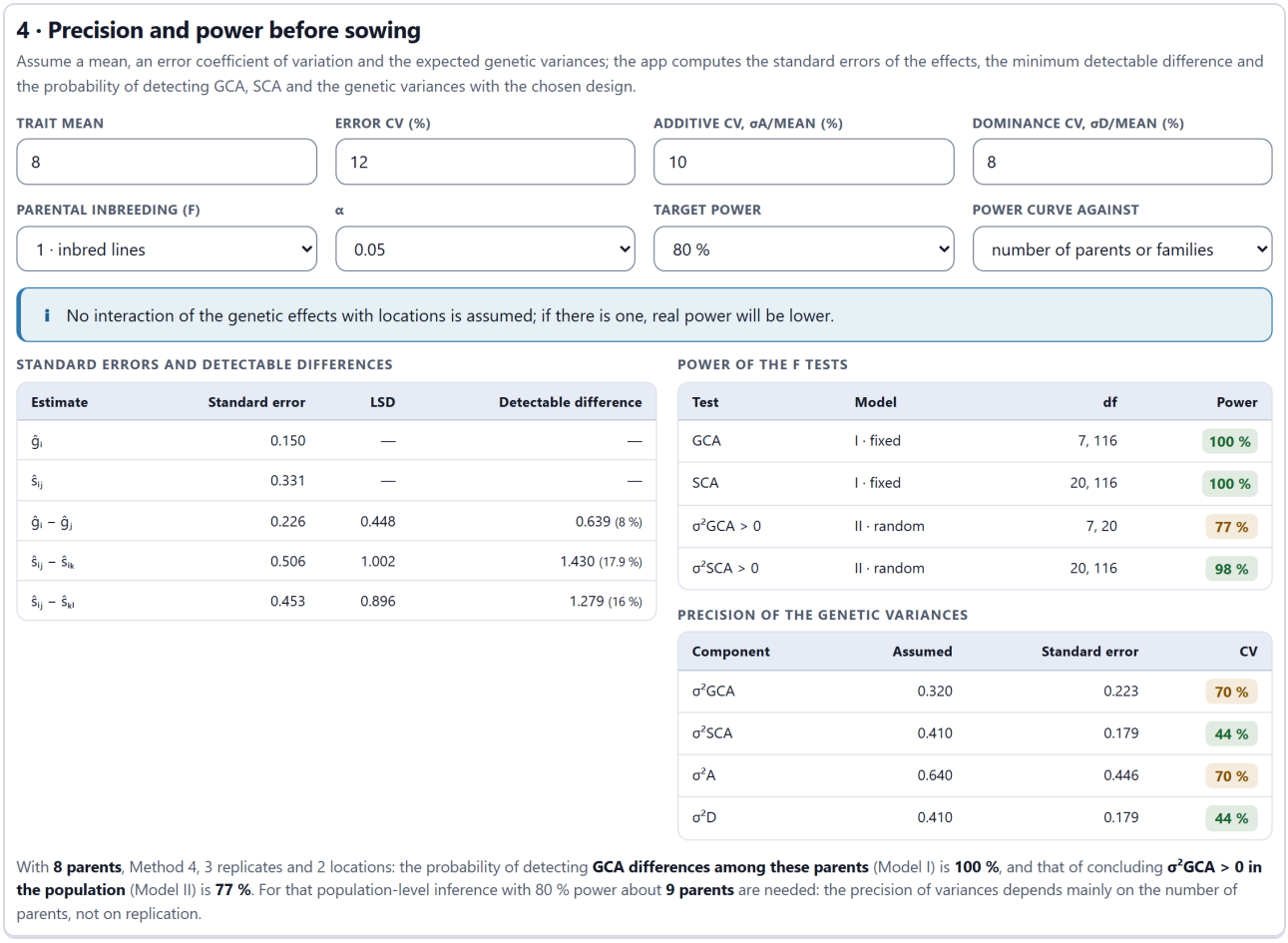
LSD = $t_{1-\alpha/2} \times SE$ · detectable difference = $(t_{1-\alpha/2} + t_{power}) \times SE$

In Model I the power of each F test comes from the noncentral F distribution; in Model II, from the central F scaled by the ratio of the expected mean squares. The precision of the genetic variances is computed from the sampling variance of the moment estimators. No interaction of the genetic effects with locations is assumed; if there is one, the real power will be lower, and the app recalls this in a message.

With the initial values, $\sigma^2_A = 0.64$ and $\sigma^2_D = 0.41$; since the lines have F = 1, $\sigma^2_{GCA} = 0.32$ and $\sigma^2_{SCA} = 0.41$.

Table 2.10. What card 4 computes in each design.

Design	Precision	Power
Griffing's methods	SE of $\hat{g}_i$, $\hat{s}_{ii}$, $\hat{s}_{ij}$ and $\hat{r}_{ij}$ and of their differences; LSD and detectable difference; SE and CV of σ^2_{GCA}, σ^2_{SCA}, σ^2_A and σ^2_D	GCA and SCA in Models I and II; in Method 1 that of Model II is approximate
Partial diallel	SE of $\hat{g}_i - \hat{g}_j$ according to the distance between the parents in the circle	GCA and SCA, Models I and II
Line × tester	Exact SE of the GCA of lines and testers and of the SCA, next to the conventional ones	Lines, testers and lines × testers, Models I and II
North Carolina designs and triple test cross	SE and CV of σ^2_A and σ^2_D	The tests of the design components (random model)
Generations	SE of the mean of each generation with its plants	—



4 · Precision and power before sowing

Assume a mean, an error coefficient of variation and the expected genetic variances; the app computes the standard errors of the effects, the minimum detectable difference and the probability of detecting GCA, SCA and the genetic variances with the chosen design.

TRAIT MEAN: 8 | ERROR CV (%): 12 | ADDITIVE CV, σA/MEAN (%): 10 | DOMINANCE CV, σD/MEAN (%): 8

PARENTAL INBREEDING (F): 1 · inbred lines | α: 0.05 | TARGET POWER: 80 % | POWER CURVE AGAINST: number of parents or families

i No interaction of the genetic effects with locations is assumed; if there is one, real power will be lower.

STANDARD ERRORS AND DETECTABLE DIFFERENCES

Estimate	Standard error	LSD	Detectable difference
$\hat{g}_i$	0.150	—	—
$\hat{s}_{ij}$	0.331	—	—
$\hat{g}_i - \hat{g}_j$	0.226	0.448	0.639 (8 %)
$\hat{s}_{ij} - \hat{s}_{ik}$	0.506	1.002	1.430 (17.9 %)
$\hat{s}_{ij} - \hat{s}_{kl}$	0.453	0.896	1.279 (16 %)

POWER OF THE F TESTS

Test	Model	df	Power
GCA	I · fixed	7, 116	100 %
SCA	I · fixed	20, 116	100 %
σ^2GCA > 0	II · random	7, 20	77 %
σ^2SCA > 0	II · random	20, 116	98 %

PRECISION OF THE GENETIC VARIANCES

Component	Assumed	Standard error	CV
σ^2GCA	0.320	0.223	70 %
σ^2SCA	0.410	0.179	44 %
σ^2A	0.640	0.446	70 %
σ^2D	0.410	0.179	44 %

With **8 parents**, Method 4, 3 replicates and 2 locations: the probability of detecting **GCA differences among these parents** (Model I) is **100 %**, and that of concluding **σ^2GCA > 0 in the population** (Model II) is **77 %**. For that population-level inference with 80 % power about **9 parents** are needed: the precision of variances depends mainly on the number of parents, not on replication.

Figure 2.7. Card 4 with the example. On the left, the standard errors and the detectable differences; on the right, the power of each F test and the precision of the genetic variances, in green when they reach the target.

- **Effects.** Two GCAs differ significantly if their difference exceeds 0.448 (the LSD), and a real difference of 0.639, 8% of the mean, would be detected with 80% probability. For the SCAs, the detectable differences are 16 to 18%.
- **Tests.** In Model I (these eight lines) GCA and SCA are detected almost surely. In Model II (a population of which the eight lines are a sample), the probability of concluding that $\sigma^2_{GCA} > 0$ is 77%, because that test has only 7 and 20 degrees of freedom.
- **Variances.** σ^2_A would be estimated with a coefficient of variation of 70% and σ^2_D of 44%: with eight parents, genetic variances are very imprecise estimates.

The text below sums it up and computes how many parents would be needed for population-level inference: about 9.

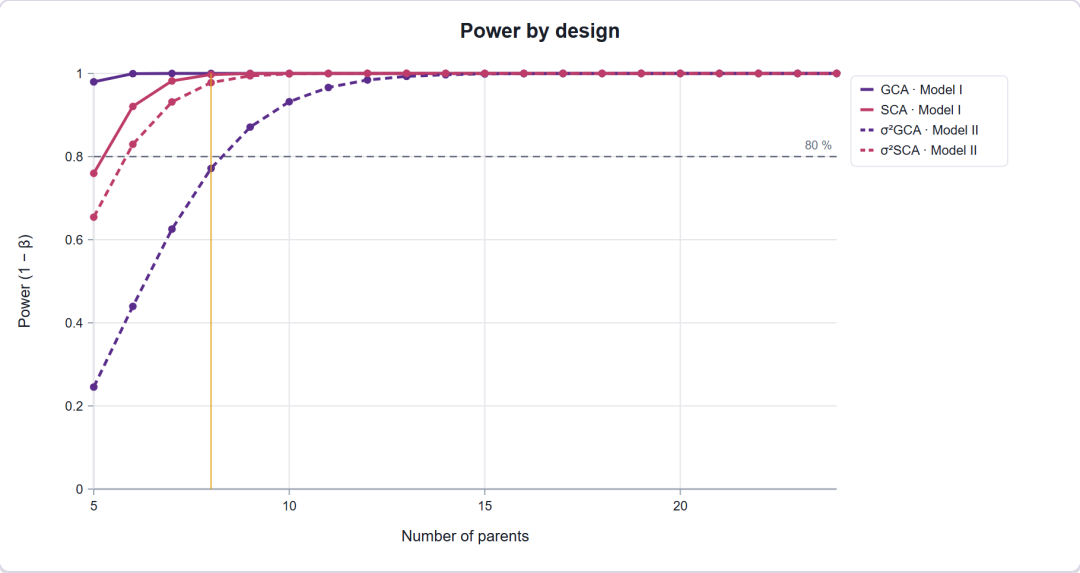


Figure 2.8. Power by number of parents, with the same number of replicates and locations. The solid lines are Model I and the dotted ones Model II; the vertical line marks the current design and the horizontal one, the target power.

IMPORTANT · PARENTS, NOT REPLICATES

If you choose the number of replicates in the menu **Power curve against**, you will see that going from 3 to 8 replicates only raises the power for σ^2_{GCA} from 77 to 83%, while one more parent takes it to 87% and two more to 93%. The precision of the variances depends above all on the number of parents, because they, not the plots, are the sample of the population.

2.7 Other designs, with their numbers

The initial values of each design are enough to see how they differ. All the following cases use three replicates, two locations and the initial assumptions of card 4.

North Carolina designs: which one estimates dominance best

Table 2.11. The three North Carolina designs with their initial values: I and II, three sets of 4 males and 4 females (per male in the I); III, 30 F_2 plants. Non-inbred plants (F = 0).

Design	Entries	Power of the tests	CV of σ^2_A	CV of σ^2_D
North Carolina I	48	males 58%; females within males 98%	79%	171%
North Carolina II	48	males 75%; females 75%; males × females 58%	48%	69%
North Carolina III	60	F_2 plants 99%; F_2 × testers 100%	39%	36%

With the same number of entries, the II estimates σ^2_A and σ^2_D much more precisely than the I, in which σ^2_D is obtained by difference and its CV exceeds 170%. The III, designed for the average degree of dominance, estimates both variances with similar precision, and the app says so in its text. Remember that these variances are estimated in a single environment; the app warns that, if locations are averaged without modeling the interaction, σ^2_A and σ^2_D are inflated by σ^2_{GE}.

Line × tester: the exact standard errors

With 20 lines, 2 testers and the lines and testers as entries, there are 40 crosses and 62 entries. The standard errors of the effects used by the app include the factors $(l-1)/l$ and $(t-1)/t$; the conventional ones omit them and overstate the uncertainty, above all when there are few testers.

Table 2.12. Exact and conventional standard errors in the 20 × 2 line × tester.

Estimate	Exact SE	Conventional SE
$\hat{g}$ of a line	0.270	0.277
$\hat{g}$ of a tester	0.062	0.088
$\hat{s}_{ij}$	0.270	0.392
$\hat{s}_{ij} - \hat{s}_{kj}$ (same tester)	0.392	0.554

In Model II, on the other hand, the power for σ^2_{GCA} of lines is only 49%, because it is tested against the interaction with 19 degrees of freedom. The app explains it: with two testers, each line's GCA is averaged over two crosses, and more testers of different genetic background give a more general GCA.

Partial diallel

With 10 parents and $s = 5$, 25 crosses are made instead of 45. The GCA of each parent is estimated from five crosses, and the precision of a difference $\hat{g}_i - \hat{g}_j$ depends on how far apart the two parents are in the circle: its standard error goes from 0.256, for neighbors, to 0.324, for opposite ones. The power for GCA in Model I is 100%, and for σ^2_{GCA} in Model II, 67%.

Generations

Table 2.13. Standard error of each generation mean, with the initial allocation of plants and with that of the button Share out to equalize precision (600 plants per location).

Generation	Initial plants	SE	Plants shared out	SE
P_1, P_2, F_1	30	0.124	73	0.079
F_2	100	0.096	146	0.079
BC_1, BC_2	100	0.086	118	0.080

The scaling tests compare combinations of means and gain precision when all of them have similar standard errors; that is why, in this design, card 4 shows this table instead of the power. With six entries and three replicates, the error has only 10 degrees of freedom per location (the app marks it in amber); a fourth replicate raises them to 15.

2.8 From the field book to Block 3

The button Continue: Data → at the end of the block leads to Block 3. There, the load card has a tile Block 2 field book that shows a summary of the current plan; in the example, 2 locations, 180 plots and a Griffing diallel. There are two ways to use it.

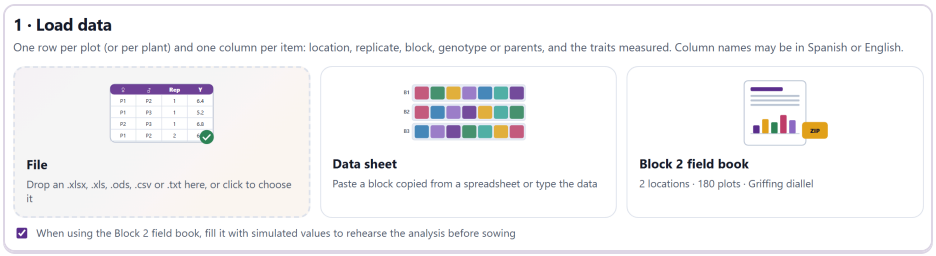


Figure 2.9. The card 1 · Load data of Block 3 with the plan of the example ready to be loaded.

1. **Before sowing, to rehearse.** With the checkbox fill it with simulated values… ticked, the field book is loaded with a column of simulated yield, with effects of parents, cross, location, replicate and block. It serves to go through the whole analysis, from the recognition of the design to the report, and to check that the experiment will give the tables you need. **These are not real data.**

2. **After harvest.** Enter the data in the empty columns of the downloaded field book and load the file in the tile File. The line that describes the design tells Block 3 which mating and field designs it is.

Like the rest of the app, Block 2 keeps nothing when the page is reloaded: download the workbook, which also keeps the random seed to regenerate the same field map.

B3

CHAPTER THREE · BLOCK 3 · DATA

Data, recognized design and field analysis

Every genetic analysis starts here. Block 3 reads the data from a file, from its own spreadsheet, from the Block 2 field book or from one of the 21 examples; works out what each column is; infers the mating design and the field design; checks the quality of the data and analyzes the trial as its design requires. The adjusted means and the error it produces are the raw material of Blocks 4 to 11.

3.1 What this block does

3.2 Loading the data

3.3 The role of each column

3.4 The recognized design

3.5 Data quality

3.6 The analysis of each environment

3.7 The combined analysis

3.8 Special cases

3.9 Downloads and moving on to the genetic blocks

3.1 What this block does

Block 3 has five cards that are worked from top to bottom. Loading the data is enough: the other four fill in by themselves and are recomputed as soon as you correct something, be it the role of a column, the design or an observation you decide to exclude.

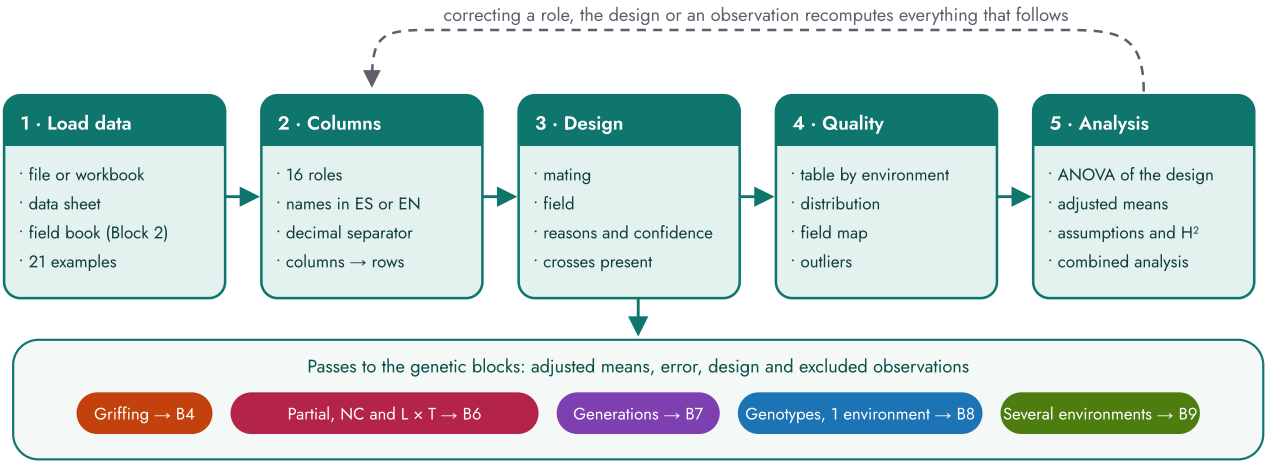


Figure 3.1. The five cards of Block 3. At the end, the button Continue leads to the block that matches the recognized design; Blocks 10 and 11 use the same data when there is a pedigree or markers.

Throughout the chapter the simulated example **diallel in an α-lattice at three locations** is used: ten parents crossed with Griffing's Method 2 (the 10 parents and their 45 crosses) plus five checks, 60 entries in all, evaluated in an α-lattice with blocks of six plots, two replicates and three locations (Chapultepec, Los Altos and Valle Verde). It is a Block 2 field book filled from a known model: $\sigma^2_{GCA} = 0.16$, $\sigma^2_{SCA} = 0.09$, $\sigma^2_{blocks} = 0.20$ and $\sigma^2_e = 0.36$ (t/ha), so that what is estimated can be compared with the truth. Data quality is illustrated with Hayman's (1954) diallel of *Nicotiana rustica*, and the data sheet with the 4 × 4 maize diallel.

3.2 Loading the data

The card 1 · Load data offers three tiles and, below them, the gallery of examples. The rule is the same for any source: **one row per plot (or per plant) and one column per item**. The first row with content holds the column names, which may be in Spanish or in English.

A file

Drag the file onto the tile File or click it to choose it. Workbooks in `.xlsx`, `.xls`, `.xlsm` and `.ods` are read, as well as `.csv`, `.tsv` and `.txt` texts. In a workbook with several sheets the one with the most data is opened and the menu Workbook sheet appears to switch to another. In a text, the separator (comma, semicolon or tab) is inferred from the first 30 lines. Empty rows are ignored, as are lines starting with `#` before the headers, except the declaration line of Block 2 (see below).

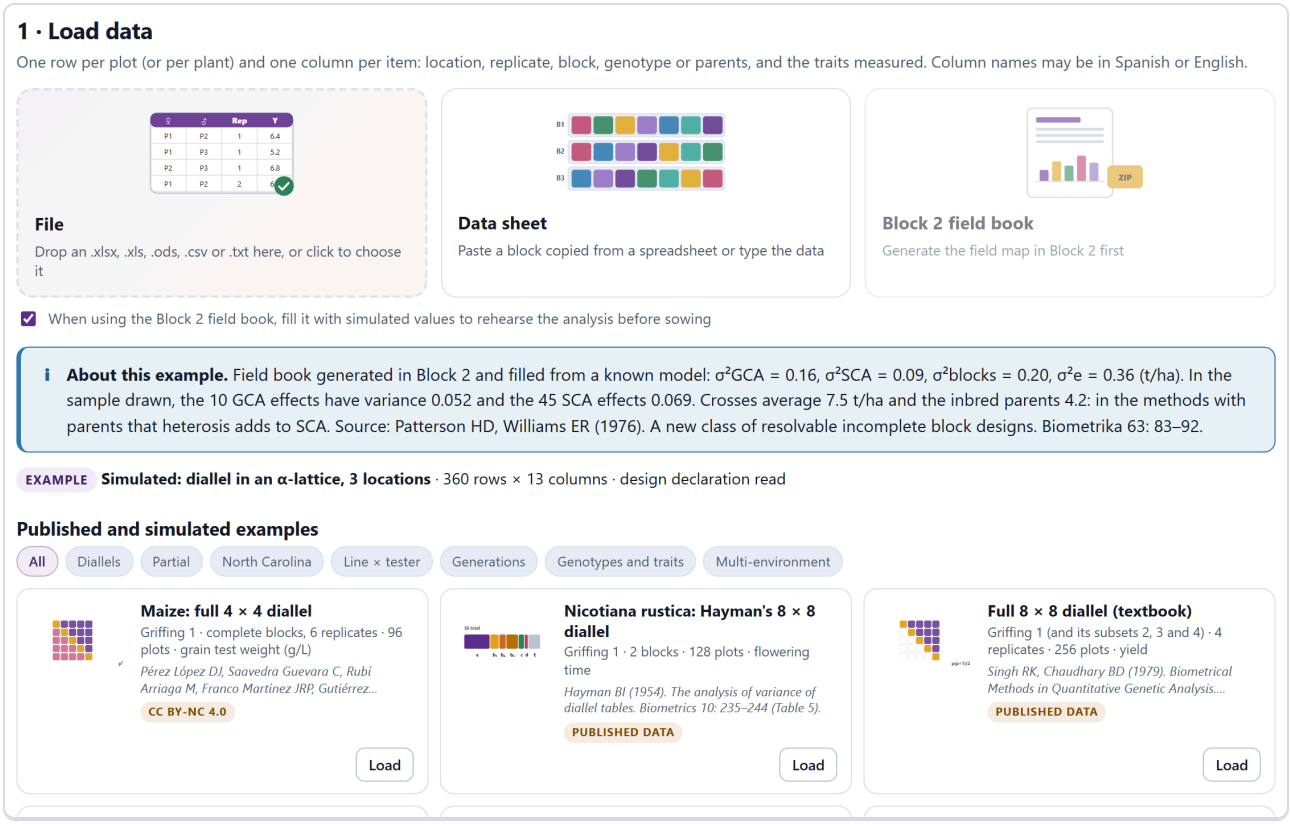


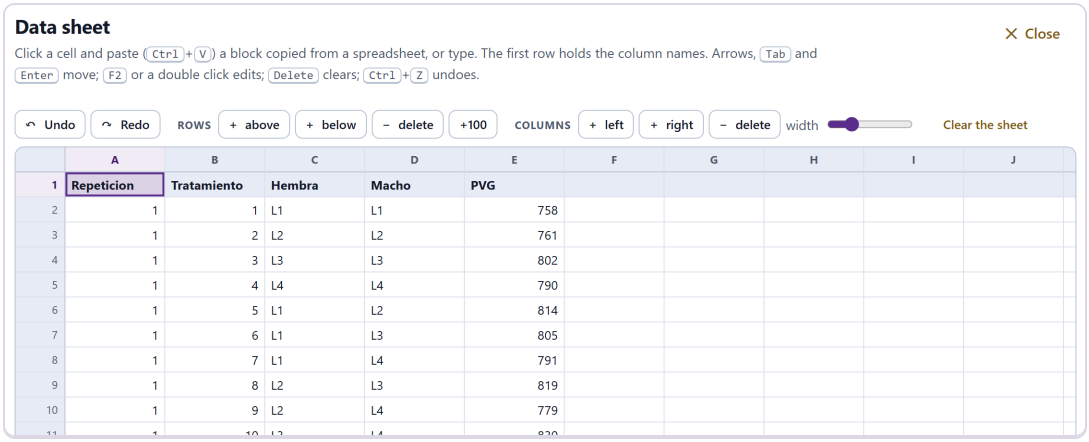
Figure 3.2. Card 1 after loading the α-lattice example. The box About this example says where the data come from and, for simulated ones, with which values they were generated; the line below sums up the source: 360 rows × 13 columns and the design declaration read.

> **NOTE · WHAT COUNTS AS A MISSING VALUE**
>
> An empty cell is a missing value, and so are `NA`, `N/A`, `NaN`, `null`, `.`, `..`, `-`, `--`, `?`, `*`, `missing`, `perdido`, `faltante`, `s/d`, `sd`, `nd`, `n.d.` and the spreadsheet errors `#N/A`, `#VALUE!` and `#DIV/0!`, in upper or lower case. If some other text appears in a variable (for example, `12.5a`), it is also taken as missing, and card 2 reports how many there were and in which row the first one is.

The data sheet

The tile Data sheet opens a spreadsheet inside the app, for anyone who does not have the file at hand or prefers to paste the data. Paste with Ctrl + V a block copied from any spreadsheet (or a text separated by commas, semicolons or tabs), or type directly. The arrows, Tab and Enter move the selection; F2 or a double click edits the cell; Delete clears it; Ctrl + Z undoes. The buttons at the top insert or delete rows and columns, add a hundred rows and adjust the width. The sheet holds up to 20,000 rows and 5,000 columns.

Below the sheet, Save as downloads its content as `.xlsx`, `.ods`, `.csv` (with comma or semicolon) or `.tsv`, with the name you type in Name. The button Use these data → sends the sheet to cards 2 to 5 just like a file. If the browser does not let you paste into the grid, open Pasting does not work in this browser? Paste into this box instead and paste into the text box that appears.



	A	B	C	D	E
1	Repeticion	Tratamiento	Hembra	Macho	PVG
2	1	1	L1	L1	758
3	1	2	L2	L2	761
4	1	3	L3	L3	802
5	1	4	L4	L4	790
6	1	5	L1	L2	814
7	1	6	L1	L3	805
8	1	7	L1	L4	791
9	1	8	L2	L3	819
10	1	9	L2	L4	779

Figure 3.3. The data sheet with the 4 × 4 maize diallel, opened with Correct in the data sheet. The first row holds the column names.

B3

IMPORTANT · THE SHEET IS SAVED IN THIS BROWSER

The sheet is saved in the browser every time it changes (the notice *saved in this browser* appears for a moment) and is restored when it is opened again, even after the page has been closed. It is the only part of the data that the app keeps: a file or an example that was loaded is lost on reloading. Since saving is local, the sheet does not travel to another computer or another browser; for that, download it. When the data come from the sheet, card 4 marks the rows of the outliers in it, with the same row number.

The Block 2 field book

The tile Block 2 field book takes the plan and the field map of Block 2 without downloading anything (Section 2.8). With the checkbox When using the Block 2 field book, fill it with simulated values… ticked it arrives with a simulated yield for practice; without it, with the yield column empty. Every Block 2 field book, including one downloaded and filled by hand, starts with a **declaration line** like this one, that of the example:

```
#BreedingPro fieldbook, mating=griffing, method=2, field=alpha, reps=2, k=6, locations=3, entries=60,
seed=4242, F=1
```

Block 3 takes from it the mating design, the method and the field design (`mating`, `method` and `field`) without having to infer them; the other items are for information. If the spreadsheet split the line over several cells, it is joined again. It is best not to delete it.

The examples

The gallery Published and simulated examples has 21 data sets. The buttons at the top filter them by family (Diallels, Partial, North Carolina, Line × tester, Generations, Genotypes and traits and Multi-environment). Each card shows the title, the design, the size, the first citation and a label: CC BY-NC 4.0, published data or simulated. When one is loaded, the box About this example gives all its sources and, for simulated ones, the true values. The examples are read exactly like a user's file; Table 3.4 sums up what Block 3 recognizes in each one.

21

KEY FACT

data sets ready for practice: 17 published and 4 simulated. The published ones come from open-access articles (CC BY-NC 4.0: non-commercial use with attribution), from open copies under a CC BY 4.0 license or from transcriptions distributed under the GPL; the six generations of *N. rustica* were rebuilt plant by plant from the published number, mean and variance of each generation. **If you publish something done with an example, cite the original work.**

3.3 The role of each column

The card 2 · Columns shows the first eight rows of the table with a menu above each column. The app proposes the role of each one from its name and its contents; you correct whichever needs it and everything is recomputed at once. Under each name you can read whether the column is numeric or text, how many distinct values it has, how many empty cells and, if it detects it, the decimal comma. A colored band groups the roles: blue for the field design, purple for the genotype and the pedigree, green for the response traits and light gray for what is ignored.

How the roles are proposed

1. **By the name.** If the name matches one in Table 3.1, the column takes that role.
2. **By the contents.** If it does not match, a column with 90% or more numbers is a response trait; the rest are ignored.
3. **One role, one column.** Except for the traits, the location and the year, each role belongs to a single column. If two claim it, the first one keeps it; for entries, the text one, which is usually the name (in the example, *Genotipo* over *Entrada*). The location and the year can be several columns, which are joined to form the environment.
4. **Without an entry column** or parent columns, the first ignored text column with two or more values becomes the entry names.

The examples come with their roles set. In your own data, check the list before going on: if the entry column is missing, the card says so in red and cards 3 to 5 do not appear until it is corrected.

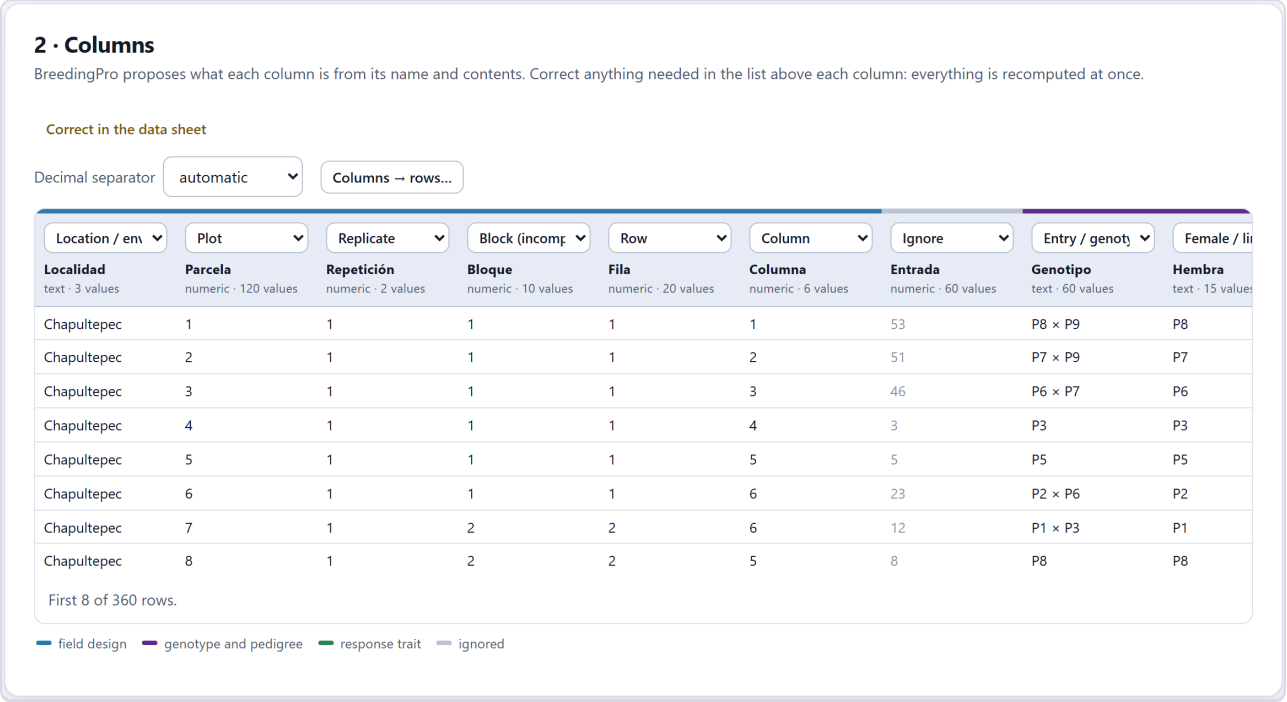


Figure 3.4. Card 2 with the α-lattice example, whose column names are in Spanish. Entrada, an order number, was left as Ignore because Genotipo already names the entries; to the right, outside the crop, come Macho, Tipo and the two traits.

Table 3.1. The 16 roles and some of the names the app recognizes. Names are compared ignoring accents, case, spaces and signs: *Rep.*, *REP* and *Repetición* are the same, and *No. parcela* equals *NoParcela*.

Role	Names recognized (examples)	What it is for
Response trait	any column with 90% or more numbers and no other role	What is analyzed; there may be several
Location / environment	localidad, ambiente, sitio, lugar, ensayo, experimento, estación, loc, env, location, environment, site, trial	Separates the environments
Year / season	año, ciclo, temporada, campaña, year, season, cycle	Joined to the location: each location × year is an environment
Replicate	repetición, rep, r, réplica, bloque completo, replicate, replication	Complete replicates
Block (incomplete)	bloque, subbloque, bloque incompleto, block, iblock, blk	Blocks within each replicate, or plain blocks
Plot	parcela, unidad experimental, plot, plot number, unit	Identifies the plot; if it repeats, the rows are plants
Row · Column	fila, hilera, rango, row, range · columna, pase, col, column, tier	Position in the field: map and row–column design
Entry / genotype	genotipo, entrada, tratamiento, variedad, cruza, híbrido, familia, clon, material, accesión, cultivar, nombre, código, genotype, entry, variety, cross	The name of the entry
Female / line / parent 1	hembra, madre, línea, progenitor 1, planta F_2, female, dam, mother, line, parent 1	Female parent, the line or the F_2 plant

Table 3.1. The 16 roles and some of the names the app recognizes. Names are compared ignoring accents, case, spaces and signs: *Rep.*, *REP* and *Repetición* are the same, and *No. parcela* equals *NoParcela*. *(continued)*

Role	Names recognized (examples)	What it is for
Male / tester / parent 2	macho, padre, probador, progenitor 2, male, sire, father, tester, parent 2	Male parent or tester
Set	conjunto, set	Sets of the North Carolina designs
Generation	generación, filial, generation	P_1, P_2, F_1, F_2, backcrosses…
Entry type	tipo, clase, categoría, tipo de entrada, type, class, category	Check, parent, reciprocal, line, tester…
Plant / individual	planta, individuo, progenie, plántula, árbol, muestra, plant, individual, tree, sample	Plant-level data
Ignore	whatever fits no role and is not numeric	Notes, dates, order numbers

IMPORTANT · AN IDENTIFICATION NUMBER IS TAKEN FOR A TRAIT

A numeric column with a name the app does not recognize (*Lot*, *Furrow*, *Key*) is proposed as a response trait and would be analyzed as if it were a character. Move it to Ignore or give it its role. The same the other way round: a trait with the name of a role, such as a column *Type* that actually measures something, must be marked as Response trait.

Parents, crosses and entry types

The pedigree can come in three ways, and the app recognizes all of them:

- **Female and male columns.** Each row gives its two parents. If there is also an entry column with numbers that correspond one to one with the pedigree, the entry is named by its pedigree (*L1 × L2*) and the number is kept; this is how Block 3 reads the 4 × 4 maize diallel of Figure 3.3.
- **The cross written in the name.** Without parent columns, names such as *A × B*, *A x B*, *A/B*, *A*B* or compact codes such as *1x3* are split into female and male, provided that at least half of the names are crosses and the parents repeat among them; a name equal to a parent is that parent. The card warns: *Cross names were split into female and male*. This is how the maize partial diallel is read.
- **A generation column** (or a type column, or the entry names themselves) with the codes P1, P2, F1, F2, F3, BC1 and BC2 (also RC1, RC2, B1, B2), RF1 for the reciprocal F_1 and BC1S and BC2S for the selfed backcrosses.

The **type of each entry** comes from the column Entry type when there is one (check, testigo or control; parent or parental; reciprocal; line; tester; F_1, cross or hybrid). If there is none, it is inferred from the pedigree: a female equal to the male is a parent, a female different from the male is an F_1, and an entry without parents in data with crosses is a check.

In the North Carolina designs and the line × tester, females and males are different plants or lines even if they share a code (*1 × 1* is a cross, not a parent): the app marks them with ♀ and ♂ and says so. If the codes repeat across sets, the set is prefixed to them (*S1·3*).

Decimal separator and reshaping the table

The menu **Decimal separator** is **automatic** by default: each column uses a decimal comma if it has more values like *12,5* or *1.234,5* than like *12.5*. Choose **point (12.5)** or **comma (12,5)** to impose it on all of them.

Some data arrive **in wide format**: one column per replicate or per environment. The button **Columns → rows...** turns them into long format. Tick the columns that are replicates of the same trait (those named *R1*, *Rep2*, *B3*, *E1* or *Loc2* come ticked), type the name of the new factor and that of the trait, and press **Stack**. Names such as *E1B2* or *Loc1_Rep2* are split into two factors.

Genotype	R1	R2	R3
A	5.2	4.9	5.6
B	6.1	6.4	5.8
...			

→

Genotype	Replicate	Yield
A	1	5.2
A	2	4.9
A	3	5.6
...		

Figure 3.5. **Columns → rows...** stacks columns R1 to R3 into a single trait with the new factor *Replicate*; the column *Genotype* is repeated in every row.

If the table is a **diallel in matrix form** (one parent per row, the same parents as column headers and a value in each cell), **Diallel table → rows** also appears, which turns it into three columns: female, male and value. **Undo the reshaping** goes back to the original table. Finally, **Correct in the data sheet** opens the loaded table in the data sheet to fix a value at its source; afterwards, **Use these data →** loads it again.

3.4 The recognized design

The card **3 · Recognized design** says which experiment the app believes you have. Six tiles sum it up: the mating design with its degree of confidence (high, medium or low; *declared* if it came in the declaration line and *chosen* if you changed it yourself), the entries with their parents and checks, the field design, the environments, the replicates with the plots per environment and the traits. Below, a list explains **why** each conclusion was reached and points out what is missing or left over.

The mating design

The app checks the rules of Table 3.2 in order and keeps the first one that holds. In a diallel, the method depends on whether there are parents (at least half of them) and reciprocals (in at least half of the pairs): with both, Method 1; parents only, 2; reciprocals only, 3; neither, 4.

Table 3.2. How the mating design is recognized, rule by rule.

#	If in the data...	then
1	three or more entries, and at least 60% of them, are generation codes	Generation means
2	there is no cross with its two parents	Genotype trial
3	the testers are P_1 and P_2 (or P_1, P_2 and F_1) and four or more plants are crossed with them	North Carolina III (with three testers, triple test cross)

Table 3.2. How the mating design is recognized, rule by rule. *(continued)*

#	If in the data...	then
4	there are no sets and the same names appear as female and as male, or there are three or more selfed parents	Diallel: Griffing's if 90% or more of the pairs are present; partial if not
5	each male has its own females (nested) and each female a single male	North Carolina I
6	there are types *line* or *tester*, the headers say line or tester, or there are up to five males and at least twice as many females	Line × tester
7	none of the above	North Carolina II (factorial)

In the partial diallel, the app checks whether each parent enters the same number *s* of crosses and in the circulant order of Kempthorne and Curnow (1961); if so it says so, and if not it warns that the analysis will be by least squares. The **confidence drops to medium** when entries of the complete design are missing, when the partial diallel is not circulant, when crosses of the factorial are missing, when the females of a North Carolina II repeat codes across males (it may be a I) or when the North Carolina III or the triple test cross have fewer than 20 plants.

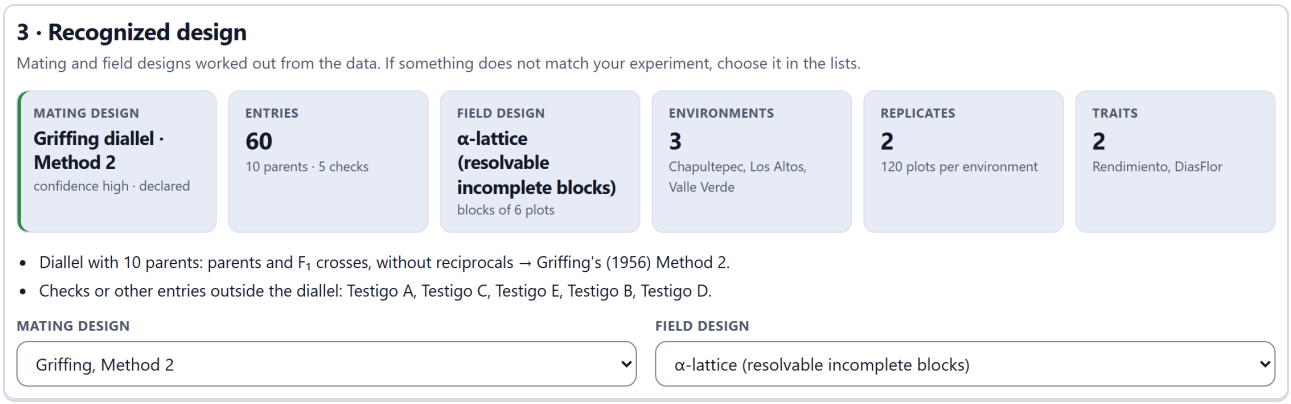


Figure 3.6. The recognized design in the α-lattice. The declaration line set Method 2 (*confidence high · declared*); the reasons say where each conclusion comes from and the two menus below allow changing it.

The field design

The field design is recognized **in each environment separately**, with the rules of Table 3.3. If the environments differ, the tile says *differs between environments* and the combined analysis uses only the replicates.

Table 3.3. How the field design is recognized in an environment.

Situation	If also...	then
Two or more replicates	a few entries are in all of them and the rest only once (less than 60% of the cells filled)	Augmented blocks
	there are more blocks than replicates and the largest block takes less than 75% of a replicate	α-lattice (resolvable incomplete blocks)
	in any other case	Randomized complete blocks

Table 3.3. How the field design is recognized in an environment. *(continued)*

Situation	If also...	then
No replicates, two or more blocks	every block has all the entries	Randomized complete blocks (the blocks act as replicates)
	there are checks in every block and the rest appears once	Augmented blocks
	the entries repeat	Incomplete blocks
	nothing repeats	Unreplicated: there is no error
Neither replicates nor blocks	the entries repeat, or there are plant-level data	Completely randomized
	there is a single value per entry	Entry means

The **row–column** design is never chosen automatically: if the plots have row and column and you want to use them in the model, choose it in the menu. The card also warns of other field problems: missing replicate × entry combinations (adjusted least squares are used), a plot that appears several times without being plant-level data (it is averaged) and a plot with two different entries, which is an error and must be corrected.

Changing what was recognized

If something does not match your experiment, change it in the menus Mating design (automatic, Griffing Methods 1 to 4, partial diallel, the three North Carolina designs, triple test cross, line × tester, generation means or genotype trial) and Field design (automatic or one of the seven designs). The change is marked as *chosen* and everything that follows is recomputed. Typical cases: a North Carolina II that is really a I because each male has its own females, or a trial with row and column that you want to analyze as rows and columns.

Crosses present, missing entries and recognized entries

The figure Crosses present in the data draws the table of the design: female × male in diallels and the North Carolina II, male × female in the I, F_2 plant × tester in the III and the triple test cross, line × tester in the line × tester. Each cell is colored by the number of plots or, with Color by, by the mean of the trait; parents carry a border, and a cross that the design expects but is not in the data is marked with an ×, with the legend *missing from the data*.

If the design is incomplete, the box Entries missing from the design lists the missing crosses, parents, reciprocals or generations. At the end, Recognized entries (folded) lists each entry with its female, its male, its type and its number of records: it is the quickest way to check that the pedigree was read correctly.

What is recognized in the examples

Table 3.4 serves as a check of the recognition and as a guide to choosing an example similar to your data.

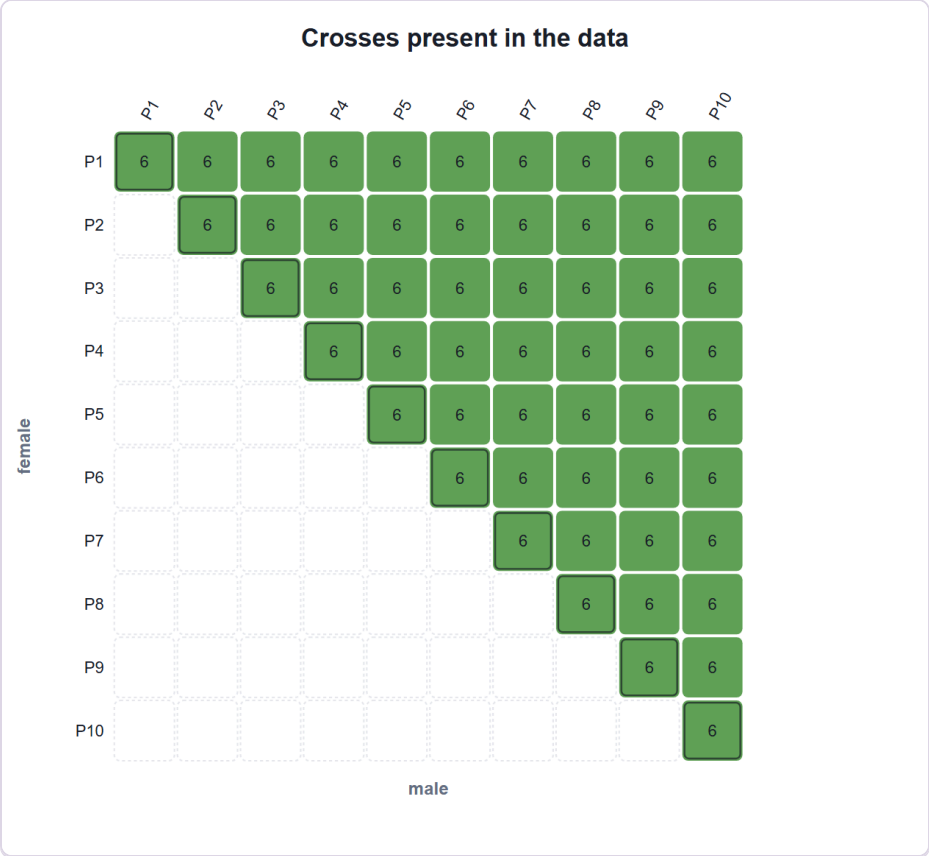


Figure 3.7. The crosses present in the α-lattice: Method 2 fills the diagonal (parents, with a border) and the upper triangle. Each cross is in 6 plots: 2 replicates × 3 locations.

Table 3.4. What Block 3 recognizes in the 21 examples. RCB: randomized complete blocks; CR: completely randomized. *Outliers*: in the first trait.

Example	Mating	Field	Env.	Outliers	Next
Maize: full 4 × 4 diallel	Griffing, Method 1	RCB	1	1	4
Nicotiana rustica: Hayman's 8 × 8 diallel	Griffing, Method 1	RCB	1	2	4
Full 8 × 8 diallel (textbook)	Griffing, Method 1	RCB	1	0	4
Diallel with parents in two environments	Griffing, Method 2	RCB	2	0	4
Maize: means of 36 single crosses	Griffing, Method 4	means	1	—	4
Maize intervarietal crosses	Griffing, Method 2	means	1	—	4
Maize: partial diallel of 8 lines	circulant partial, $s = 5$	RCB	1	0	6
North Carolina I	North Carolina I	RCB, by plant	1	0	6
Potato: North Carolina II	North Carolina II (♀ and ♂)	RCB	1	0	6
North Carolina III	North Carolina III, medium confidence	RCB	1	0	6
Simulated: triple test cross	triple test cross	RCB	1	0	6
Line × tester with parents	line × tester	RCB	1	0	6
Maize: 13 hybrids and 15 traits	genotype trial	RCB	1	1	8
Maize: 13 hybrids in 4 environments	genotype trial	RCB	4	1	9

Table 3.4. What Block 3 recognizes in the 21 examples. RCB: randomized complete blocks; CR: completely randomized. *Outliers*: in the first trait. *(continued)*

Example	Mating	Field	Env.	Outliers	Next
Oat: 10 genotypes in 14 environments	genotype trial	RCB	14	2	9
Potato: 28 clones in 6 localities	genotype trial	RCB	6	2	9
Means of 17 genotypes in 12 environments	genotype trial	means	12	—	9
Simulated: inbred lines with a pedigree	genotype trial	RCB	3	0	9
Simulated: diallel in an α-lattice	Griffing, Method 2 (declared)	α-lattice	3	0	4
Nicotiana rustica: six generations	generation means	CR, each plant a unit	1	0	7
Simulated: generation means, plant by plant	generation means	RCB, by plant	1	0 (1 to check)	7

3.5 Data quality

The card **4 · Data quality** checks one trait at a time; the menu **Trait** in the corner switches from one to another. It has four parts: a summary by environment, the distribution, the field map and the list of outliers. The summary gives, for each environment (and for all together if there are several), the records, the missing values, the excluded ones, the mean, the standard deviation, the coefficient of variation, the minimum, the maximum and the skewness.

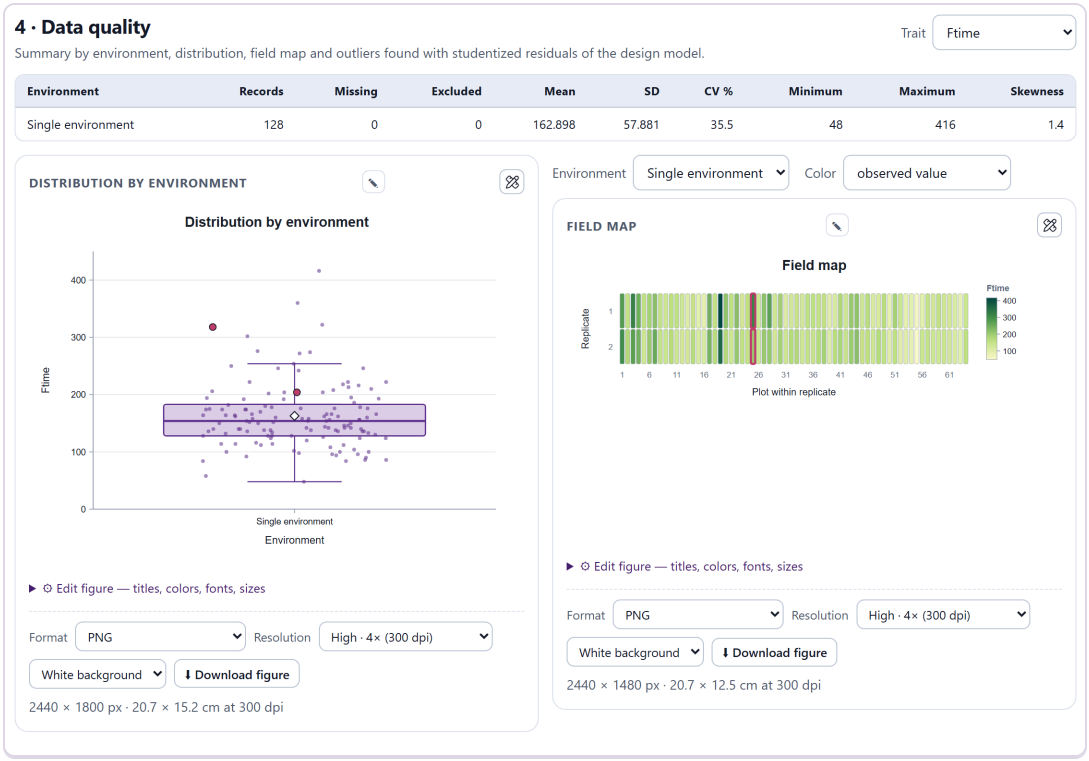


Figure 3.8. Card 4 with Hayman's diallel (flowering time). The red dots are the two plots of D × A flagged as outliers; the red frame on the map marks their position, the same in both replicates.

The **distribution by environment** is a box plot with the observations on top (they can be hidden) and a diamond at the mean; the values flagged as outliers are painted red. The **field map** puts each plot in its place: by row and column if the data have them and, if not, in the order of the rows within each replicate. The menu Color switches between the observed value and the standardized residual of the design model; the latter, on a diverging scale, reveals patches in the land (clustered residuals of the same sign) that the blocks did not absorb. Suspect plots carry a red border in both.

Outliers

Each observation is compared with what the design model predicts (for example, replicate + entry in complete blocks), by means of its externally studentized residual. Two levels: **outlier**, if the Bonferroni test gives $p < 0.05$, and **check**, if $|t| \geq 3.5$ without reaching that threshold. The table gives the environment, the entry, the replicate or block, the value, the fitted value, t, p and the rows of the file where it is.

> **IMPORTANT · WITH TWO REPLICATES OUTLIERS COME IN PAIRS**
>
> In complete blocks with two replicates, the two residuals of an entry are equal and of opposite sign: if one plot is abnormal, its partner is flagged with the same $|t|$. That is what happens with D × A ($t = -4.68$ and $+4.68$). Statistics cannot tell which of the two is wrong; the field book can.

Outliers

"Outlier": Bonferroni test on externally studentized residuals (Cook & Weisberg 1982), p < 0.05. "Check": |t| ≥ 3.5. Exclude only values with a proven error (typing, measurement or damaged plot): the whole block is recomputed.

Exclude	Level	Environment	Entry	Rep/block	Value	Fitted	t	p (Bonferroni)	Rows
☐	outlier	Single environment	D × A	2	204	262.055	−4.68	0.0021	90
☐	outlier	Single environment	D × A	1	318	259.945	4.68	0.0021	26

Figure 3.9. The two outliers of Hayman's diallel: the cross D × A flowered at 204 days in replicate 2 and at 318 in replicate 1, when the model expected about 260.

> **THEORY IN BRIEF · STUDENTIZED RESIDUALS AND BONFERRONI**
>
> $$t_i = e_i \,/\, [s_{(i)}\sqrt{(1 - h_{ii})}] \quad \cdot \quad p_{\text{Bonferroni}} = \min(1,\, n \cdot p_i)$$
>
> e_i is the residual of plot i, h_{ii} its leverage and $s_{(i)}$ the error standard deviation estimated **without** that plot, so that a very odd value does not inflate the scale by which it is judged. t_i follows a t with the error degrees of freedom minus one; since the n plots are tested at once, the p value is multiplied by n (Cook and Weisberg, 1982). Thus, in a trial of 128 plots, fewer than 0.05 false outliers in all are expected by pure chance, not six.

The checkbox Exclude removes the observation from **all** the analyses: those of this block, those of the genetic blocks and the report, where the excluded rows appear in gray. Use it only when there is a proven error (in data entry, in measurement or a damaged plot), not because a value gets in the way. Restore every excluded observation brings them all back at once, and the note below says how many are out.

3.6 The analysis of each environment

The card **5 · Field-design analysis** fits to each environment the model that matches its design (Table 3.5) and delivers the adjusted means of the entries. The menu **Environment** picks the environment or, if there are several, the **Combined analysis** (Section 3.7), which is what is shown when the data are loaded.

With sets (North Carolina I and II), the model adds set and replicate within set. In the α-lattice and incomplete blocks, the analysis of variance is intra-block (sequential sums of squares, with the entries adjusted for blocks), but the means are estimated with blocks as random effects by restricted maximum likelihood (REML; Patterson and Thompson, 1971), which recovers the inter-block information. If the REML fit is not possible, the means stay intra-block and the card says so.

Table 3.5. The model of each field design. *y* is the value of the plot (the mean of its plants when there are plant-level data).

Design	Model	Adjusted means
Completely randomized	$y = \mu + \text{entry} + e$	least squares
Randomized complete blocks	$y = \mu + \text{replicate} + \text{entry} + e$; Tukey's (1949) test for non-additivity	least squares
α-lattice	$y = \mu + \text{replicate} + \text{block(replicate)} + \text{entry} + e$; entries adjusted for blocks	REML, random blocks
Incomplete blocks	$y = \mu + \text{block} + \text{entry} + e$	REML, random blocks
Augmented blocks	$y = \mu + \text{block} + \text{entry} + e$, fixed blocks estimated from the checks (Federer, 1956)	least squares, adjusted for block
Rows and columns	$y = \mu + \text{replicate} + \text{row(rep.)} + \text{column(rep.)} + \text{entry} + e$	REML, random rows and columns
Entry means	one value per entry; the error comes from outside (Section 3.8)	the observed means

The tiles

The tile **Model** gives the design and how the means were obtained; **Grand mean**, how many plots (or plants, or means) went in. The **Coefficient of variation** comes with the error mean square (MSE) and its degrees of freedom; the **LSD (5 %)**, with the average standard error of a difference between two means. **Trial heritability** gives that of Cullis, Smith and Coombes (2006) and the standard one. In designs with incomplete blocks there are also **Efficiency vs complete blocks** and the block variance (or those of rows and columns) with that of the error. The colored edge of some tiles sums up the reading: green if the value is good, amber if it is acceptable and red if it is poor.

At Chapultepec, the 120 plots give a mean of 6.961 t/ha, a CV of 8.2% (MSE = 0.3264 with 41 df) and an LSD of 1.255. The entries, adjusted for blocks, differ clearly ($F = 11.92$ with 59 and 41 df). The estimated block variance, 0.1462, and that of the error, 0.3254, are close to the values used in the simulation (0.20 and 0.36); with only 20 blocks per location, the former is imprecise. The efficiency of 119% means that with complete blocks about 19% more replicates would have been needed to compare the entries with the same precision; without recovering the inter-block information (intra-block analysis) the gain would be only 4%.

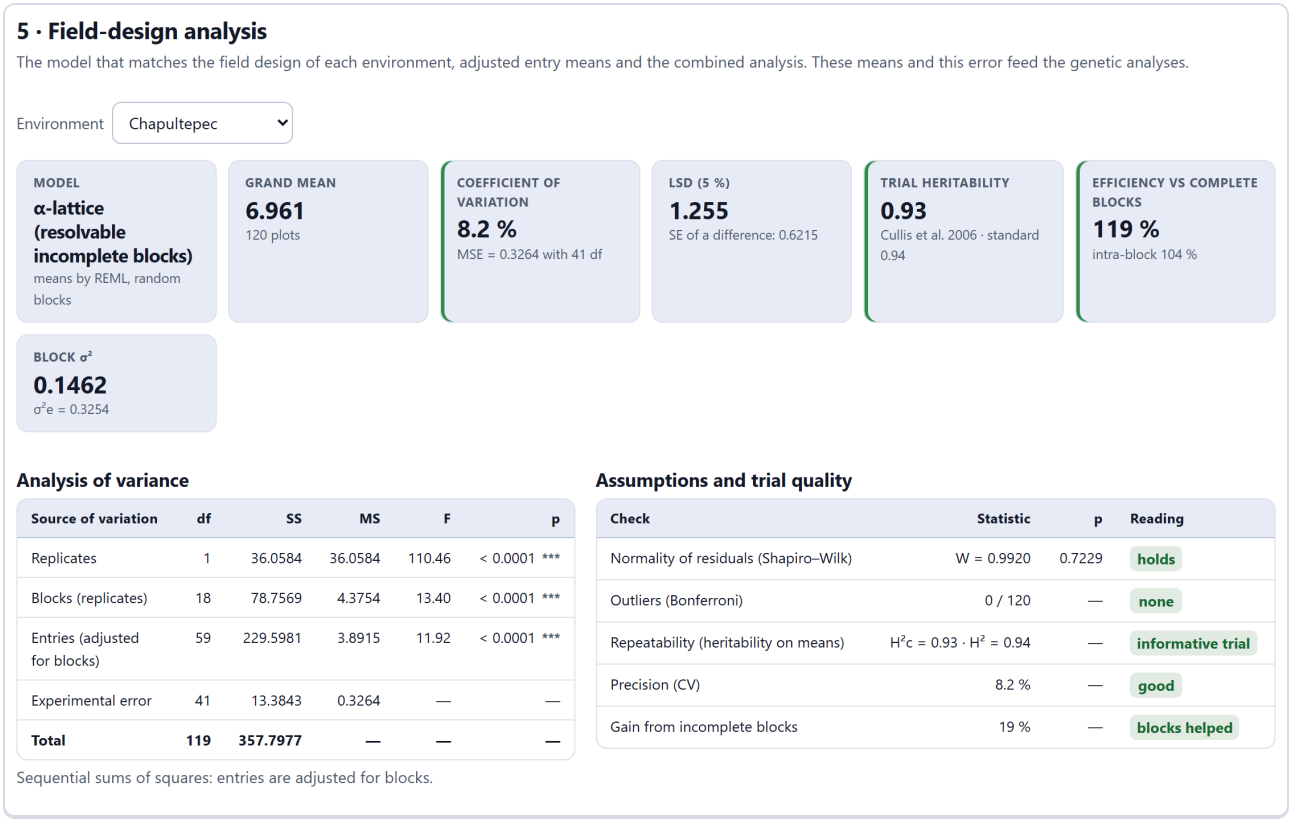


Figure 3.10. Card 5 at Chapultepec, one of the three locations of the α-lattice: summary tiles, intra-block analysis of variance and table of assumptions and trial quality.

THEORY IN BRIEF · TRIAL HERITABILITY AND EFFICIENCY

$$\text{standard } H^2 = \sigma^2_g / (\sigma^2_g + \sigma^2_e/r) \quad \cdot \quad H^2_{Cullis} = 1 - \bar{v}_\Delta / (2\sigma^2_g) \quad \cdot \quad \text{efficiency} = SED^2_{RCB} / SED^2_{design}$$

Entries are taken as random and checks as fixed (at least five entries that are not checks are needed); r is the harmonic mean of the replicates, $\bar{v}_\Delta$ the average prediction error variance of a difference and SED the average standard error of a difference. Cullis's heritability also holds with unbalanced data or incomplete blocks. Both are a **repeatability**: they say how well the trial ranks the entries. Narrow-sense heritability, with σ^2_A, is estimated in Blocks 4 to 8.

Assumptions and trial quality

To the right of the analysis of variance, a table checks the assumptions and gives a reading of each test, with the rules of Table 3.6. Tukey's test is only done in balanced randomized complete blocks, and the gain from incomplete blocks only in the α-lattice and in rows and columns.

Table 3.6. The tests of the assumptions table and how they are read.

Test	Fine	To check
Normality of residuals (Shapiro–Wilk)	$p \geq 0.05$: *holds*	$p < 0.05$: *check the Q–Q plot*
Replicate × entry additivity (Tukey, 1 df)	$p \geq 0.05$: *additive*	$p < 0.05$: *non-additive: try a transformation*
Outliers (Bonferroni)	none	some: *review in card 4*
Repeatability (H^2_c)	≥ 0.5: *informative trial*	0.3 to 0.5: *moderate*; < 0.3: *low: little discrimination*
Precision (CV)	≤ 12%: *good*	12 to 20%: *acceptable*; > 20%: *low*
Gain from incomplete blocks	≥ 5%: *blocks helped*	< 5%: *little gain*

At Chapultepec everything comes out fine: normal residuals ($W = 0.9920$, $p = 0.7229$), no outliers, $H^2_c = 0.93$, a CV of 8.2% and a gain of 19% from the blocks. Hayman's diallel is the opposite case: non-normal residuals ($p = 0.0011$) and non-additivity ($p = 0.0006$), which usually indicates that the trait should be transformed (for example, with the logarithm when the variance grows with the mean). The CV limits are only a guide: they depend on the crop and the trait.

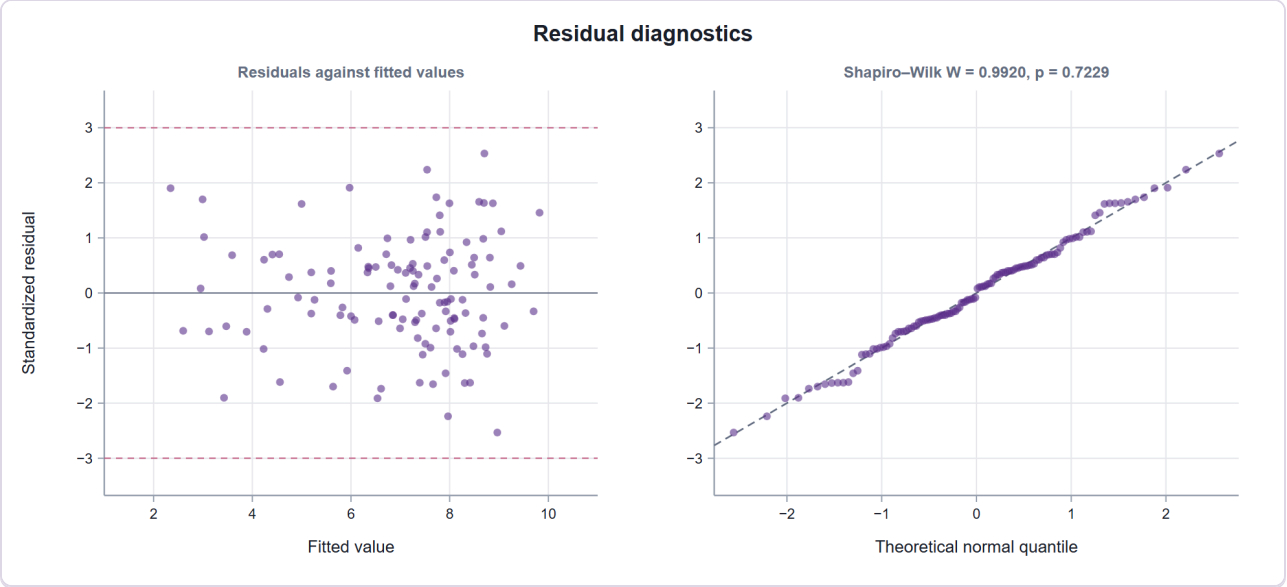


Figure 3.11. The residual diagnostics of Chapultepec. On the left, no pattern against the fitted values and within ±3; on the right, the quantiles follow the normal line.

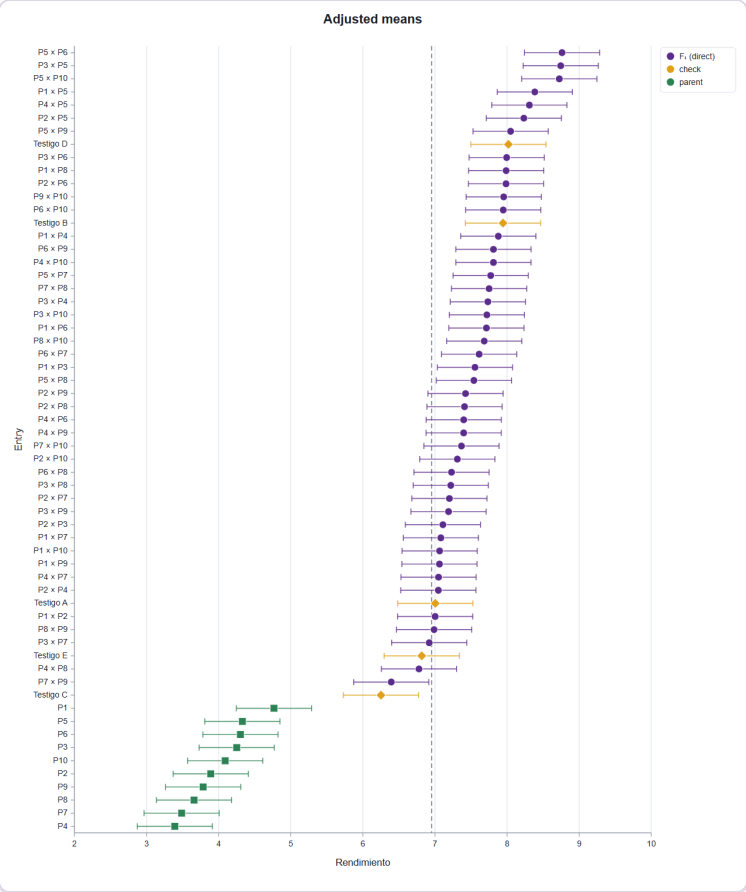


Figure 3.12. The adjusted means of the combined analysis (Section 3.7): the 45 crosses outperform the ten parents and the checks fall among them. The dashed line is the grand mean.

The adjusted means

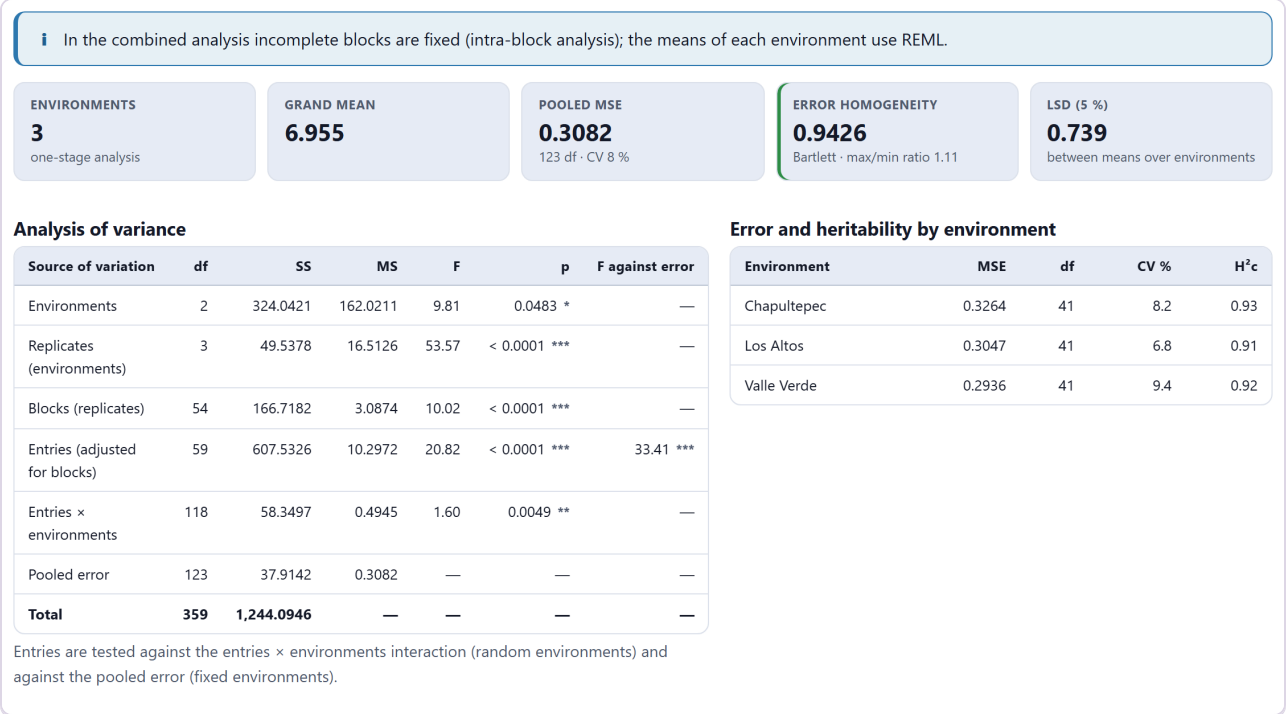
Below comes the table of **Adjusted means**, sorted from highest to lowest, with the rank, the entry, its type, the number of plots, the mean and its standard error. A dagger (†) marks the means that the design cannot estimate because the entry is not connected with the others. The figure of means (Figure 3.12, from the combined analysis) shows them with 95% confidence intervals and colors by type; its panel allows sorting them by name, limiting the number of entries (60 by default) and removing the intervals. On screen the table shows up to 300 rows; the download includes them all.

3.7 The combined analysis

With two or more environments, the menu **Environment** offers the **Combined analysis**. It is fitted in one stage, with all the plots:

$$y = \mu + \text{environment} + \text{replicate(environment)}\ [+\ \text{block(replicate)}] + \text{entry} + \text{entry} \times \text{environment} + e$$

The block term enters, as a fixed effect, only if all the environments are α-lattices. Environments are tested against replicates within environments, and entries **in two ways**: against the entry × environment interaction, which is the right test if the environments are a random sample of the possible ones (column *F*), and against the pooled error, which is right if the environments are fixed (column F against error).



i In the combined analysis incomplete blocks are fixed (intra-block analysis); the means of each environment use REML.

ENVIRONMENTS	GRAND MEAN	POOLED MSE	ERROR HOMOGENEITY	LSD (5 %)
3	6.955	0.3082	0.9426	0.739
one-stage analysis		123 df · CV 8 %	Bartlett · max/min ratio 1.11	between means over environments

Analysis of variance

Source of variation	df	SS	MS	F	p	F against error
Environments	2	324.0421	162.0211	9.81	0.0483 *	—
Replicates (environments)	3	49.5378	16.5126	53.57	< 0.0001 ***	—
Blocks (replicates)	54	166.7182	3.0874	10.02	< 0.0001 ***	—
Entries (adjusted for blocks)	59	607.5326	10.2972	20.82	< 0.0001 ***	33.41 ***
Entries × environments	118	58.3497	0.4945	1.60	0.0049 **	—
Pooled error	123	37.9142	0.3082	—	—	—
Total	**359**	**1,244.0946**	—	—	—	—

Entries are tested against the entries × environments interaction (random environments) and against the pooled error (fixed environments).

Error and heritability by environment

Environment	MSE	df	CV %	H^2c
Chapultepec	0.3264	41	8.2	0.93
Los Altos	0.3047	41	6.8	0.91
Valle Verde	0.2936	41	9.4	0.92

Figure 3.13. The combined analysis of the three locations of the α-lattice. The table on the right sums up the error and heritability of each environment.

The tiles give the number of environments and whether the analysis was in one or two stages, the grand mean, the pooled MSE, the **error homogeneity** with Bartlett's (1937) test and the ratio of the largest to the smallest MSE, and the LSD between means over all environments. In the example, the errors of the three locations are homogeneous (p = 0.9426, ratio 1.11), so combining them poses no problem. The entry × environment interaction is significant (F = 1.60, p = 0.0049) but small compared with the differences between entries (F = 20.82 against the interaction): the ranking of the crosses changes little from one location to another. The cross P5 × P6 has the highest mean, 8.7622 t/ha (Figure 3.12).

> **IMPORTANT · WHEN THE ERRORS ARE NOT HOMOGENEOUS**
>
> If Bartlett's test gives $p < 0.05$, the card warns that the combined F tests are approximate. Such data call for a model with one error variance per environment, which is in Block 9 (two-stage analysis with REML).

If the one-stage model would have more than 1,400 parameters (many entries in many environments), the analysis is done **in two stages** (Möhring and Piepho, 2009): first the adjusted means of each environment and then an analysis of the entry × environment table, whose error is the average variance of the adjusted means, on the plot scale. The card says so and states how many parameters the full model would have had. Entries that are not in every environment are left out of that second stage; in one stage, if there are empty entry × environment combinations, the overall means come from the model without interaction.

3.8 Special cases

Plant-level data

When several rows share the same plot (or there is a column Plant / individual), the data are plant-level. Each plot is summarized by the mean of its plants and the analysis of variance is done with those means; the row Between plants within plots is added separately, on the plant scale, with the sampling error (Steel, Torrie and Dickey, 1997). The note below the table recalls that the sums of squares of the plots are on the mean scale (they are multiplied by the number of plants to bring them to the plant scale) and tests the experimental error against the sampling error. In the North Carolina I example, with three plants per plot, the between-plants row has 48 df and an MS of 0.074; $F = 4.33$ ($p = 0.0002$) says that the plots differ more than the plants explain: there is a genuine plot error.

Plants without plots

In generation-mean experiments it is common to measure single plants, without replicates, blocks or plots. Then **each plant is an experimental unit** and the plants of each generation are its replicates: the design is completely randomized and the replicates tile gives the median number of plants per entry. With the six generations of *N. rustica* (160 plants) the result is a CV of 12%, an MSE of 1.8503 with 154 df, an LSD of 0.884 and a trial heritability of 0.98.

Entry means and external error

With a single value per entry, the data are means and the error cannot be estimated. Card 3 then shows the box Error of the original experiment: for each trait, the error mean square, its degrees of freedom, its scale and the replicates (on the plot scale, the MS is divided by the replicates). With it there are standard errors, LSD and tests in the genetic blocks; without it, only means. The mean examples bring their error: 21.05 with 2,558 df on the mean scale in the 36 maize crosses (LSD = 12.723), and 1.8 with 576 df on the plot scale, with four replicates, in the 17 genotypes (0.45 for the means).

Augmented blocks

In the augmented design, the checks are repeated in every block and the new entries appear once. The blocks are fixed and are estimated from the checks; each new entry is adjusted for the effect of its block (Federer, 1956). With two or more checks, the entries sum of squares is split into Checks and New entries and new vs. checks, and the note gives that of blocks eliminating entries.

Other cases

- **A design the data do not support** is replaced by a simpler one, with a warning: a lattice without real incomplete blocks becomes complete blocks, and so do rows and columns without those columns.
- **Entries without replicates and without repeated checks**: the design is *unreplicated* and there is no error; the app marks it in red.
- **Without degrees of freedom for the error**, or with fewer than three plots with data, the environment is not analyzed and the card explains why.

3.9 Downloads and moving on to the genetic blocks

At the foot of card 5 there are three downloads, which include **all** the traits and all the environments, not just what is on screen. Each figure also has its own download panel (Section I.6).

Table 3.7. The downloads of Block 3.

Button	Content
Results (.xlsx)	Five sheets: *Means* (by trait and environment, and combined); *ANOVA* (all sources, with df, SS, MS, *F* and *p*); *Quality* (design, plots, mean, CV, MSE, error df, LSD, Cullis and standard H^2, σ^2_g, block σ^2, *p* of Shapiro–Wilk and of Tukey, outliers); *Data* (the table with the column *Excluded*); *Design* (mating, method, parents, field design, environments and source)
Adjusted means (.csv)	Trait, environment, entry, type, plots, adjusted mean and standard error
Cleaned data (.csv)	The original table with a column *Excluded* (1 if the observation was left out)

In addition, at the end of card 5, How is each design analyzed? sums up the methods of Sections 3.5 to 3.7 with their references, ready for the materials and methods section of an article; the report of Block 12 includes them in full prose.

The button Continue at the end changes according to the recognized mating design (Table 3.8); in the example it reads Continue: Griffing →. ← Plan goes back to Block 2. The other blocks are always one click away in the list of blocks on the left.

Table 3.8. Where the button Continue leads.

Recognized design	Next block
Griffing diallel (complete)	4 Griffing diallels (and then 5)
Partial diallel, North Carolina designs, triple test cross, line × tester	6 Mating designs
Generation means	7 Generations
Genotype trial in one environment	8 Genetic parameters and selection
Genotype trial in several environments	9 Genotype × environment

What passes to the following blocks is the table with its roles, the records with their pedigree, the mating and field designs, the excluded observations, the external error and, for each trait, the analysis of this card (adjusted means, MSE and degrees of freedom), which is computed the first time a block asks for it. Any change in Block 3 (a role, the design, an exclusion) is reflected in them when they are opened again. As in the rest of the app, the loaded data and the results are not kept when the page is reloaded; only the data sheet is kept in the browser.

RULE OF THUMB · BEFORE MOVING ON TO THE GENETIC BLOCKS

If a column was left with a wrong role or as Ignore…	**then** correct it in card 2: everything else depends on it.
If the mating design has medium or low confidence…	**then** read the reasons and check Recognized entries and the figure of crosses present.
If there are outliers…	**then** compare them with the field book and exclude only proven errors.
If normality or additivity fail…	**then** consider transforming the trait before loading it again.

The genetic blocks use exactly what remains in Block 3: its data, its design, the excluded observations and the external error.

B4

CHAPTER FOUR · BLOCK 4 · GRIFFING

B4

Griffing diallels

The diallel is the classic design for choosing parents: you cross a set of lines among themselves and separate what each one passes on to all its crosses (general combining ability) from what appears only in particular combinations (specific combining ability). Block 4 runs Griffing's (1956) analysis with its four methods and two models, in one or in several environments, and helps you choose the best parents and crosses.

4.1 The four methods and the two models

4.2 Data and model

4.3 The analysis of variance of the diallel

4.4 The effects

4.5 Variance components and genetic parameters

4.6 Best parents and crosses

4.7 The four methods on the same data

4.8 Several environments

4.9 Special cases, downloads and moving on to Block 5

4.1 The four methods and the two models

Griffing (1956) distinguished four ways of running a diallel with p parents, depending on whether the parents themselves and the reciprocal crosses are evaluated, and two ways of interpreting the results, depending on whether the parents matter in their own right or represent a population.

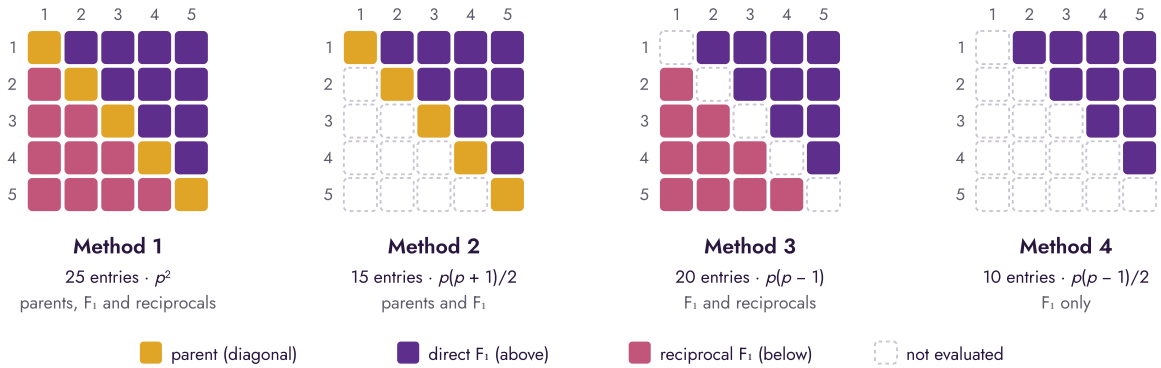


Figure 4.1. The diallel cells each method uses, with five parents. Rows are the female and columns the male. Block 4 recognizes the method in the data and lets you analyze the same data with any of the methods they contain.

Model I treats the parents as fixed: they are the ones of interest, and what you are after are their effects (which one passes on more, which cross stands out). **Model II** treats them as a random sample from a population: what matters are the variances of that population, from which the additive variance, the dominance variance and the heritabilities are derived. Both models fit the same equation; the F tests and the interpretation change.

THEORY IN BRIEF · GRIFFING'S MODEL

$$\bar{x}_{ij} = \mu + g_i + g_j + s_{ij} + r_{ij} + \bar{e}_{ij} \quad \cdot \quad s_{ij} = s_{ji} \quad \cdot \quad r_{ij} = -r_{ji} = m_i - m_j + n_{ij}$$

The mean of the cross $i \times j$ is the grand mean plus the general combining ability (GCA) of each parent, g_i and g_j, plus the specific combining ability (SCA) of the pair, s_{ij}, plus the reciprocal effect r_{ij}, which exists only in Methods 1 and 3. The reciprocal effect splits into a maternal part, $m_i - m_j$, which depends only on which parent was the female, and a non-maternal part, n_{ij} (Cockerham, 1963). Griffing's restrictions ($\Sigma g_i = 0$ and the SCA effects of each parent summing to zero) make the effects read as deviations from the average of the set.

Throughout the chapter you will use the **full 4 × 4 maize diallel** of Pérez López et al. (2020): four lines, L1 to L4, with their 12 direct and reciprocal crosses and the lines themselves, in complete blocks with six replications; the trait is grain test weight (PVG, in g/L). It is a Method 1 diallel, with everything the block can show, and its published results let you check the app. Section 4.8 uses the simulated α-lattice of Chapter 3 for the analysis across several environments.

4.2 Data and model

Block 4 does not read files: it takes the adjusted means of each entry and the error of each environment from Block 3, with any observations you excluded there. If the loaded data are not a complete Griffing diallel, the card tells you which block analyzes them (partial diallels, the North Carolina designs and line × tester in Block 6; generations in Block 7; genotype trials in Block 8 or 9). The card **1 · Data and model** summarizes the data in one line and offers the controls in Table 4.1.

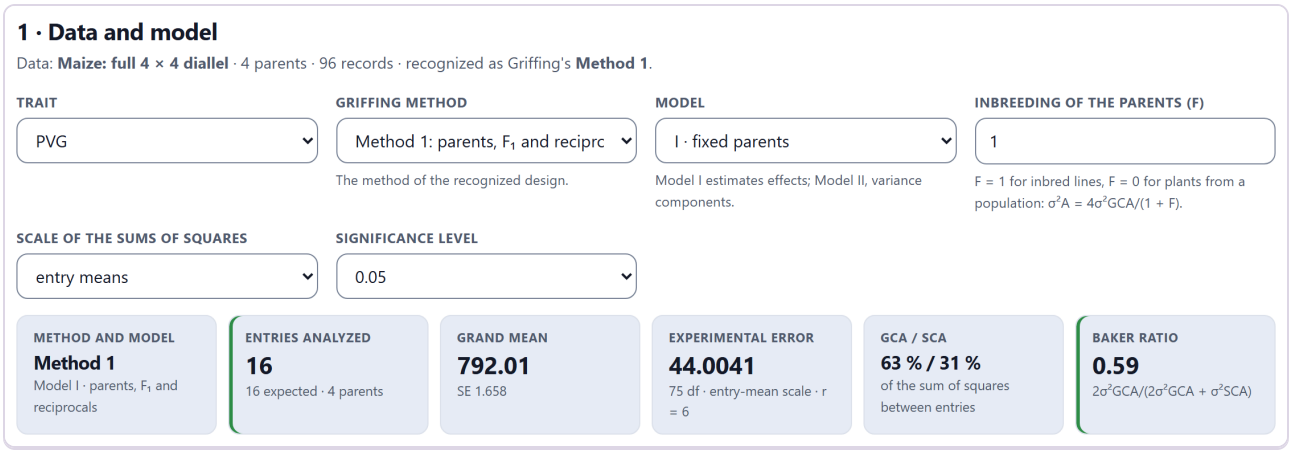


Figure 4.2. Card 1 with the 4 × 4 maize diallel. The tiles summarize the method, the entries, the grand mean, the error, the split of the sum of squares between GCA and SCA, and the Baker ratio.

Table 4.1. The controls of card 1.

Control	Options	What it is for
Trait	those of Block 3	The whole block is recomputed for the chosen trait
Griffing method	those the data contain	Analyze a subset of the entries, as if the experiment had been designed with that method; the recognized one is marked
When reciprocals are dropped	direct cross · average of both orientations	Only in Methods 2 and 4 with data that include reciprocals: which value represents each pair
Model	I · fixed parents · II · random parents	Which *F* tests are shown first and how the components are read
Environments	fixed · random	With several environments: what each effect is tested against (Section 4.8)
Inbreeding of the parents (F)	from 0 to 1; 1 by default	Converts σ^2_{GCA} and σ^2_{SCA} into σ^2_A and σ^2_D: 1 for inbred lines, 0 for plants from a population
Scale of the sums of squares	entry means · plots	Changes only the presentation; the *F* tests are the same
Significance level	0.05 · 0.01 · 0.10	For the least significant differences and the readings

With the maize data, the block analyzes the 16 entries expected in Method 1, with a grand mean of 792.01 g/L and an error of 44.0041 (the error mean square from Block 3 divided by the six replications) with 75 degrees of freedom. GCA accounts for 63% of the sum of squares between entries and SCA for 31%; the rest are reciprocal effects. The entries tile turns amber if any entry of the complete design is missing.

> **NOTE · A METHOD OTHER THAN THE DESIGN**
>
> With data from a Method 1 diallel you can request Methods 2, 3 or 4: the block takes only the entries that method uses (without reciprocals, without parents, or without either), repeats the Block 3 field analysis with them and estimates its own error. This gives you what an experiment designed with that method would have produced. Section 4.7 makes the comparison of all four at once.

4.3 The analysis of variance of the diallel

The card **2 · Analysis of variance of the diallel** partitions the variation between entries into GCA, SCA and reciprocal effects, and the latter into maternal and non-maternal. It has two *F* columns: the one for the chosen model, with its *p* value and its denominator, and the one for the other model, on the right, so you can compare without switching models.

2 · Analysis of variance of the diallel

Partition of the entries into GCA, SCA and reciprocal effects (and these into maternal and non-maternal).

Source of variation	df	SS	MS	F · Model I	p	Denominator	F · Model II
Entries	**15**	**7,970.9149**	**531.3943**	—	—	—	—
General combining ability (GCA)	3	5,027.066	1,675.6887	38.08	< 0.0001 ***	Experimental error	4.10 ns
Specific combining ability (SCA)	6	2,452.599	408.7665	9.29	< 0.0001 ***	Experimental error	9.29 ***
Reciprocal effects	6	491.25	81.875	1.86	0.0988 ns	Experimental error	1.86 ns
Maternal	3	40.125	13.375	0.30	0.8224 ns	Experimental error	0.30 ns
Non-maternal	3	451.125	150.375	3.42	0.0216 *	Experimental error	3.42 *
Experimental error	75	3,300.3038	44.0041	—	—	—	—

Sums of squares on the entry-mean scale; the error is MS_{error}/r = 44.0041 with 75 df.

Figure 4.3. The analysis of variance of the 4 × 4 maize diallel. GCA and SCA are highly significant; the reciprocal effects are not significant as a whole, but their non-maternal part is (p = 0.0216).

Table 4.2. What each source is tested against, by model, in one environment.

Source	Model I (fixed parents)	Model II (random parents)
GCA	experimental error	SCA (approximate in Method 1)
SCA	experimental error	experimental error
Reciprocal, maternal and non-maternal	experimental error	experimental error

In the maize data, GCA has F = 38.08 against the error (Model I), but only F = 4.10 against SCA (Model II), which is not significant with 3 and 6 degrees of freedom: with four parents there is very little information to speak about the variance of a population. That is the underlying difference between the models. Under Model I the question is whether these four lines differ (they clearly do); under Model II, whether the population they came from has additive variance, and four lines are not enough to say so.

> **VERIFIED · THE PUBLISHED VALUES**
>
> With Scale of the sums of squares set to plots, the sums of squares are those of the source article: 30,162.40 for GCA, 14,715.59 for SCA, 2,947.50 for reciprocals, 240.75 for maternal and 2,706.75 for non-maternal effects, with F = 38.08, 9.29, 1.86, 0.30 and 3.42. The GCA effects and their standard errors (2.0311) also match to the last decimal.

With missing entries, the sums of squares are sequential, in the order GCA, SCA and reciprocals, and the note below the table says so. With several environments, the table adds the environments and the interactions of each source with them, and the collapsed panel Analysis within each environment repeats the analysis environment by environment (Section 4.8).

B4

4.4 The effects

The card 3 · Effects presents the estimates with their standard error and their t test. On the left, the GCA effects sorted from highest to lowest; on the right, the standard errors and least significant differences for each kind of comparison, because not all differences have the same precision: comparing two SCA effects that share a parent is not the same as comparing two with no parent in common.

3 · Effects

General combining ability

#	Parent	$\hat{g}_i$	SE	t	p
1	L3	20.0104	2.0311	9.85	< 0.0001 ***
2	L4	−0.1146	2.0311	−0.06	0.9552 ns
3	L1	−6.0729	2.0311	−2.99	0.0038 **
4	L2	−13.8229	2.0311	−6.81	< 0.0001 ***

Comparisons and least significant differences

Difference	SE	LSD (0.05)
Between two GCA effects ($\hat{g}_i - \hat{g}_j$)	3.3168	6.6074
Between two SCA effects sharing a parent ($\hat{s}_{ij} - \hat{s}_{ik}$)	5.7448	11.4443
Between two SCA effects with no parent in common ($\hat{s}_{ij} - \hat{s}_{kl}$)	4.6906	9.3442
Differences involving an $\hat{s}_{ii}$	6.6336	13.2147
Between two non-maternal effects ($\hat{n}_{ij} - \hat{n}_{kl}$)	4.9936	9.9477
Between two maternal effects ($\hat{m}_i - \hat{m}_j$)	3.3168	6.6074

Each kind of difference has its own standard error; with unbalanced data the average is reported.

Figure 4.4. The GCA effects of the maize lines and the least significant differences. L3 passes on to its crosses 20 g/L more than the average; L2, 13.8 less. Two GCA effects differ if they are more than 6.61 g/L apart.

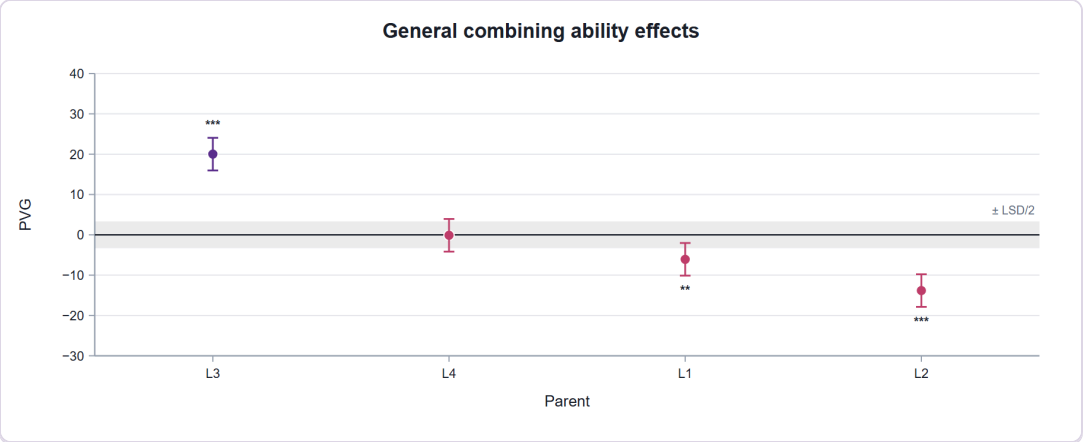


Figure 4.5. The figure of GCA effects, with 95% confidence intervals, significance asterisks and the gray band of ± LSD/2 around zero. Positive effects are purple and negative ones pink; in the editor you can switch to bars, sort by name and remove the band or the intervals.

Specific combining ability is shown as a color matrix: above the diagonal, the SCA of each cross; on the diagonal, that of each parent ($\hat{s}_{ii}$, in Methods 1 and 2); below, when they exist, the reciprocal effects. Red is positive and blue negative, with the significance asterisks next to the value. The full SCA table, with standard error and p, is collapsed below the figure.

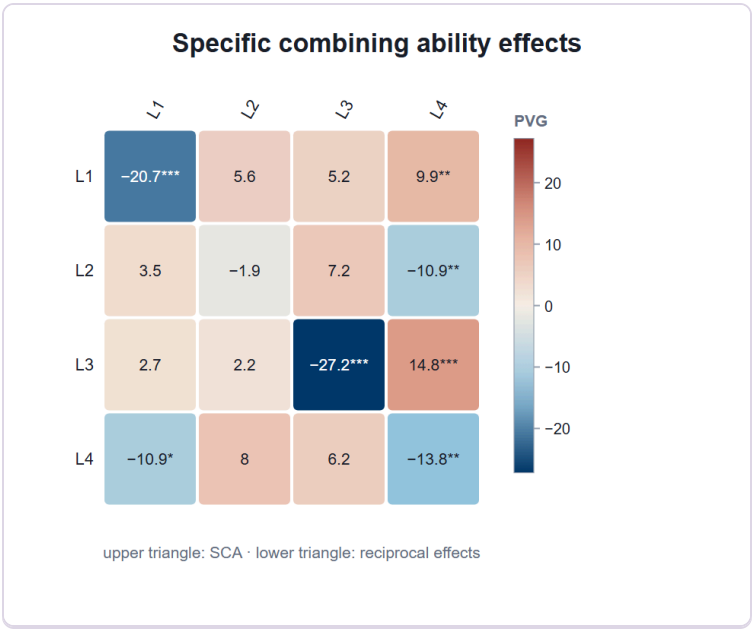


Figure 4.6. SCA (above) and reciprocal effects (below) of the maize diallel. The best specific combination is L3 × L4 ($\hat{s}$ = 14.8); the diagonal values, all negative, say that each line performs worse as a parent than its GCA predicts.

With reciprocals (Methods 1 and 3), the panel of **reciprocal, maternal and non-maternal effects** also appears: on the left, the maternal effect of each parent, m_i; on the right, for each pair, the reciprocal effect r_{ij} and its non-maternal part n_{ij}, each with its test.

Reciprocal, maternal and non-maternal effects

$\hat{r}_{ij} = \hat{m}_i - \hat{m}_j + \hat{n}_{ij}$: the maternal part depends only on which parent was the female (Cockerham 1963).

Parent	$\hat{m}_i$	SE	t	p
L1	−1.1875	2.0311	−0.58	0.5605 ns
L2	1.6875	2.0311	0.83	0.4087 ns
L3	0.3125	2.0311	0.15	0.8781 ns
L4	−0.8125	2.0311	−0.40	0.6903 ns

Pair (female × male)	$\hat{r}_{ij}$	SE	p	$\hat{n}_{ij}$	p
L1 × L2	3.5	4.6906	0.4579 ns	6.375	0.0584 ns
L1 × L3	2.6667	4.6906	0.5714 ns	4.1667	0.2129 ns
L1 × L4	−10.9167	4.6906	0.0226 *	−10.5417	0.0022 **
L2 × L3	2.25	4.6906	0.6329 ns	0.875	0.7926 ns
L2 × L4	8	4.6906	0.0922 ns	5.5	0.1014 ns
L3 × L4	6.1667	4.6906	0.1926 ns	5.0417	0.1327 ns

Figure 4.7. Maternal and reciprocal effects of the maize diallel. No parent has a maternal effect; the only difference between reciprocals is in the pair L1 × L4 (r = −10.92, p = 0.0226), and it is non-maternal.

READING RULE · WHICH EFFECT IS USEFUL FOR WHAT

High, significant GCA ($\hat{g}_i$)	The parent passes its value on to all its crosses: useful for **forming populations and synthetics** and for choosing the parents of a program.
High, significant SCA ($\hat{s}_{ij}$)	The pair performs better than its GCA effects predict: useful for **hybrids** of that combination, not for choosing parents.
Maternal effect (m_i)	Which parent is the female matters: use the parent in the favorable direction (cytoplasm, endosperm, seed size).
Non-maternal reciprocal (n_{ij})	The direction matters only in that pair; confirm it before drawing conclusions.

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4.5 Variance components and genetic parameters

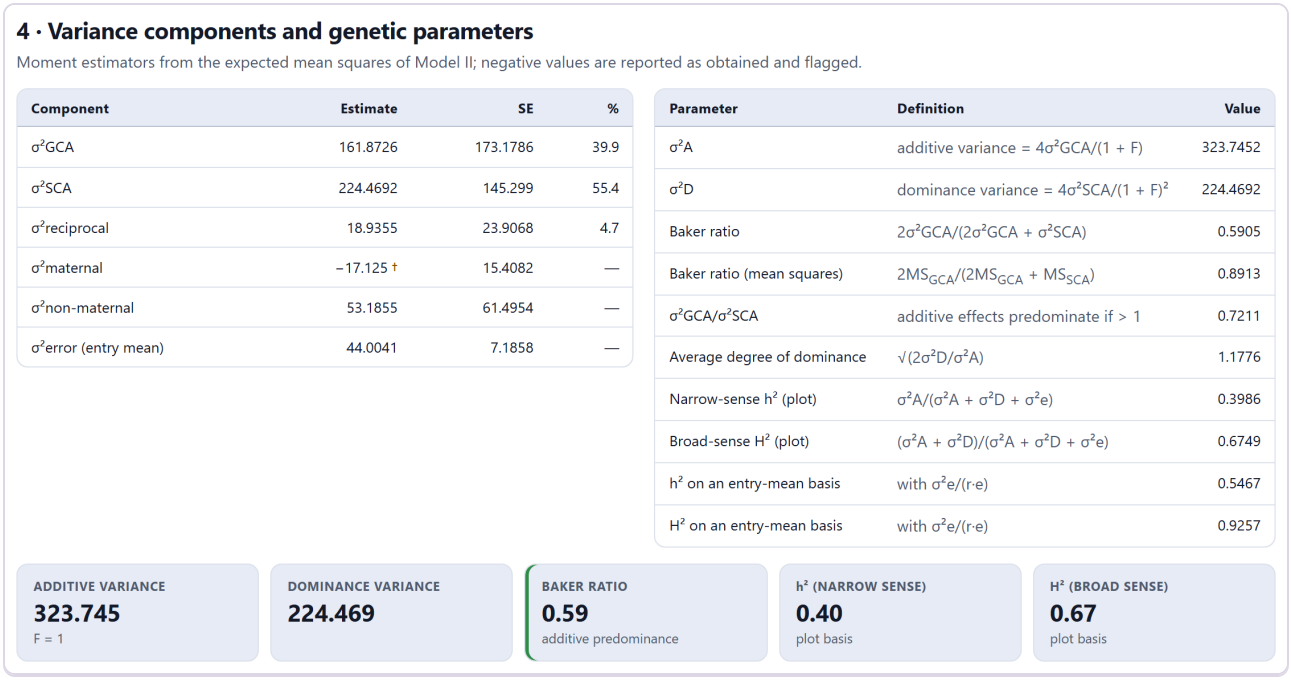
The card 4 · Variance components and genetic parameters estimates the Model II variances from Griffing's expected mean squares (moment estimators), with their standard error and their percentage of the genetic variance. On the right it converts them into genetic parameters, and below it summarizes them in five tiles and a bar chart with intervals.

THEORY IN BRIEF · FROM COMPONENTS TO PARAMETERS

$\sigma^2_A = 4\sigma^2_{GCA}/(1+F)$ · $\sigma^2_D = 4\sigma^2_{SCA}/(1+F)^2$ · Baker ratio $= 2\sigma^2_{GCA}/(2\sigma^2_{GCA} + \sigma^2_{SCA})$

average degree of dominance $= \sqrt{(2\sigma^2_D/\sigma^2_A)}$ · $h^2 = \sigma^2_A/(\sigma^2_A + \sigma^2_D + \sigma^2_e)$ · $H^2 = (\sigma^2_A + \sigma^2_D)/(\sigma^2_A + \sigma^2_D + \sigma^2_e)$

The conversions assume no epistasis, genes in linkage equilibrium and parents with the stated inbreeding F (Griffing, 1956; Wright, 1985). The Baker (1978) ratio runs from 0 to 1: close to 1, GCA predicts the crosses well and additive action dominates. Heritabilities are given on a plot basis and on an entry-mean basis (with σ^2_e divided by the replications and the environments). The standard errors of the components come from the variance of the mean squares, $2MS^2/df$.



4 · Variance components and genetic parameters

Moment estimators from the expected mean squares of Model II; negative values are reported as obtained and flagged.

Component	Estimate	SE	%
σ^2GCA	161.8726	173.1786	39.9
σ^2SCA	224.4692	145.299	55.4
σ^2reciprocal	18.9355	23.9068	4.7
σ^2maternal	−17.125 †	15.4082	—
σ^2non-maternal	53.1855	61.4954	—
σ^2error (entry mean)	44.0041	7.1858	—

Parameter	Definition	Value
σ^2A	additive variance = $4\sigma^2$GCA/(1 + F)	323.7452
σ^2D	dominance variance = $4\sigma^2$SCA/(1 + F)^2	224.4692
Baker ratio	$2\sigma^2$GCA/($2\sigma^2$GCA + σ^2SCA)	0.5905
Baker ratio (mean squares)	$2MS_{GCA}/(2MS_{GCA} + MS_{SCA})$	0.8913
σ^2GCA/σ^2SCA	additive effects predominate if > 1	0.7211
Average degree of dominance	√($2\sigma^2$D/σ^2A)	1.1776
Narrow-sense h^2 (plot)	σ^2A/(σ^2A + σ^2D + σ^2e)	0.3986
Broad-sense H^2 (plot)	(σ^2A + σ^2D)/(σ^2A + σ^2D + σ^2e)	0.6749
h^2 on an entry-mean basis	with σ^2e/(r·e)	0.5467
H^2 on an entry-mean basis	with σ^2e/(r·e)	0.9257

Figure 4.8. The components and parameters of the maize diallel. $\sigma^2_{maternal}$ comes out negative and carries a dagger: it is reported as obtained.

In the maize data, σ^2_A = 323.75 and σ^2_D = 224.47; the Baker ratio is 0.59, with additive predominance, and the average degree of dominance is 1.18. Narrow-sense heritability is 0.40 on a plot basis and 0.55 on an entry-mean basis. But the standard errors tell you how far to trust those numbers: σ^2_{GCA} = 161.87 ± 173.18 is not different from zero. With four parents the variances are imprecise, and the message boxes remind you of it.

IMPORTANT · WHAT THE CARD'S WARNINGS SAY

- **Negative components:** they are moment estimates and are reported as obtained, marked with †. They indicate a component close to zero or a model that does not describe the data well. If σ^2_{GCA} or σ^2_{SCA} is negative, the ratios and heritabilities are not computed.
- **Model I:** the parents were chosen on purpose, so σ^2_{GCA} and σ^2_{SCA} describe only this set. Reading them as the σ^2_A and σ^2_D of a population requires the parents to be a random sample of it.
- **Methods 1 and 2:** with the parents included, the components describe the descendant population; for the ancestral population from which the parents were taken, use Methods 3 or 4 (Kuehl et al., 1968; Wright, 1985).

4.6 Best parents and crosses

The card 5 · Best parents and crosses ranks parents and crosses according to what you are after: the Objective menu chooses the highest value (yield, grain weight) or the lowest value (days to flowering, height, incidence of a disease), and Show limits the table of crosses to 10, 20 or 50 rows, or all.

5 · Best parents and crosses

Objective: the highest value | Show: 20

Parents by their GCA

#	Parent	$\hat{g}_i$	p	Reading
1	L3	20.01	< 0.0001 ***	favorable
2	L4	−0.115	0.9552 ns	no difference
3	L1	−6.073	0.0038 **	unfavorable
4	L2	−13.823	< 0.0001 ***	unfavorable

Crosses evaluated

#	Cross	Mean	$\hat{\mu} + \hat{g}_i + \hat{g}_j$	$\hat{s}_{ij}$	p
1	L3 × L4	826.667	811.906	14.76	0.0002 ***
2	L1 × L3	811.167	805.948	5.219	0.1635 ns
3	L2 × L3	805.417	798.198	7.219	0.0553 ns
4	L3 (parent)	804.833	812.021	−27.198	< 0.0001 ***
5	L1 × L4	795.75	785.823	9.927	0.0091 **
6	L4 (parent)	778	791.896	−13.781	0.0071 **
7	L1 × L2	777.667	772.115	5.552	0.1385 ns
8	L2 × L4	767.167	778.073	−10.906	0.0043 **
9	L2 (parent)	762.5	778.187	−1.865	0.7089 ns
10	L1 (parent)	759.167	785.937	−20.698	< 0.0001 ***

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Figure 4.9. The maize parents by their GCA, with a reading in words, and the crosses evaluated with their mean, the prediction from GCA ($\mu + \hat{g}_i + \hat{g}_j$) and their SCA. The parents appear among the crosses with their $\hat{s}_{ij}$.

The **Reading** column marks each parent as *favorable*, *unfavorable* or *no difference*, according to the sign of its GCA and its significance in the desired direction. In the table of crosses, the difference between the observed mean and the one predicted from the GCA effects is the SCA: L3 × L4 averages 826.67 g/L, 14.76 more than L3 and L4 predict.

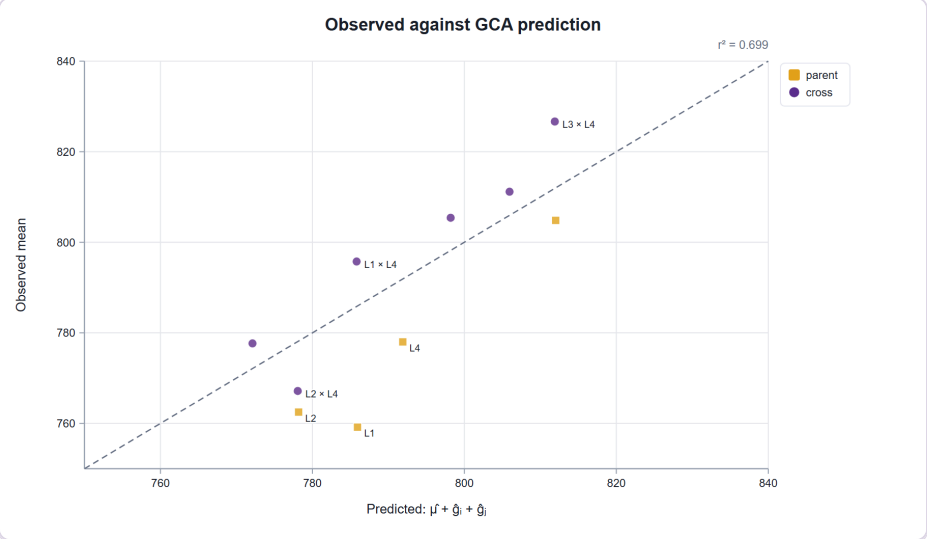

Figure 4.10. Observed mean against the mean predicted from GCA. Points above the diagonal have positive SCA; the parents (squares), all below it, perform worse than their GCA effects predict. In the maize data, GCA explains 70% of the variation between entries ($r^2 = 0.699$).

At the end of the card, a reading summarizes the results in words. With the maize data it says that the parent with the best GCA is L3, that the crosses with the highest value are L3 × L4, L1 × L3 and L2 × L3, that the outstanding specific combinations are L3 × L4 and L1 × L4 (of interest for hybrids, not for choosing parents), and that there is a significant reciprocal effect, so it pays to watch which parent is used as the female.

If crosses of the design are missing, the table **Crosses not evaluated: prediction from GCA** also appears, with $\mu + \hat{g}_i + \hat{g}_j$ and its standard error for each cross that was not made. It is the average expected value of the combination, without its SCA, which cannot be known without evaluating it.

4.7 The four methods on the same data

The card **6 · The four methods on the same data** analyzes the data with each method they contain. Each method uses its own subset of entries and estimates its own error, as if the experiment had been designed that way. The table gives the number of entries, the error, the sums of squares of GCA, SCA and reciprocals, the components, the Baker ratio and what each method estimates; the figure below compares the sums of squares or the components.

6 · The four methods on the same data

Each method uses a subset of the entries and estimates its own error, as if the experiment had been designed that way. What changes between methods is not only precision: the reference population changes too (with parents it describes the descendant population; with F_1 only, the ancestral one).

Method	Entries	Error df	Error MS	GCA SS	SCA SS	Reciprocal SS	σ^2GCA	σ^2SCA	Baker	What it estimates
Method 1	16 / 16	75	44.004	5,027.07 ***	2,452.6 ***	491.25 ns	161.873	224.469	0.59	descendant population; includes maternal effects
Method 2	10 / 10	45	28.434	3,234.18 ***	1,876.38 ***	—	127.555	284.296	0.47	descendant population; no reciprocal effects
Method 3	12 / 12	55	46.325	4,430.44 ***	402.46 *	491.25 ns	318.896	77.453	0.89	ancestral population; with maternal effects
Method 4	6 / 6	25	26.815	2,342.96 ***	183.25 *	—	344.681	64.81	0.91	ancestral population; the most efficient per cross

Figure 4.11. The four methods with the maize data. Without the parents (Methods 3 and 4), σ^2_{SCA} drops from 224–284 to 65–77 and the Baker ratio rises from 0.47–0.59 to 0.89–0.91.

The comparison teaches something a single analysis hides. In Methods 1 and 2, the parents enter the analysis with their own very negative SCA effects ($\hat{s}_{ii}$): the lines perform much worse than their crosses (inbreeding depression or, seen from the crosses, heterosis), and Griffing's model loads that difference onto SCA. That is why σ^2_{SCA} comes out larger and the Baker ratio smaller. Among the crosses alone (Methods 3 and 4), GCA explains almost everything. None of the results is wrong: they describe different populations, the descendant one with parents and the ancestral one without them.

4

KEY FACT · THE FOUR METHODS, VERIFIED

With the 8 × 8 diallel from the book by Singh and Chaudhary (1979), Block 4 reproduces exactly, in all four methods, the sums of squares, the errors, the Model II components and the eight GCA effects of an independent implementation. Those data give a negative σ^2_{GCA} in all four methods, the typical case in which the ratios are not computed.

4.8 Several environments

With data from several environments, Block 4 performs a two-stage analysis: it takes the adjusted means of each environment from Block 3 and analyzes the entries × environments table with the pooled error. The analysis of variance table adds the environments and the interaction of each source with them (GCA × environments, SCA × environments and, in Methods 1 and 3, reciprocals × environments). The Environments control decides what the effects are tested against: with **fixed** environments, against the error; with **random** environments, each effect against its interaction with the environments. Under Model II with several environments, GCA is tested with a quasi-*F* and Satterthwaite degrees of freedom (Singh, 1973).

2 · Analysis of variance of the diallel

Partition of the entries into GCA, SCA and reciprocal effects (and these into maternal and non-maternal).

Source of variation	df	SS	MS	F · Model I	p	Denominator	F · Model II
Environments	2	143.0184	71.5092	356.05	< 0.0001 ***	Experimental error	356.05 ***
Entries	**54**	**349.164**	**6.466**	—	—	—	—
General combining ability (GCA)	9	21.5244	2.3916	11.91	< 0.0001 ***	Experimental error	0.31 ns
Specific combining ability (SCA)	45	327.6397	7.2809	36.25	< 0.0001 ***	Experimental error	29.32 ***
Entries × environments	**108**	**32.9551**	**0.3051**	—	—	—	—
GCA × environments	18	10.6025	0.589	2.93	0.0003 ***	Experimental error	2.37 **
SCA × environments	90	22.3526	0.2484	1.24	0.1450 ns	Experimental error	1.24 ns
Experimental error	108	21.6905	0.2008	—	—	—	—

Sums of squares on the entry-mean scale; the error is 0.2008 with 108 df. The means come from an incomplete-block design and are less precise than MS_{error}/r: the error of this analysis is the average variance of the adjusted means, 27 % larger than the intra-block mean square (Möhring & Piepho 2009). The error is pooled over the 3 environments (108 df); the blocks of each environment were already removed in Block 3.

Figure 4.12. The simulated α-lattice (Method 2) at three locations. GCA changes between locations (GCA × environments, p = 0.0003); SCA does not (p = 0.145). The note explains the error of the second stage.

NOTE · THE ERROR WHEN THERE ARE INCOMPLETE BLOCKS

The adjusted means of an α-lattice are less precise than MS_{error}/r, so the error of the second stage is the average variance of the adjusted means, on the plot scale (Möhring and Piepho, 2009): in the example, 0.2008 instead of 0.1583. With balanced complete blocks the two are equal.

The figure GCA in each environment shows, for each parent, its effect at each location. Parallel lines indicate stable GCA; crossing lines, parents whose value depends on the environment. In the example, P5 is the best at Los Altos and Valle Verde and the second best at Chapultepec, while P9 goes from 0.15 at Los Altos to −0.55 at Valle Verde.

Because the data are simulated, the block compares the estimates with the truth: below the GCA table, a note gives the correlation between the $\hat{g}_i$ and the true effects (0.863). With Method 2, σ^2_{GCA} comes out negative (−0.145) and σ^2_{SCA} very large (2.34): the crosses average 7.5 t/ha and the inbred parents 4.2, and that heterosis is added to SCA, as in Section 4.7. With Method 4, using only the crosses, the components come close to the simulated values (Table 4.3).

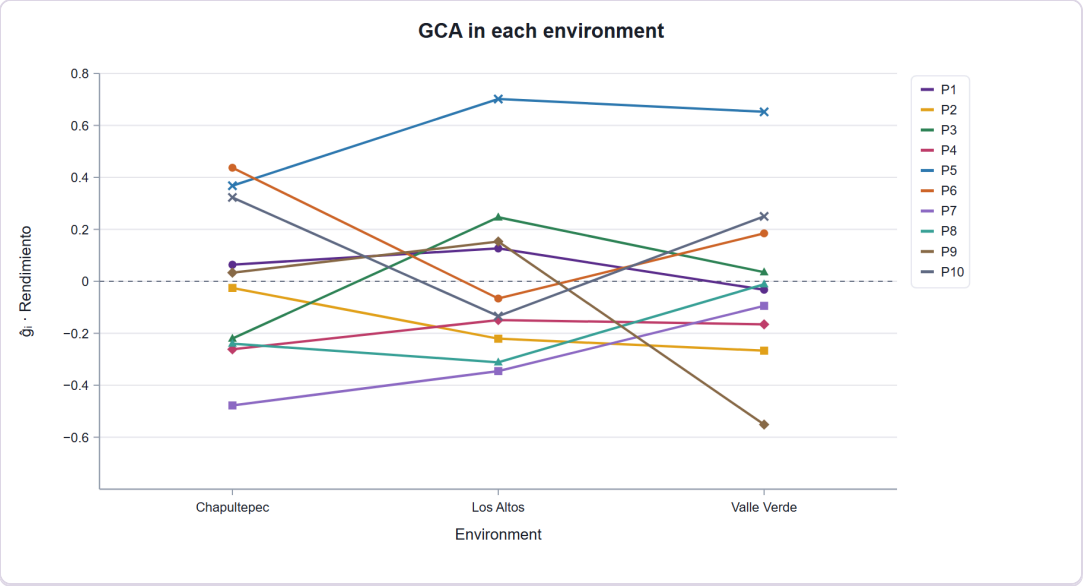

Figure 4.13. The GCA of the ten parents at each location of the simulated α-lattice.

Table 4.3. Components of the simulated α-lattice with Method 4 and the values used to generate the data. σ^2_{GCA} is compared with the variance of the ten effects drawn, not with that of the model (0.16).

Component	Estimate ± SE	True
σ^2_{GCA}	0.080 ± 0.051	0.052
σ^2_{SCA}	0.037 ± 0.032	0.069
$\sigma^2_{GCA \times environment}$	0.037 ± 0.024	0.029
$\sigma^2_{SCA \times environment}$	0.063 ± 0.053	0

With ten parents, the GCA and SCA variances are estimated with few degrees of freedom and their standard errors are of the same order as the estimates themselves; this is the limitation of any small diallel, not of the program.

4.9 Special cases, downloads and moving on to Block 5

Means with an external error

If Block 3 received entry means with the error of the original experiment (Section 3.8), Block 4 analyzes those means with that error. This is the case of Griffing's (1956) example, 36 single crosses of maize from nine lines (Method 4) with an error of 21.05 with 2,558 degrees of freedom on the mean scale: GCA has F = 110.49 and SCA 16.13; σ^2_{GCA} = 283.76 and σ^2_{SCA} = 318.39, with a Baker ratio of 0.64.

Incomplete or unbalanced designs

The block estimates the effects by least squares weighted by the number of plots, with Griffing's restrictions imposed exactly. With complete, balanced data it reproduces Griffing's formulas; with missing crosses or unequal replications it remains valid and the standard errors adjust to the data at hand. Because Griffing's model has as many parameters as cells, something has to be assumed about what is missing, and the warnings in card 1 say what: SCA = 0 is assumed for the crosses that were not made, and non-maternal effect = 0 for the pairs evaluated in only one direction. With entries that have different numbers of plots, the components use the harmonic mean of the replications.

Downloads and the next block

At the end of card 6, Assumptions, formulas and references summarizes the model, the restrictions, the estimation and the references, for the methods section of a paper. There are two downloads: Results (.xlsx), with six sheets (analysis of variance, effects, components, parameters, crosses with their mean, prediction and SCA, and comparison of methods), and Effects (.csv), with the grand mean and all the GCA, SCA, maternal, reciprocal and non-maternal effects with their standard error, t and p. Each figure has its own download panel.

The Continue: Hayman–Jinks → button takes you to Block 5, which analyzes the same diallel from another angle: the Wr–Vr graph, Hayman and Jinks's genetic parameters, and Gardner and Eberhart's analysis for varieties and heterosis.

B5

CHAPTER FIVE · BLOCK 5 · HAYMAN–JINKS

Hayman–Jinks and Gardner–Eberhart

The same diallel can be read in two other ways. Hayman and Jinks's reading asks about gene action: how much of the variation is additive and how much is due to dominance, whether dominance is partial or overdominance, and which parents carry the dominant alleles. Gardner and Eberhart's reading, designed for crosses between varieties, splits heterosis into average, variety and specific heterosis. Block 5 does both, with the Wr–Vr graph, and checks on your data that both agree with Griffing.

5.1 Three readings of a diallel

5.2 Data and assumptions

5.3 Hayman's partition

5.4 The genetic components

5.5 The Wr–Vr graph and the order of dominance

5.6 Gardner and Eberhart: varieties and heterosis

5.7 The equivalences

5.8 Special cases, downloads and the next block

5.1 Three readings of a diallel

Griffing (Chapter 4) describes each parent by its combining ability. Hayman and Jinks go deeper and describe the genes, with an additive–dominance model and a set of assumptions. Gardner and Eberhart describe varieties and the heterosis of their crosses. All three split the same variation between entries, so their sums of squares match; what changes is the question and the interpretation.

Table 5.1. The question each analysis of Block 5 answers and what it needs.

Analysis	Question	Needs
Hayman's (1954) partition	Are there additive effects, mean dominance, dominance that varies between parents, specific dominance and reciprocal effects?	Full table with reciprocals (or a half table, with Morley Jones, 1965)
Hayman and Jinks components	How much do the additive variance (D) and the dominance variance (H_1, H_2) weigh? Is there overdominance? Do dominant alleles predominate?	Homozygous parents included in the experiment
Wr–Vr graph	Does the additive–dominance model hold? Which parents carry more dominant genes?	At least five or six parents for the tests to have power
Gardner and Eberhart (1966), Analyses II and III	How does each variety perform, how much average heterosis is there, which varieties contribute heterosis and which crosses have specific heterosis?	The varieties (parents) and their crosses
Heterosis of every cross	By how much does each cross exceed the mean of its parents and the better of them?	Parents and crosses

The chapter uses two published examples. For Hayman and Jinks, Hayman's (1954) 8 × 8 diallel of *Nicotiana rustica* (eight lines, full table with reciprocals, two blocks, days to flowering), the classic case of the Wr–Vr graph. For Gardner and Eberhart, the crosses among six maize varieties of Lonnquist and Gardner (1961), a half diallel with the varieties as parents. Both are reproduced exactly with the published values.

5.2 Data and assumptions

Like Block 4, Block 5 takes the adjusted means and the error from Block 3. The card **1 · Data and assumptions** summarizes the data in one line (parents, kind of table and E, the variance of an entry mean), shows the warnings and offers the controls in Table 5.2. The tiles summarize how the variation splits among additive effects, dominance and reciprocals, the main Hayman and Jinks ratios and the regression of Wr on Vr.

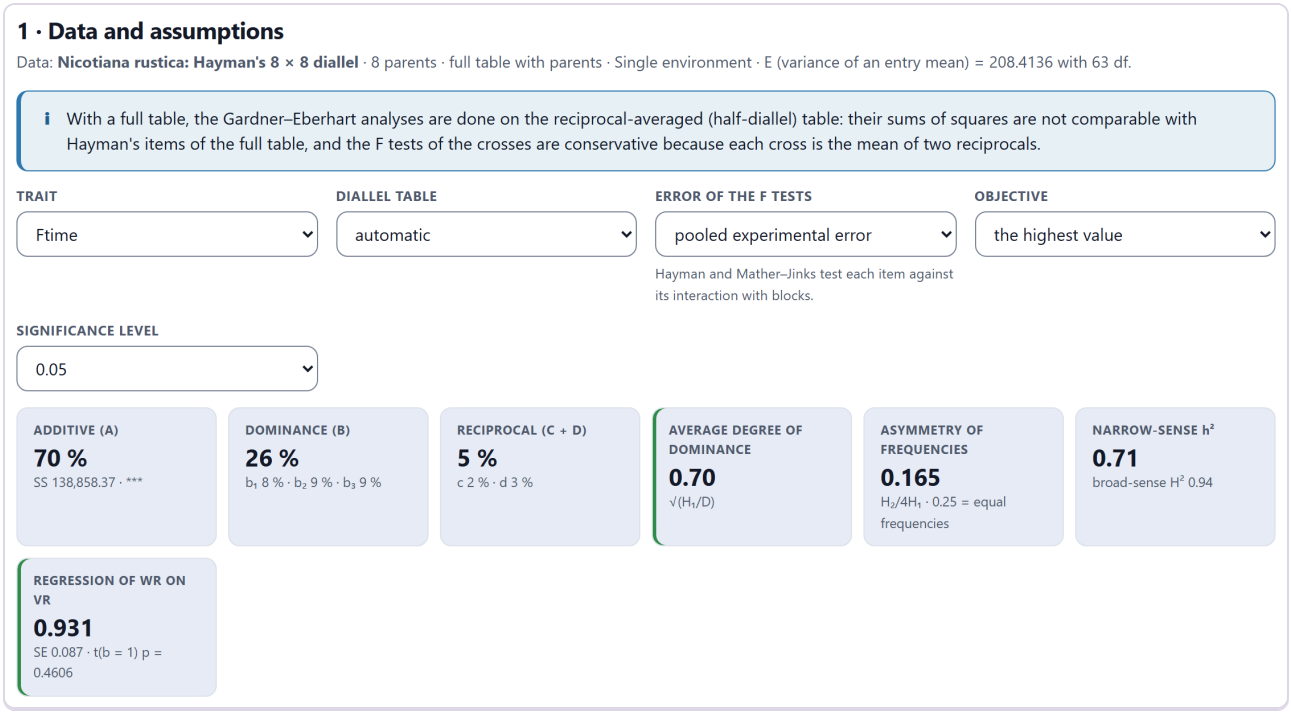

Figure 5.1. Card 1 with Hayman's diallel. Additive effects account for 70% of the variation between entries; dominance for 26%; reciprocals for 5%.

B5

Table 5.2. The controls of card 1.

Control	Options	What it is for
Trait	those of Block 3	The whole block is recomputed for the chosen trait
Environment	each environment · mean of all	With several environments: these analyses are done on a single diallel table
Diallel table	automatic · full · half	Use the table with reciprocals (Hayman) or one cross per pair (Morley Jones)
Error of the F tests	pooled experimental error · interaction of each item with blocks	The second is the test of Hayman and of Mather and Jinks; it needs complete blocks
Error corrections (half diallel)	derived for single crosses · classical	Only in a half diallel: how the error is subtracted from H_1, H_2 and h^2 (Section 5.4)
Objective	the highest value · the lowest value	Which parent counts as the better one in the heterosis and how the crosses are sorted
Significance level	0.05 · 0.01 · 0.10	For the readings and the intervals

The assumptions, one by one

Hayman and Jinks's additive–dominance model assumes diploid segregation, homozygous parents, no reciprocal effects and no epistasis, at most two alleles per locus, and genes distributed independently among the parents. The assumptions table tests those the data allow and says which ones they do not.

Assumptions of the analysis

The additive–dominance model of Hayman and Jinks assumes: diploid segregation, homozygous parents, no reciprocal effects, no epistasis, no multiple alleles and genes independently distributed among the parents.

Assumption or test	Statistic	p	Reading
No reciprocal effects (items c and d)	F(c) = 2.31 · F(d) = 1.41	0.0367 *	reciprocal effects exist: averaging them biases the components
Additive–dominance model: slope b = 1	b = 0.931 ± 0.087, t = 0.79	0.4606 ns	b = 1 is not rejected
Uniformity of Wr and Vr (Hayman's t^2 test)	t^2 = 0.249	0.6357 ns	holds
Slope different from zero (dominance present)	t = 10.65	< 0.0001 ***	yes
Independent distribution of genes (item b_2)	F = 11.70	< 0.0001 ***	dominance differs between arrays
Mean dominance different from zero (item b_1)	F = 73.88	< 0.0001 ***	dominance present
Arrays for Wr − Vr over blocks	F = 5.04	0.0244 *	suggests epistasis
Arrays for Wr + Vr over blocks	F = 17.16	0.0007 ***	dominance present
Homozygous parents and diploid segregation	—	—	not tested with these data
No multiple alleles	—	—	needs F_2 or backcrosses

Figure 5.2. The assumptions with Hayman's diallel. There are reciprocal effects (p = 0.037), the slope does not depart from 1 and dominance is clear; the arrays differ in Wr − Vr across blocks (p = 0.024), a sign of epistasis.

With these data the tests do not fully agree, and that is normal. The slope of Wr on Vr (b = 0.931 ± 0.087) does not depart from 1 and the t^2 test of uniformity is passed, but the analysis of Wr − Vr across blocks does find differences between arrays; moreover, the reciprocals are significant, so averaging them, as the model requires, biases the components slightly. The prudent reading is to use the components as an approximate description and to rely on the scaling tests of Block 7 or on the triple test cross of Block 6 if epistasis matters.

5.3 Hayman's partition

The card **2 · Hayman's partition** divides the variation between entries into six items (Table 5.3). With the full table all six are used; with a half diallel, the first four, computed with the method of Morley Jones (1965), which matches Gardner and Eberhart's partition. The last row, the variation among parents, is the one Walters and Gale (1977) propose for testing gene effects when there are reciprocal effects, because item a tests them only if there is no dominance.

Table 5.3. Hayman's items and what they measure. p: parents.

Item	df	Measures	Equivalent to
a	$p - 1$	Additive effects: differences between the arrays (rows) of the table	GCA of Griffing's Method 1
b_1	1	Mean dominance: the F_1s against the parents	average heterosis ($\bar{h}$)
b_2	$p - 1$	Dominance that differs between arrays: dominant alleles concentrated in some parents	variety heterosis (h_j)
b_3	$p(p - 3)/2$	Specific dominance of each cross	SCA of Method 3 (full table) or Method 4

Table 5.3. Hayman's items and what they measure. *p*: parents. *(continued)*

Item	df	Measures	Equivalent to
c	$p - 1$	Maternal effects	maternal effects of Method 1
d	$(p - 1)(p - 2)/2$	Remaining reciprocal differences	non-maternal effects of Method 1

2 · Hayman's partition

Full table: Hayman's (1954) partition into a, b_1, b_2, b_3, c and d. Sums of squares on the entry-mean scale (multiply by r = 2 for the plot scale).

Item	df	SS	MS	F	p	MS of item × blocks	F against blocks
a · additive effects	7	138,858.3672	19,836.9096	95.18	< 0.0001 ***	1,430.8371	13.86 **
b_1 · mean dominance (F_1 vs parents)	1	15,398.4425	15,398.4425	73.88	< 0.0001 ***	389.8225	39.50 ns
b · dominance (total)	**28**	**51,119.2227**	**1,825.6865**	**8.76**	**< 0.0001** ***	—	—
b_2 · dominance differing between arrays	7	17,076.3516	2,439.4788	11.70	< 0.0001 ***	257.6064	9.47 **
b_3 · specific dominance	20	18,644.4286	932.2214	4.47	< 0.0001 ***	162.0595	5.75 ***
c · maternal effects	7	3,369.5625	481.3661	2.31	0.0367 *	517.9107	0.93 ns
d · remaining reciprocal differences	21	6,186.4375	294.5923	1.41	0.1467 ns	342.125	0.86 ns
Among parents (genic effects, Walters & Gale 1977)	7	71,473.2187	10,210.4598	48.99	< 0.0001 ***	—	—

The F tests use the pooled experimental error (63 df). The sum of the item × blocks interactions is the blocks × entries error of the design. The last row is the sum of squares among parents, which Walters & Gale (1977) propose for testing the genic effects when reciprocal effects exist: item a tests genic effects only if there is no dominance.

Figure 5.3. Hayman's partition of days to flowering in *N. rustica*. The last two columns test each item against its interaction with blocks; with two blocks, those tests have very few degrees of freedom (b_1, with $F = 39.5$, does not reach significance).

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With the Error of the F tests menu set to interaction of each item with blocks, the *F* tests in the table become those of the last two columns. These interactions are computed only with complete blocks in one environment; their sum is the blocks × entries error from Block 3. The figure below the table shows the percentage of the variation each item accounts for, or its mean squares.

VERIFIED · THE PUBLISHED VALUES

On the plot scale (twice as large, with two blocks), the items are those of Hayman (1954): a = 277,716.73, b_1 = 30,796.89, b_2 = 34,152.70, b_3 = 37,288.86, c = 6,739.13 and d = 12,372.88, with $F = 95.18$ for a, 2.31 for c and 1.41 for d. The partition of a half diallel reproduces table 1 of Morley Jones (1965).

5.4 The genetic components

The card 3 · Genetic components estimates the six parameters of Hayman and Jinks's model with their standard error and their *t* test (with $p - 2$ degrees of freedom), and on the right the ratios read from them. Below, five tiles and a reading in words. The card appears only if the experiment includes the parents.

3 · Genetic components

D measures additive variance, H_1 and H_2 dominance variance (H_2 with the imbalance of frequencies), F the relation between dominant alleles and frequencies, h^2 overall dominance and E the error. The t tests use Hayman's standard error with p − 2 degrees of freedom.

Component	Estimate	SE	t	p
E (environmental)	208.4136	117.3928	1.78	0.1262 ns
D (additive)	10,002.0462	352.1785	28.40	< 0.0001 ***
F (allele frequencies)	6,768.2211	832.1653	8.13	0.0002 ***
H_1 (dominance)	4,907.8447	809.6064	6.06	0.0009 ***
H_2 (dominance, asymmetry)	3,234.5458	704.357	4.59	0.0037 **
h^2 (overall dominance)	6,645.6376	472.3717	14.07	< 0.0001 ***

Ratio or parameter	Definition	Value	Reading
Average degree of dominance	$\sqrt{(H_1/D)}$	0.7005	partial to complete dominance
Asymmetry of frequencies at dominant loci	$H_2/4H_1$	0.1648	unequal frequencies
Dominant / recessive alleles	KD/KR	2.8685	excess of dominant alleles
Groups of genes showing dominance	h^2/H_2	2.0546	—
σ^2A	$\frac{1}{2}(D + H_1 - H_2 - F)$	2,453.562	—
σ^2D	$\frac{1}{4}H_2$	808.6364	—
Narrow-sense h^2	Mather & Jinks form	0.707	—
Broad-sense H^2	Mather & Jinks form	0.9399	—

D (ADDITIVE)	H_1 (DOMINANCE)	F	h^2/H_2	E
10,002.046	**4,907.845**	**6,768.221**	**2.05**	**208.414**
SE 352.178 · ***	SE 809.606 · ***	more dominant alleles	groups of genes showing dominance	variance of an entry mean

D = 10,002.046 and H_1 = 4,907.845: the additive part exceeds dominance, with an average degree of dominance of 0.70. $H_2/4H_1$ = 0.165 (0.25 when allele frequencies are equal). KD/KR = 2.87: dominant alleles predominate among the parents. h^2/H_2 = 2.05: at least 2 groups of genes with dominance in the same direction. Heritability: narrow-sense h^2 0.71 and broad-sense H^2 0.94 (Mather & Jinks form, entry-mean basis).

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Figure 5.4. The components of Hayman's diallel. D dominates over H_1; the average degree of dominance is 0.70 and dominant alleles predominate (KD/KR = 2.87).

THEORY IN BRIEF · THE COMPONENTS AND THEIR RATIOS

$$\hat{D} = V_0L_0 - E \cdot \hat{F} = 2V_0L_0 - 4W_0L_{01} - 2(p-2)E/p \cdot \hat{H}_1 = V_0L_0 - 4W_0L_{01} + 4V_1L_1 - (3p-2)E/p$$
$$\hat{H}_2 = 4V_1L_1 - 4V_0L_1 - 2E \cdot \hat{h}^2 = 4(ML_1 - ML_0)^2 - 4(p-1)E/p^2$$

V_0L_0 is the variance of the parents; V_1L_1, the mean of the array variances; W_0L_{01}, the mean of their covariances with the parents; V_0L_1, the variance of the array means; and $ML_1 - ML_0$, the difference between the mean of the crosses and that of the parents. D is the additive variation; H_1, the dominance variation; H_2, the dominance variation weighted by allele frequencies; F, the covariance between additive and dominance effects (positive if dominant alleles predominate); h², the net effect of dominance summed over loci; and E, the error of an entry mean (Hayman, 1954; Jinks, 1954). From them come $\sqrt{(H_1/D)}$, the average degree of dominance (greater than 1: overdominance); $H_2/4H_1$, which equals 0.25 if the alleles have equal frequencies; KD/KR, the ratio of dominant to recessive alleles; and h^2/H_2, the approximate number of groups of genes showing dominance.

In *N. rustica*, D = 10,002 ± 352 and H_1 = 4,908 ± 810: the additive part exceeds dominance, and the average degree of dominance, 0.70, indicates partial dominance. $H_2/4H_1$ = 0.165 says that the alleles do not have equal frequencies; a positive F and KD/KR = 2.87, that the parents carry more dominant than recessive alleles; and h^2/H_2 = 2.05, that at least two groups of genes show dominance in the same direction. Narrow-sense heritability, in Mather and Jinks's form, is 0.71, and broad-sense heritability, 0.94.

> **IMPORTANT · THE CORRECTIONS FOR A HALF DIALLEL**
>
> Hayman's classical error corrections assume that each cross is the mean of two reciprocals. In a half diallel each cross was evaluated **only once**, so its variance is twice as large and those corrections fall short: in simulations with no genetic variation at all, with E = 4 and five parents, the classical ones give H_1 = 6.52 instead of zero. That is why the block uses by default corrections derived for single crosses ($H_1 - (5p - 4)E/p$, $H_2 - 4E + 4(p - 1)E/p^2$ and $h^2 - 4(p^2 - 1)E/p^3$), checked by simulation, and leaves the classical ones in the menu for when the reciprocals have truly been averaged. The card's note says which were used.

5.5 The Wr–Vr graph and the order of dominance

The card **4 · Wr–Vr graph and order of dominance** plots each parent as a point: Vr is the variance of its array (its row of the diallel table) and Wr, the covariance of that array with the parents. If the additive–dominance model holds, the points fall on a line of slope 1, inside the limiting parabola $Wr^2 = Vr \cdot V_0L_0$. Chapter 1 shows with the diallel lab how the graph changes with each kind of dominance and epistasis.

READING RULE · THE WR–VR GRAPH

Slope that does not depart from 1	The additive–dominance model is adequate; if it departs, there is epistasis or genes correlated between parents.
Intercept positive · zero · negative	Partial dominance · complete dominance · overdominance.
Distance from the origin of each point	Near: the parent carries mostly dominant alleles; far: recessive ones.
Correlation between the parent's value and Wr + Vr	Positive: dominant alleles decrease the trait; negative: they increase it.

The second figure plots the parents by Wr + Vr (from most dominant to most recessive) against their own value. In *N. rustica* the correlation is 0.74: the parents with more dominant alleles flower earlier, so earliness is dominant. The line predicts that a completely dominant parent would flower at 131 days and a completely recessive one at 407.

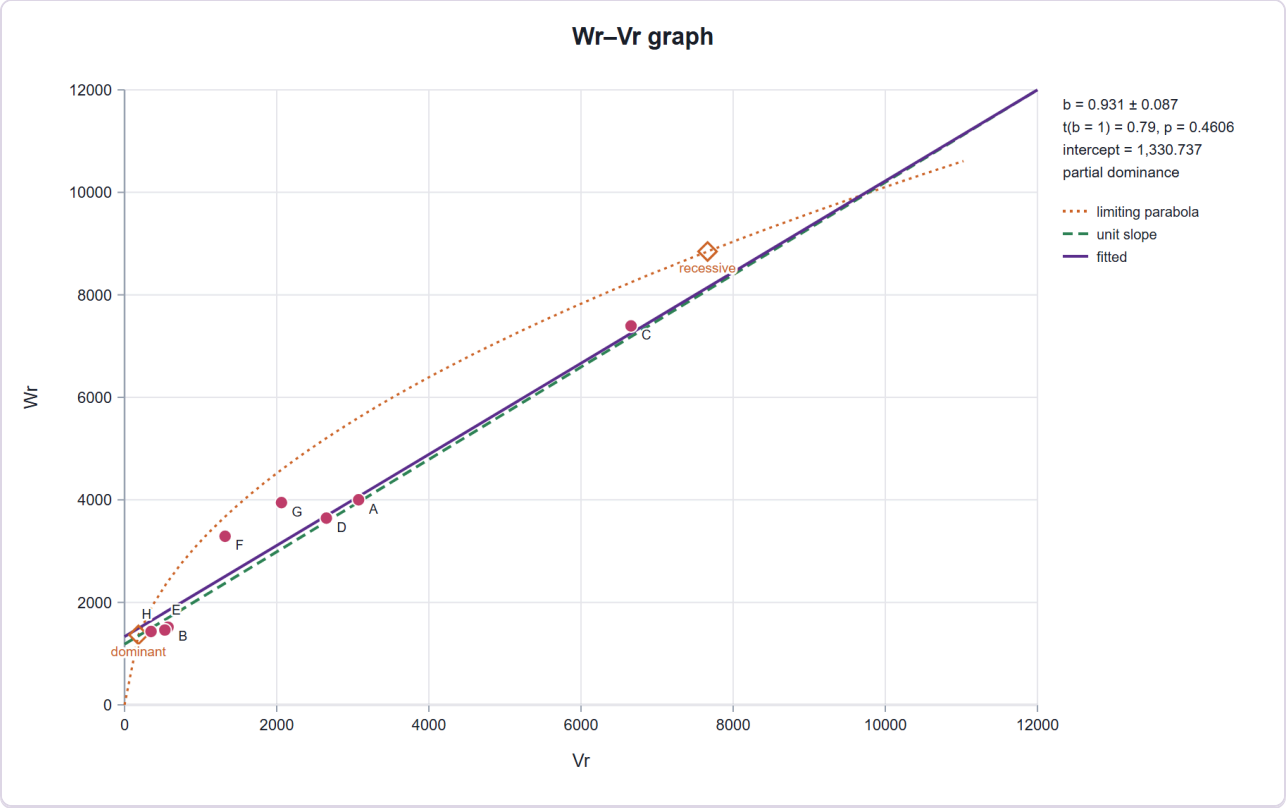

Figure 5.5. The Wr–Vr graph of days to flowering. The fitted line (purple) almost coincides with the unit-slope line (green) and cuts the Wr axis above the origin: partial dominance. H, E and B, near the origin, carry more dominant genes; C, far from it, more recessive ones. The diamonds mark where a parent with all dominant or all recessive alleles would fall.

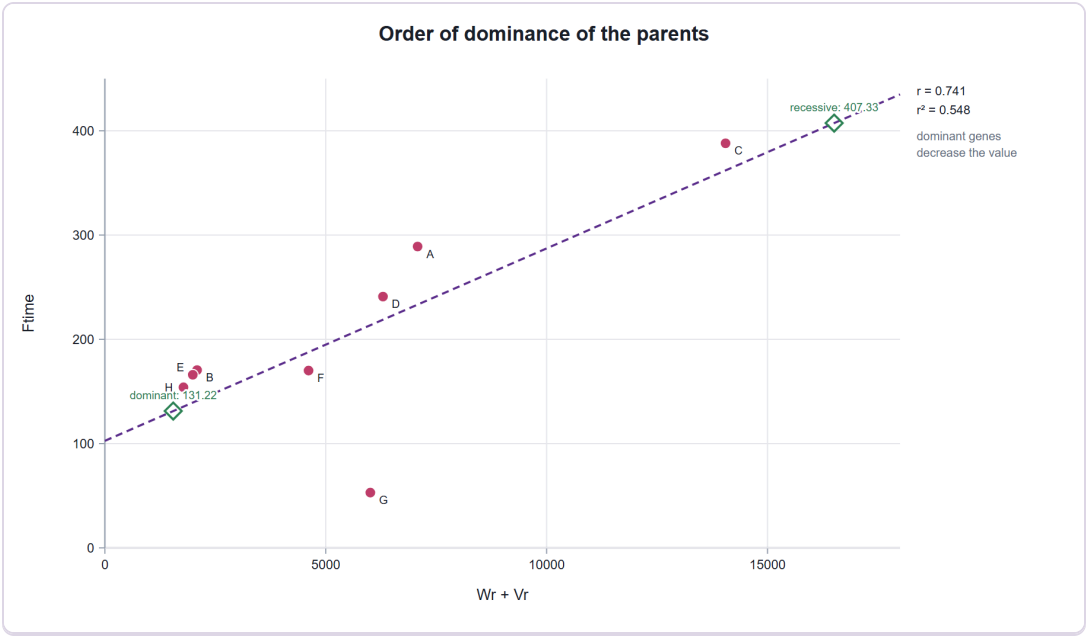

Figure 5.6. The order of dominance: days to flowering of each parent against Wr + Vr, with the predicted extremes (diamonds). G, very early but with intermediate values of Wr + Vr, departs from the line.

Regression of Wr on Vr

Statistic	Value	Reading
Slope b	0.931 ± 0.087	—
t (b = 0)	10.65 · p < 0.0001	dominance present
t (b = 1)	0.79 · p = 0.4606	additive–dominance model adequate
Intercept	1,330.7373	partial dominance
t^2 test (uniformity of Wr and Vr)	0.249 · p = 0.6357	holds
r (parent value, Wr + Vr)	0.741 · r^2 = 0.548	dominant genes decrease the value
Completely dominant parent (prediction)	131.217	—
Completely recessive parent (prediction)	407.333	—

Arrays

Parent	Own value	Vr	Wr	Wr + Vr	Wr − Vr	Order of dominance
A	289	3,076.892	4,002.15	7,079.041	925.258	7
B	170.5	571.496	1,515.987	2,087.482	944.491	3
C	388	6,657.722	7,392.431	14,050.153	734.709	8
D	241	2,652.875	3,642.732	6,295.607	989.857	6
E	166	530.115	1,459.962	1,990.077	929.847	2
F	170	1,321.392	3,289.489	4,610.881	1,968.097	4
G	53	2,062.567	3,946.549	6,009.116	1,883.982	5
H	154	348.143	1,430.857	1,779	1,082.714	1

Figure 5.7. The regression of Wr on Vr with its reading, and the table of arrays: each parent's own value, Vr, Wr, Wr + Vr, Wr − Vr and its place in the order of dominance (1 = the most dominant, H).

Hayman's t^2 test compares the variance of Wr with that of Vr; if it is significant, the assumptions fail. With complete blocks, the collapsed panel **Analysis of variance of Wr − Vr and Wr + Vr over blocks** repeats the graph in each block: arrays not significant for Wr − Vr indicate that the model is adequate, and arrays significant for Wr + Vr, that dominance is present.

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5.6 Gardner and Eberhart: varieties and heterosis

Gardner and Eberhart's (1966) Analysis II describes each cross between two varieties as the mean of its varieties plus a heterosis, which in turn splits into average heterosis ($\bar{h}$), the heterosis of each variety (h_j) and the specific heterosis of the pair ($s_{jj'}$). Analysis III separates the varieties per se (sp_j) from their combining ability among the crosses (g_j), and $g_j = \frac{1}{2}v_j + h_j$ holds. The card **5 · Gardner & Eberhart** does both, with their estimates, a figure of the variety and heterosis effects, and the heterosis of every cross.

5 · Gardner & Eberhart: varieties, heterosis and Analysis III

Analysis II describes each cross as the mean of its varieties plus average heterosis, variety heterosis and specific heterosis. Analysis III separates the parents per se from the combining ability among crosses; the identity $\hat{g}_j = \frac{1}{2}\hat{v}_j + \hat{h}_j$ holds.

Analysis II

Source	df	SS	MS	F	p
Average heterosis ($\bar{h}$)	1	115.4404	115.4404	16.26	0.0002 ***
Variety effects (v_j)	5	234.23	46.846	6.60	< 0.0001 ***
Variety heterosis (h_j)	5	59.72	11.944	1.68	0.1527 ns
Specific heterosis ($s_{jj'}$)	9	63.781	7.0868	1.00	0.4515 ns

Analysis III

Source	df	SS	MS	F	p
Average heterosis ($\bar{h}$)	1	115.4404	115.4404	16.26	0.0002 ***
Parents (sp_j)	5	55.975	11.195	1.58	0.1804 ns
GCA among crosses (g_j)	5	237.975	47.595	6.70	< 0.0001 ***
Specific heterosis ($s_{jj'}$)	9	63.781	7.0868	1.00	0.4515 ns

Figure 5.8. Analyses II and III with the six maize varieties of Lonnquist and Gardner (1961). Average heterosis is highly significant; variety and specific heterosis are not.

In this example the average heterosis is $\bar{h} = 5.19 \pm 1.29$ ($p = 0.0002$): the crosses yield on average 5.2 units more than the mean of their varieties ($\mu_v = 92.45$). Only variety K2 contributes a significant heterosis of its own ($\hat{h} = 4.50$, $p = 0.011$), and only the cross B × G has specific heterosis ($\hat{s} = 4.81 \pm 2.06$). In Analysis III, K2 is also the variety with the best combining ability among the crosses ($\hat{g} = 6.375$, $p < 0.0001$): ½ × 3.75 + 4.50. All these figures are the published ones.

> **IMPORTANT · VARIETIES OR LINES**
>
> Gardner and Eberhart's analysis is made for open-pollinated varieties; Hayman and Jinks's assumes homozygous parents. With varieties as parents, the Hayman and Jinks components (here, an average degree of dominance of 2.49, which would suggest overdominance) should not be read as genetic parameters. With a full table, Gardner and Eberhart are computed on the table with the reciprocals averaged; the warning in card 1 reminds you of it.

Heterosis of every cross

Against the mid-parent and against the better parent, with its standard error and t test (from the experimental error, not from the selection of the better parent). Block 7 adds generations, inbreeding depression and checks.

Cross	Cross F_1	Mid-parent	Heterosis	%	SE	p	Better parent	Heterosis (%)	p
G × K2	103.3	92.05	11.25	12.2	3.263	0.0010 **	96.2	7.4	0.0644 ns
H × K2	105.4	93.95	11.45	12.2	3.263	0.0009 ***	96.2	9.6	0.0176 *
B × G	101.3	92.25	9.05	9.8	3.263	0.0074 **	96.6	4.9	0.2172 ns
K × K2	101.6	93.75	7.85	8.4	3.263	0.0193 *	96.2	5.6	0.1570 ns
H × M	98.8	91.35	7.45	8.2	3.263	0.0260 *	91.7	7.7	0.0644 ns
K2 × M	100.7	93.6	7.1	7.6	3.263	0.0335 *	96.2	4.7	0.2371 ns
B × K2	102.7	96.4	6.3	6.5	3.263	0.0583 ns	96.6	6.3	0.1107 ns
G × H	92.7	89.8	2.9	3.2	3.263	0.3777 ns	91.7	1.1	0.7916 ns
B × H	97.1	94.15	2.95	3.1	3.263	0.3696 ns	96.6	0.5	0.8949 ns
H × K	94.1	91.5	2.6	2.8	3.263	0.4288 ns	91.7	2.6	0.5266 ns
K × M	93.5	91.15	2.35	2.6	3.263	0.4743 ns	91.3	2.4	0.5615 ns

Figure 5.9. The heterosis of every cross relative to the mean of its parents and to the better of them, with its standard error and its test. Six crosses significantly exceed the mean of their parents; only H × K2 exceeds the better parent.

The heterosis table does not depend on any model: it compares means with the exact variance of each contrast. The test against the better parent does not correct for having chosen it as the better one, and the header warns about it. The Objective menu defines which parent is the better one: with days to flowering or height, the lowest value.

Below the table, a bar chart shows the heterosis of every cross as a percentage, in solid color if it is significant, relative to the mid-parent or to the better parent. Block 7 does the same with the F_1 of a generation-means experiment and adds the inbreeding depression of the F_2.

5.7 The equivalences

The card **6 · Equivalences between the methodologies** computes all three on your data and checks, row by row, that the sums of squares that should match do match. It is not an ornament: if a row says "no", something is wrong in the data or in the computation, and that is how an error was found and fixed during the validation of the block.

6 · Equivalences between the methodologies

The three methodologies split the same variation between entries in different ways. All three are computed here on your data and every identity is checked: if something does not match, the note says so.

Source of variation	Hayman / Morley Jones	Griffing	Gardner & Eberhart	Do they match?
Full table (with reciprocals)	—	—	—	—
Additive effects	a: **138,858.3672**	Method 1 GCA: **138,858.3672**	—	yes
Total dominance	b = b_1 + b_2 + b_3: **51,119.2227**	Method 1 SCA: **51,119.2227**	—	yes
Maternal effects	c: **3,369.5625**	Method 1 maternal: **3,369.5625**	—	yes
Remaining reciprocal differences	d: **6,186.4375**	Method 1 non-maternal: **6,186.4375**	—	yes
Specific dominance	b_3: **18,644.4286**	Method 3 SCA: **18,644.4286**	—	yes
Reciprocal-averaged table (half diallel)	—	—	—	—
Additive effects / varieties	a: **100,042.8875**	Method 2 GCA: **100,042.8875**	varieties (v_j): **100,042.8875**	yes
Total dominance / heterosis	b: **36,670.8**	Method 2 SCA: **36,670.8**	$\bar{h}$ + h_j + s: **36,670.8**	yes
Average heterosis	b_1: **13,687.5045**	—	$\bar{h}$: **13,687.5045**	yes
Variety heterosis	b_2: **13,661.0813**	—	h_j: **13,661.0813**	yes
Specific heterosis	b_3: **9,322.2143**	Method 4 SCA: **9,322.2143**	s: **9,322.2143**	yes
GCA among crosses	—	Method 4 GCA: **42,230.75**	Analysis III g_j: **42,230.75**	yes
Parents per se	among parents: **71,473.2187**	—	sp_j: **71,473.2187**	yes

Gardner & Eberhart identity: b_2 = SS(parents) + GCA(Method 4) − GCA(Method 2) = 13,661.0812 against b_2 = 13,661.0813 ✓. Sums of squares are on the entry-mean scale. The effects do not coincide between methodologies even when the sums of squares do: Griffing's Method 2 includes the parents in GCA, Gardner & Eberhart's Analysis III does not.

Figure 5.10. The equivalences with Hayman's diallel: with the full table, a, b, c, d and b_3 are the GCA, SCA, maternal and non-maternal effects of Method 1 and the SCA of Method 3; with the reciprocals averaged, the Morley Jones items are the sources of Analysis II and of Griffing's Methods 2 and 4.

The sums of squares match, but the effects do not: the GCA of Griffing's Method 2 includes the parents, and the g_j of Analysis III does not. That is why the same parent can rank differently in Block 4 and in Block 5 without either being wrong. Below the table, Gardner and Eberhart's identity b_2 = SS(parents) + GCA(Method 4) − GCA(Method 2) is also checked.

5.8 Special cases, downloads and the next block

- **No parents** (Methods 3 and 4): there are no Hayman and Jinks components and no Gardner and Eberhart analysis, because they need the parents per se; the partition is still computed.
- **Missing entries:** Wr, Vr and the components need the full table; card 1 warns how many are missing.

- **Few parents:** with fewer than six, the Wr–Vr regression has three degrees of freedom or fewer, and the block warns that the tests have little power.
- **Several environments:** the Environment menu chooses one or the mean of all. With incomplete blocks, E is the variance of the adjusted means, as in Block 4 (Section 4.8).

The downloads are Results (.xlsx), with six sheets (Hayman's items, components and ratios, arrays, Gardner and Eberhart, heterosis and the diallel table), and Diallel table (.csv), the parent × parent table of means. Assumptions, formulas and references, at the end of card 6, gathers the formulas and the sources.

Block 5 is the last one specific to diallels. Its Continue button takes you to Block 8 (genetic parameters and selection among the crosses) or, with several environments, to Block 9 (genotype × environment); Block 6 is for the other mating designs.

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CHAPTER SIX · BLOCK 6 · MATING DESIGNS

Mating designs

Diallels are not the only way to estimate genetic variances. Comstock and Robinson's North Carolina designs, Kearsey and Jinks's triple test cross, line × tester and partial diallels organize the crosses differently, almost always to evaluate more families with less work. Block 6 analyzes the six designs with a single engine: it computes the expected mean squares of the actual experiment, chooses what to test each source against and converts the components into σ^2_A, σ^2_D and heritabilities.

6.1 Six designs, one idea

6.2 Data and assumptions

6.3 The analysis of variance

6.4 The effects

6.5 Components and genetic parameters

6.6 North Carolina I

6.7 North Carolina III and the triple test cross

6.8 Line × tester

6.9 Partial diallel

6.10 Several environments, downloads and the next block

6.1 Six designs, one idea

All these designs rest on the same principle: relatives resemble each other in proportion to the genes they share. Half sibs (one parent in common) share a quarter of the additive variance; full sibs, half of the additive variance and a quarter of the dominance variance. Each design organizes the crosses so that the analysis of variance separates those two covariances.

$$\mathrm{Cov(HS)} = \tfrac{1}{4}(1 + F)\sigma^2_A \quad \cdot \quad \mathrm{Cov(FS)} = \tfrac{1}{2}(1 + F)\sigma^2_A + \tfrac{1}{4}(1 + F)^2\sigma^2_D$$

Table 6.1. The designs of Block 6, what they estimate and the chapter example that illustrates each.

Design	How the crosses are made	Estimates	Example
North Carolina I	Each male with several females of its own (nested)	σ^2_A from the males; σ^2_D from the females within males	2 sets × 3 males × 2 females
North Carolina II	Every male with every female of its set (factorial)	σ^2_A from males and from females; σ^2_D from their interaction	potato: 2 sets of 5 × 5
North Carolina III	Each F_2 plant with the two parents of the F_1	σ^2_A from the sums; σ^2_D from the differences	16 F_2 plants in 4 sets
Triple test cross	Like design III, plus the F_1 as a third tester	The same and, in addition, epistasis	simulated: 30 F_2 plants
Line × tester	Each line with each tester; optionally, the parents	GCA of lines and testers; SCA; heterosis	5 lines × 3 testers
Partial diallel	Each parent with s of the others (circulant)	GCA and SCA as in a diallel, with fewer crosses	maize: 8 lines, $s = 5$

Block 3 recognizes the design and decides which column is the female, the male, the tester or the F_2 plant. If the data come from none of these designs, card 1 says where they are analyzed: complete diallels in Blocks 4 and 5; generations in Block 7.

6.2 Data and assumptions

The card **1 · Data and model** takes the plot values from Block 3 (with the field analysis that matches the design and any observations you excluded), summarizes the experiment in one line and gives the main results in tiles. The table **What the design assumes**, below, says what each design assumes and what happens if the assumption fails.

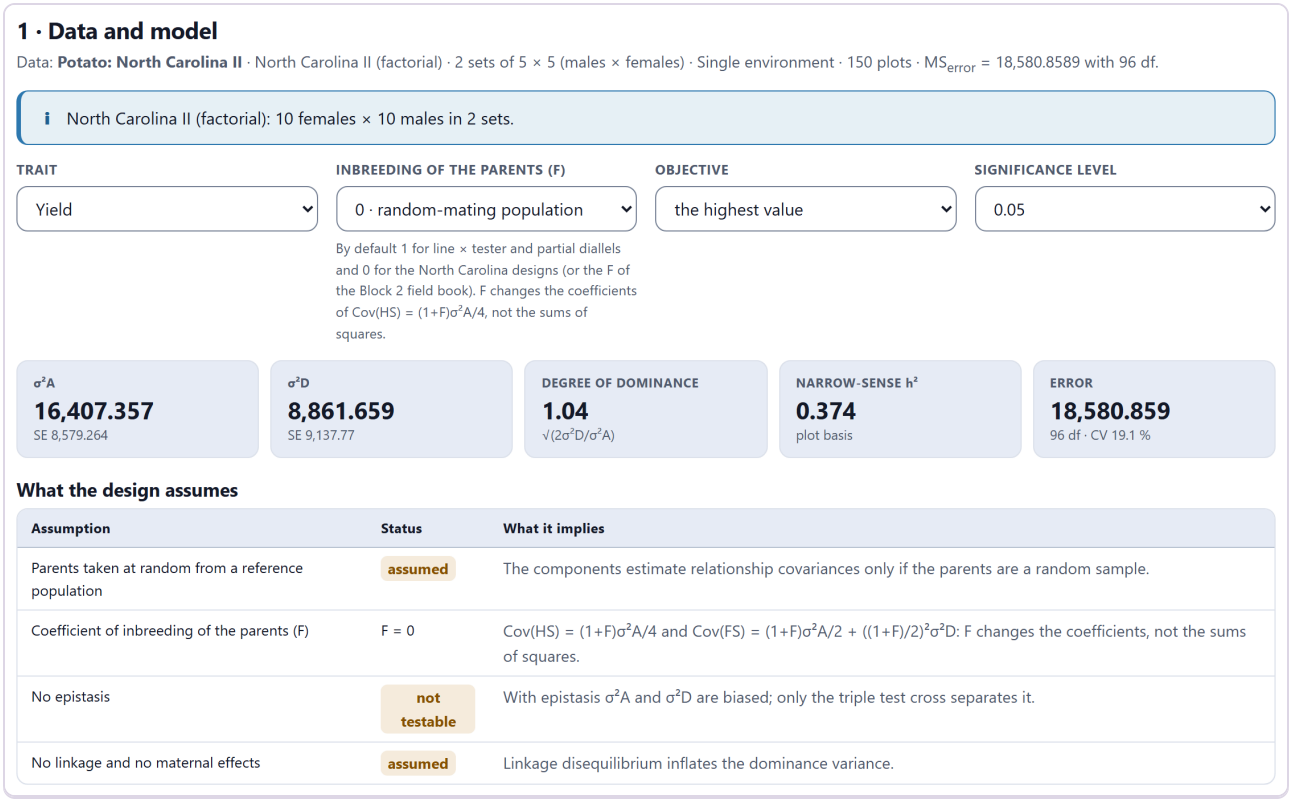

Figure 6.1. Card 1 with the potato North Carolina II: two sets of five males × five females, three replications and 150 yield plots. The tiles give σ^2_A, σ^2_D, the degree of dominance, the heritability and the error.

Table 6.2. The controls of card 1.

Control	What it is for
Trait	The whole block is recomputed for the chosen trait.
Environment	With several environments, one of them or all together, which adds the environments and their interactions to the model (Section 6.10).
Inbreeding of the parents (F)	0, 0.5, 0.75 or 1. By default, 1 for line × tester and partial diallels (inbred lines) and 0 for the North Carolina designs and the triple test cross (plants from a population); if the Block 2 field book declared another value, that one is used. F changes the coefficients of the covariances, not the sums of squares.
F_1 tester (L_3)	Only in the triple test cross: which of the three testers is the F_1 of the other two. The app guesses it from the name.
Objective · Significance level	The favorable direction for ranking heterosis, and the level of the tests.

! IMPORTANT · THE VALUE OF F CHANGES Σ² A AND Σ² D

With the same mean squares, $\sigma^2_A = 4\text{Cov(HS)}/(1+F)$ and $\sigma^2_D = 4\sigma^2_{SCA}/(1+F)^2$. Going from F = 0 to F = 1 divides σ^2_A by 2 and σ^2_D by 4. In the line × tester example, with F = 0 σ^2_D would be 1,679 and the degree of dominance 8.76; with F = 1, which is correct for inbred lines, $\sigma^2_D = 420$ and the degree of dominance 6.19. Always check this control.

6.3 The analysis of variance

The card **2 · Analysis of variance of the mating design** fits the model of the design to the plot values: the field strata (sets, replications, blocks) and the partition of the entries specific to each design. Besides df, sums of squares and mean squares, the table gives the **expected mean square** of each source, what it is tested against and its *p* value.

2 · Analysis of variance of the mating design

Source of variation	df	Sum of squares	Mean square	Expected MS	F	Against	p
Sets	1	847,836.4359	847,836.4359	$\sigma^2e + 15\sigma^2_m + 15\sigma^2_f + 3\sigma^2_{mf}$	5.25	quasi-F (12.9 df)	0.0395 *
Replicates (within set and environment)	4	144,345.1536	36,086.2884	σ^2e	1.94	error	0.1097 ns
Males	8	861,052.5036	107,631.563	$\sigma^2e + 15\sigma^2_m + 3\sigma^2_{mf}$	4.27	Males × females	0.0014 **
Females	8	527,022.5938	65,877.8242	$\sigma^2e + 15\sigma^2_f + 3\sigma^2_{mf}$	2.61	Males × females	0.0254 *
Males × females	32	807,267.3062	25,227.1033	$\sigma^2e + 3\sigma^2_{mf}$	1.36	error	0.1295 ns
Experimental error	96	1,783,762.4535	18,580.8589	—	—	—	—
Total	149	4,971,286.4467	—	—	—	—	—

The expected mean squares do not come from a table: the coefficient of every random effect is the trace $tr(A_tZ_kZ_k')/df_t$ of this analysis, so they hold with missing cells too. Every source is tested against the mean square whose expectation differs only by the term being tested; when there is none, a combination with Satterthwaite degrees of freedom is used.

Figure 6.2. The analysis of variance of the North Carolina II. Males and females are tested against their interaction; the interaction, against the error; the sets, with a quasi-*F* with 12.9 degrees of freedom.

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THEORY IN BRIEF · EXPECTED MEAN SQUARES COMPUTED, NOT COPIED

Textbooks give tables of expected mean squares for each balanced design. Block 6 computes them for the experiment you have: the coefficient of each variance in the mean square of each source is a trace, $tr(A_tZ_kZ_k')/df_t$, obtained by fitting each column of random effects as if it were the response. With balanced data it reproduces the textbook coefficients (in Figure 6.2, 15 = $r{\cdot}f$ and 3 = r); with missing crosses or unequal replications the coefficients stop being integers and remain exact. Each source is tested against the mean square whose expectation differs only in the term being tested; if there is none, against a combination of mean squares with Satterthwaite degrees of freedom (quasi-*F*).

Below the table, the figure **Contribution of every source** shows what part of the variation between entries corresponds to each term of the mating design.

6.4 The effects

The card **3 · Effects of parents and crosses** estimates the effect of each parent (its general combining ability) and that of each cross (the specific one), with its standard error and its *t* test. In the North Carolina II, males and females have their own table and the interaction another, plus a dot plot of the general effects and a color matrix of the interaction, set by set.

3 · Effects of parents and crosses

Effects of the males (GCA)

Male	Set	Effect	SE	t	p
♂1	1	−130.2355	31.4798	−4.14	< 0.0001 ***
♂2	1	158.1712	31.4798	5.02	< 0.0001 ***
♂3	1	−5.7168	31.4798	−0.18	0.8563 ns
♂4	1	−6.7701	31.4798	−0.22	0.8302 ns
♂5	1	−15.4488	31.4798	−0.49	0.6247 ns
♂6	2	−2.7207	31.4798	−0.09	0.9313 ns
♂7	2	−12.2853	31.4798	−0.39	0.6972 ns
♂8	2	81.9413	31.4798	2.60	0.0107 *
♂9	2	21.2773	31.4798	0.68	0.5007 ns
♂10	2	−88.2127	31.4798	−2.80	0.0061 **

Effects of the females (GCA)

Female	Set	Effect	SE	t	p
♀1	1	36.3345	31.4798	1.15	0.2513 ns
♀2	1	18.3459	31.4798	0.58	0.5614 ns
♀3	1	35.2272	31.4798	1.12	0.2659 ns
♀4	1	10.4479	31.4798	0.33	0.7407 ns
♀5	1	−100.3555	31.4798	−3.19	0.0019 **
♀6	2	−64.888	31.4798	−2.06	0.0420 *
♀7	2	−41.7627	31.4798	−1.33	0.1878 ns
♀8	2	122.7473	31.4798	3.90	0.0002 ***
♀9	2	13.232	31.4798	0.42	0.6752 ns
♀10	2	−29.3287	31.4798	−0.93	0.3538 ns

Figure 6.3. The effects of the ten potato males and ten females. Because males and females share numbers but are different plants, the app marks them with ♂ and ♀.

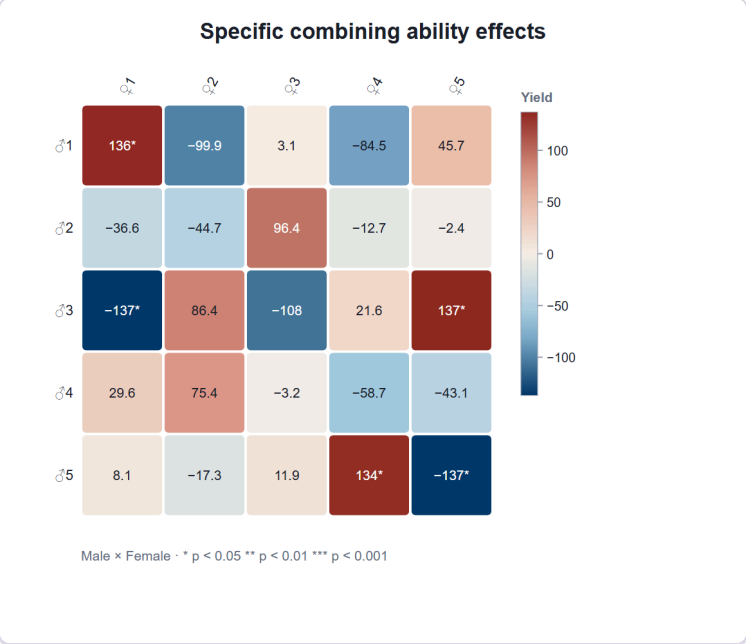

Figure 6.4. The males × females interaction effects of the first set. In red, the crosses that perform better than their parents predict; in blue, those that perform worse.

The effects are estimated by least squares with the restrictions of the model (the effects in each set sum to zero), so their standard error is that of the full contrast: for a line, $\sigma^2(1/rt - 1/rlt)$ and not σ^2/rt, as some textbooks print. In line × tester, the standard error of a line GCA is 2.464, not 2.755.

6.5 Components and genetic parameters

The card **4 · Variance components and genetic parameters** solves for the components from the expected mean squares (on the left) and converts them into covariances of relatives, σ^2_A, σ^2_D, degree of dominance and heritabilities (on the right). Each component is a combination of mean squares, so its standard error comes from them: $\mathrm{Var}(\Sigma c_i M_i) = \Sigma c_i^2 \cdot 2M_i^2/df_i$.

4 · Variance components and genetic parameters

Components of the model

Component	Estimate	SE	% of the total	F test of its source
σ^2_m · Males	5,493.6306	3,612.2715	18.9	0.0014 **
σ^2_f · Females	2,710.0481	2,235.8169	9.3	0.0254 *
σ^2_{mf} · Males × females	2,215.4148	2,284.4424	7.6	0.1295 ns
σ^2_{error} · Experimental error	18,580.8589	2,681.916	64.1	—

The standard errors are asymptotic: $\mathrm{Var}(\Sigma c_i M_i) = \Sigma c_i^2 \cdot 2M_i^2/df_i$. The test of a component being zero is the F test of its source in the analysis of variance, not the ratio of the estimate to its standard error.

Genetic parameters

Parameter	Definition	Value	SE	Reading
Cov(half sibs)	$(1+F)\sigma^2A/4$	4,101.8394	2,144.816	
Cov(full sibs)	$(1+F)\sigma^2A/2 + ((1+F)/2)^2\sigma^2D$	10,419.0935	4,481.5205	
σ^2A	$4\cdot Cov(HS)/(1+F)$	16,407.3574	8,579.2638	additive variance
σ^2D	$4\sigma^2_{mf}/(1+F)^2$	8,861.6592	9,137.7697	dominance variance
Average degree of dominance	$\sqrt{(2\sigma^2D/\sigma^2A)}$	1.0393	—	average overdominance
Narrow-sense h^2	σ^2A/σ^2P (plot basis)	0.3742	—	
Broad-sense H^2	$(\sigma^2A + \sigma^2D)/\sigma^2P$	0.5763	—	
h^2 on a family-mean basis	$\sigma^2A/(\sigma^2A + \sigma^2D + MS_e/re)$	0.5215	—	

ℹ σ^2A from males 21,974.523 and from females 10,840.192: a large difference means maternal effects or different samples.

Figure 6.5. The components and parameters of the potato data. The test that a component equals zero is the *F* of its source: σ^2_m is clearly greater than zero (p = 0.0014) even though its standard error is large.

In the potato data, σ^2_A = 16,407 ± 8,579 and σ^2_D = 8,862 ± 9,138, with an average degree of dominance of 1.04 and a heritability of 0.37 on a plot basis and 0.52 on a family-mean basis. The North Carolina II gives two estimates of σ^2_A, one from the males (21,975) and another from the females (10,840), and the app averages them; when they differ a lot, as here, think of maternal effects or of males and females not coming from the same population.

> **IMPORTANT · NEGATIVE COMPONENTS**
>
> A negative component is reported as obtained: it usually indicates that the true variance is close to zero or that there are few replications. A negative σ^2_D is taken as zero in the denominator of narrow-sense h^2, and with it the degree of dominance, H^2 and the Baker ratio are not computed; with a negative σ^2_A no heritability is computed. The card's warning explains it each time.

6.6 North Carolina I

In the nested design each male is crossed with females that are not crossed with any other male. The variance among males is the half-sib covariance, and that of females within males is the full-sib covariance minus the half-sib one: $\sigma^2_A = 4\sigma^2_m/(1 + F)$ and $\sigma^2_D = 4(\sigma^2_f - \sigma^2_m)/(1 + F)^2$. Males are tested against females within males, and the latter against the error.

Source of variation	df	Sum of squares	Mean square	Expected MS	F	Against	p
Sets	1	0.178	0.178	$\sigma^2e + 4\sigma^2_m + 2\sigma^2_f$	0.10	Males	0.7718 ns
Replicates (within set and environment)	2	0.9965	0.4982	σ^2e	4.66	error	0.0371 *
Males	4	7.3904	1.8476	$\sigma^2e + 4\sigma^2_m + 2\sigma^2_f$	6.89	Females within males	0.0198 *
Females within males	6	1.6083	0.2681	$\sigma^2e + 2\sigma^2_f$	2.51	error	0.0956 ns
Experimental error	10	1.0691	0.1069	—	—	—	—
Between plants within plot	48	3.5533	0.074	—	—	—	—
Total	23	11.2422	—	—	—	—	—

Figure 6.6. The North Carolina I of the example, with three plants measured per plot: the last row is the sampling error among plants, on the plant scale.

In the example the males differ ($F = 6.89$, $p = 0.020$), but the females within males do not ($p = 0.096$), and σ^2_D comes out negative (-1.26 ± 1.39): with six males and two females per male, dominance cannot be estimated well. The assumptions table recalls a limitation of the design itself: because each female has only one male, maternal effects stay inside the female component and overestimate σ^2_D when they exist. With per-plant data, heritability is also given on a plant basis, separating the sampling error among plants from the plot error.

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6.7 North Carolina III and the triple test cross

In the North Carolina III each F_2 plant is crossed with the two parents of the F_1, L_1 and L_2. For each plant, the sum of its two crosses reflects its additive value and the difference, its dominance: the variance of the sums estimates σ^2_A and that of the differences, $2\sigma^2_D$, both after subtracting the error of a pair (Comstock and Robinson, 1952). The triple test cross adds the F_1 as a third tester, L_3, and the contrast $L_1 + L_2 - 2L_3$, which equals zero if there is no epistasis (Kearsey and Jinks, 1968).

THEORY IN BRIEF · SUMS, DIFFERENCES AND EPISTASIS

$\mathrm{Var}(L_1 + L_2) - 2MS_e/r = \sigma^2_A$ · $\tfrac{1}{2}[\mathrm{Var}(L_1 - L_2) - 2MS_e/r] = \sigma^2_D$ · $D = 2\sigma^2_A$, $H = 4\sigma^2_D$, degree of dominance $\sqrt{(H/D)}$

A locus of the F_2 crossed with the two parents gives differences a − d, a and a + d with frequencies ¼, ½ and ¼, so the variance of the differences is $H/2 = 2\sigma^2_D$. The correlation between sums and differences tells you toward which parent dominance goes. In the triple test cross, the mean of the contrast $L_1 + L_2 - 2L_3$ tests overall epistasis (type i) and its variation among individuals, epistasis of types j and l.

Source of variation	df	Sum of squares	Mean square	Expected MS	F	Against	p
Sets	3	2.795	0.9317	$\sigma^2e + 4\sigma^2_i$	0.53	F_2 plants	0.6682 ns
Replicates (within set and environment)	4	3.205	0.8012	σ^2e	1.10	error	0.3762 ns
F_2 plants	12	20.97	1.7475	$\sigma^2e + 4\sigma^2_i$	2.40	error	0.0278 *
Testers	1	1.1025	1.1025	$\sigma^2e + 2\sigma^2_{it}$	0.47	Plants × testers	0.5046 ns
Testers × sets	3	0.8275	0.2758	$\sigma^2e + 2\sigma^2_{it}$	0.12	Plants × testers	0.9476 ns
Plants × testers	12	27.965	2.3304	$\sigma^2e + 2\sigma^2_{it}$	3.20	error	0.0055 **
Experimental error	28	20.405	0.7288	—	—	—	—
Total	63	77.27	—	—	—	—	—

Figure 6.7. The North Carolina III of the example: 16 F_2 plants in four sets, crossed with P1 and P2. The plants are tested against the error, and so is the plants × testers interaction: the testers are fixed.

The testers are two fixed lines, not a sample, so the interaction of each plant with them does not enter the expected mean square of the plants (Comstock and Robinson's restricted model). That is why the plants are tested against the error ($F = 2.40$, $p = 0.028$), and the plant component, 0.255, is exactly $\sigma^2_A/4 = 1.019/4$. The plants × testers interaction is significant ($p = 0.006$): there is dominance, with $\sigma^2_D = 0.80 \pm 0.49$ and a degree of dominance of 1.25.

The card **5 · Sums, differences and epistasis** brings those computations together. With the simulated triple test cross from the example gallery (eight loci without epistasis, D = 11.52 and H = 8; $\sigma^2_A = 5.76$, $\sigma^2_D = 2$), the block estimates $\sigma^2_A = 4.97 \pm 2.20$ and $\sigma^2_D = 1.86 \pm 0.94$, with a degree of dominance of 0.87 (true value, 0.83).

5 · Sums, differences and epistasis

Quantity	Value	Test	What it measures
Variance of the sums	8.2564	29 df	$\sigma^2A + 2MS/r$
Variance of the differences	7.0092	29 df	$2\sigma^2D + 2MS/r$
Error of a pair (2MS/r)	3.2849	178 df	subtracted from both variances
Mean of the differences	10.1657	t = 21.03 · p < 0.0001	additive difference between the two testers
Sum–difference correlation	−0.5517	p = 0.0013	dominance toward P1

Figure 6.8. Sums and differences of the simulated triple test cross. The mean of the differences measures the additive difference between the testers; the correlation between sums and differences (−0.55, $p = 0.001$) says that dominance goes toward P1.

NOTE · THE ORDER OF THE TESTERS

The epistasis contrast makes sense only if L_3 is the F_1 of L_1 and L_2. The app orders the testers by name (the F_1 last); if the names do not say so, choose the right one in **F_1 tester (L_3)**. With the wrong order, data without epistasis would show it as highly significant.

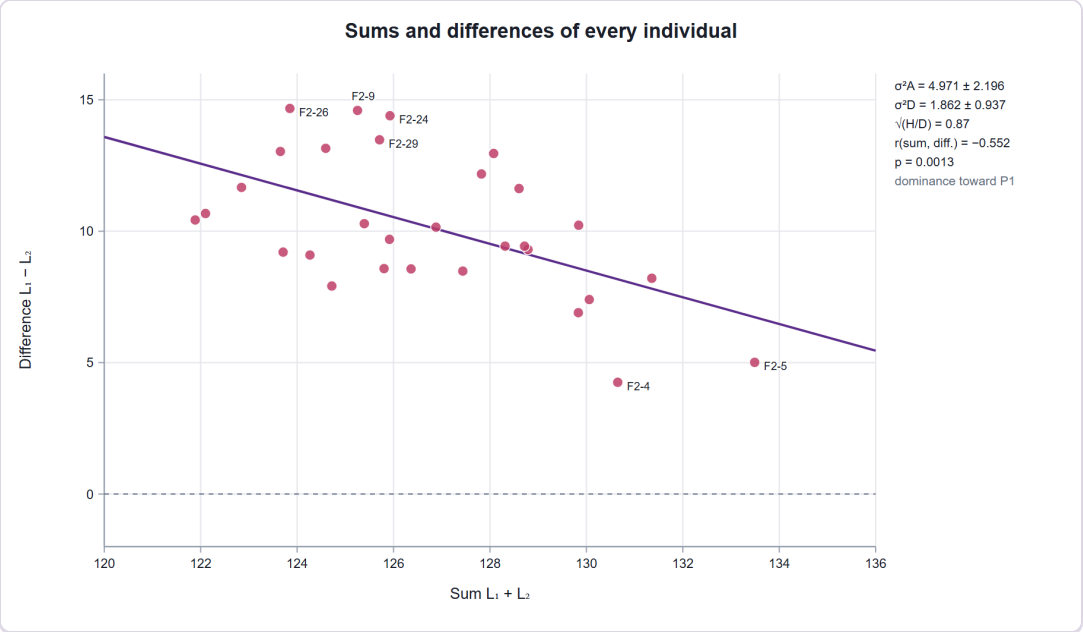

Figure 6.9. Each F_2 individual by its sum and its difference. The negative slope indicates that individuals with more alleles that increase the value have smaller differences: dominance goes toward the high-value parent.

Test of epistasis (Kearsey & Jinks 1968)

Source	df	Sum of squares	Mean square	F	p
Total epistasis	30	84.9113	2.8304	0.57	0.9630 ns
· overall (i)	1	0.0714	0.0714	0.01	0.9043 ns
· among individuals (j and l)	29	84.8399	2.9255	0.59	0.9510 ns

The contrast $L_1 + L_2 - 2L_3$ of every individual is zero without epistasis. Mean = 0.069 ± 0.5731 (t = 0.12, p = 0.9043): no overall epistasis is detected; among individuals p = 0.9510: it does not differ between individuals. With epistasis, σ^2A and σ^2D are biased.

Figure 6.10. The test of epistasis. In the simulated data without epistasis, neither the mean of the contrast (p = 0.90) nor its variation among individuals (p = 0.95) is significant, as it should be. Below, a figure shows the contrast of each individual with its interval.

6.8 Line × tester

Line × tester (Kempthorne, 1957) crosses each line with a few testers. If the parents were also evaluated, the entries are split into parents, crosses and the contrast between both groups; the crosses, into lines, testers and their interaction. The GCA of lines and testers comes from their effects, and the SCA from the interaction.

In the example, the lines differ in their GCA (L3, with +23.8, is the best; L4, with −13.9, the worst) and so do two testers (T7, +6.4; T8, −6.7), but with five lines and three testers their variances do not reach significance against the interaction, which dominates. The tester variance comes out negative, and σ^2_A, computed from the average of the line and tester variances, ends up at 21.9 ± 197: almost all the variation among crosses is specific. The warning in card 4 explains why: a few testers chosen on purpose do not estimate the half-sib covariance of a population.

Source of variation	df	Sum of squares	Mean square	Expected MS	F	Against	p
Replicates (within set and environment)	3	83.0001	27.6667	$\sigma^2 e$	0.30	error	0.8225 ns
Parents against crosses	1	53.6629	53.6629	$\sigma^2 e + 4.17\sigma^2_l + 6.96\sigma^2_t + 1.39\sigma^2_{lt}$	0.03	quasi-F (11.1 df)	0.8663 ns
Among parents	7	6,299.6202	899.9457	$\sigma^2 e$	9.88	error	< 0.0001 ***
Lines	4	10,318.3614	2,579.5903	$\sigma^2 e + 12\sigma^2_l + 4\sigma^2_{lt}$	1.46	Lines × testers	0.3008 ns
Testers	2	1,718.9258	859.4629	$\sigma^2 e + 20\sigma^2_t + 4\sigma^2_{lt}$	0.49	Lines × testers	0.6324 ns
Lines × testers	8	14,162.3672	1,770.2959	$\sigma^2 e + 4\sigma^2_{lt}$	19.44	error	< 0.0001 ***
Experimental error	66	6,010.298	91.0651	—	—	—	—
Total	91	38,646.2355	—	—	—	—	—

Figure 6.11. Five lines × three testers plus eight parents. Lines and testers are tested against their interaction; the interaction is highly significant ($F = 19.44$). The parents versus crosses contrast carries a quasi-F because the mean of the crosses contains the random effects of lines and testers.

5 · Heterosis of every cross

Cross	Mean	Mid-parent	Heterosis	%	p	Over the better parent	%	p
L3 × T8	117.735	78.035	39.7	50.9	< 0.0001 ***	35.735	43.6	< 0.0001 ***
L3 × T7	111.54	82.515	29.025	35.2	< 0.0001 ***	20.58	22.6	0.0033 **
L5 × T8	94.82	68.05	26.77	39.3	< 0.0001 ***	12.82	15.6	0.0618 ns
L1 × T7	107.64	88.303	19.337	21.9	0.0015 **	16.68	18.3	0.0160 *
L3 × T6	100.885	87.23	13.655	15.7	0.0225 *	0.495	0.5	0.9417 ns
L4 × T6	105.795	96.015	9.78	10.2	0.0989 ns	5.405	5.4	0.4260 ns
L5 × T7	81.935	72.53	9.405	13	0.1123 ns	−9.025	−9.9	0.1857 ns
L5 × T6	84.15	77.245	6.905	8.9	0.2416 ns	−16.24	−16.2	0.0189 *
L2 × T7	97.64	94.61	3.03	3.2	0.6058 ns	−0.62	−0.6	0.9271 ns
L2 × T8	85.65	90.13	−4.48	−5	0.4460 ns	−12.61	−12.8	0.0661 ns
L1 × T6	68.55	93.018	−24.468	−26.3	< 0.0001 ***	−31.84	−31.7	< 0.0001 ***

Figure 6.12. With the parents in the experiment, card 5 gives the heterosis of every cross relative to the mean of its parents and to the better of them. L3 × T8 exceeds the mean of its parents by 51% and the better parent by 44%.

6.9 Partial diallel

In a partial diallel each parent is crossed with s of the others, usually in the circulant order of Kempthorne and Curnow (1961) that Block 2 prepares. Block 6 fits $y_{ij} = \mu + g_i + g_j + s_{ij}$ by least squares on the crosses that were made, so it works for the circulant arrangement and for any incomplete set of crosses, and it gives the GCA of each parent and the SCA of each cross as in a Method 4 diallel.

Here neither GCA ($p = 0.085$) nor SCA ($p = 0.52$) is significant as a whole, although two lines stand out: line 2 ($\hat{g} = +0.59$, $p = 0.009$) and line 7 ($\hat{g} = -0.49$, $p = 0.028$). σ^2_{SCA} comes out negative, so there is no Baker ratio or degree of dominance; with $F = 1$, $\sigma^2_A = 0.13 \pm 0.12$ and the heritability is 0.13 on a plot basis.

Source of variation	df	Sum of squares	Mean square	Expected MS	F	Against	p
Replicates (within set and environment)	3	17.0744	5.6915	$\sigma^2 e$	6.85	error	0.0005 ***
General combining ability	7	13.1418	1.8774	$\sigma^2 e + 17.14\sigma^2_{GCA} + 4\sigma^2_{SCA}$	2.42	Specific combining ability	0.0852 ns
Specific combining ability	12	9.297	0.7747	$\sigma^2 e + 4\sigma^2_{SCA}$	0.93	error	0.5223 ns
Experimental error	57	47.3843	0.8313	—	—	—	—
Total	79	86.8975	—	—	—	—	—

Figure 6.13. The partial diallel of eight maize lines with $s = 5$ (20 crosses and four replications). The coefficient of σ^2_{GCA}, 17.14 = 120/7, is not an integer: the trace computation obtains it without tables.

6.10 Several environments, downloads and the next block

With data from several environments, the Environment menu lets you analyze one of them or all together. In the second case the model adds the environments and the interaction of each genetic term with them (males × environments, females × environments, lines × testers × environments…), and the expected mean squares and the tests are recomputed by traces for that model; σ^2_A and σ^2_D then discount their interaction with the environments.

The card 6 · Assumptions, formulas and references summarizes the model and formulas of each design with their references. Results (.xlsx) downloads the analysis of variance, the components, the parameters, the effects and, depending on the design, the sums and differences or the heterosis. Each figure has its own download panel.

The Continue button takes you to Block 8 (genetic parameters, correlations and selection indices among the families) or, with several environments, to Block 9 (genotype × environment). ← Data goes back to Block 3.

B7

CHAPTER SEVEN · BLOCK 7 · GENERATIONS

Generations and heterosis

A cross between two pure lines and its descendants—the F_1, the F_2 and the backcrosses—are enough to find out how a trait is inherited. Their means tell whether the genes act additively, with dominance or with epistasis; their variances separate the additive part from the dominance part and from the environment. Block 7 runs Mather's scaling tests, Cavalli's joint scaling test with the sequence of models, the components D, H, F and E with their heritabilities, the minimum number of effective factors, heterosis and inbreeding depression.

7.1 The generations and what they expect

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7.6 Generation variances: D, H, F and E

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7.1 The generations and what they expect

Every generation mixes the three genotypes of each locus—AA, Aa and aa—in known proportions. That is why its mean is a fixed combination of a few parameters—the mean m, the additive effects [d], the dominance effects [h] and the three two-locus epistases [i], [j] and [l]—and its variance, another combination of D, H, F and E. With six generations or more, those combinations can be solved.

Table 7.1. The generations that Block 7 recognizes, the names it accepts in the generation column, their expected mean in the F∞ metric and their expected variance without epistasis or linkage. P_1 is the parent that carries the alleles that increase the value.

Generation	Names accepted	Expected mean	Expected variance
P_1, P_2	P1, P2	m ± [d] + [i]	E
F_1 and its reciprocal	F1; F1R, RF1, F1rec	m + [h] + [l]	E
F_2	F2	m + ½[h] + ¼[l]	½D + ¼H + E
F_3	F3	m + ¼[h] + $^1/_{16}$[l]	¾D + $^3/_{16}$H + E
BC_1 ($F_1 \times P_1$)	RC1, BC1, B1, retrocruza1	m + ½[d] + ½[h] + ¼[i] + ¼[j] + ¼[l]	¼D + ¼H − ½F + E
BC_2 ($F_1 \times P_2$)	RC2, BC2, B2, retrocruza2	m − ½[d] + ½[h] + ¼[i] − ¼[j] + ¼[l]	¼D + ¼H + ½F + E
Selfed BCs	BC1S, B11, BS1 (and those of P_2)	m ± ½[d] + ¼[h] + ¼[i] ± ⅛[j] + $^1/_{16}$[l]	½D + $^3/_{16}$H ∓ ¼F + E

Block 3 recognizes the generations design when the generation column (or the name of the entries) has at least three of these codes and they are the majority of the entries. The data can be plants measured one by one, with or without plots and blocks, or one value per plot. The scaling tests use P_1, P_2, F_1, F_2 and the backcrosses; the models, all the generations present. The coefficients of the F_3 and of the selfed backcrosses were derived from the genotypic frequencies and checked by simulation.

7.2 Data and controls

The **1 · Generations and model** card takes the data from Block 3 (without the observations you have excluded), summarizes in one line the generations, the number of plants or plots and the metric, and gathers the main results in tiles: the heterosis over the mid-parent, the standard heterosis if there is a check, [d] and [h] of the chosen model, the degree of dominance, the narrow-sense heritability (or D, if it comes out negative) and the minimum number of effective factors. Those that cannot be computed do not appear.

There are six controls:

- **Trait** and, if there are several environments, **Environment**.
- **Metric of the parameters**: Mather and Jinks's F∞ or Hayman's F_2 (Section 7.5).

- **Variance of every mean**: automatic from the design (Section 7.3), or forced to the plot error or to the variance among individuals.
- **Model in tables and figure**: the one chosen by the sequence or the perfect-fit solution with the six parameters.
- **Objective**: the highest or the lowest value. It only defines which is the better parent for heterosis; for days to flowering or plant height, where low values are sought, you should change it.
- **Significance level**: 0.05, 0.01 or 0.10. It decides which model "fits" and which parameters count as significant.

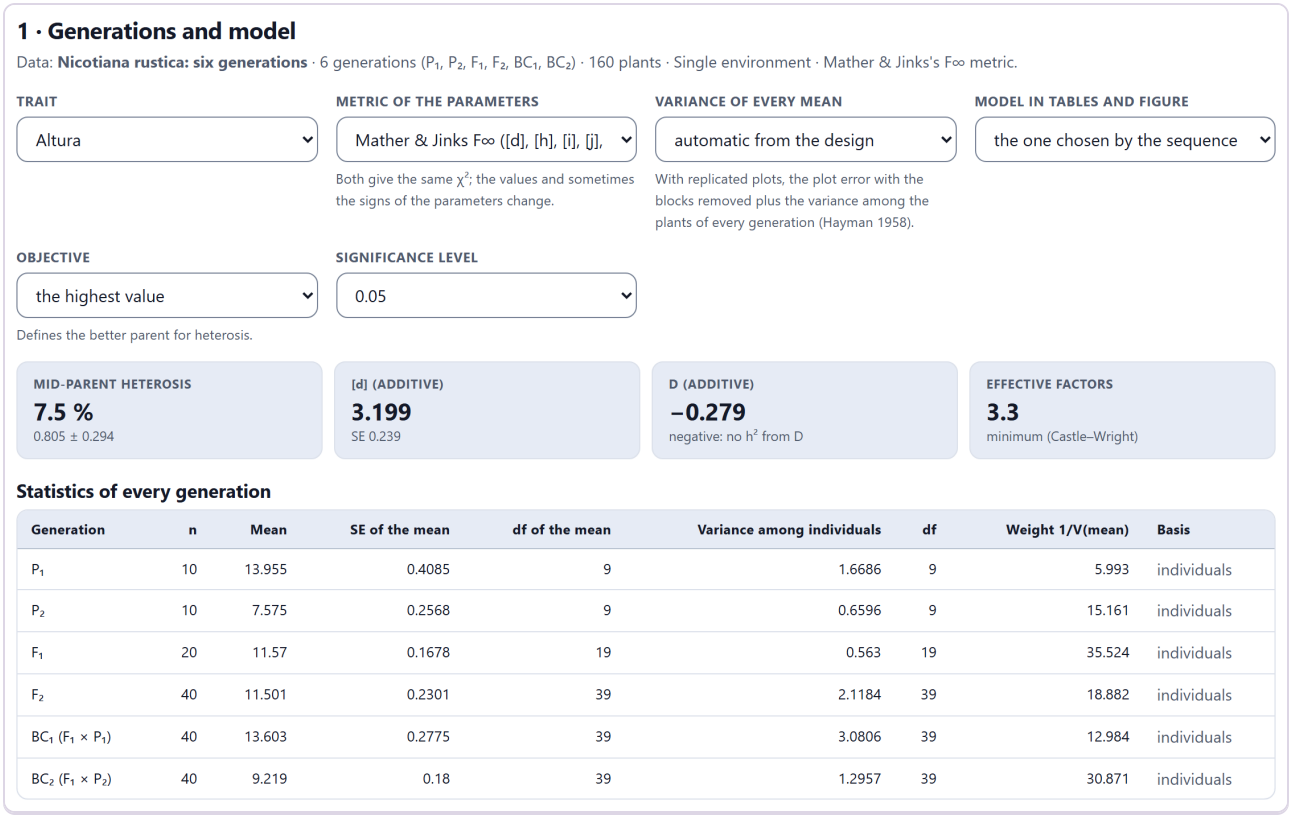


Figure 7.1. Card 1 with the *Nicotiana rustica* example: six generations and 160 plants measured one by one. D comes out negative and the tile says so instead of giving a heritability.

The **Statistics of every generation** table is the basis of everything else: number of plants, mean, standard error and degrees of freedom of the mean, variance among individuals with its degrees of freedom, the weight 1/V(mean) and the basis of that weight. The *N. rustica* example comes from Gale, Mather and Jinks (1977), who published the number of plants, the mean and the variance of each generation; the 160 plants were rebuilt with exactly those values, which are all these analyses use.

7.3 The variance of every mean

All the tests of the block compare means, and each mean is weighted by its precision. Where its variance comes from depends on how the experiment was planted, and in automatic mode the Variance of every mean menu distinguishes three cases.

- **Plants randomized one by one** (the *N. rustica* example, or one plant per plot): $V(\bar{y}) = s^2/n$, the variance among individuals divided by their number. The basis reads individuals.
- **Replicated plots of several plants**: the mean of a plot varies because of the plot error, common to all generations, and because of the plants it contains, which vary much more in an F_2 than in a parent. The basis reads means of *r* plots.
- **One value per plot**: the residual mean square of the plot means, the same for all generations. Without plants there is no variance among individuals, so the block computes neither D, H, F, E nor effective factors, and warns you.

$V(\bar{y}_g) = (\sigma^2_p + s^2_g/k)/r$ · $\sigma^2_p = MS_{res} - \text{mean}(s^2/k)$ · MS_{res}: plot means adjusted for generation and block

Here s^2_g is the variance among the plants of the same plot of generation *g*, *k* the number of plants per plot and *r* the number of plots. The plot error σ^2_p comes from the residual mean square of the plot means once the generation and block effects are removed, minus the part the plants already contribute (Hayman, 1958). Since it is a combination of mean squares, the degrees of freedom of each mean are Satterthwaite's (1946) and are not integers.

Generation	n	Mean	SE of the mean	df of the mean	Variance among individuals	df	Weight 1/V(mean)	Basis
P_1	30	60.3467	0.6671	10.2	4.0664	27	2.247	means of 3 plots
P_2	30	44.79	0.6426	8.9	3.1071	27	2.421	means of 3 plots
F_1	30	54.06	0.6635	10.0	3.9236	27	2.272	means of 3 plots
F_2	120	50.905	0.6381	8.7	11.7389	117	2.456	means of 3 plots
BC_1 ($F_1 \times P_1$)	60	57.665	0.6472	9.2	6.5697	57	2.387	means of 3 plots
BC_2 ($F_1 \times P_2$)	60	48.9867	0.6945	12.1	10.372	57	2.073	means of 3 plots
Testigo	30	50.8133	0.6517	12.5	4.3531	27	2.355	means of 3 plots

Figure 7.2. The simulated example of the gallery: six generations and a check in three blocks, with one plot of each per block (10 plants of the parents, the F_1 and the check, 40 of the F_2 and 20 of each backcross). The generation means have standard errors of 0.64 to 0.69 with 8.7 to 12.1 degrees of freedom; the last row is the check, which only enters the standard heterosis (Section 7.7).

In the simulated example, the residual mean square of the 18 plot means is 1.303 with 10 degrees of freedom; the plants contribute 0.375, so $\sigma^2_p = 0.928$. A mixed model fitted by restricted maximum likelihood, with a plot effect and one residual variance per generation, gives practically the same variances of the means (0.445 against 0.445 in P_1, 0.407 against 0.408 in the F_2).

> **IMPORTANT · THE BLOCKS DO NOT ENTER THE VARIANCE OF THE MEANS**
>
> Every generation is in all blocks, so a good block raises all the means equally and the effect cancels out in any comparison among them. If the variance of each mean were computed from the plot means as they come, with the blocks inside, the standard errors would grow for no reason and the tests would lose almost all their power. In 1,000 simulations like the example, the true variance of scaling test C was 4.96; the block's calculation estimated it at 4.96 on average, whereas with the blocks inside it would have been 15.9.

The menu lets you force the basis. plot error (blocks removed) applies the formula above whenever there is more than one plot per generation; among individuals / n uses s^2/n, which is correct with plants randomized one by one but ignores the plot error when there are plots: the standard errors come out too small and any difference looks significant. If the generations do not all have the same basis, a warning points it out.

7.4 Scaling tests

Mather's (1949) scaling tests are contrasts among means that equal zero when the additive–dominance model is enough. Each one combines different generations and is therefore sensitive to a different kind of epistasis; the last column of the table tells what each test estimates in the F∞ metric.

$$A = 2BC_1 - P_1 - F_1 \quad \cdot \quad B = 2BC_2 - P_2 - F_1 \quad \cdot \quad C = 4F_2 - 2F_1 - P_1 - P_2$$
$$D = 2F_2 - BC_1 - BC_2 \quad \cdot \quad D' = P_1 + P_2 + 2F_2 - 4F_3$$

2 · Scaling tests

Test	Value	SE	t	df	p	What it estimates
$A = 2BC_1 - P_1 - F_1$	1.681	0.7093	2.37	45.4	0.0221 *	−½[i] + ½[j] − ½[l]
$B = 2BC_2 - P_2 - F_1$	−0.707	0.4729	−1.49	52.4	0.1409 ns	−½[i] − ½[j] − ½[l]
$C = 4F_2 - 2F_1 - P_1 - P_2$	1.334	1.0921	1.22	62.8	0.2265 ns	−2[i] − [l]
$D = 2F_2 - BC_1 - BC_2$	0.18	0.5668	0.32	77.6	0.7517 ns	−½[i]

1 of 4 tests departs from zero (A): the additive–dominance model is not enough and epistasis has to be included. D tests [i] alone, C − 4D estimates −[l] and A − B tests [j].

Figure 7.3. The scaling tests of *N. rustica*. Only A departs from zero; the note below recalls that D tests [i] alone, that C − 4D estimates −[l] and that A − B tests [j].

In *N. rustica*, A = 1.681 ± 0.709 (t = 2.37, p = 0.022) and the other three are not significant. A single significant test is enough to say that the additive–dominance model falls short. Since A − B = [j] equals 2.39 and D = −½[i] is almost zero, suspicion falls on additive × dominance epistasis, which the joint test confirms. D' appears when there is an F_3.

The variance of each test is $\Sigma c^2 V(\bar{y})$, with the squared coefficients. Its degrees of freedom are Satterthwaite's for that combination: 45.4 for A, fewer than the 67 obtained by adding those of its three generations, because a third of the variance of A comes from P_1, which has only 9 degrees of freedom. Below the table, a figure shows each test with its confidence interval; the significant bars are drawn in solid color.

7.5 Joint scaling test

Cavalli's (1952) joint scaling test fits the means of all generations at once by weighted least squares, with weight $1/V(\bar{y})$. If the model describes the means, the weighted sum of squared deviations follows a χ^2 with as many degrees of freedom as generations minus parameters.

3 · Generation means: joint scaling test

Sequence of models

Model	df	χ^2	p	Fits?	Parameters significant
m, [d], [h]	3	9.438	0.0240 *	no	all
m, [d], [h], [i]	2	8.899	0.0117 *	no	not all
m, [d], [h], [l]	2	9.307	0.0095 **	no	not all
m, [d], [h], [i], [j]	1	0.128	0.7204 ns	yes	not all
m, [d], [h], [i], [l]	1	8.506	0.0035 **	no	not all
m, [d], [h], [i], [j], [l]	0	perfect fit	—	—	not all
m, [d], [i], [j] (without the non-significant parameters) chosen	2	0.146	0.9294 ns	yes	all

Chosen model

Parameter	Meaning	Estimate	SE	t	p
m	mean of all the pure lines the cross can give	11.5661	0.1174	98.52	< 0.0001 ***
[d]	additive effects (½ of the difference between parents)	3.1992	0.2393	13.37	< 0.0001 ***
[i]	additive × additive epistasis	−0.7799	0.2767	−2.82	0.0048 **
[j]	additive × dominance epistasis	2.3046	0.7689	3.00	0.0027 **

Observed and expected

Generation	Observed mean	Expected by the model	Difference	Contribution to χ^2
P_1	13.955	13.9854	−0.0304	0.0055
P_2	7.575	7.587	−0.012	0.0022
F_1	11.57	11.5661	0.0039	0.0005
F_2	11.501	11.5661	−0.0651	0.0801
BC_1 ($F_1 \times P_1$)	13.603	13.5469	0.0561	0.0409
BC_2 ($F_1 \times P_2$)	9.219	9.1954	0.0236	0.0172

$\chi^2 = 0.1464$ with 2 df ($p = 0.9294$): the model describes the means.

Figure 7.4. Card 3 with *N. rustica*. The model m, [d], [h] is rejected; m, [d], [h], [i], [j] fits, but [h] is not significant, and removing it leaves m, [d], [i], [j], the chosen one ($\chi^2_2 = 0.146$, $p = 0.93$). On the right, each observed mean, the one expected by the model and its contribution to the χ^2.

VERIFIED · THE PUBLISHED VALUES

The model m, [d], [h] with the data of Gale, Mather and Jinks (1977) gives $\chi^2_3 = 9.438$ with $P = 2.40\%$ and m = 11.0036 ± 0.1914, [d] = 3.6349 ± 0.1924, [h] = 0.5937 ± 0.2813, the values of the paper; the second data set of the same paper gives $\chi^2_3 = 12.5731$, also the published one.

The block fits six models of increasing complexity and chooses the simplest one that is not rejected and has all its parameters significant; if none achieves this, it takes the first one that is not rejected and removes the non-significant parameters one at a time, starting with the least significant, as long as the fit holds (Jayasekara and Jinks, 1976).

The chosen model says that the height of *N. rustica* in this cross depends on additive effects ([d] = 3.20 ± 0.24), additive × additive epistasis ([i] = −0.78 ± 0.28) and, above all, additive × dominance epistasis ([j] = 2.30 ± 0.77); dominance alone is not significant. The figure below places the observed means with their intervals next to the expected ones, with the mid-parent line as a reference.

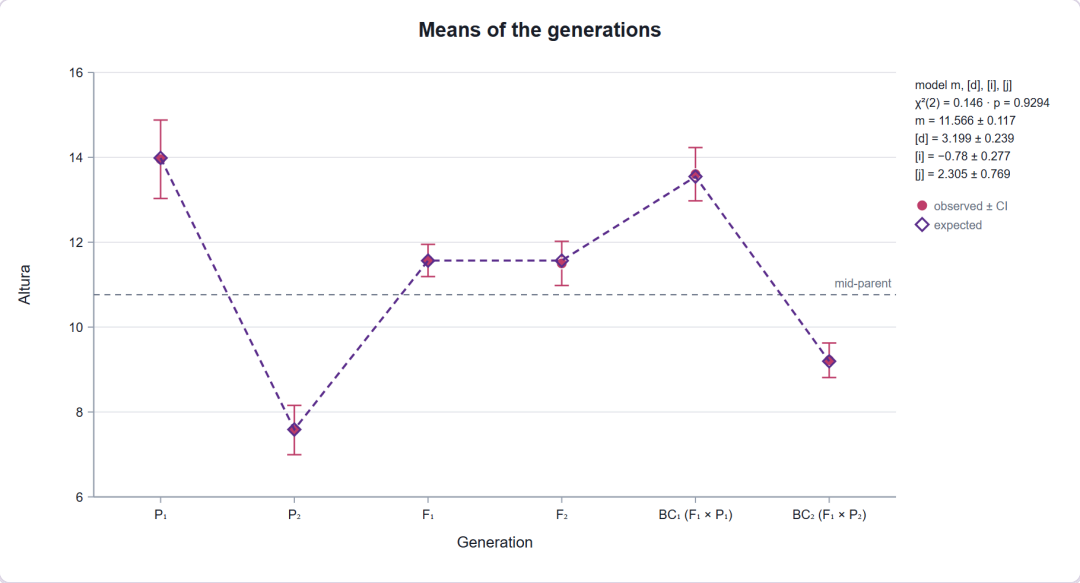


Figure 7.5. Observed means (circles, with confidence interval) and means expected by the chosen model (diamonds). On the right, the model and its estimates.

With Model in tables and figure set to perfect fit, the tables and the figure show the six-parameter solution, which with six generations leaves no degrees of freedom for the χ^2: only the *t* tests of each parameter remain. If the model includes [h] and [l], the note classifies the epistasis as complementary (same sign) or duplicate (opposite signs), after Jinks and Jones (1958). The classification only makes sense if both parameters are significant, and it depends on the metric.

Table 7.2. The perfect-fit solution of *N. rustica* in the two metrics. The χ^2 of any model is the same in both; the parameters are not.

Metric	m	d	h	i	j	l
Mather and Jinks F∞: [d], [h]…	11.125	3.190	1.059	−0.360	2.388	−0.614
Hayman F_2: d, h…	11.501	4.384	0.445	−0.360	1.194	−0.614

Both metrics describe the same thing from a different origin: the F∞ measures from the mean of the pure lines the cross can produce, and Hayman's from the F_2. You go from one to the other with $m_{F\infty} = m - \frac{1}{2}h + \frac{1}{4}l$, [d] = d − j, [h] = h − l, [i] = i, [j] = 2j and [l] = l. Many papers that cite Hayman (1958) actually use the Jinks and Jones convention, with [j] = 2j; with Hayman's metric, the note gives that value and the full equivalence. When comparing with a paper, check the metric first.

With plot means. When the variance of each mean is estimated with few degrees of freedom, as in an experiment with three blocks, the χ^2 rejects too often: in simulations without epistasis, about 9% of the time at 5%. Then the p of each model is that of the Welch–James approximation (Johansen, 1980), which compares χ^2/c with an F whose degrees of freedom depend on those of each variance; with it, the rejections drop to about 6%. With individual plants, Cavalli's χ^2 is kept, which is the one in the papers.

Sequence of models

Model	df	χ^2	p	Fits?	Parameters significant
m, [d], [h]	3	12.464	0.0195 *	no	not all
m, [d], [h], [i] chosen	2	4.914	0.1123 ns	yes	all
m, [d], [h], [l]	2	10.201	0.0160 *	no	not all
m, [d], [h], [i], [j]	1	4.019	0.0524 ns	yes	not all
m, [d], [h], [i], [l]	1	0.726	0.4008 ns	yes	all
m, [d], [h], [i], [j], [l]	0	perfect fit	—	—	not all

Chosen model

Parameter	Meaning	Estimate	SE	t	p
m	mean of all the pure lines the cross can give	48.7536	1.3523	36.05	< 0.0001 ***
[d]	additive effects (½ of the difference between parents)	7.9774	0.416	19.17	< 0.0001 ***
[h]	dominance effects	5.678	1.815	3.13	0.0018 **
[i]	additive × additive epistasis	4.0029	1.4568	2.75	0.0060 **

Observed and expected

Generation	Observed mean	Expected by the model	Difference	Contribution to χ^2
P_1	60.3467	60.734	−0.3873	0.3371
P_2	44.79	44.7792	0.0108	0.0003
F_1	54.06	54.4317	−0.3717	0.3138
F_2	50.905	51.5926	−0.6876	1.1611
BC_1 ($F_1 \times P_1$)	57.665	56.5821	1.0829	2.7995
BC_2 ($F_1 \times P_2$)	48.9867	48.6047	0.382	0.3025

χ^2 = 4.9144 with 2 df (p = 0.1123; Welch–James p, F with 24.7 denominator df): the model describes the means. With plot means the variances of the means have few degrees of freedom and the χ^2 of Cavalli rejects too often; the p of every model comes from the Welch–James approximation (Johansen 1980).

Figure 7.6. The simulated example, whose true model is m = 50, [d] = 8, [h] = 6, [i] = 3, [j] = 0 and [l] = −2. The sequence chooses m, [d], [h], [i] (p = 0.11 by Welch–James) with estimates close to the true ones.

In the simulated example, [d] = 7.98 ± 0.42, [h] = 5.68 ± 1.82 and [i] = 4.00 ± 1.46, against the true 8, 6 and 3. The dominance × dominance epistasis ([l] = −2) is too small to detect with three blocks: the model with [l] also fits, but the sequence keeps the simpler one. The scaling tests already announced it: C = −9.64 ± 3.02 and D = −4.84 ± 1.59 are significant, and both contain [i].

7.6 Generation variances: D, H, F and E

The 4 · Generation variances: D, H, F and E card fits the variances among individuals of each generation to their expected values (Table 7.1) by weighted least squares. D = Σd^2 is the additive variance, H = Σh^2 the dominance variance, F = Σdh the association between them (positive when dominance points toward P_1; then the backcross to P_1 varies less than the other) and E the environmental variance. The parents and the F_1, which do not segregate, estimate E; the F_2 and the backcrosses, the rest. Each variance is weighted by df/($2V^2$), with V its fitted value, and the weights are recomputed until they no longer change (Mather and Jinks).

4 · Generation variances: D, H, F and E

Components

Component	Meaning	Estimate	SE	t
D	additive variance (Σd^2); σ^2A = ½D in an F_2	13.0721	7.6711	1.70
H	dominance variance (Σh^2); σ^2D = ¼H in an F_2	6.0151	11.3013	0.53
F	association between additive and dominance effects (Σdh)	3.8023	2.2998	1.65
E	environmental variance among individuals	3.699	0.5812	6.36

Heritability and dominance

Parameter	Definition	Value	Reading
σ^2A (F_2 basis)	½D	6.5361	
σ^2D (F_2 basis)	¼H	1.5038	
Average degree of dominance	√(H/D)	0.6783	partial dominance
Narrow-sense h^2	½D/(½D + ¼H + E)	0.5568	
Broad-sense H^2	(½D + ¼H)/(½D + ¼H + E)	0.6849	
Warner's (1952) narrow-sense h^2	[2V(F_2) − V(BC_1) − V(BC_2)]/V(F_2)	0.5568	uses three variances only
Broad-sense H^2 from F_2	[V(F_2) − E]/V(F_2)	0.6849	

Figure 7.7. The components of the simulated example (true values: D = 10, H = 6, F = 2, E = 4) and, on the right, what is derived from them: σ^2_A and σ^2_D in the F_2, the average degree of dominance and four heritabilities.

In the simulated example, D = 13.07 ± 7.67, H = 6.02 ± 11.30, F = 3.80 ± 2.30 and E = 3.70 ± 0.58; the fit of the variances is not rejected ($\chi^2_2 = 0.53$, $p = 0.77$). The standard errors of D and H are large: separating the additive variance from the dominance variance with six variances requires many segregating plants. From them come an average degree of dominance $\sqrt{(H/D)} = 0.68$ (partial dominance; the true value is 0.77), a narrow-sense h^2 of 0.557 and a broad-sense H^2 of 0.685, both in the F_2 and per plant.

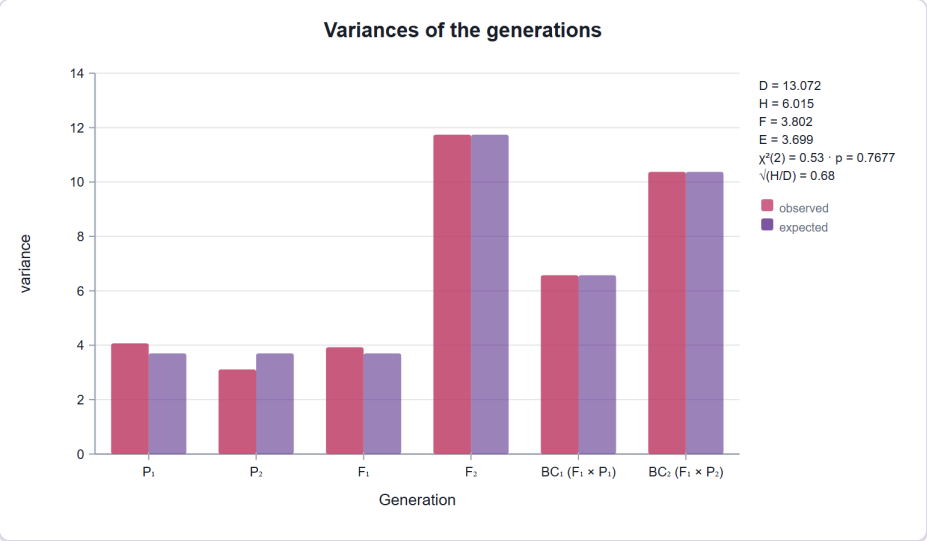


Figure 7.8. Observed and expected variances of the simulated example. The parents and the F_1 share the expected value E; the F_2 and the backcrosses are fitted exactly.

Warner's (1952) h^2, [2V(F_2) − V(BC_1) − V(BC_2)]/V(F_2), uses only three variances. With the six basic generations it coincides exactly with the narrow-sense h^2, because D, H and F are fitted with no residual to the F_2 and the backcrosses; it differs when there are more generations, such as the F_3. The broad-sense H^2 from the F_2, [V(F_2) − E]/V(F_2), is the other direct reading.

IMPORTANT · NEGATIVE COMPONENTS

In *N. rustica*, D = −0.28 ± 2.44 and F = −1.78 ± 0.76, and the fit of the variances is borderline (χ^2_2 = 5.41, p = 0.067). With a negative D, neither the degree of dominance nor the heritabilities that use D and H are computed; Warner's h² (−0.066) and the H² from the F_2 (0.596) are, because they use the variances directly. A negative D usually indicates few plants, linkage or epistasis.

7.7 Heterosis and inbreeding depression

The 5 · Heterosis, inbreeding depression and effective factors card compares the F_1 with the mean of its parents and with the better of them, and the F_2 with the F_1. The difference is tested, not the percentage, with its standard error and Satterthwaite degrees of freedom; the percentage is given to help read it.

5 · Heterosis, inbreeding depression and effective factors

Measure	Difference	SE	%	t	df	p
Mid-parent heterosis	0.805	0.2939	7.5	2.74	28.1	0.0106 *
Better-parent heterosis	−2.385	0.4416	−17.1	−5.40	12.1	0.0002 ***
Inbreeding depression ($F_1 - F_2$)	0.069	0.2848	0.6	0.24	57.9	0.8094 ns

Potence ratio (F_1 − mid-parent)/(½|$P_1 - P_2$|) = 0.252 ± 0.101: weak dominance toward the higher parent. Without epistasis the F_2 should keep half of the F_1 heterosis: 11.167 was expected and 11.501 observed. With a check in the data (an entry that is not a generation, such as the commercial variety to be replaced), the table adds the standard heterosis.

Figure 7.9. The heterosis of *N. rustica*. The F_1 exceeds the mean of its parents by 7.5%, but falls 17.1% below the taller one; the F_2 loses almost no height relative to the F_1.

In the example, the heterosis over the mid-parent is 0.805 ± 0.294 (p = 0.011) and over the better parent, −2.385 ± 0.442 (p = 0.0002): the F_1 is intermediate, closer to the tall parent. The potence ratio, (F_1 − MP)/(½|$P_1 - P_2$|) = 0.252 ± 0.101, sums it up: 0 is intermediate inheritance, ±1 complete dominance toward one parent and more than 1, apparent overdominance. The inbreeding depression, $F_1 - F_2$ = 0.069, is not significant; without epistasis the F_2 would keep half the heterosis (11.167), and 11.501 was observed, another sign of the epistasis found by the joint test.

THEORY IN BRIEF · HETEROSIS IS NOT DOMINANCE

In the F∞ metric, the heterosis over the mid-parent is F_1 − MP = [h] + [l] − [i]. A positive additive × additive epistasis raises both parents and subtracts heterosis; a negative [l] (duplicate epistasis) does too. In the simulated example, with [h] = 6, [i] = 3 and [l] = −2, the expected heterosis is only 1: 1.49 ± 0.81 was observed, which is not significant even though there is clear dominance. When the chosen model includes [h], the note of the card gives this calculation with its estimates (5.68 − 4.00 = 1.68 in the simulated example).

Below the note, a figure draws each measure with its interval, as a difference or as a percentage. If you choose the lowest value in Objective, the better parent becomes the one with the lower value and the useful heterosis is the negative one.

Standard heterosis

For a breeding program, the practical question is not whether the F_1 beats its parents, but whether it beats the variety or hybrid grown today. That is **standard heterosis**: F_1 − check, as a percentage of the check mean. To compute it, you only need to plant the check in the same trial and put it in the data as one more entry that is not a generation: with its name in the entries column, or written in the generation column itself (`Testigo`, `H-520`), which Block 3 reads as a check. With several checks, the table adds one row for each.

standard heterosis = $\bar{F}_1$ − check · SE = $\sqrt{[V(\bar{F}_1) + V(\text{check})]}$ · % = 100 × difference/check

The variances of both means come from the same plot model of Section 7.3, but fitted with the check, so the block effects and the plot error are those of the whole trial. The checks do not enter the scaling tests, the models or the heterosis over the parents, which remain the same as without them.

Measure	Difference	SE	%	t	df	p
Mid-parent heterosis	1.4917	0.8091	2.8	1.84	10.0	0.0951 ns
Better-parent heterosis	−6.2867	0.9408	−10.4	−6.68	10.3	< 0.0001 ***
Inbreeding depression ($F_1 - F_2$)	3.155	0.9206	5.8	3.43	9.4	0.0070 **
Standard heterosis (against Testigo)	3.2467	0.9138	6.4	3.55	12.4	0.0038 **

Potence ratio (F_1 − mid-parent)/(½|P_1 − P_2|) = 0.192 ± 0.105: weak dominance toward the higher parent. Without epistasis the F_2 should keep half of the F_1 heterosis: 53.314 was expected and 50.905 observed. In terms of the model, the F_1 heterosis is [h] + [l] − [i] = 1.675. Standard heterosis compares the F_1 with the check (Testigo: 50.8133) as a percentage of its mean; it uses the plot error of the whole trial, checks included, and the checks do not enter the scaling tests or the models.

Figure 7.10. The heterosis of the simulated example. The F_1 does not beat its better parent, but it does beat the check, a uniform variety with a true mean of 51, by 3.25 ± 0.91 (6.4%, p = 0.0038).

In the simulated example, the F_1 (54.06) falls 6.29 below the tall parent and only 1.49 above the mean of its parents, without significance; against the check (50.81) it gains 3.25 with 12.4 degrees of freedom. The three readings do not contradict each other: they answer different questions. The heterosis over the parents speaks of gene action; the standard heterosis, of whether the F_1 is worth it compared with what is already grown.

VERIFIED · INDEPENDENT RECONSTRUCTION

With the check of the simulated example, the difference, its standard error, the Satterthwaite degrees of freedom (12.404) and the p value agree to 10^{-8} with a reconstruction that fits the plot means by least squares. A mixed model fitted by restricted maximum likelihood, with random plots and one among-plant variance for each entry, gives 3.247 ± 0.919 against 3.247 ± 0.914.

7.8 Minimum number of effective factors

How many genes separate the two parents? The Castle–Wright formula compares the difference between them with the segregation variance σ^2_S, the genetic variance that appears when they are crossed. If all the increasing alleles are in one parent and the genes have equal effects, $n_E = (\bar{P}_2 - \bar{P}_1)^2/(8\sigma^2_S)$. Lande (1981) proposed four ways of estimating σ^2_S from different generations, each with its standard error.

Minimum number of effective factors

Segregation variance	$\sigma^2 s$	Effective factors	SE
$V(F_2) - V(F_1)$	1.5554	3.27	1.17
$V(F_2) - [½V(F_1) + ¼V(P_1) + ¼V(P_2)]$	1.2549	4.05	1.79
$V(BC_1) + V(BC_2) - [V(F_1) + ½V(P_1) + ½V(P_2)]$	2.6492	1.92	0.69

$n_E = (\bar{P}_2 - \bar{P}_1)^2/(8\sigma^2 s)$ with a parental difference of −6.38 (Castle–Wright, as corrected by Lande 1981 with its standard error). Cockerham's (1986) correction, which removes the sampling error of the means from the numerator and uses one $\sigma^2 s$ from all the variances: $\sigma^2 s$ = 1.4135, E = 1.0931 and n_E = 3.58. It is a minimum: dominance, epistasis, linkage and unequal effects between loci all make it smaller than the true number of genes. The leading factor cannot account for more than $1/\sqrt{n_E}$ of the difference.

Figure 7.11. The minimum number of effective factors of *N. rustica*. The estimate with $2V(F_2) - V(BC_1) - V(BC_2)$ does not appear because that variance, which is ½D, comes out negative.

In the example, the three estimates range from 1.9 to 4.1 factors, and Cockerham's (1986) correction, which removes from the numerator the sampling error of the parental means and combines all the variances into a single $\sigma^2 s$, gives 3.58. It is a minimum: dominance, epistasis, linkage and genes of unequal effect make it smaller than the real number. Even so, it says something useful: the gene of largest effect cannot explain more than $1/\sqrt{n_E}$ of the difference between the parents, here about half.

> ✓ VERIFIED · THE PUBLISHED VALUES
>
> With the tomato fruit-weight statistics used by Lande (1981, table 3), the block gives 9.78 ± 0.65, 10.03 ± 0.66, 9.12 ± 1.15 and 11.14 ± 0.77 factors; the paper publishes 9.8 ± 0.6, 10.0 ± 0.7, 9.1 ± 1.1 and 11.1 ± 0.8.

7.9 Several environments, downloads and next step

With data from several environments, the Environment menu lets you analyze one or all together. Together, the blocks of each environment count as different blocks and the interaction of the generations with the environments stays in the plot error; with plants without plots, each environment acts as a plot.

The 6 · Assumptions, formulas and references card summarizes the assumptions (homozygous parents, diploid inheritance, no maternal effects, no linkage between the interacting genes, an adequate scale) with the formulas and the references. If the scaling tests fail, before adding epistasis it is worth trying a transformation of the trait: the epistasis may be a scale effect. Results (.xlsx) downloads all the tables of the block, each on its own sheet, and every figure has its download panel.

Since Blocks 8 to 11 work with genotypes and not with generations, Continue takes you to Block 12, the report; ← Data returns to Block 3.

B8

CHAPTER EIGHT · BLOCK 8 · SELECTION

Genetic parameters and selection

Before selecting you need to know how much of what you see is genetic, how the traits relate to one another and how much can be gained. Block 8 computes the heritability of each trait with its exact interval, the coefficients of variation and the genetic advance; the genotypic, phenotypic and environmental correlations with path analysis; six selection indices and the genotypes they choose; and the expected gain of the recurrent-selection methods.

8.1 What the block does

8.2 Genetic parameters of every trait

8.3 Heritability and selection intensity

8.4 Several environments

8.5 Correlations

8.6 Path analysis

8.7 Selection indices

8.8 The genotypes selected

8.9 Recurrent-selection methods

8.10 Downloads and next step

8.1 What the block does

Block 8 works with genotypes evaluated in a trial: lines, hybrids, clones, families or the crosses of a diallel treated as entries. It takes from Block 3 the field analysis of each trait, with the appropriate design (complete blocks, α-lattice, augmented blocks or only means with an external error), and reduces it to the mean squares and cross products that the genetic parameters need. The generations of a cross are analyzed in Block 7, not here.

Table 8.1. The five cards of Block 8 and what they give.

Card	What it gives
1 · Genetic parameters of every trait	σ^2_G, σ^2_{GE}, σ^2_e, plot-basis and entry-mean heritability with its exact interval, coefficients of variation and genetic advance
2 · Correlations and path analysis	Genotypic, phenotypic and environmental correlations with their standard errors; direct and indirect effects on one trait
3 · Selection indices	Smith–Hazel, base, restricted, proportional gains, desired gains and eigen, with the expected gain in each trait
4 · Genotypes selected	The ranking of the genotypes with the chosen index, the rank sum or Elston's index, and the realized gain
5 · Recurrent-selection methods	The gain per cycle and per year of seven methods from σ^2_A, σ^2_D and the error

The thread of the chapter is the trial of 13 maize hybrids from the examples gallery: three replications in complete blocks and 15 plant and ear traits (data distributed by its author under a free license). The correlations, paths and indices use five of them: plant height (PH), ear height (EH), grain weight per ear (KW), thousand-kernel weight (TKW) and kernels per ear (NKE). The screenshots use a selected proportion of 20%.

8.2 Genetic parameters of every trait

The 1 · Genetic parameters of every trait card has three controls: Proportion selected (from 1 to 50%), Selection intensity (that of an infinite population or the exact one for the genotypes of the trial) and Significance level, which sets the confidence interval. With several environments, Environment also appears, with the option of combining them all (Section 8.4).

The components come from the expected mean squares of the random model: $\sigma^2_G = (MS_G - MS_e)/r$, with its standard error. From them two broad-sense heritabilities are read: that of a plot and that of a genotype mean, which is the one that matters when you select on means of r replications.

$$H^2_{plot} = \sigma^2_G/(\sigma^2_G + \sigma^2_e) \quad \cdot \quad H^2_{means} = \sigma^2_G/(\sigma^2_G + \sigma^2_e/r) = 1 - MS_e/MS_G \quad \cdot \quad GA = i\cdot\sigma_P\cdot H^2$$

Grain weight per ear (KW) has $\sigma^2_G = 180.97 \pm 115.24$ and $\sigma^2_e = 280.41$: its heritability is 0.392 per plot and 0.659 on means of three replications, with an interval from 0.135 to 0.887. Cob length (CL), grain percentage (PERK) and cob weight (CW) exceed 0.88; ear length (EL), at 0.126, has almost no genetic variance in these hybrids (p = 0.37).

1 · Genetic parameters of every trait

Data: **Maize: 13 hybrids and 15 traits** · 13 genotypes · 15 traits · Single environment · proportion selected 20 % → intensity 1.3998.

PROPORTION SELECTED: 20 % | SELECTION INTENSITY: infinite population, i = φ(x)/p | SIGNIFICANCE LEVEL: 0.05

i Some lower limits of the Knapp interval are negative: they are shown as they are, because truncating them at zero would change their coverage.

Trait	Mean	σ²G	σ²e	H² plot	H² means	95 % CI (Knapp)	H² Cullis (REML)	CVg %	CVp %	CVe %	GA	GA %	p (F of genotypes)
PH	2.1666	0.0171 ± 0.0119	0.0328	0.343	0.61 ± 0.195	0.009 – 0.871	0.610	6.03	7.72	8.35	0.1429	6.60	0.0239 *
EH	1.078	0.005 ± 0.0074	0.0338	0.129	0.308 ± 0.346	−0.759 – 0.771	0.308	6.56	11.83	17.05	0.0549	5.10	0.2136 ns
EP	0.4958	0.0001 ± 0.0004	0.0022	0.064	0.171 ± 0.415	−1.107 – 0.725	0.171	2.47	5.97	9.42	0.0071	1.43	0.3341 ns
EL	14.668	0.0472 ± 0.1801	0.9838	0.046	0.126 ± 0.437	−1.221 – 0.710	0.126	1.48	4.18	6.76	0.1079	0.74	0.3733 ns
ED	47.8727	5.3688 ± 2.5333	2.4302	0.688	0.869 ± 0.066	0.667 – 0.957	0.869	4.84	5.19	3.26	3.0234	6.32	< 0.0001 ***
CL	28.4487	4.269 ± 1.9402	1.415	0.751	0.901 ± 0.05	0.747 – 0.967	0.901	7.26	7.65	4.18	2.7446	9.65	< 0.0001 ***
CD	15.7703	0.2396 ± 0.1939	0.6336	0.274	0.532 ± 0.234	−0.190 – 0.845	0.532	3.10	4.26	5.05	0.4996	3.17	0.0550 ns
CW	20.7801	18.4583 ± 8.5924	7.5408	0.710	0.88 ± 0.06	0.695 – 0.960	0.880	20.68	22.04	13.21	5.6421	27.15	< 0.0001 ***
KW	146.8108	180.9719 ± 115.2435	280.4088	0.392	0.659 ± 0.17	0.135 – 0.887	0.659	9.16	11.28	11.41	15.2917	10.42	0.0119 *
NR	15.7846	1.1809 ± 0.6664	1.2711	0.482	0.736 ± 0.132	0.329 – 0.913	0.736	6.88	8.02	7.14	1.3049	8.27	0.0027 **
NKR	30.4	2.1467 ± 2.0777	7.7978	0.216	0.452 ± 0.274	−0.392 – 0.819	0.452	4.82	7.17	9.19	1.3793	4.54	0.1012 ns
CDED	0.5946	0.001 ± 0.0004	0.0004	0.718	0.884 ± 0.058	0.706 – 0.962	0.884	5.20	5.53	3.26	0.0407	6.85	< 0.0001 ***
PERK	87.6461	3.1367 ± 1.401	0.8671	0.783	0.916 ± 0.042	0.786 – 0.972	0.916	2.02	2.11	1.06	2.3723	2.71	< 0.0001 ***
TKW	317.7178	840.9388 ± 491.7661	1,018.4304	0.452	0.712 ± 0.144	0.269 – 0.905	0.712	9.13	10.81	10.04	34.2623	10.78	0.0045 **
NKE	467.8974	1,982.3207 ± 1,158.9308	2,398.7815	0.452	0.713 ± 0.144	0.270 – 0.905	0.713	9.52	11.27	10.47	52.6103	11.24	0.0045 **

H^2 means = $\sigma^2G/\sigma^2\bar{P} = 1 - MS_e/MS_G$, with $\sigma^2\bar{P} = MS_G/r$; its standard error by the delta method and the exact interval of Knapp, Stroup & Ross (1985). CVg, CVp and CVe after Burton & DeVane (1953), with CVp on the entry-mean basis. Genetic advance GA = $i \cdot \sigma P \cdot H^2$ (Johnson, Robinson & Comstock 1955) on the same basis as H^2, with i = 1.3998.

Figure 8.1. The parameters of the 15 maize traits. Each row gives the variance components, the two heritabilities, Knapp's interval, Cullis's heritability, the coefficients of variation, the genetic advance and the *F* test of genotypes.

The interval is the exact one of Knapp, Stroup and Ross (1985), which uses the *F* distribution of the ratio of mean squares instead of assuming normality. Its lower limit can be negative when the genetic variance is small; it is shown as it comes out, because truncating it at zero would change its coverage, and a warning reminds you. The genetic, phenotypic and environmental coefficients of variation (CVg, CVp and CVe) follow Burton and DeVane (1953); CVp is given on an entry-mean basis, like the heritability. The genetic advance GA (Johnson, Robinson and Comstock, 1955) is how much the mean would rise if the chosen proportion were selected on its phenotypic mean: for KW, 15.29 g, 10.4%.

B8

VERIFIED · TWO INDEPENDENT CALCULATIONS

The components, heritabilities, intervals and coefficients of variation of the 15 traits, in one environment and in four, equal those of an independent reconstruction to within 3.5×10^{-10}, and those of KW agree with another restricted maximum likelihood implementation (σ^2_G = 180.97188, σ^2_e = 280.40877). In 3,000 simulations with $H^2 = 0.75$, Knapp's 95% interval covered the true value 95.07% of the time.

The H² Cullis (REML) column is the heritability of Cullis, Smith and Coombes (2006), which Block 3 computes from the average variance of the differences between means. In balanced complete blocks it equals H^2 on means; with incomplete blocks, where the moment-based components are approximate, it is the recommended one, and a warning says so. With only means and an external error, the variance among means plays the role of MS_G.

8.3 Heritability and selection intensity

Below the table, a figure shows the entry-mean heritability of each trait with its interval, over the usual classes: low below 0.3 and high above 0.6 (the gray band). Intervals that go below −1 end in an arrow at the edge.

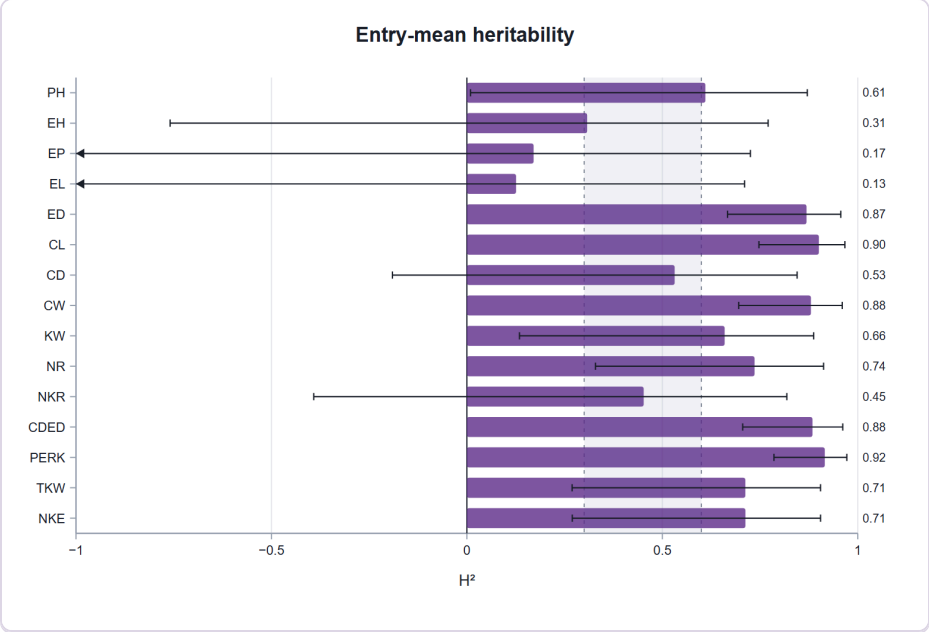


Figure 8.2. Entry-mean heritability of the 15 traits. With 13 genotypes and three replications the intervals are wide: that of KW goes from 0.13 to 0.89.

The selection intensity i is the superiority of the selected individuals in standard deviations. For an infinite population, $i = \varphi(x_p)/p$, where p is the proportion selected. With few genotypes the real intensity is smaller: the exact for N of M genotypes option uses the expected mean of the N best of M normal values. The folded box Exact and finite selection intensity gives the table of values and compares, for the loaded trial, the exact intensity with the approximations of Burrows (1972) and Bulmer (1980).

Table 8.2. Selection intensity of an infinite population. Some books call it k and call the percentage selected the "intensity".

Proportion selected	1%	2%	5%	10%	15%	20%	30%	50%
Intensity i	2.6652	2.4209	2.0627	1.7550	1.5544	1.3998	1.1590	0.7979

With 13 hybrids and 20% selected, 3 genotypes are chosen: the exact intensity is 1.2273, against 1.3183 for an infinite population with the same fraction (3 of 13). To choose one of 20 genotypes, the app gives 1.8675 exact, 1.8434 by Burrows and 1.8880 by Bulmer, the values of the example in Walsh and Lynch (2018).

8.4 Several environments

With data from several environments, the **Environment** menu lets you analyze one or **all combined**. In the combined analysis the table adds σ^2_{GE}, the genotype × environment interaction, and the entry-mean heritability becomes $H^2 = \sigma^2_G/(\sigma^2_G + \sigma^2_{GE}/e + \sigma^2_e/re) = 1 - MS_{GE}/MS_G$, with e environments: the genotypes are tested against their interaction with the environments, not against the error.

The contrast with a single environment is the lesson of Figure 8.3. KW goes from $H^2 = 0.659$ to 0.204: its σ^2_G across environments is 30.3 ± 66.6 and its σ^2_{GE}, 343.0. What looked genetic in one location was, to a large extent, the response of each hybrid to that location. With a negative σ^2_G the heritability is reported as is, but the genetic advance is taken as zero and the card's warning lists the affected traits. Block 9 analyzes that interaction in depth.

! Negative σ²G in EH, EP, EL, CL, CW, NR, CDED, PERK, TKW, NKE: heritability is reported but the genetic advance is taken as zero.

i Some lower limits of the Knapp interval are negative: they are shown as they are, because truncating them at zero would change their coverage.

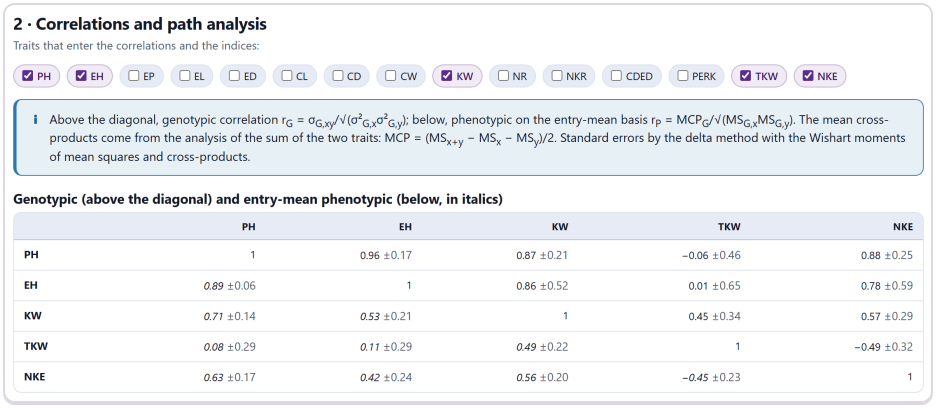
Trait	Mean	σ²G	σ²GE	σ²e	H² plot	H² means	95 % CI (Knapp)	CVg %	CVp %	CVe %	GA	GA %	p (F of genotypes)
PH	2.4848	0.0005 ± 0.0061	0.0425 ± 0.0118	0.0224	0.007	0.035 ± 0.455	−1.247 – 0.670	0.86	4.58	6.03	0.0056	0.23	0.4386 ns
EH	1.3433	−0.0029 ± 0.0038	0.0332 ± 0.0095	0.0207	−0.057	−0.407 ± 0.663	−2.276 – 0.519	—	6.29	10.72	0	0.00	0.7308 ns
EP	0.5368	−0.0002 ± 0.0001	0.0011 ± 0.0004	0.0015	−0.068	−0.712 ± 0.807	−2.987 – 0.415	—	2.84	7.28	0	0.00	0.8403 ns
EL	15.163	−0.0335 ± 0.0896	0.4969 ± 0.2089	1.0842	−0.022	−0.185 ± 0.559	−1.760 – 0.595	—	2.81	6.87	0	0.00	0.6071 ns
ED	49.5373	0.5568 ± 0.6415	2.8221 ± 0.8779	2.5946	0.093	0.377 ± 0.294	−0.452 – 0.787	1.51	2.45	3.25	0.6409	1.29	0.1343 ns
CL	29.0109	−0.7643 ± 0.3427	4.3313 ± 1.1535	1.6523	−0.146	−1.676 ± 1.261	−5.231 – 0.085	—	2.33	4.43	0	0.00	0.9646 ns
CD	15.9703	0.0292 ± 0.0995	0.4546 ± 0.1833	0.9027	0.021	0.134 ± 0.408	−1.017 – 0.704	1.07	2.92	5.95	0.0876	0.55	0.3504 ns
CW	24.8141	−3.3828 ± 2.2475	24.2908 ± 6.6946	12.0174	−0.103	−0.916 ± 0.903	−3.463 – 0.345	—	7.74	13.97	0	0.00	0.8864 ns
KW	172.9389	30.291 ± 66.6111	342.9643 ± 112.7742	386.8087	0.040	0.204 ± 0.375	−0.853 – 0.728	3.18	7.04	11.37	3.4823	2.01	0.2854 ns
NR	16.1231	−0.0805 ± 0.1594	1.0384 ± 0.3823	1.6448	−0.031	−0.255 ± 0.591	−1.922 – 0.571	—	3.49	7.95	0	0.00	0.6507 ns
NKR	32.25	0.1871 ± 0.7248	2.9633 ± 1.3684	7.8508	0.017	0.118 ± 0.416	−1.054 – 0.699	1.34	3.90	8.69	0.2082	0.65	0.3648 ns
CDED	0.586	−0.0002 ± 6.4353e-5	0.0009 ± 0.0002	0.0004	−0.174	−3.24 ± 1.999	−8.875 – −0.449	—	1.32	3.56	0	0.00	0.9949 ns
PERK	87.435	−0.6986 ± 0.2337	3.2808 ± 0.8662	1.1595	−0.187	−3.201 ± 1.98	−8.783 – −0.436	—	0.53	1.23	0	0.00	0.9947 ns
TKW	338.6663	−70.9993 ± 155.1593	1,217.6649 ± 361.8742	918.4442	−0.034	−0.229 ± 0.579	−1.862 – 0.580	—	5.20	8.95	0	0.00	0.6352 ns
NKE	511.6436	−359.7191 ± 204.5736	1,643.3576 ± 654.2458	3,167.9854	−0.081	−1.142 ± 1.01	−3.987 – 0.268	—	3.47	11.00	0	0.00	0.9209 ns

H^2 means = $\sigma^2G/\sigma^2\bar{P} = 1 - MS_{GE}/MS_G$, with $\sigma^2\bar{P} = MS_G/re$; its standard error by the delta method and the exact interval of Knapp, Stroup & Ross (1985). CVg, CVp and CVe after Burton & DeVane (1953), with CVp on the entry-mean basis. Genetic advance GA = $i{\cdot}\sigma\bar{P}{\cdot}H^2$ (Johnson, Robinson & Comstock 1955) on the same basis as H^2, with $i = 1.3998$.

Figure 8.3. The same 13 hybrids in four environments. The interaction with the environment exceeds the genetic variance in every trait, and ten of them have a negative σ^2_G.

8.5 Correlations

The **2 · Correlations and path analysis** card begins with the boxes of the traits that enter the correlations and the indices; when a trial is loaded, the first six are ticked. The table brings together two matrices: above the diagonal, the genotypic correlations; below, in italics, the entry-mean phenotypic ones, each with its standard error.



2 · Correlations and path analysis

Traits that enter the correlations and the indices:

☑ PH ☑ EH ☐ EP ☐ EL ☐ ED ☐ CL ☐ CD ☐ CW ☑ KW ☐ NR ☐ NKR ☐ CDED ☐ PERK ☑ TKW ☑ NKE

ℹ Above the diagonal, genotypic correlation $r_G = \sigma_{G,xy}/\sqrt{(\sigma^2_{G,x}\sigma^2_{G,y})}$; below, phenotypic on the entry-mean basis $r_P = MCP_G/\sqrt{(MS_{G,x}MS_{G,y})}$. The mean cross-products come from the analysis of the sum of the two traits: MCP = $(MS_{x+y} - MS_x - MS_y)/2$. Standard errors by the delta method with the Wishart moments of mean squares and cross-products.

Genotypic (above the diagonal) and entry-mean phenotypic (below, in italics)

	PH	EH	KW	TKW	NKE
PH	1	0.96 ±0.17	0.87 ±0.21	−0.06 ±0.46	0.88 ±0.25
EH	*0.89* ±0.06	1	0.86 ±0.52	0.01 ±0.65	0.78 ±0.59
KW	*0.71* ±0.14	*0.53* ±0.21	1	0.45 ±0.34	0.57 ±0.29
TKW	*0.08* ±0.29	*0.11* ±0.29	*0.49* ±0.22	1	−0.49 ±0.32
NKE	*0.63* ±0.17	*0.42* ±0.24	*0.56* ±0.20	*−0.45* ±0.23	1

Figure 8.4. The correlations of the five chosen traits. Above the diagonal, genotypic; below, entry-mean phenotypic.

The covariances are obtained from the analysis of the sum of each pair of traits, in the same design as the traits: the mean cross product is MCP = $(MS_{x+y} - MS_x - MS_y)/2$, and from it the genotypic, phenotypic and environmental covariances follow just as the variances follow from the mean squares. The standard errors are computed by the delta method, with the moments of the mean squares and cross products. The environmental (residual) correlations are in a folded box.

In the maize, the two heights go almost together ($r_G = 0.96$) and both are associated with grain weight per ear (0.87 and 0.86): the tall hybrids yield more per ear. Thousand-kernel weight and kernels per ear are opposed (−0.49): ears with many kernels have smaller ones, the classic trade-off between yield components.

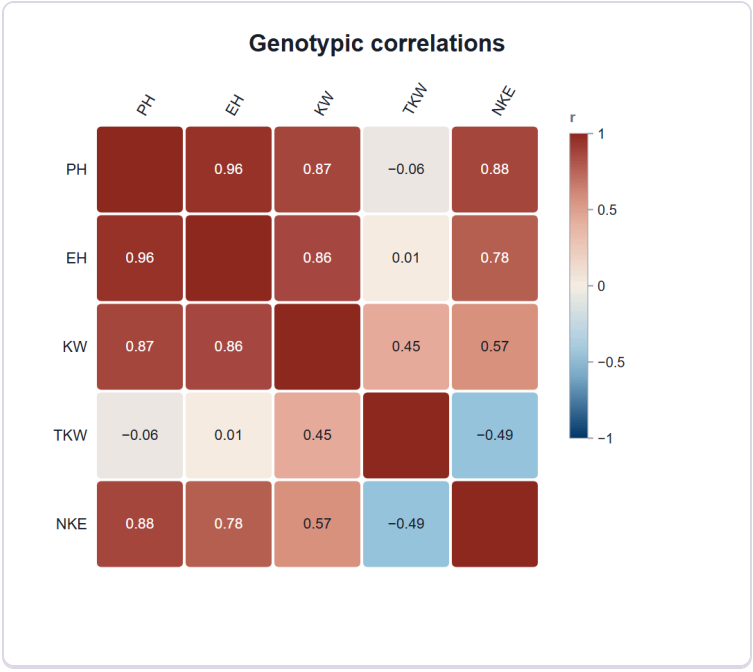


Figure 8.5. The genotypic correlations as a heat map. The **Figure** menu switches to the phenotypic or environmental ones.

With few genotypes, a genotypic correlation can fall outside [−1, 1]: moment estimators are not bounded when σ^2_G is small. The app does not truncate it; it marks it and warns you. If σ^2_G is negative, the genotypic correlation does not exist and the cell is left empty. Records missing any of the chosen traits are left out, so that all the covariances use the same plots.

8.6 Path analysis

Wright's path analysis splits the correlation of each trait with a dependent trait into a direct effect and indirect effects through the others: $r_{iy} = p_i + \Sigma r_{ij} p_j$. The direct effects solve $\mathbf{p} = \mathbf{R}^{-1}\mathbf{r}$. You choose the dependent trait, the basis (genotypic or phenotypic correlations) and, if needed, a ridge constant k (Carvalho and Cruz, 1996).

Path analysis

DEPENDENT TRAIT: KW | CORRELATIONS: phenotypic | RIDGE CONSTANT K: 0

Explanatory trait	Direct effect	via PH	via EH	via TKW	via NKE	r with KW	VIF
PH	**0.0421**	—	0.0062	0.0715	0.5907	0.7106	10.19
EH	**0.007**	0.0376	—	0.097	0.3932	0.5348	6.25
TKW	**0.9033**	0.0033	0.0007	—	−0.4222	0.4852	1.85
NKE	**0.9375**	0.0265	0.0029	−0.4068	—	0.5602	3.62

ℹ R^2 = 0.9971, residual effect $\sqrt{(1 - R^2)}$ = 0.0541. Condition number of the correlation matrix 41.1 (weak collinearity). Every correlation splits as $r_{iy} = p_i + \Sigma\, r_{ij} p_j$.

Figure 8.6. Paths of grain weight per ear (KW) with phenotypic correlations. In bold, the direct effects; on the right, the total correlation and the variance inflation factor (VIF).

The result reads almost like an identity: grain weight per ear is the number of kernels times the weight of each one, and NKE and TKW have direct effects of 0.94 and 0.90, with R^2 = 0.997. Their total correlations with KW are smaller (0.56 and 0.49) because each carries a negative indirect effect through the other (−0.41 and −0.42). Plant height, on the other hand, is correlated with KW (0.71) almost only through kernels per ear (0.59): its direct effect is 0.04. The condition number, 41.1, indicates weak collinearity.

Path analysis

DEPENDENT TRAIT: KW | CORRELATIONS: genotypic | RIDGE CONSTANT K: 0

× R^2 = 1.044 is above 1: the genotypic correlations of KW with the explanatory traits are not compatible with those among them (the full matrix is not positive definite), and the direct effects are meaningless. Use the phenotypic correlations, drop the traits with a small $\sigma^2 G$ or increase the ridge constant.

Figure 8.7. The same paths with genotypic correlations. R^2 would come out greater than 1, and the app shows the warning instead of the table.

IMPORTANT · WHEN PATHS MAKE NO SENSE

The direct effects only mean something if the correlation matrix is possible, that is, positive definite. With genotypic correlations from few genotypes this often fails, in two ways: among the explanatory traits there are impossible correlations (outside [−1, 1], or incompatible with one another), or the correlations with the dependent trait do not agree with those among the explanatory traits, and R^2 comes out greater than 1. In the maize the second happens: $R^2 = 1.044$. In both cases the app does not show the table and tells you what to do: use the phenotypic correlations, remove traits with small σ^2_G or increase the ridge constant. With strong collinearity (condition number above 100 or large VIFs), the direct effects are unstable even if the matrix is valid.

8.7 Selection indices

A selection index combines the traits into a single value, $I = \mathbf{b'y}$, to choose the genotypes that most improve the aggregate genetic value $H = \mathbf{w'g}$, where $\mathbf{w}$ are the economic weights. The 3 · Selection indices card begins with a table to define the objective: the direction of each trait, its economic weight, whether it must remain unchanged (restricted index) and the desired gain. The default weights are $\pm 1/\sigma_P$: the same importance for one phenotypic standard deviation of each trait.

All the indices work on an entry-mean basis, with the phenotypic matrix $\mathbf{P} = \mathrm{MCP}_G/r$ and the genotypic matrix $\mathbf{G}$. For each one, the table gives σ_I, the response in H, the correlation ρ_{HI} between the index and the aggregate value, the heritability of the index, its efficiency relative to Smith–Hazel and the expected gain in each trait, $\Delta G = i \cdot \mathbf{Gb}/\sigma_I$.

Table 8.3. The indices of Block 8.

Index	Coefficients	When to use it
Smith–Hazel	$\mathbf{b} = \mathbf{P}^{-1}\mathbf{Gw}$	The optimum for H if **P**, **G** and **w** are well estimated
Base (Williams, 1962)	$\mathbf{b} = \mathbf{w}$	When **G** and **P** are unreliable
Restricted (Kempthorne and Nordskog, 1959)	Smith–Hazel with $\Delta G = 0$ in the marked traits	To improve without moving certain traits
Proportional gains (Mallard, 1972; Tallis, 1985)	Smith–Hazel with the gains in the requested proportion	When the proportion among gains matters, with its θ
Desired gains (Pesek and Baker, 1969)	$\mathbf{b} = \mathbf{P}^{-1}\mathbf{G}(\mathbf{G'P}^{-1}\mathbf{G})^{-1}\mathbf{d}$	Without economic weights: the gain of each trait is requested
Eigen, ESIM (Cerón-Rojas and colleagues, 2006)	$\mathbf{Gb} = \lambda^2\mathbf{Pb}$ with the largest λ^2	Without weights: the most heritable combination

In the example, Smith–Hazel achieves $\rho_{HI} = 0.778$ and the base index, 0.576 (efficiency 0.740). With plant height restricted, the index loses almost nothing (efficiency 0.999): height barely changed with Smith–Hazel (+0.006 m).

3 · Selection indices

Direction and economic weight of every trait; tick those that must not change (restricted index) or type the gain you want (proportional and desired-gains indices).

Trait	Direction	Economic weight	Restrict (no change)	Desired gain	σF
PH	decrease	-5.976	☑	—	0.1673
EH	decrease	-7.84	☐	—	0.1275
KW	increase	0.06036	☐	—	16.5663
TKW	increase	0.02911	☐	—	34.3572
NKE	increase	0.01896	☐	—	52.7439

! G was not compatible with P: 1 of 5 canonical heritabilities (eigenvalues of G on the scale where P = I) were outside [0, 1] (−5.947) and were brought into that range before computing the indices. It usually happens with few genotypes or with nearly redundant traits (a ratio of two others, for instance).

i Entry-mean matrices: P = MCP$_G$/r and G = σ$_G$. Smith–Hazel b = P^{-1}Gw; response R = i·σ$_I$ and gain per trait ΔG = i·Gb/σ$_I$ with i = 1.3998. For the base index the response in H is i·w'Gw/√(w'Pw). The default weights are ±1/σP (equal importance per phenotypic standard deviation) and can be changed.

i The eigen index (ESIM) solves Gb = λ²Pb for the largest λ² through the Cholesky factor of P; it uses no economic weights: its accuracy for its own target is λ = 0.923, and the table gives its correlation with the same H as the other indices.

Comparison of indices

Index	σ$_I$	Response in H	ρ$_{HI}$	Index h²	Efficiency against Smith–Hazel	ΔG PH	ΔG EH	ΔG KW	ΔG TKW	ΔG NKE
Smith–Hazel	0.7779	1.089	0.778	0.696	1.000	0.0057	0.0032	8.9154	33.4232	−19.1525
Base index	1.7351	0.8061	0.576	0.332	0.740	0.003	−0.0043	6.154	20.3123	−9.092
Restricted	0.7775	1.0883	0.778	0.700	0.999	0	0.0003	8.4407	33.6528	−21.0121
Eigen index (ESIM)	0.1155	0.4466	0.319	0.853	0.410	−0.1436	−0.0696	−8.2159	21.405	−57.2198

Figure 8.8. The objective of the example: lower plant and ear height and raise grain weight, thousand-kernel weight and kernels per ear, with plant height restricted. Below, the comparison of the indices.

Both indices raise TKW by 33 g and KW by 8 to 9 g, but lower kernels per ear by 19 to 21: because of the negative correlation between TKW and NKE, the index prefers heavier kernels to more kernels. The eigen index, which does not use the weights, seeks something else (its accuracy relative to its own objective is λ = 0.923); relative to the objective of the weights, ρ_{HI} = 0.319.

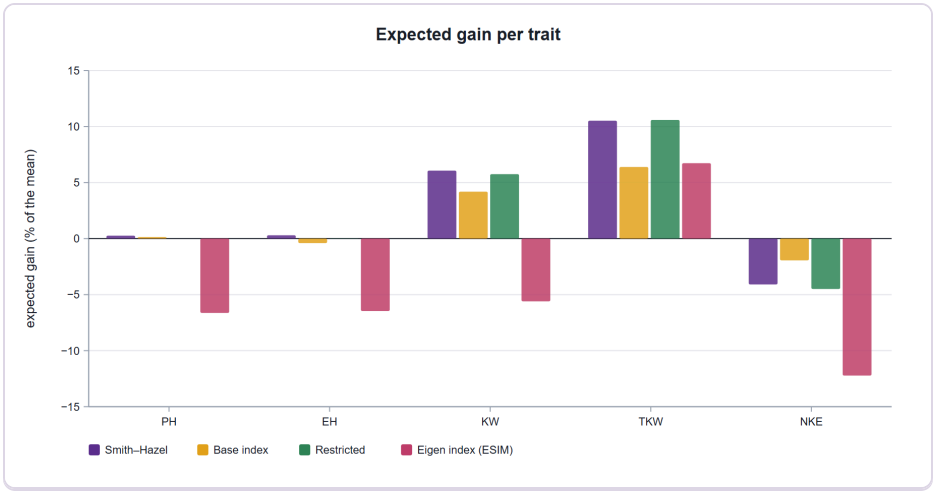


Figure 8.9. Expected gain in each trait, as a percentage of its mean, with each index.

When you type a desired gain for any trait, the proportional-gains and desired-gains indices appear. If θ, which measures whether the proportional-gains index moves the traits in the requested direction, comes out negative or zero, a warning says so (Itoh and Yamada, 1987).

THEORY IN BRIEF · G HAS TO FIT INSIDE P

The genetic variance of any combination of traits cannot exceed its phenotypic variance. On the scale in which **P** is the identity, the eigenvalues of **G** are canonical heritabilities and must lie between 0 and 1. With few genotypes or with nearly redundant traits they fall outside: in the example, one equals −5.95. If this is not corrected, Smith–Hazel gives accuracies greater than 1. The app brings those heritabilities into the interval [0, 1] before computing the indices, the idea of Hayes and Hill (1981), and says how many it changed; if **P** itself is not positive definite, it first bends it with Schaeffer's (2014) method. A trait with $\sigma^2_G \leq 0$ keeps its phenotypic information, but its genotypic row and column are set to zero.

The folded box **Coefficients of every index and Smith–Hazel detail** gives the coefficients **b** of all the indices and, for Smith–Hazel, the share of the economic value contributed by each trait and the accuracy that would be lost if it were removed.

▼ Coefficients of every index and Smith–Hazel detail

Index	b PH	b EH	b KW	b TKW	b NKE
Smith–Hazel	0.0105	−1.9067	−0.0378	0.0427	0.017
Base index	−5.976	−7.84	0.0604	0.0291	0.019
Restricted	−0.1716	−1.7859	−0.0385	0.0429	0.017
Eigen index (ESIM)	−0.8044	0.594	−0.0033	0.0029	0.0005

Trait	Weight w	Coefficient b	Expected gain	% of the economic value	% lost if dropped
PH	−5.976	0.0105	0.0057	−3.13	0.00
EH	−7.84	−1.9067	0.0032	−2.28	0.78
KW	0.0604	−0.0378	8.9154	49.42	0.09
TKW	0.0291	0.0427	33.4232	89.35	0.64
NKE	0.019	0.017	−19.1525	−33.35	0.22

Figure 8.10. Coefficients of the indices and Smith–Hazel detail. TKW contributes 89% of the economic value and KW 49%; NKE subtracts 33%. Removing any of the five traits costs less than 1% of accuracy.

A coefficient can have the opposite sign to its weight: that of KW is negative even though you want to raise grain weight, because TKW and NKE already carry that information and the index corrects what KW adds. That is why the coefficients are not interpreted one by one; what matters are the expected gains.

VERIFIED · THE PUBLISHED VALUES

With the matrices of the book by Cerón-Rojas and Crossa (2018), the Smith–Hazel index gives **b** = (0.555, −1.063, 1.087), R = 19.21 and ρ = 0.894; the restricted and proportional-gains indices reproduce its examples (R = 6.87 and 16.717, θ = 6.213). The maize indices, including the eigen index measured against the same H, agree with an independent reconstruction, and none of 2,000 random indices exceeds the accuracy of Smith–Hazel.

8.8 The genotypes selected

The 4 · Genotypes selected card ranks the genotypes with the index chosen in Rank with: any of those of the previous card, the rank sum of Mulamba and Mock (1978) or Elston's (1963) multiplicative index, which need neither **P** nor **G**. As many genotypes are selected as the chosen proportion indicates; the checks do not enter the selection.

4 · Genotypes selected

RANK WITH

Smith–Hazel

#	Genotype	Index value	PH	EH	KW	TKW	NKE
1	**H1** selected	1.0406	2.1973	1.0953	156.8165	368.7676	425.1333
2	**H7** selected	0.7931	2.1773	1.1467	155.6554	358.0459	440.6667
3	**H2** selected	0.7275	2.154	1.0467	160.3076	345.9272	466.3333
4	H3	0.5169	2.0433	0.9529	138.9867	336.0535	421
5	H8	0.4283	2.1	1.0913	141.4307	347.7625	407.3333
6	H5	0.3313	2.0947	0.9593	161.1175	305.4973	536.4667
7	H13	0.2146	2.5987	1.3127	181.6017	319.627	578.8667
8	H4	0.0217	2.0493	0.9427	143.4543	306.7286	474.2
9	H6	−0.1552	2.1533	1.1513	137.0672	328.0381	419.5333
10	H9	−0.5744	2.028	0.9073	122.4679	294.3591	419.7333
11	H11	−0.7216	2.1013	1.1027	140.7153	288.6308	487.7333
12	H12	−1.0877	2.43	1.2933	148.3617	283.6332	516.8667
13	H10	−1.535	2.038	1.0113	120.5581	247.2608	488.8

Selection differential and realized gain

Trait	Overall mean	Mean of the selected	Differential DS	DS %	H^2	Gain DS·H^2	Gain %
PH	2.1666	2.1762	0.0097	0.45	0.610	0.0059	0.27
EH	1.078	1.0962	0.0183	1.69	0.308	0.0056	0.52
KW	146.8108	157.5932	10.7823	7.34	0.659	7.1101	4.84
TKW	317.7178	357.5802	39.8624	12.55	0.712	28.3984	8.94
NKE	467.8974	444.0444	−23.853	−5.10	0.713	−16.997	−3.63

3 of 13 genotypes are selected. These gains are empirical (realized selection differential times the entry-mean heritability), not the predictions of the index in the previous card; checks do not enter the selection.

Figure 8.11. The 13 hybrids ranked with Smith–Hazel. With 20%, H1, H7 and H2 are selected. Below, the selection differential and the realized gain in each trait.

The three chosen hybrids have ears with heavy kernels (358 g per thousand kernels against 318 for the mean) and average height. H13, the one with the highest grain weight per ear (181.6 g), ranks seventh because it is the tallest (2.60 m plant and 1.31 m ear height): the index charges for the height you wanted to lower. The table below gives, for each trait, the selection differential and the realized gain SD·H^2: KW rises 7.1 g (4.8%) and TKW 28.4 g (8.9%), while NKE drops 17.0 kernels.

These gains are empirical: the differential of the genotypes actually chosen, multiplied by the heritability. They are not the predictions of the index, which assume a normal, infinite population; with 13 genotypes it is normal for them to differ.

8.9 Recurrent-selection methods

The 5 · Expected gain of the recurrent-selection methods card compares seven recurrent-selection methods in an outcrossing population with the general formula $G_c = i \cdot c \cdot Cov/\sigma_P$ (Sprague and Eberhart, 1977; Fehr, 1987): the covariance between the unit evaluated and the unit recombined, over the phenotypic standard deviation of the selection unit, with the parental control c (½ if only one parent is selected, 2 if selfed seed of the selected plants is recombined).

5 · Expected gain of the recurrent-selection methods

σ²A	σ²D	σ²AE (additive × environment)	σ²DE (dominance × environment)
68	42	70	42
σ²e (plot error)	**σ² (between plots, phenotypic)**	**σ²w (between plants within plots)**	**replicates r**
96	46	700	2
environments t	**F of the parents of the lines**	**recombination seasons**	**PHENOTYPIC SELECTION IN GRIDS**
3	0.875	3	yes (no σ² between plots)

Back to the starting values

Method	c	Seasons per cycle	Gain per cycle	Per year: 1 season	2 similar seasons	2 different seasons	3 seasons
Phenotypic, one parent selected after flowering	0.5	1 M	**1.965**	1.965	3.93	1.965	1.965
Modified ear-to-row, one parent	0.5	1 M	**2.394**	2.394	4.788	2.394	2.394
Half-sib, recombining remnant seed	1	2 MA	**4.788**	2.394	4.788	4.788	4.788
Half-sib, recombining selfed seed	2	3 AMA	**9.575**	3.192	6.383	4.788	9.575
Full-sib	1	2 MA	**6.86**	3.43	6.86	6.86	6.86
$S_{0:1}$ lines	1	3 AMA	**10.834**	3.611	7.223	5.417	10.834
Selfed lines from parents with inbreeding F (F = 0.875)	1	8 AAAAMAAA	**16.211**	2.026	4.053	4.053	5.404

Gain per cycle $G_c = i \cdot c \cdot Cov/\sigma_P$ (Sprague & Eberhart 1977; Fehr 1987, Tables 17-1 and 17-2) with i = 1.755: c is the parental control (½ when only the female is selected, 2 when selfed seed of the selected plants is recombined). Every method is a sequence of seasons: M needs the evaluation environment and A can go in an off-season; the gain per year comes from laying the sequence, cycle after cycle, on the calendar. The starting values are those of Fehr's example (1987, Table 17-9, maize yield in q/ha) and should be replaced by your own.

Figure 8.12. Card 5 with the values of Fehr's (1987) example for maize yield and 10% selected. Each method is a sequence of seasons: M in the evaluation environment, A in any season.

The input values are σ^2_A, σ^2_D, their interactions with the environment, the plot error, the phenotypic variance among plots and the variance among plants within plots, with the replications, the environments, the inbreeding of the parents of the lines and the recombination seasons. If Block 6 estimated σ^2_A and σ^2_D, the card takes them from there; the other starting values are those of Fehr's example and you should replace them with your own. Back to the starting values restores them.

The gain per year results from fitting the sequence of each method, cycle after cycle, into the calendar: one season a year, two equal seasons, two different ones (one in the evaluation environment and an off-season one) or three. Per cycle, the methods with selfing gain more (16.2 the parent lines with F = 0.875, 10.8 the $S_{0:1}$), but they need more seasons; per year, with two different seasons, full-sib selection gains the most (6.86).

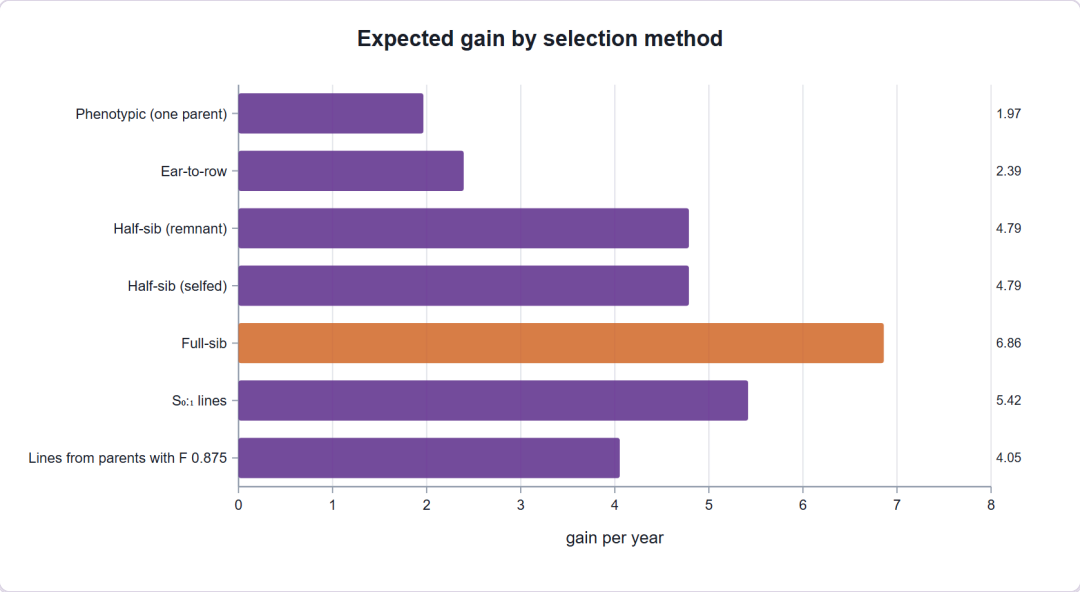


Figure 8.13. The gain per year of each method with two different seasons; in another color, the largest. The figure panel switches to the gain per cycle.

> **VERIFIED · THE PUBLISHED VALUES**
>
> With 10% selected, the gains per cycle are 1.97, 2.39, 4.79, 9.58, 6.86, 10.83 and 16.21 q/ha; Fehr (1987, table 17-9) publishes 2.0, 2.4, 4.8, 9.6, 6.8, 10.8 and 16.2 with k = 1.75 (the app uses 1.7550). The gains per year with one, two equal, two different and three seasons agree with those of the same table.

It is a prediction. The real response is smaller if there is epistasis, the Bulmer effect (selection reduces the additive variance), drift in small populations or interaction with the environment that was not estimated.

8.10 Downloads and next step

The 6 · Assumptions, formulas and references card summarizes the formulas of each card with their references. Results (.xlsx) downloads the parameters, the correlation matrices and **P** and **G**, the indices, the selection ranking and the recurrent-selection methods, each on its own sheet. Every figure has its download panel.

Continue takes you to Block 9 (genotype × environment) if there are several environments, or to Block 10 (BLUP) if there is only one. The back button returns to the block that analyzed the design: Hayman–Jinks for Griffing diallels, Mating designs for the North Carolina designs, the triple test cross, line × tester and partial diallels, and Data for genotype trials.

B9

CHAPTER NINE · BLOCK 9 · GENOTYPE × ENVIRONMENT

Genotype × environment interaction

A genotype that wins in one location can lose in another. Block 9 measures that interaction and describes it in every way the literature uses: the analysis of variance of the table of means with a two-stage model for unequal errors, the regressions on the environmental index, the stability variances, superiority and reliability, the rank statistics, AMMI with its stability indices and the views of the GGE biplot, plus a summary that compares them.

9.1 What the block does

9.2 The table and its analysis of variance

9.3 Unequal errors: the two-stage analysis

9.4 Regression on the environmental index

9.5 Stability variances, superiority and reliability

9.6 Ranks and yield–stability

9.7 AMMI

9.8 The GGE biplot

9.9 Summary of the statistics

9.10 Downloads and next step

9.1 What the block does

Block 9 needs the same genotypes evaluated in two or more environments: locations, years or planting dates. From Block 3 it takes the adjusted mean of each genotype in each environment, with the field design of that environment, and the error of each one; with that table it computes everything else. If the data have a single environment, card 1 explains how to mark the column that distinguishes them.

Table 9.1. The cards of Block 9.

Card	What it gives
1 · Genotype × environment table	Analysis of variance of the table, heat map and two-stage analysis with the error of each environment
2 · Regression on the environmental index	Finlay and Wilkinson, Eberhart and Russell, Perkins and Jinks
3 · Stability variances, superiority and reliability	Wricke, Shukla, Lin and Binns, Annicchiarico
4 · Rank statistics and yield–stability	Huehn with the tests of Nassar and Huehn; Kang's YS_i
5 · AMMI	Interaction axes with the Gollob and Cornelius tests; ASV, YSI, WAAS and WAASY; winners per environment
6 · GGE biplot	Which won where, mean vs stability, ideal genotype and environment, discriminativeness, comparison of two genotypes
7 · Summary of the statistics	All the statistics with the rank of each genotype and their rank correlations

The thread of the chapter is the oat trial from the gallery: 10 genotypes in 14 environments with three replications in complete blocks (420 plots), with grain yield (GY). The AMMI section uses the classic example of 28 potato clones in six locations in Peru, and the checks, the table of 17 genotypes in 12 environments with which Kang (1993) illustrated his statistic.

9.2 The table and its analysis of variance

The 1 · Genotype × environment table card has three controls: the Trait; the Best value, which you change to the lowest for days to flowering or disease incidence and which reorients all the statistics that depend on "the best"; and how empty cells are filled, by EM-AMMI with the additive model or with one or two interaction axes (Gauch and Zobel, 1990).

The analysis of variance is that of the table of means multiplied by r, on a plot scale, with the pooled error of all environments (0.0967 with 252 degrees of freedom). It has two tests for the genotypes: against the interaction ($F = 5.41$), if the environments are a sample of the region where the recommendation will be made, and against the error ($F = 14.93$), if they are the only ones of interest. The interaction is highly significant ($F = 2.76$, $p < 0.0001$): the ranking of the genotypes changes from one environment to another. That the environments take most of the sum of squares is normal and says nothing about the interaction.

1 · Genotype × environment table

TRAIT: GY | BEST VALUE: the highest (yield) | EMPTY CELLS: EM WITH: additive + 1 axis (AMMI1)

GY: 10 genotypes × 14 environments. Pooled error on the plot scale: MS = 0.0967 with 252 df, r = 3 (harmonic mean).

GENOTYPES	ENVIRONMENTS	SS OF ENVIRONMENTS	SS OF GENOTYPES	SS OF G × E	CV
10	14	86.3 %	4.0 %	9.6 %	11.63 %

Analysis of variance

Source	df	SS	MS	F	p	F against error	% of the SS
Environments	13	279.5736	21.5057	62.325 / rep	< 0.0001 ***	—	86.3
Replicates within environments	28	9.6615	0.3451	3.569	< 0.0001 ***	—	—
Genotypes	9	12.995	1.4439	5.411 / G×A	< 0.0001 ***	14.933 · < 0.0001 ***	4
Genotypes × environments	117	31.2196	0.2668	2.760	< 0.0001 ***	—	9.6
Pooled error	252	24.3667	0.0967	—	—	—	—

Analysis of the table of means multiplied by r (plot scale). Genotypes are tested against the interaction when the environments sample the target region, and against the pooled error when they are the only ones of interest; environments, against replicates within environments.

Figure 9.1. Card 1 with the oat trial. The environments account for 86% of the sum of squares, the genotypes 4% and the interaction almost 10%.

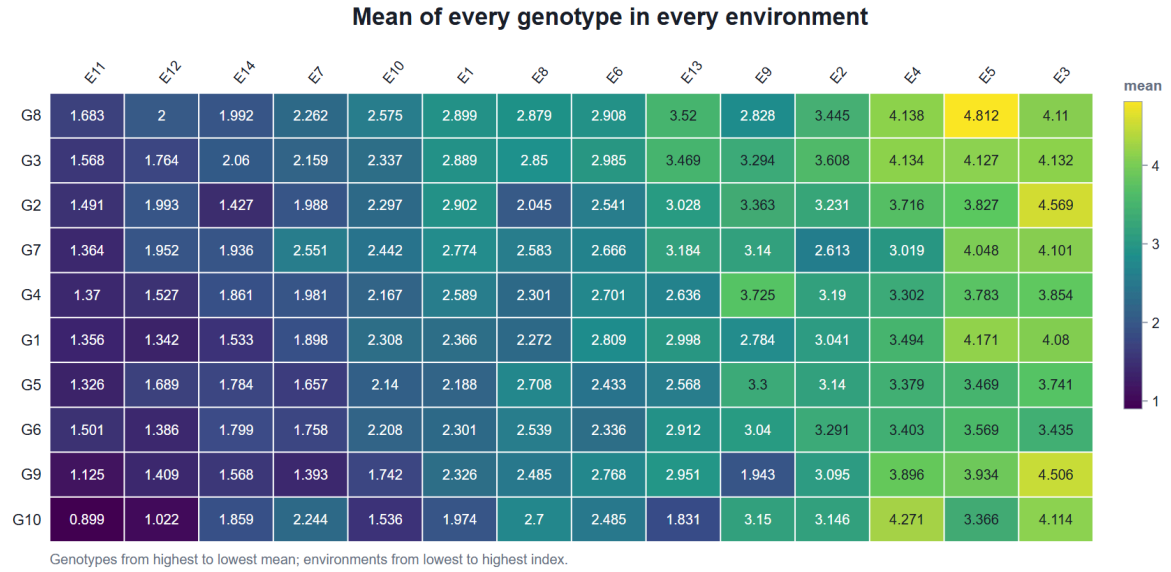


Figure 9.2. The table as a heat map, with the genotypes from highest to lowest mean and the environments from lowest to highest index. Its panel switches it to the interaction (the residuals of the additive model).

The folded box Means of every genotype in every environment has the full table; the estimated cells carry an asterisk. In the analysis of variance, the interaction loses one degree of freedom for each estimated cell.

9.3 Unequal errors: the two-stage analysis

The analysis of the table assumes that all environments have the same error, which almost never happens. Below the analysis of variance, the Error variance of every environment and two-stage analysis table gives the error mean square of each environment with Bartlett's test, and fits a two-stage model (Smith, Cullis and Thompson, 2005; Möhring and Piepho, 2009): each mean carries the variance of its own environment and the interaction is random, with σ^2_{GE} estimated by restricted maximum likelihood.

Error variance of every environment and two-stage analysis

Environment	Mean	Error MS	df	r	CV %
E1	2.521	0.144	18	3	15.05
E2	3.18	0.1785	18	3	13.29
E3	4.064	0.1789	18	3	10.41
E4	3.675	0.1377	18	3	10.1
E5	3.911	0.0664	18	3	6.59
E6	2.663	0.0586	18	3	9.09
E7	1.989	0.091	18	3	15.17
E8	2.536	0.0278	18	3	6.57
E9	3.057	0.1109	18	3	10.9
E10	2.175	0.0267	18	3	7.51
E11	1.368	0.105	18	3	23.68
E12	1.609	0.0535	18	3	14.38
E13	2.91	0.0994	18	3	10.83
E14	1.782	0.0753	18	3	15.4

Bartlett's test of equal error variances: χ^2 = 36.822 with 13 df, p = 0.0004; largest/smallest ratio = 6.71. Two-stage model: the mean of every genotype in every environment carries the variance of its own environment (known from stage one) and the interaction is random. σ^2GE (REML) = 0.0541; likelihood ratio for σ^2GE = 0: 127.262, p < 0.0001 (mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$). Wald test of equal means: F = 5.989 with 9 and 117 df (approximate), p < 0.0001. Mean standard error of a difference: 0.1092.

Figure 9.3. The error of each oat environment. The largest is 6.7 times the smallest and Bartlett's test rejects equality (p = 0.0004); the note gives the result of the two-stage model.

In the oat trial, σ^2_{GE} = 0.0541 and the likelihood ratio declares it different from zero (127.3, $p < 0.0001$, with the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ that corresponds to a variance on the boundary). The Wald test of equal means gives F = 5.99. The folded box below compares the two-stage means with the simple ones: here G7 and G2 swap third and fourth place. With very different errors among environments, the differences can be larger; in the potato, for example, a different clone takes first place.

9.4 Regression on the environmental index

The environmental index I_j is the mean of the environment minus the grand mean: it measures how good each environment was. The **2 · Regression on the environmental index** card fits, for each genotype, the line of its mean in each environment on that index (Finlay and Wilkinson, 1963). The slope b tells how the genotype responds to the improvement of the environment (b = 1, like the average; greater, it takes more advantage of good environments; smaller, it holds up better in poor ones) and the deviations from the line, S²d, whether that response is predictable (Eberhart and Russell, 1966).

2 · Regression on the environmental index

STANDARD ERROR OF b

the genotype's own deviations

Genotype	Mean	b ± SE	t (b = 1)	p	S²d	F (S²d = 0)	p	R²	Reading
G8	3.004	1.034 ± 0.087	0.40	0.6993 ns	0.0382	2.19	0.0129 *	0.922	significant deviations: unpredictable response
G3	2.955	1.031 ± 0.046	0.67	0.5150 ns	−0.0128	0.60	0.8396 ns	0.977	stable in the sense of Eberhart & Russell
G2	2.744	1.054 ± 0.094	0.58	0.5722 ns	0.0497	2.54	0.0035 **	0.913	significant deviations: unpredictable response
G7	2.741	0.819 ± 0.098	−1.84	0.0904 ns	0.0579	2.80	0.0013 **	0.852	significant deviations: unpredictable response
G4	2.642	0.937 ± 0.082	−0.77	0.4581 ns	0.0298	1.93	0.0319 *	0.917	significant deviations: unpredictable response
G1	2.604	1.063 ± 0.057	1.09	0.2959 ns	−0.0014	0.96	0.4919 ns	0.966	stable in the sense of Eberhart & Russell
G5	2.537	0.887 ± 0.067	−1.70	0.1143 ns	0.009	1.28	0.2306 ns	0.937	stable in the sense of Eberhart & Russell
G6	2.534	0.861 ± 0.062	−2.25	0.0439 *	0.003	1.09	0.3651 ns	0.942	responds less than average: adapted to poor environments
G9	2.51	1.192 ± 0.116	1.65	0.1252 ns	0.0938	3.91	< 0.0001 ***	0.897	significant deviations: unpredictable response
G10	2.471	1.122 ± 0.150	0.81	0.4332 ns	0.1773	6.50	< 0.0001 ***	0.823	significant deviations: unpredictable response

Environmental index I_j = environment mean − grand mean. $b = \Sigma Y_{ij} I_j / \Sigma I_j^2$ (Finlay & Wilkinson 1963); β = b − 1 is the parameter of Perkins & Jinks (1968). The standard error of b comes from the genotype's own deviations (12 df). S²d = deviation MS − MS_e/r (Eberhart & Russell 1966), tested with $F = MS_{dev}/(MS_e/r)$ with 12 and 252 df. Because the index contains the genotype itself, that F has expectation (g − 1)/g without interaction and the test is somewhat conservative with few genotypes.

Figure 9.4. The regression of the 10 oat genotypes, from highest to lowest mean. A bar in the margin marks those with significant b or S²d; the last column reads them in words.

G3 is the example of stability in the Eberhart and Russell sense: high mean (2.955), b = 1.031 ± 0.046 and S²d not different from zero, with R² = 0.977. G8, the one with the highest mean, responds like the average (b = 1.034), but its deviations are significant (p = 0.013): it wins almost always, though not always for the same reason. G6 is the only one with b significantly less than 1 (0.861, p = 0.044), adapted to poor environments, and G10 the most unpredictable (S²d = 0.177).

The **Standard error of b** menu offers three sources: the genotype's own deviations (12 degrees of freedom here), the pooled deviations or the experimental error. The folded box **Eberhart and Russell's analysis of variance** partitions the interaction into the heterogeneity of slopes (not significant here, p = 0.13) and the deviations of each genotype.

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With the 17 genotypes in 12 environments of Kang (1993), the sums of squares of genotypes, environments and interaction are 186.9082, 2,284.0116 and 493.2518, the heterogeneity of slopes 44.8576 and the deviations 448.3942; the slopes of genotypes A to E, 0.8617, 1.1505, 0.9755, 1.2230 and 0.9641; and their S²d, the ecovalences, the Shukla variances and the Lin and Binns superiority are also the published ones.

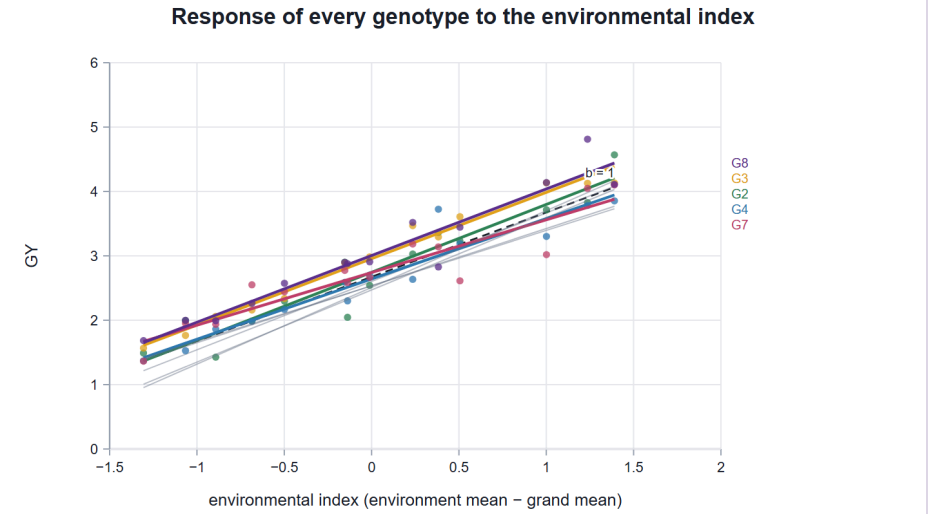

Figure 9.5. The response lines of the genotypes, with the five with the highest mean highlighted and their observed means; the dashed line is the average response (*b* = 1).

Figure 9.6. Mean against *b*. The quadrants separate broad adaptation (high mean, *b* close to 1) from specific adaptation to good or poor environments; the diamonds have a significant S²d.

NOTE · TWO CORRECTIONS TO THE CLASSIC CALCULATION

The deviations of all genotypes add up to zero in each environment, so the pooled deviations have $(g-1)(e-2)$ degrees of freedom, 108 in the oat trial, and not the $g(e-2) = 120$ of Eberhart and Russell; the note of the table also gives the mean square with its calculation. Moreover, since the environmental index contains the genotype itself, the F of S²d has expectation $(g-1)/g$ without interaction: with few genotypes the test is somewhat conservative.

9.5 Stability variances, superiority and reliability

The 3 · Stability variances, superiority and reliability card begins with two measures of how much each genotype contributes to the interaction. Wricke's (1962) ecovalence, W_i, is the sum of its squared residuals from the additive model, with its percentage of the total. Shukla's (1972) stability variance, σ^2_i, is the same information in the form of a variance, which can be tested against the error; s^2_i is what remains of it after removing the regression on the index.

3 · Stability variances, superiority and reliability

Ecovalence and stability variance

Genotype	Mean	W_i	W_i %	σ^2_i	F	p	s^2_i	p
G3	2.955	0.7249	2.32	0.0364	0.38	0.9764 ns	0.0408	0.9540 ns
G1	2.604	1.2194	3.91	0.0839	0.87	0.5879 ns	0.0836	0.5840 ns
G6	2.534	1.8064	5.79	0.1403	1.45	0.1364 ns	0.1003	0.4147 ns
G5	2.537	1.8441	5.91	0.144	1.49	0.1216 ns	0.1227	0.2370 ns
G4	2.642	2.3434	7.51	0.192	1.99	0.0224 *	0.2008	0.0190 *
G8	3.004	2.5689	8.23	0.2137	2.21	0.0097 **	0.2322	0.0059 **
G2	2.744	3.032	9.71	0.2582	2.67	0.0016 **	0.2753	0.0011 **
G7	2.741	4.1623	13.33	0.3669	3.79	< 0.0001 ***	0.3061	0.0003 ***
G9	2.51	5.5627	17.82	0.5015	5.19	< 0.0001 ***	0.4405	< 0.0001 ***
G10	2.471	7.9553	25.48	0.7316	7.57	< 0.0001 ***	0.7537	< 0.0001 ***

Wricke's (1962) ecovalence $W_i = r\,\Sigma_j (Y_{ij} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot j} + \bar{Y}_{\cdot\cdot})^2$, and its percentage of the total. Shukla's (1972) stability variance $\sigma^2_i = [g(g-1)W_i - \Sigma W]/[(g-1)(g-2)(e-1)]$: it estimates the interaction plus error variance of that genotype (its average is the interaction MS) and is tested with $F = \sigma^2_i/MS_e$ with 13 and 252 df. s^2_i is the same variance after removing the regression on the environmental index (12 df). Plot scale; divide by r for the means scale.

Figure 9.7. Ecovalence and stability variances, from lowest to highest σ^2_i. G3 contributes 2.3% of the interaction and G10, 25.5%.

Below, the Lin and Binns (1988) superiority, P_i, measures the distance of each genotype to the best one in each environment: the smaller, the better. It combines mean and stability, and the table splits it into its genetic and interaction fractions and computes it separately in favorable and unfavorable environments. Annicchiarico's (1992) reliability, W_i, is the relative performance (as a percentage of the mean of each environment) that the genotype exceeds with probability $1-\alpha$; the risk α is chosen above the table.

Superiority and reliability

ANNICCHIARICO'S RISK α

0.25

Genotype	Mean	P_i	genetic part %	P_i favorable	P_i unfavorable	W_i (Annicchiarico)	favorable	unfavorable
G3	2.955	**0.041**	47.5	0.0723	0.0175	**107.98**	105.78	109.81
G8	3.004	**0.0412**	27	0.0882	0.0059	**107.41**	101.87	112.51
G2	2.744	**0.1256**	66.5	0.1494	0.1077	**94.49**	100.74	90.71
G7	2.741	**0.1485**	57.1	0.318	0.0214	**96.05**	88.91	103.35
G1	2.604	**0.1691**	89.2	0.2278	0.1251	**91.51**	94.76	89.21
G4	2.642	**0.1725**	75.6	0.2888	0.0853	**93.71**	91.04	96.22
G6	2.534	**0.238**	80.4	0.377	0.1337	**90.38**	90.49	89.99
G5	2.537	**0.24**	78.9	0.3825	0.1332	**90.23**	89.53	90.45
G9	2.51	**0.2909**	71	0.3897	0.2168	**82.25**	85.26	80.61
G10	2.471	**0.3437**	67.6	0.4747	0.2454	**77.53**	82.38	73.56

Superiority of Lin & Binns (1988): $P_i = \Sigma_j (Y_{ij} - M_j)^2/(2e)$, with M_j the highest mean of environment j; it is split into a genetic fraction, $e(\bar{Y}_{i\cdot} - \bar{M})^2/(2e)$, and an interaction one, and computed apart in the 6 favorable environments (index ≥ 0) and the 8 unfavorable ones. Reliability of Annicchiarico (1992): W_i = mean − $z_{1-\alpha}$·SD of the relative performance $100 \cdot Y_{ij}/\bar{Y}_{j\cdot}$ with α = 0.25 (z = 0.6745): the relative performance the genotype exceeds with probability 1 − α.

Figure 9.8. Superiority and reliability, from lowest to highest P_i. G3 and G8 almost tie in superiority, by different paths: G3 by being stable, G8 by its mean.

The table shows things the mean alone does not say. The superiority of G8 comes mostly from the interaction (only 27% is genetic): it is almost unbeatable in unfavorable environments (P_i = 0.006) and less so in favorable ones (0.088). With α = 0.25, G3 exceeds 108% of the mean of each environment with 75% probability, and G8 reaches 112.5% in the unfavorable ones. The reliability is only computed when the highest value is the best.

9.6 Ranks and yield–stability

Huehn's (1979, 1990) statistics work with the rank of each genotype in each environment, not with its values, and therefore do not depend on normality or on extreme values. $S^{(1)}$ is the mean difference between the ranks of the genotype in each pair of environments and $S^{(2)}$, their variance, both with the data corrected for the genotype effect; $S^{(3)}$ and $S^{(6)}$ use the uncorrected data and combine yield and stability. The Z tests of Nassar and Huehn (1987) mark the unstable genotypes with an asterisk.

Kang's (1993) yield–stability statistic, YS_i, adds two ratings. The yield rating is the rank by the mean, adjusted with the least significant difference from the grand mean (+1 to +3 if it exceeds it, −1 to −3 if it falls below). The stability rating penalizes a significant Shukla variance (−2, −4 or −8 at 10, 5 or 1%). The genotypes with YS_i greater than the average are selected.

4 · Rank statistics and yield–stability

Huehn's ranks

Genotype	Mean	$S^{(1)}$	$Z^{(1)}$	$S^{(2)}$	$Z^{(2)}$	$S^{(3)}$	$S^{(6)}$
G3	2.955	2.7033	1.772	5.2967	1.930	1.831	1.525
G4	2.642	2.9121	0.749	6.2692	0.868	11.2	4.4
G8	3.004	2.967	0.552	6.5714	0.623	5.732	1.951
G5	2.537	3.3626	0.020	8.5714	0.023	16.804	6.471
G7	2.741	3.3626	0.020	8.3077	0.001	15.306	4.776
G1	2.604	3.4286	0.082	8.5714	0.023	16.581	6.065
G6	2.534	3.4505	0.113	8.7253	0.050	13.982	5.228
G2	2.744	3.6264	0.530	9.6044	0.406	17	4.773
G9	2.51	3.6813	0.724	9.6319	0.423	18.276	6.414
G10	2.471	4.5714	8.046 *	16.8791	16.476 *	32.276	9.655

Huehn's (1979, 1990) ranks: $S^{(1)}$ is the mean absolute difference between the ranks of the genotype in every pair of environments and $S^{(2)}$ their variance, with data corrected for the genotype effect ($Y_{ij} - \bar{Y}_{i\cdot} + \bar{Y}_{\cdot\cdot}$); $S^{(3)}$ and $S^{(6)}$ use the uncorrected data, divided by the mean rank, and combine yield and stability. Rank 1 = lowest value. Tests of Nassar & Huehn (1987): $Z = (S - E)^2/V \sim \chi^2_1$ with $E(S^{(1)}) = 3.300$, $V = 0.2009$, $E(S^{(2)}) = 8.250$, $V = 4.5194$; * = unstable ($p < 0.05$). Sum of $Z^{(1)} = 12.608$ ($p = 0.2464$) and of $Z^{(2)} = 20.822$ ($p = 0.0224$), χ^2 with 10 df: they test whether the genotypes differ in stability.

Figure 9.9. Huehn's ranks, from lowest to highest $S^{(1)}$. Only G10 is significantly unstable; the sum of the $Z^{(2)}$ says that the genotypes differ in stability ($p = 0.022$).

Kang's yield–stability (YS_i)

ONE-SIDED LSD AT

5 %

Genotype	Mean	Rank	LSD adjustment	σ^2_i	Stability rating	YS_i	Selected
G3	2.955	9	+3	0.0364	0	**12**	✓
G8	3.004	10	+3	0.2137	−8	**5**	✓
G1	2.604	5	−1	0.0839	0	**4**	✓
G5	2.537	4	−2	0.144	0	**2**	✓
G2	2.744	8	+1	0.2582	−8	**1**	✓
G4	2.642	6	−1	0.192	−4	**1**	✓
G6	2.534	3	−2	0.1403	0	**1**	✓
G7	2.741	7	+1	0.3669	−8	**0**	—
G9	2.51	2	−2	0.5015	−8	**−8**	—
G10	2.471	1	−2	0.7316	−8	**−9**	—

Kang (1993): the rank by the mean (1 = lowest) is adjusted with the LSD = $t_{0.95}\cdot\sqrt{(2MS_e/re)} = 0.112$ (+1, +2 or +3 when the mean exceeds the grand mean by 0, 1 or 2 LSD; −1, −2 or −3 when below) and Shukla's instability is penalized: −2, −4 or −8 when its F is significant at 10, 5 or 1 %. The genotypes with YS_i above its average (0.9) are selected.

Figure 9.10. Kang's YS_i. G3 comes first (12) for its high mean without penalty; G8, the one with the highest mean, loses 8 points for its instability and comes second. Seven genotypes are selected.

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With Kang's (1993) table, the YS_i of genotypes A, B, C and D are 14, 3, 12 and 3, and A, C, E, G, H, J, N and O are selected, as in the paper. To achieve this, two values equal to 12 significant digits are treated as a tie: with rounded data, two genotypes can have the same mean and differ only in the last bit.

9.7 AMMI

AMMI (additive main effects and multiplicative interaction; Gollob, 1968; Gauch, 1988) fits the grand mean and the effects of genotypes and environments, and decomposes the interaction into singular values: a few axes (IPCA) summarize almost all the pattern and the rest is noise. The **5 · AMMI** card decides how many axes to retain, computes the stability indices derived from them and predicts the best genotypes of each environment. This section uses the 28 potato clones in six locations.

5 · AMMI

NUMBER OF AXES BY THE TEST: Cornelius' residual F (recommended) | AXES FOR WAAS: by the chosen test | WEIGHT OF THE MEAN IN WAASY (%): 50

Interaction axes

Axis	Singular value	SS	%	cumulative %	df (Gollob)	MS	F	p
IPCA1	66.7548	13,368.5954	56.26	56.26	31	431.245	11.645	< 0.0001 ***
IPCA2	46.2874	6,427.5799	27.05	83.31	29	221.6407	5.985	< 0.0001 ***
IPCA3	27.337	2,241.9398	9.44	92.75	27	83.0348	2.242	0.0005 ***
IPCA4	18.5075	1,027.5785	4.32	97.07	25	41.1031	1.110	0.3287 ns
IPCA5	15.2327	696.1012	2.93	100	23	30.2653	0.817	0.7094 ns

Model	SS left	df	MS	F_R	p
AMMI0	23,761.7949	135	176.0133	4.753	< 0.0001 ***
AMMI1	10,393.1995	104	99.9346	2.699	< 0.0001 ***
AMMI2	3,965.6195	75	52.8749	1.428	0.0191 *
AMMI3	1,723.6798	48	35.91	0.970	0.5341 ns
AMMI4	696.1012	23	30.2653	0.817	0.7094 ns

Gollob (1968): 3 axes significant at 5 %. The F_R of Cornelius et al. (1992) tests what is left of the interaction after n axes: AMMI3 is enough. Gollob's test is liberal —in simulations without interaction it declares the first axis significant in more than a third of the cases—, which is why F_R is the default rule (Piepho 1995). WAAS and WAASY use 3 axes.

Figure 9.11. The interaction axes of the potato. Above, each axis with Gollob's test; below, Cornelius's residual test, which tests what remains of the interaction after *n* axes.

The first axis captures 56.3% of the interaction and the first two, 83.3%. Gollob's test declares three axes significant; the residual test of Cornelius, Seyedsadr and Crossa (1992) says that after two axes there is still interaction left (p = 0.019) and after three there is not (p = 0.53): AMMI3 is enough.

IMPORTANT · GOLLOB'S TEST IS LIBERAL

In 10,000 simulated tables without any interaction, Gollob's *F* declared the first axis significant 38% of the time, when it should do so 5% of the time; the residual *F* kept its level. That is why the residual test is the default rule of the Number of axes by the test menu (Piepho, 1995), with Gollob's as an option. Axes for WAAS lets you set the number by hand.

With the retained axes, four indices are computed, in the Stability of the genotypes table: ASV, the distance of each genotype to the origin on the first two axes, weighted by their sum of squares (Purchase and colleagues, 2000); YSI, the sum of the ranks by ASV and by the mean (Farshadfar, 2008); WAAS, the average of the absolute values of its scores on the retained axes, weighted by the share of the interaction each one explains (Olivoto and colleagues, 2019); and WAASY, which rescales the mean and WAAS from 0 to 100 and combines them with the weight you choose.

WAAS separates two kinds of high-mean clones (Figure 9.14). 319.20 has the third-highest mean (38.8), but the largest interaction after Desiree (WAAS = 2.54): it yields a lot in some locations and little in others. 364.21 and 427.7 have somewhat lower means (34.1 and 36.2) and much more stability, and that is why they head WAASY. 402.7 is the most stable of all (WAAS = 0.10), but with a low mean. The Best genotypes of every environment table, at the end of the card, gives the three best of each location predicted by AMMI3; in La Molina 2003 (LM-03), the winner is precisely 319.20.

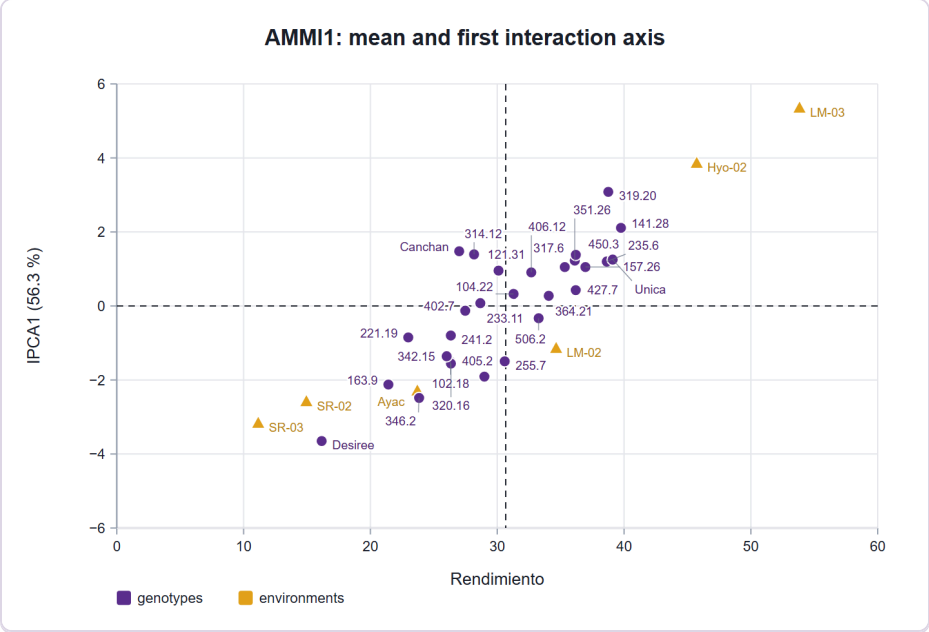

Figure 9.12. AMMI1: the mean of each clone and each location against its first interaction axis. Clones near zero on the vertical axis are stable; those that share a sign with a location have a positive interaction with it.

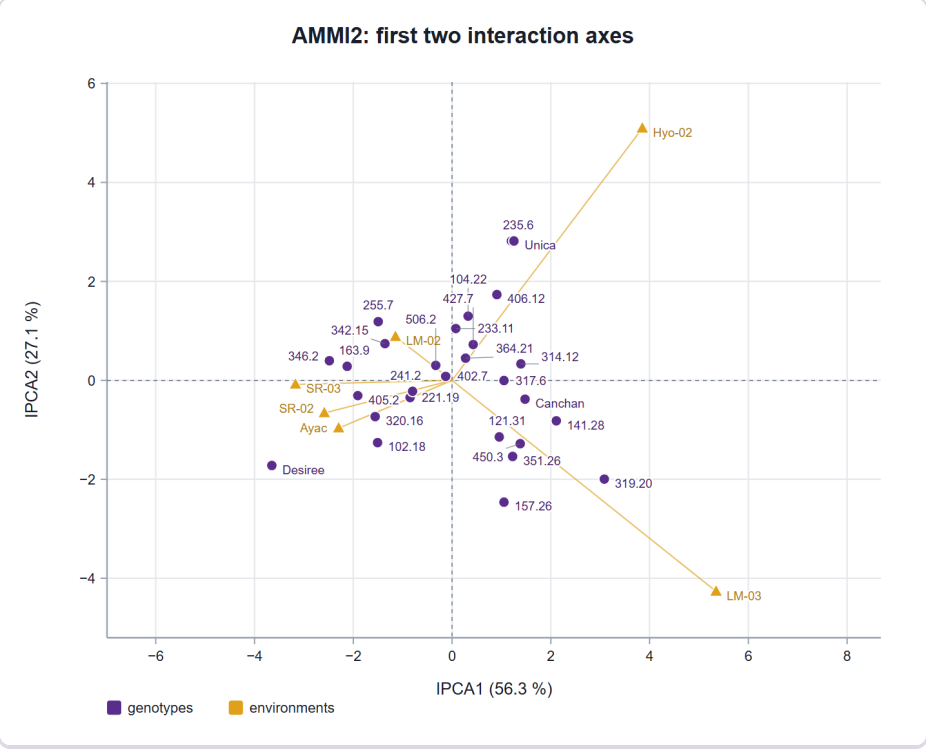

Figure 9.13. AMMI2: the first two axes. San Ramón (SR-02, SR-03) and Ayacucho lie together on the negative side of the first axis; Huancayo (Hyo-02) and La Molina 2003 (LM-03) point in opposite directions on the second: the clones that win in one are not those that win in the other.

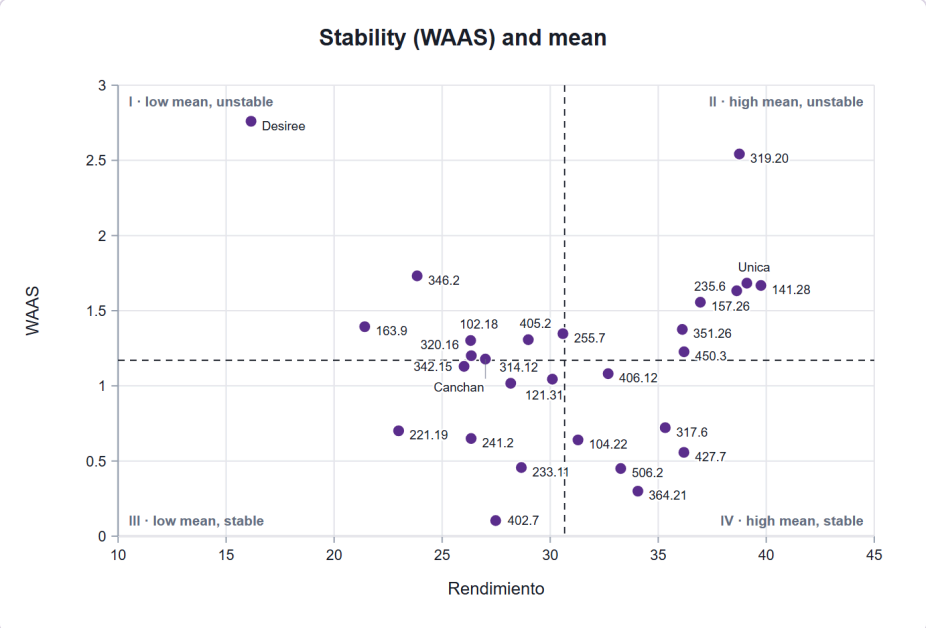

Figure 9.14. WAAS against the mean. Quadrant IV (high mean, stable) holds the desirable clones: 364.21, 427.7 and 506.2 head WAASY.

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The sums of squares of the five potato axes are 13,368.60, 6,427.58, 2,241.94, 1,027.58 and 696.10; Gollob's *F* values, 11.645, 5.985, 2.242, 1.110 and 0.817; and the residual ones, 4.753, 2.699, 1.428 and 0.970, the values of the classic example. The ASV and WAAS indices agree with an independent implementation; WAAS differs in the fourth decimal place from another that rounds the percentages of the axes to one decimal.

Best genotypes of every environment

Environment	Index	1st	2nd	3rd
SR-03	−19.516	**255.7** 16.352	405.2 15.946	235.6 15.633
SR-02	−15.718	**141.28** 20.412	351.26 19.842	405.2 18.925
Ayac	−6.967	**141.28** 31.583	351.26 31.341	319.20 28.363
LM-02	3.975	**Unica** 48.668	157.26 46.557	506.2 45.122
Hyo-02	15.061	**235.6** 73.616	Unica 72.178	141.28 60.49
LM-03	23.165	**319.20** 86.796	141.28 77.493	157.26 76.515

Values predicted by AMMI3: the main effects plus 3 interaction axes; environments from lowest to highest index.

Figure 9.15. The three best clones of each location according to AMMI3, from the poorest location to the best.

9.8 The GGE biplot

The GGE biplot (Yan and colleagues, 2000; Yan and Kang, 2003) decomposes the environment-centered table, which contains the effect of the genotypes plus their interaction (G + G×E), and draws genotypes as points and environments as vectors. The 6 · GGE biplot card has three controls: the View, the Scaling of the environments (unscaled, or divided by their standard deviation so that they weigh the same) and the Singular-value partition, which by default is the one suited to each view.

In the oat trial, the first two components explain 41.7% and 22.7% of G + G×E. If together they explain less than 60%, a warning recommends reading the biplot with caution, because it leaves out much of the pattern.

G8 wins in nine of the 14 environments; G9 in E3 and E4, G7 in E7 and E14, and G4 in E9. The table below the figure lists the environments of each winner. Those groups are possible mega-environments if the pattern repeats in other years; with a single year, they may be just noise. The mean vs stability view projects each genotype onto the average-environment axis (the arrow): the projection approximates its mean and the distance to the axis, its instability.

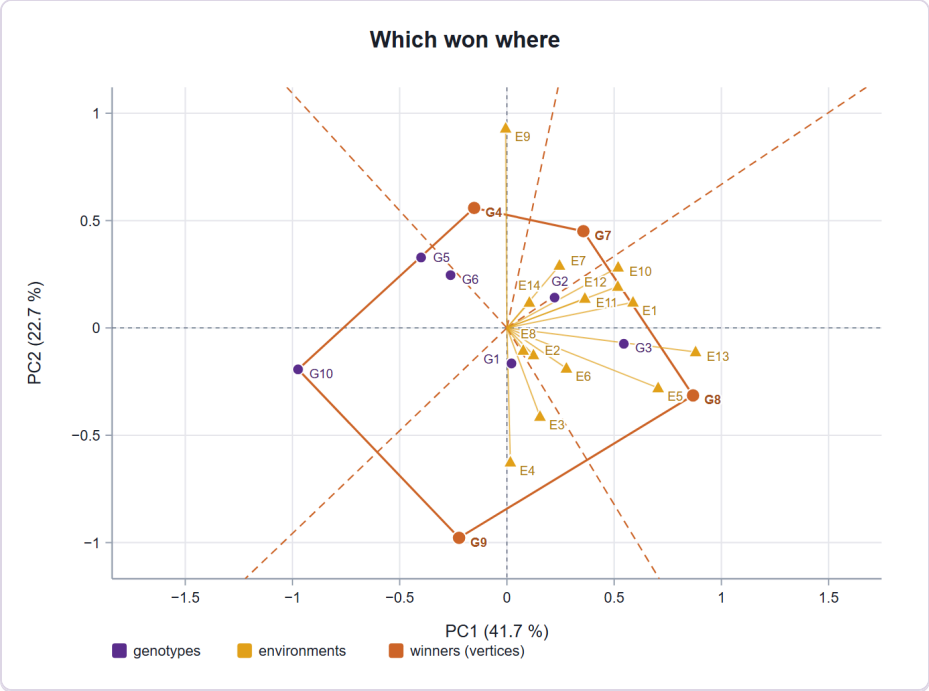

Figure 9.16. Which won where. The polygon joins the outermost genotypes; the lines perpendicular to its sides divide the plane into sectors, and the genotype at the vertex wins in the environments of its sector.

Table 9.2. The views of the GGE biplot and the question each one answers.

View	Question	Partition
Which won where	Which genotype wins in each group of environments? Are there mega-environments?	symmetric
Mean vs stability	Which genotypes yield more and which are stable?	genotype-focused
Genotypes against the ideal one	Which comes closest to the ideal genotype (high and stable)?	genotype-focused
Discriminativeness vs representativeness	Which environments discriminate the genotypes well and which represent the region?	environment-focused
Environments against the ideal one	Which environment is best for testing?	environment-focused
Compare two genotypes	In which environments does each of the two win?	symmetric

In the environment views, the length of the vector measures how much the environment discriminates among genotypes and the angle with the average-environment axis, how much it resembles the others. A long environment with a large angle is useful for discarding unstable genotypes, not for choosing broadly adapted ones. The `compare two genotypes` view adds two menus to choose them and a table that says, for each environment, which one the biplot favors and whether it agrees with the observed means.

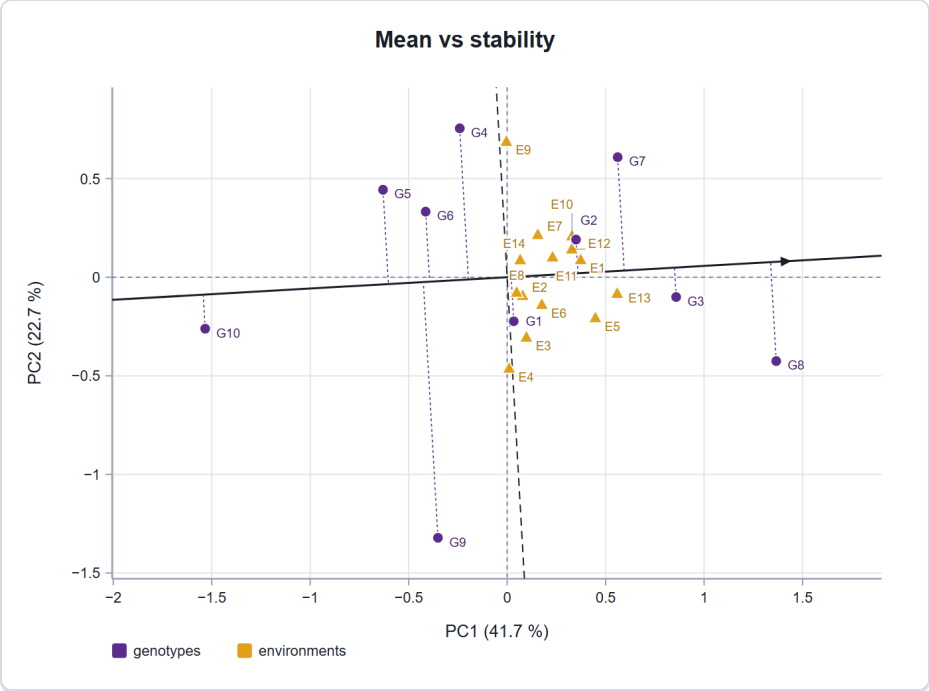

Figure 9.17. Mean vs stability. G8 and G3 have the largest projections; G3, moreover, hardly departs from the axis. G9 and G4 are the least stable.

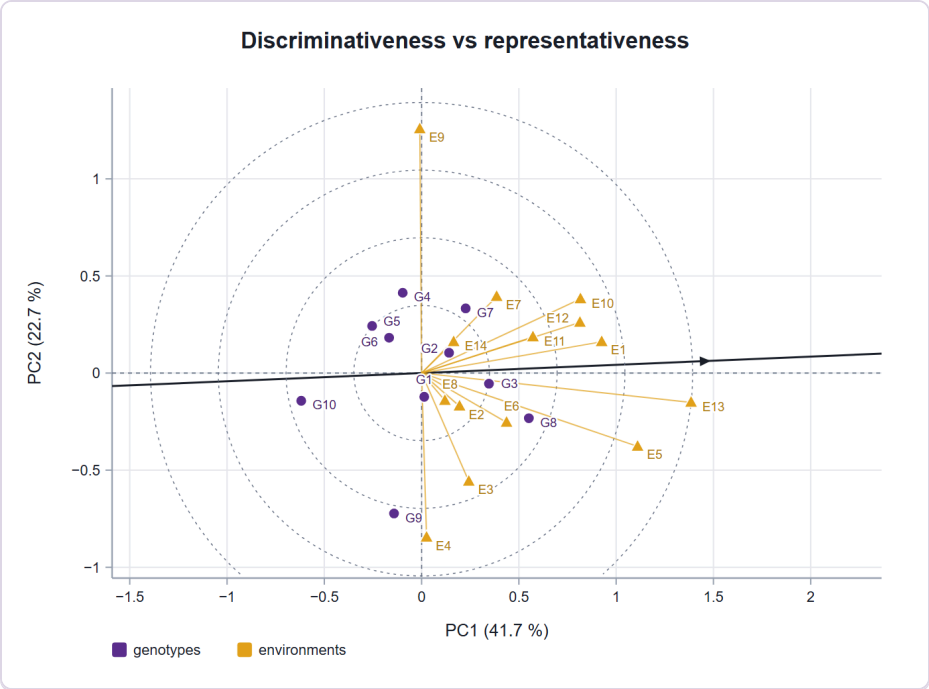

Figure 9.18. Discriminativeness and representativeness. E13 and E5, long and close to the average-environment axis, are good test environments; E9 discriminates but, almost perpendicular to the axis, does not represent the others.

9.9 Summary of the statistics

The 7 · Summary of the statistics card brings together in one table the 16 statistics of each genotype with its rank in parentheses (1 = best) and the mean rank. The mean rank serves to organize the discussion, not to decide: it adds without weights statistics that measure different things, and some repeat the same information.

7 · Summary of the statistics

Genotype	Mean	b	S²d	R²	W	σ²	Pi	Wi (Annicchiarico)	S1	S2	S3	S6	YSi	ASV	WAAS	WAASY	Mean rank
G3	2.955 (2)	1.031 (1)	−0.013 (1)	0.977 (1)	0.725 (1)	0.036 (1)	0.041 (1)	107.975 (1)	2.703 (1)	5.297 (1)	1.831 (1)	1.525 (1)	12 (1)	0.113 (1)	0.098 (1)	95.465 (1)	**1.06**
G8	3.004 (1)	1.034 (2)	0.038 (6)	0.922 (5)	2.569 (6)	0.214 (6)	0.041 (2)	107.411 (2)	2.967 (3)	6.571 (3)	5.732 (2)	1.951 (2)	5 (2)	0.574 (6)	0.346 (6)	79.067 (2)	**3.5**
G1	2.604 (6)	1.063 (5)	−0.001 (2)	0.966 (2)	1.219 (2)	0.084 (2)	0.169 (5)	91.512 (6)	3.429 (6)	8.571 (5.5)	16.581 (6)	6.065 (7)	4 (3)	0.346 (4)	0.161 (2)	57.069 (4)	**4.22**
G4	2.642 (5)	0.937 (4)	0.03 (5)	0.917 (6)	2.343 (5)	0.192 (5)	0.173 (6)	93.712 (5)	2.912 (2)	6.269 (2)	11.2 (3)	4.4 (3)	1 (6)	0.594 (7)	0.348 (7)	44.935 (6)	**4.81**
G2	2.744 (3)	1.054 (3)	0.05 (7)	0.913 (7)	3.032 (7)	0.258 (7)	0.126 (3)	94.494 (4)	3.626 (8)	9.604 (8)	17 (8)	4.773 (4)	1 (6)	0.249 (2)	0.26 (4)	61.955 (3)	**5.25**
G6	2.534 (8)	0.861 (8)	0.003 (3)	0.942 (3)	1.806 (3)	0.14 (3)	0.238 (7)	90.378 (7)	3.451 (7)	8.725 (7)	13.982 (4)	5.228 (6)	1 (6)	0.265 (3)	0.232 (3)	44.587 (7)	**5.31**
G5	2.537 (7)	0.887 (6)	0.009 (4)	0.937 (4)	1.844 (4)	0.144 (4)	0.24 (8)	90.23 (8)	3.363 (4.5)	8.571 (5.5)	16.804 (7)	6.471 (9)	2 (4)	0.43 (5)	0.281 (5)	40.8 (8)	**5.81**
G7	2.741 (4)	0.819 (9)	0.058 (8)	0.852 (9)	4.162 (8)	0.367 (8)	0.149 (4)	96.047 (3)	3.363 (4.5)	8.308 (4)	15.306 (5)	4.776 (5)	0 (8)	0.663 (8)	0.393 (8)	50.429 (5)	**6.28**
G9	2.51 (9)	1.192 (10)	0.094 (9)	0.897 (8)	5.563 (9)	0.502 (9)	0.291 (9)	82.254 (9)	3.681 (9)	9.632 (9)	18.276 (9)	6.414 (8)	−8 (9)	0.983 (9)	0.561 (9)	14.643 (9)	**8.94**
G10	2.471 (10)	1.122 (7)	0.177 (10)	0.823 (10)	7.955 (10)	0.732 (10)	0.344 (10)	77.535 (10)	4.571 (10)	16.879 (10)	32.276 (10)	9.655 (10)	−9 (10)	1.225 (10)	0.692 (10)	0 (10)	**9.81**

In brackets, the rank of every genotype by every statistic (1 = best: highest mean, b closest to 1, least instability, highest Annicchiarico W_i, YS_i and WAASY). The mean rank summarizes, without weights, statistics that measure different things: it orders the discussion, it does not decide by itself.

Figure 9.19. The oat summary, sorted by mean rank. G3 is first by almost every statistic (mean rank 1.06); G8, the one with the highest mean, comes second (3.5), and G10 last (9.81).

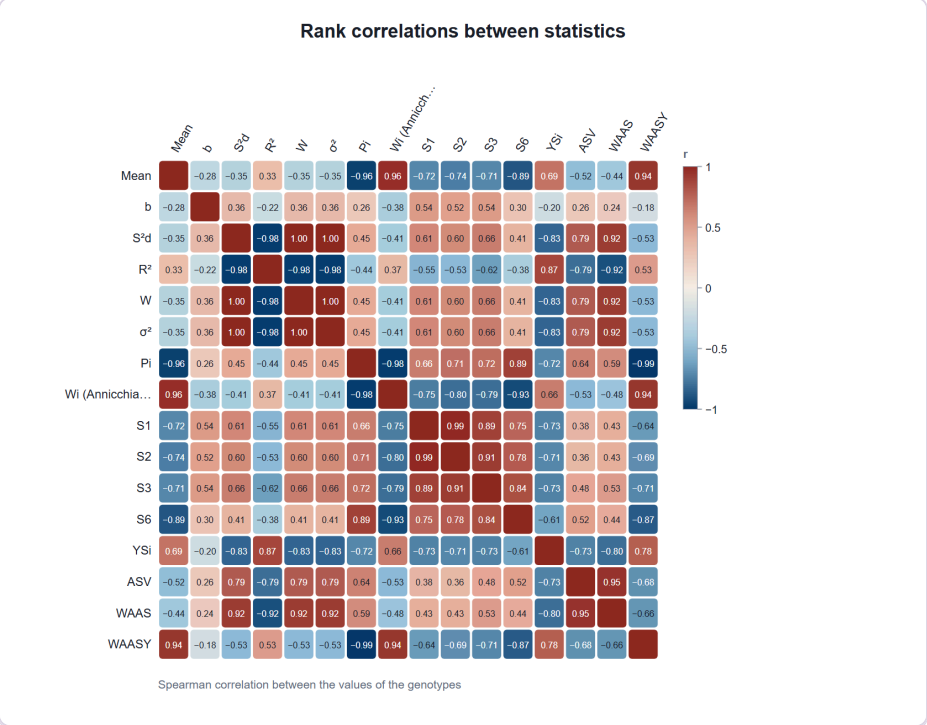

Figure 9.20. Spearman correlations among the oat statistics.

The statistics fall into two families (Lin, Binns and Lefkovitch, 1986; Becker and Léon, 1988): those of static stability, in which the genotype yields the same in all environments, and the dynamic ones, in which it follows the average pattern of the environments; those that combine mean and stability (P_i, YS_i, WAASY, $S^{(3)}$, $S^{(6)}$) form a separate group. In the oat trial (Figure 9.20), the mean goes with P_i, the reliability and WAASY, and the interaction measures (S^2d, W, σ^2, ASV, WAAS) form another group. W_i and σ^2_i always rank alike, because σ^2_i is a linear function of W_i. Static stability is rarely desirable for yield: a genotype that does not change does not take advantage of good environments.

9.10 Downloads and next step

The 8 · Assumptions, formulas and references card summarizes the models, the formulas and the references. Results (.xlsx) downloads, each on its own sheet, the genotype × environment table with the estimated cells, the analysis of variance, the regressions, the variances, the ranks, the AMMI axes and scores, the GGE biplot coordinates and the two-stage analysis. Every figure has its download panel.

Continue takes you to Block 10 (BLUP and mixed models with relationships) and ← Selection returns to Block 8, which with several environments computes the heritability with the genotype × environment interaction.

B10

CHAPTER TEN · BLOCK 10 · BLUP AND CROSSBREEDING

Mixed models, BLUP and crossbreeding

Relationship says how alike two individuals are, and with it a record also tells you about the individual's parents, its sibs and its progeny. Block 10 reads the pedigree, computes inbreeding and the relationship matrix, predicts breeding values with the animal or the sire model, estimates the variance components by REML, predicts the genotypes of a trial with their relationships and, for crossbreeding, estimates Dickerson's parameters and predicts crosses, rotations and composites.

B10

10.1 What the block does

10.2 Records and pedigree

10.3 The animal model

10.4 Sires, repeated records and other effects

10.5 BLUP of the genotypes of a trial

10.6 Crossbreeding

10.7 Downloads and next step

10.1 What the block does

Unlike the previous blocks, Block 10 nearly always works with data of its own: records of individuals with their sire and dam in cards 1 and 2, and means of breeds and their crosses in card 4. Only card 3 uses the trial loaded in Block 3, to which it adds the pedigree from card 1. It serves animals and plants alike: it accepts selfing, parents that are already pure lines and lines derived through several generations of selfing.

Table 10.1. The cards of Block 10.

Card	What it gives
1 · Records and pedigree	Reading and checking of the pedigree, inbreeding, Mendelian sampling variance and relationship matrix A
2 · Animal model: breeding values (BLUP) and components (REML)	Animal or sire model, with permanent environment, maternal effect and other random factors; breeding values with their reliability; genetic trend
3 · BLUP of the trial genotypes (Block 3) with the pedigree	Prediction of the entries of a trial with independent genotypes, with their relationships, or with an additive and a non-additive part
4 · Crossbreeding: Dickerson's parameters and prediction of systems	Direct and maternal breed effects, heterosis and recombination loss; prediction of crosses, rotations and composites
5 · Assumptions, formulas and references	The models, their formulas and the references

The thread of the chapter is a simulated sheep flock: 439 lambs born over three years, progeny of 60 rams and 320 ewes, weighed at weaning. It was simulated with $\sigma^2_a = 30$ and $\sigma^2_e = 70$ ($h^2 = 0.30$), a sex effect and a year effect, and each year's replacements were drawn at random from the lambs, with no selection. The examples of Mrode (2014) and Schaeffer (2019) serve for the checks; plants come in with the pedigree of the 60 lines of the simulated trial of Block 3, and crossbreeding with a simulated three-breed experiment and four published examples.

10.2 Records and pedigree

Card **1 · Records and pedigree** expects one row per record with the individual, its sire and its dam; an unknown parent is written 0 or NA, or left empty. Parents without a row of their own are added as founders: in the flock there are 170, the 20 rams and 150 ewes of the first year. **Records (file)** reads .xlsx, .xls, .xlsm, .ods, .csv, .tsv and .txt files; **Separate pedigree (optional)** adds a second file with ancestors that have no records, so that the relationships reach further back. Seven examples load ready-made data (Table 10.2).

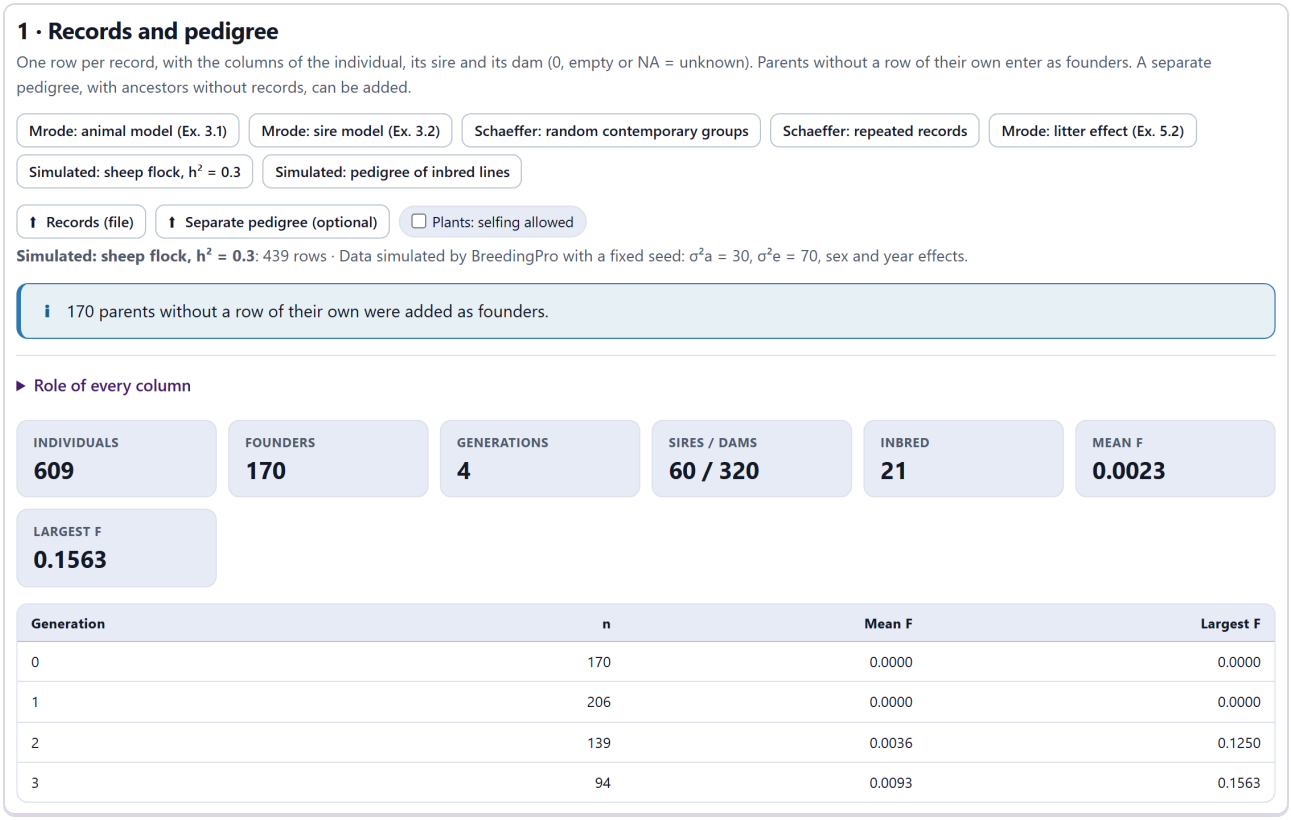

Figure 10.1. Card 1 with the simulated flock: 609 individuals in four generations, 21 of them inbred, with a largest F of 0.1563.

Table 10.2. The examples of cards 1 and 2.

Example	Data	Shows
Mrode: animal model (Ex. 3.1)	5 calves with records, 8 individuals; $\sigma^2_a = 20$ and $\sigma^2_e = 40$	The animal model
Mrode: sire model (Ex. 3.2)	The same data, $\sigma^2_s = 5$ and $\sigma^2_e = 55$	The sire model
Schaeffer: random contemporary groups	15 records, 29 individuals; fixed year, random contemporary group	Other random factors
Schaeffer: repeated records	22 records of 12 individuals	The permanent environment
Mrode: litter effect (Ex. 5.2)	10 piglets from three litters, 15 individuals	The litter as a random effect
Simulated: sheep flock, $h^2 = 0.3$	439 lambs, 609 individuals	REML and the thread of the chapter
Simulated: pedigree of inbred lines	8 parents and 60 F_6 lines	Selfing; card 3

The folded box **Role of every column** has a menu for each column with the roles of Table 10.3. The app proposes them from the name: Animal, ID, Individual or Genotype is the individual; Sire, Father or Bull, the sire; Dam or Mother, the dam; CG, Litter or Group, a random effect; a numeric column with few integer values, a fixed effect, and one with many values, the response trait. In plants, the seed parent goes in as the dam and the pollen parent as the sire.

Table 10.3. The column roles.

Role	What it is for
Individual	The identifier; it is repeated if the individual has several records
Sire (or parent 1) · Dam (or parent 2)	The parents
Response trait	The traits analyzed; card 2 picks one
Fixed effect (factor)	Sex, year, herd, season or type of birth
Covariate	Age at weighing or birth weight, with a linear regression
Random effect (factor)	Contemporary group or litter, with its own variance
Generations of selfing	The *t* generations of each line since the cross of its parents
Founder inbreeding	The F of a founder: 1 for a pure line
Birth year (trend)	Only for the genetic trend
Ignore	The column is left out

The card checks the pedigree before using it and warns about each problem: an individual that appears twice with different parents (the first row is kept), an individual that is its own parent (that parent is deleted), a loop in which someone turns out to be its own ancestor, and the same individual as sire and as dam, which is normal in monoecious plants but not in animals. It then sorts the pedigree so that parents come before their progeny and gives a summary, the inbreeding by generation and, when the pedigree has up to 60 individuals, the relationship matrix A as a heat map.

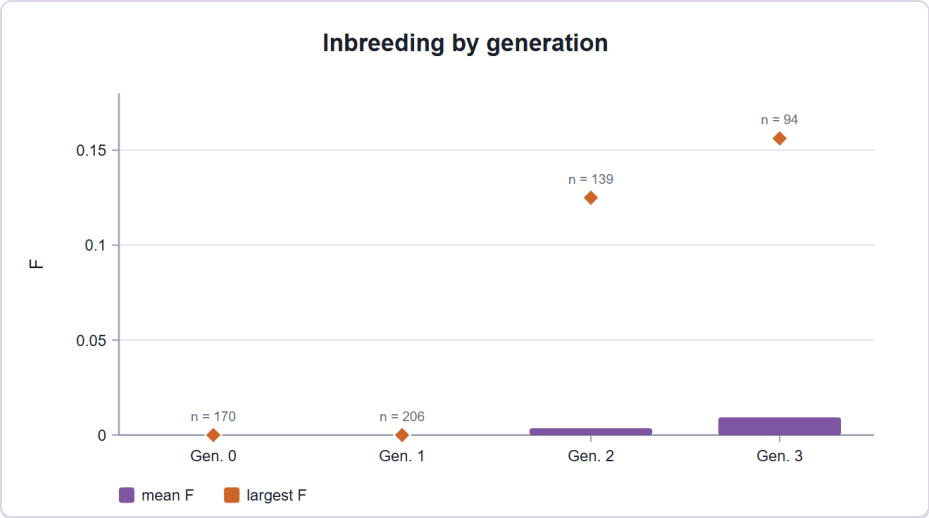

Figure 10.2. Inbreeding of the flock by generation: the bars are the mean F and the diamonds, the largest. Inbreeding appears in the second generation, from half-sib matings.

The folded box **Individuals, inbreeding and Mendelian sampling** lists each individual with its generation, its selfings, its F and its b, the Mendelian sampling variance relative to σ^2_a; an asterisk marks the added founders. Inbreeding comes from the path vectors of Meuwissen and Luo (1992), and the inverse A^{-1} is built directly with the rules of Henderson (1976) and Quaas (1976): for each individual, $1/b$ is added to its diagonal, $-1/(2b)$ to the cells with its parents and $1/(4b)$ to the cell of the parents with each other. So the model never needs to invert A.

$b = ½ - ¼(F_s + F_d)$ with both parents known, $¾ - ¼F_s$ with one and 1 with none; $\log|A| = \Sigma \log b$

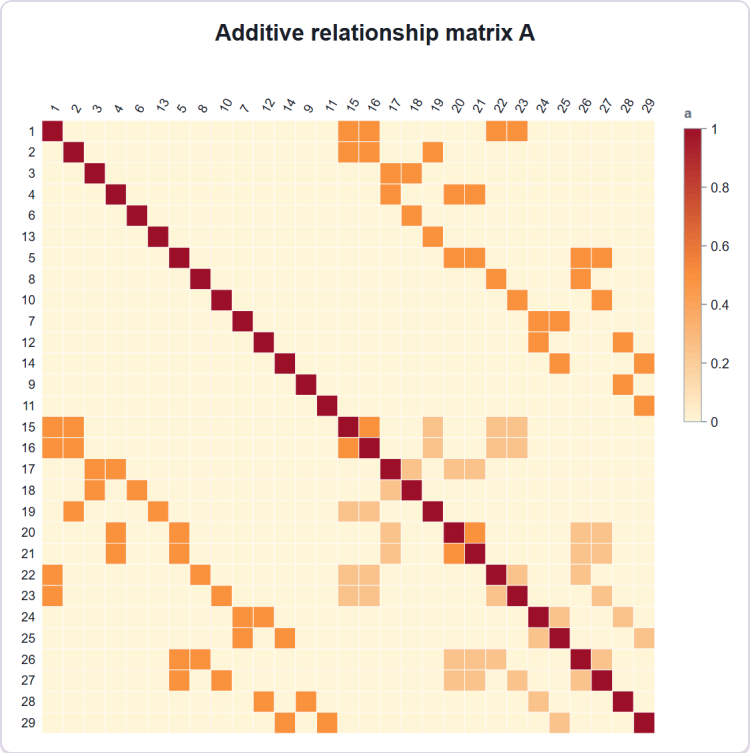

Figure 10.3. The A matrix of Schaeffer's example with contemporary groups: 29 individuals, the 14 founders first. Parent and offspring and the full sibs 15 and 16 have a relationship of 0.5; half sibs, such as 22 and 23, 0.25.

In plants you tick **Plants: selfing allowed**, and an individual with the same sire and dam is a self and not an error. The lines example brings the pedigree of the 68 entries of the simulated trial of Block 3: 8 parents that are pure lines (founder inbreeding = 1) and 60 lines, five from each of 12 single crosses, with five generations of selfing. A line with t selfings since the cross of its parents s and d keeps the relationships of its F_1 with everyone else and has

$F = 1 - (½)^t(1 - f_{sd})$, with f_{sd} the coancestry coefficient of its parents; its $b = (1 + F) - ¼(a_{ss} + a_{dd} + 2a_{sd})$ enters Quaas's rules unchanged.

With pure, unrelated parents, $f_{sd} = 0$ and each F_6 line has $F = 1 - 1/32 = 0.9688$; the mean F of the 68 entries is 0.9724 (Figure 10.4).

Simulated: pedigree of inbred lines: 68 rows · The pedigree of the simulated line trial of Block 3: 8 inbred parents and 60 F_6 lines from 12 crosses.

▸ Role of every column

INDIVIDUALS	FOUNDERS	GENERATIONS	SIRES / DAMS	INBRED	MEAN F
68	8	2	7 / 7	68	0.9724

LARGEST F 1.0000

Generation	n	Mean F	Largest F
0	8	1.0000	1.0000
1	60	0.9688	0.9688

Figure 10.4. The pedigree of the lines: 8 founders with F = 1 and 60 lines with F = 0.9688.

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In Schaeffer's seven-animal pedigree (2019, tables 2.9 to 2.14), the F values are 0, 0, 0, 0, 0.125, 0.25 and 0.28125, $b_7 = 0.40625$, and A and A^{-1} match the published ones; in Mrode's (2014, table 2.1), $F_5 = F_6 = 0.125$ and $A \cdot A^{-1}$ is the identity to within 10^{-12}. For the 68 lines, the app's A^{-1} is the inverse of the A computed separately by the tabular method, to within 1.8×10^{-15}.

10.3 The animal model

Card **2 · Animal model: breeding values (BLUP) and components (REML)** has three menus: the **Trait**; the **Model**, **animal (every individual)** or **sire**, and the **Variances**, **estimate them by REML** or **give them (known)**. Two checkboxes add the **Permanent environment (repeated records)** and the **Maternal genetic effect**. The line beneath says which fixed and random effects go in; they are changed in the role table of card 1. **Fit the model** solves.

$\mathbf{y} = X\mathbf{b} + Z\mathbf{a} + \mathbf{e}$, with $\text{var}(\mathbf{a}) = A\sigma^2_a$ and $\text{var}(\mathbf{e}) = I\sigma^2_e$; Henderson's mixed-model equations with $A^{-1}\sigma^2_e/\sigma^2_a$ added to the block of **a**; PEV = $\sigma^2_e C^{aa}$

The animal model gives a breeding value to every individual in the pedigree, whether or not it has records. The equations are solved exactly, with dense matrices, up to 2,600 unknowns; beyond that, the app says so and suggests trimming the pedigree to the ancestors of the individuals with records. The components are estimated by restricted maximum likelihood with average-information iterations (Gilmour, Thompson and Cullis, 1995), a few EM steps at the start and bounds at zero.

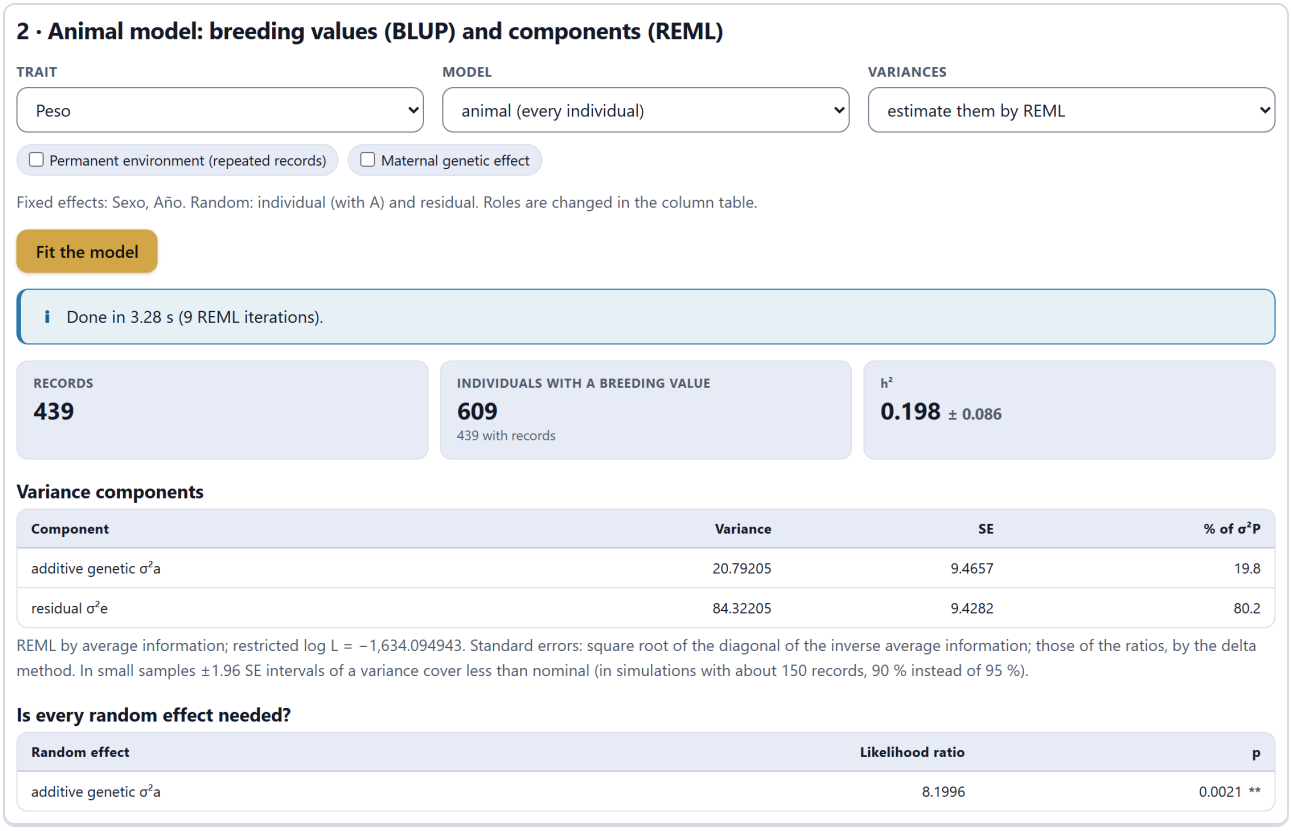


Figure 10.5. Card 2 with the flock: REML converged in 9 iterations. σ^2_a = 20.79 ± 9.47 and σ^2_e = 84.32 ± 9.43, with h^2 = 0.198 ± 0.086; the likelihood ratio says that σ^2_a is not zero.

The estimated heritability, 0.198, lies 1.2 standard errors from the true one, 0.30: with 439 records of a single weighing, the additive variance is estimated with little precision. The note warns about it: in small samples, the ±1.96 standard-error interval of a variance covers less than it claims (in simulations with about 150 records, 90% instead of 95%). The table **Is every random effect needed?** compares the restricted likelihood with and without each effect; the ratio, 8.20, is compared with the $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ mixture of a variance on the boundary (Self and Liang, 1987) and gives p = 0.0021. That comparison holds only between models with the same fixed effects.

The folded box **Fixed effects (least-squares means)** gives the mean of each level with its standard error. In the flock, males weigh 31.49 ± 0.96 and females 30.03 ± 0.96; the years, 30.63, 30.16 and 31.48.

The breeding value is a deviation from the base of the pedigree, the founders, whose expected mean is 0. It combines the individual's own record with those of all its relatives, in the proportion their relationship dictates. A0282 has the highest record in the table (64.26), yet comes second; A0294 is third with only 46.20 because it is the sire of the first, A0386, and has more progeny: its reliability, 0.377, is the highest of the 12 rows. The 170 founders without records also receive a breeding value, from their progeny alone: ram S4 is 16th (4.34, with reliability 0.389).

SEP = $\sqrt{\text{PEV}}$; reliability $r^2 = 1 - \text{PEV}/[(1 + F)\sigma^2_a]$; accuracy $r = \sqrt{r^2}$

Breeding values

ORDER: highest breeding value | ROWS: 25

Rank	Individual	Sire	Dam	F	Records	Own mean	Breeding value	SEP	Reliability r^2	Accuracy r
1	A0386	A0294	A0297	0.0000	1	54.29	**7.1979**	3.8357	0.292	0.541
2	A0282	A0142	A0037	0.0000	1	64.26	**7.0382**	3.8202	0.298	0.546
3	A0294	A0057	A0049	0.0000	1	46.20	**6.3799**	3.5995	0.377	0.614
4	A0049	S17	D37	0.0000	1	47.22	**5.5466**	3.7995	0.306	0.553
5	A0085	S4	D64	0.0000	1	51.97	**5.4571**	3.8583	0.284	0.533
6	A0249	A0157	A0059	0.0000	1	50.10	**5.2018**	3.8343	0.293	0.541
7	A0132	S18	D98	0.0000	1	53.22	**5.1346**	3.7801	0.313	0.559
8	A0184	S17	D137	0.0000	1	53.34	**4.6885**	3.9168	0.262	0.512
9	A0268	A0137	A0191	0.0000	1	37.79	**4.6405**	3.6992	0.342	0.585
10	A0191	S20	D140	0.0000	1	45.25	**4.5121**	3.8214	0.298	0.546
11	A0181	S15	D135	0.0000	1	56.17	**4.5091**	3.8224	0.297	0.545
12	A0059	S5	D45	0.0000	1	42.99	**4.4832**	3.8158	0.300	0.547

Figure 10.6. The first 12 breeding values of the flock. The menus choose the order (highest or lowest breeding value, highest reliability or identifier) and how many rows are shown (25, 50, 100 or all).

The formula of several textbooks, $1 - \mathrm{PEV}/\sigma^2_a$, overstates the reliability of inbred individuals; the app includes $1 + F$. In the flock, reliabilities range from 0.05 to 0.47: with a heritability of 0.2, a single record says little. Because the flock is simulated, the note gives the correlation between the estimated and the true breeding value of the 609 individuals: 0.619.

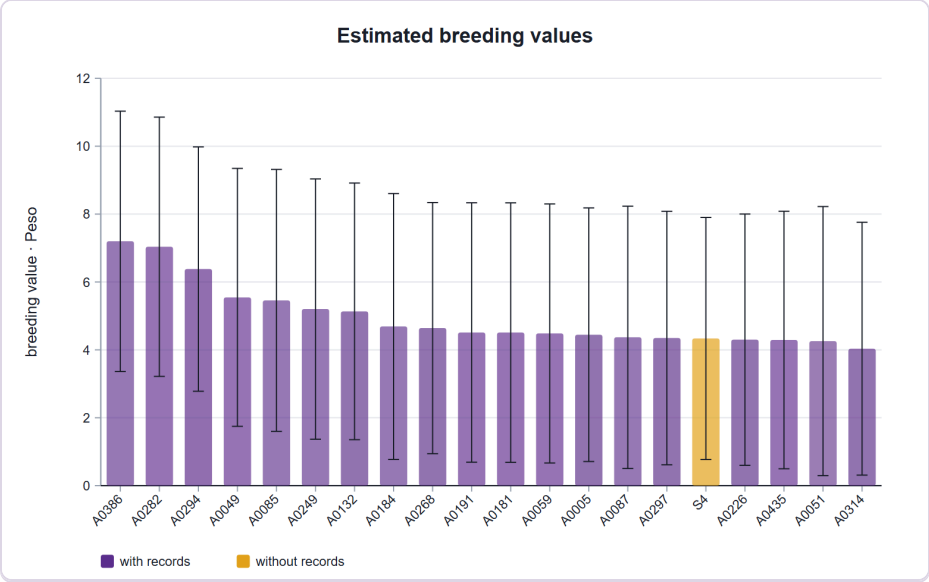


Figure 10.7. The 20 best breeding values with ± SEP. Color separates individuals with records from those without, and intensity shows accuracy. Its panel switches the view to value against accuracy.

The whiskers of Figure 10.7 are wide: the SEP of almost 4 kg is as large as the differences among the top places. Even so, the breeding value is the best bet: it already discounts that uncertainty, and choosing those with the highest breeding values gives the greatest expected response even if the exact order of the first few is uncertain.

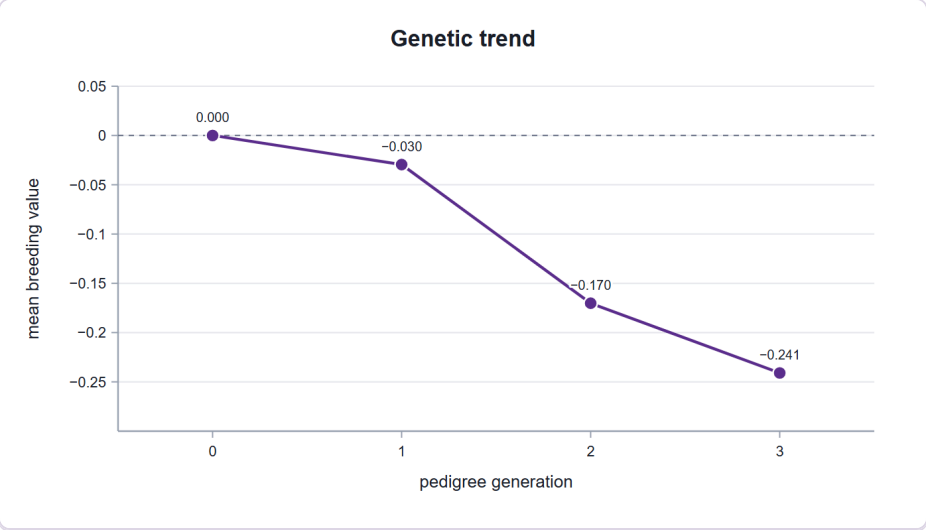


Figure 10.8. The genetic trend of the flock: the mean breeding value by pedigree generation. Because the replacements were chosen at random, there is no real trend; −0.24 in the third generation is small next to a SEP of 3.8.

The Trend by menu groups by pedigree generation, by birth year or by any fixed or random factor. In a program with selection, the slope of that line is the genetic response achieved.

> **VERIFIED · THE PUBLISHED VALUES**
>
> With Mrode's example 3.1 (2014) and $\sigma^2_a = 20$, $\sigma^2_e = 40$, the sex effects are 4.3585 (males) and 3.4044 (females), and the breeding values of animals 1 to 8, 0.0984, −0.0188, −0.0411, −0.0087, −0.1857, 0.1769, −0.2495 and 0.1826, equal to the published ones to within 10^{-7}; animal 1 has PEV 18.84 and reliability 0.058. The REML of the flock agrees with an independent maximization of the restricted likelihood (log L = −1,634.0949).

B10

10.4 Sires, repeated records and other effects

The sire model. With Model: sire, the random effect is the transmitting ability of each sire, half its breeding value, with the relationships of the sires and their ancestors. Its variance is $\sigma^2_s = \frac{1}{4}\sigma^2_a$, so $h^2 = 4\sigma^2_s/\sigma^2_p$, and the column EBV (2 × TA) gives twice the transmitting ability so that it reads on the scale of the animal model. The records and mean columns become those of each sire's progeny, and the records of individuals with no known sire do not contribute to the sire effect (the app counts them in a notice). It is a smaller model than the animal model, but it ignores the dams and assumes that their mates are a random sample.

Repeated records. The checkbox Permanent environment (repeated records) adds an effect of each individual, with no relationships, that is repeated in all its records: what an individual has of good or bad from its history and not from its genes. Repeatability is $t = (\sigma^2_a + \sigma^2_{pe})/\sigma^2_P$. If no individual has more than one record, the effect is not estimable and the app leaves it out with a notice.

Figure 10.9. Schaeffer's repeated-records example with its variances: $h^2 = 0.209$, repeatability 0.302 and contemporary group 0.116 of the phenotypic variance.

In that example, the breeding value table adds the column Permanent environment. Animal 21, with two records averaging 83.50, has the highest breeding value (6.86) and a permanent environment of 3.05: part of what it yields above average does not pass to its progeny.

The maternal effect. The checkbox Maternal genetic effect adds the maternal breeding value of each individual's dam, also with A, with no covariance with the direct effect (Willham, 1963). The tiles give m^2 and the total heritability $(\sigma^2_a + \tfrac{1}{2}\sigma^2_m)/\sigma^2_P$, and the table, the Maternal value of each individual. If the direct–maternal covariance exists, the total heritability is $(\sigma^2_a + 1.5\sigma_{am} + 0.5\sigma^2_m)/\sigma^2_P$ (Willham, 1972), and the note warns about it.

Other random factors. Each column with the role Random effect (factor) enters with its own variance, independent of the others, and its tile gives the fraction of the phenotypic variance it explains: the contemporary group in Schaeffer's examples and the litter in Mrode's ($\sigma^2_a = 20$, litter 15 and $\sigma^2_e = 65$, with $h^2 = 0.20$ and litter 0.15). A random contemporary group pays off when there are many small groups: with the group fixed, in a group of two or three animals the group effect takes almost the whole record; with the group random, that effect shrinks toward zero and the record says more about the animal.

Given or estimated variances. With give them (known) a box appears for each variance; it is used when they come from a larger population or from the literature, and the solutions are those of Henderson's equations with those values, without standard errors or tests. With few records, such as Mrode's five calves, REML cannot estimate anything useful; with the flock it can.

VERIFIED · THE PUBLISHED VALUES

Mrode's sire model (2014, Ex. 3.2): sex 4.3357 and 3.3820, and transmitting abilities of sires 1, 3 and 4 equal to 0.0220, 0.0140 and −0.0430. Schaeffer (2019), table 3.1: years 94.82 and 87.14, contemporary groups ±2.0653 and ±3.6688, and breeding values of animals 25, 28 and 17 equal to 7.8180, 7.3074 and −6.4880. Tables 6.1 and 6.2: breeding values of animals 7, 11 and 21 equal to 1.3535, 3.2348 and 6.8624, with permanent environments of −0.0910, 2.5214 and 3.0512. Schaeffer prints reliabilities of 0.2442 and 0.1980 computed with rounded values; the exact ones are 0.24414 and 0.1986.

10.5 BLUP of the genotypes of a trial

Card **3 · BLUP of the trial genotypes (Block 3) with the pedigree** joins the trial loaded in Block 3 with the pedigree from card 1; the entries must carry the same name in both. The notice at the top counts how many do (here, all 68). Without a trial in Block 3, the card asks you to load one; if no name matches, it says so, and with the simulated lines trial it suggests loading their pedigree. That pedigree has no response trait, so card 2 warns that it serves only for card 3.

The **Genetic effect** menu has three models: **independent genotypes (K = I)**, the usual BLUP; **additive with the pedigree (K = A)**, in which relatives share information, and **additive (A) + non-additive (I)**, with one part inherited through the pedigree and another specific to each genotype (Oakey and colleagues, 2006). Environments and replicates within environments are fixed and the genotype × environment interaction is random; with a single environment, only the replicates are fixed. The predicted value is the mean of the fixed effects plus the genotypic BLUP, and the heritability is the generalized one of Cullis, Smith and Coombes (2006).

$H^2 = 1 - \bar{v}/(2\sigma^2_G)$, with $\bar{v}$ the mean PEV of a difference between two entries; with A, $\sigma^2_G = (1 + \bar{F})\sigma^2_a$, and with A + I σ^2_i is added to it

Table 10.4. The three models with the simulated lines (true values: $\sigma^2_a = 0.3$, $\sigma^2_i = 0.1$, $\sigma^2_{GE} = 0.1$ and $\sigma^2_e = 0.4$).

Model	σ^2_a	σ^2_i or σ^2_G	σ^2_{GE}	Cullis H^2	Correlation with the truth
K = I	—	$\sigma^2_G = 0.386$	0.060	0.799	0.897
K = A	0.333	—	0.061	0.889	0.896
A + I	0.091	$\sigma^2_i = 0.239$	0.060	0.819	0.899

With K = A, the tile **log L: A against I** reads 1.386: the two models have the same parameters, so the higher log L says that the relationships fit better. The two σ^2_G are not comparable: with K = A it refers to the population of the founders, $(1 + \bar{F})\sigma^2_a = 0.657$; with K = I, to the 68 entries as they are, 0.386, which come from only 8 parents.

3 · BLUP of the trial genotypes (Block 3) with the pedigree

ℹ 68 of 68 entries of the trial are in the pedigree.

TRAIT: Rendimiento

GENETIC EFFECT: additive with the pedigree (K = A)

Fit

ENTRIES	PLOTS	CULLIS H^2	σ^2G AMONG ENTRIES	LOG L: A AGAINST I
68	408	0.889	0.6574	1.386 (larger = A fits better)

Component	Variance	SE
additive genetic σ^2a (with A)	0.33332	0.0773
genotype × environment σ^2GE	0.06072	0.0422
residual σ^2e	0.46101	0.0460

Model: environments and replicates within environments fixed, genotype × environment random, genotypes random with the relationships A of the pedigree. Predicted value = mean of the fixed effects (6.5275) + genotypic BLUP. Generalized H^2 of Cullis, Smith & Coombes (2006) = 1 − $\bar{v}/(2\sigma^2G)$ with $\bar{v}$ = 0.1453, the mean PEV of a difference; $\sigma^2G = (1 + \bar{F})\sigma^2a$ with $\bar{F}$ = 0.9724.

Simulated data: correlation with the true genotypic value 0.896 for the BLUP and 0.897 for the observed mean.

Figure 10.10. The simulated lines with K = A: σ^2_a = 0.333 (the true one is 0.3), Cullis H^2 0.889, and a higher log L than with K = I.

Rank	Entry	Parents	F	Plots	Observed mean	Additive	Predicted value	SEP	Reliability
1	L24	P5 × P6	0.9688	6	8.1040	1.2656	**7.7931**	0.4019	0.754
2	L57	P4 × P8	0.9688	6	7.8662	1.1208	**7.6483**	0.4019	0.754
3	L51	P3 × P7	0.9688	6	7.9468	1.0132	**7.5407**	0.4019	0.754
4	L31	P6 × P8	0.9688	6	7.6815	1.0020	**7.5295**	0.4019	0.754
5	L34	P6 × P8	0.9688	6	7.5173	0.8758	**7.4033**	0.4019	0.754
6	P8	—	1.0000	6	7.4557	0.8094	**7.3369**	0.3617	0.804
7	L32	P6 × P8	0.9688	6	7.3993	0.7851	**7.3126**	0.4019	0.754
8	L58	P4 × P8	0.9688	6	7.3027	0.6875	**7.2150**	0.4019	0.754
9	L21	P5 × P6	0.9688	6	7.3233	0.6653	**7.1928**	0.4019	0.754
10	L19	P3 × P4	0.9688	6	7.3710	0.6424	**7.1699**	0.4019	0.754

Figure 10.11. The 10 best entries with K = A. L51 has the second-best observed mean, but comes third: its family (P3 × P7) is worse than that of L57 (P4 × P8).

For each entry, the table gives its parents, its F, its plots, the observed mean, the additive part, the predicted value, the SEP and the reliability; with A + I it adds the non-additive part. Parent P8 is sixth with a reliability of 0.804, higher than that of the lines (0.754), because it has 15 descendant lines that tell about it.

Here the trial is balanced and the heritability is high, so BLUP and the observed mean rank almost the same (correlations with the truth of 0.896 and 0.897). BLUP wins when plots are missing, when the entries have different numbers of replicates or when the heritability is low; the relationships win when there are families with few plots each.

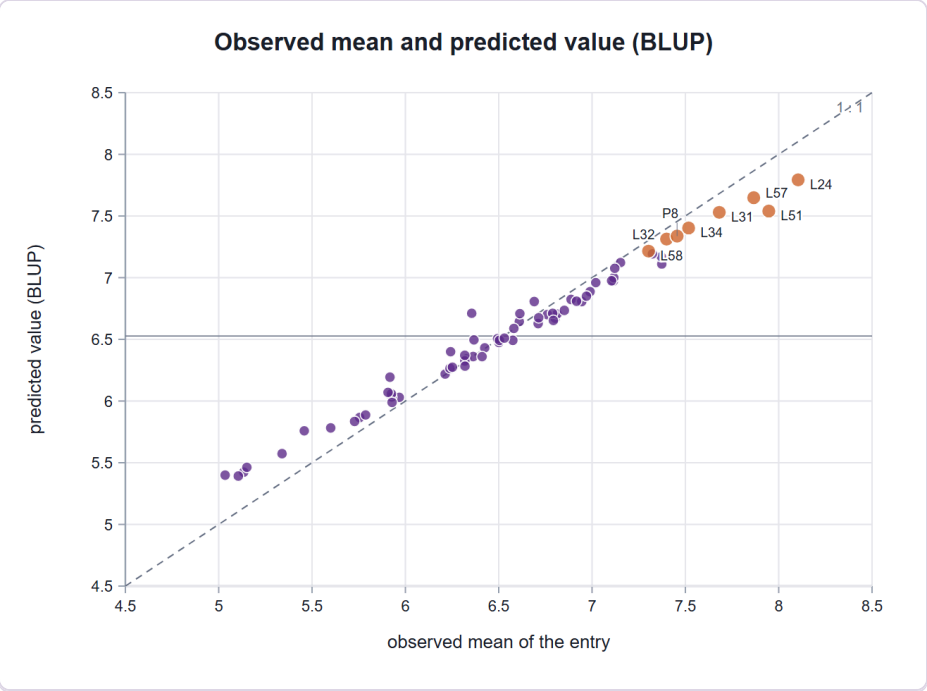


Figure 10.12. Predicted value against observed mean. BLUP shrinks the extreme means toward the mean of the fixed effects (the horizontal line); the dashed diagonal is 1 : 1 and the 10 best are highlighted.

IMPORTANT · ADDITIVE AND NON-ADDITIVE IN A SINGLE TRIAL

With A + I, the test of the additive part is significant ($\chi^2 = 3.24$, $p = 0.036$), but the split is not reliable: $\sigma^2_a = 0.091$ and $\sigma^2_i = 0.239$ for true values of 0.3 and 0.1. Separating the two parts takes many families with varied relationships; with 12 crosses they are confounded, and you should also look at the additive-only model, as the card's note says.

10.6 Crossbreeding

Card **4 · Crossbreeding: Dickerson's parameters and prediction of systems** works with group means. Each group is written as a cross, sire first: A, A×B, C×(A×B), (A×B)×(A×B). For prediction you can add rot(A,B) and rot(A,B,C), the two- and three-breed rotations at equilibrium averaged over the cycle, and comp(A,B,C), the composite with equal parts after random mating; comp(A:0.5,B:0.25,C:0.25) accepts other proportions.

mean of a group = $\mu + \Sigma\, p_k g^{I}_k + \Sigma\, p_{dam,k} g^{M}_k + h^{I}H^{I} + h^{M}H^{M} + r^{I}R^{I}$, with $h^{I} = 1 - \Sigma\, p_{sire,k} p_{dam,k}$, h^{M} the h^{I} of the dam and $r^{I} = (h_{sire} + h_{dam})/4$

B10

It is Dickerson's model (1969, 1973): g^{I} and g^{M} are the direct and maternal additive effects of each breed, H^{I} and H^{M} the heterosis of the individual and that of its dam, and R^{I} the recombination loss, what is lost when the favorable gene combinations of each breed are broken up. The **What to do** menu has three options: estimate the parameters from group means, predict with known parameters, or predict from breed means and heterosis in percent. Five examples cover them (Table 10.5).

Table 10.5. The crossbreeding examples.

Example	Data	What it does
Simulated: three breeds	16 groups of 30 animals: purebreds, F_1, backcrosses, F_2 and three-breed crosses	Estimates all the parameters
Simulated: four-breed diallel	4 purebreds and their 12 reciprocal F_1	Heterosis of each pair of breeds
Two-breed diallel (van der Werf)	The 4 means of a two-breed diallel	Exact fit
Boer × local goats (Tesema et al. 2023)	Published parameters	Predicts with parameters
Heterosis in percent (Buchanan & Northcutt)	Means of three breeds, heterosis of 4.7 and 4.2%	Predicts in percent

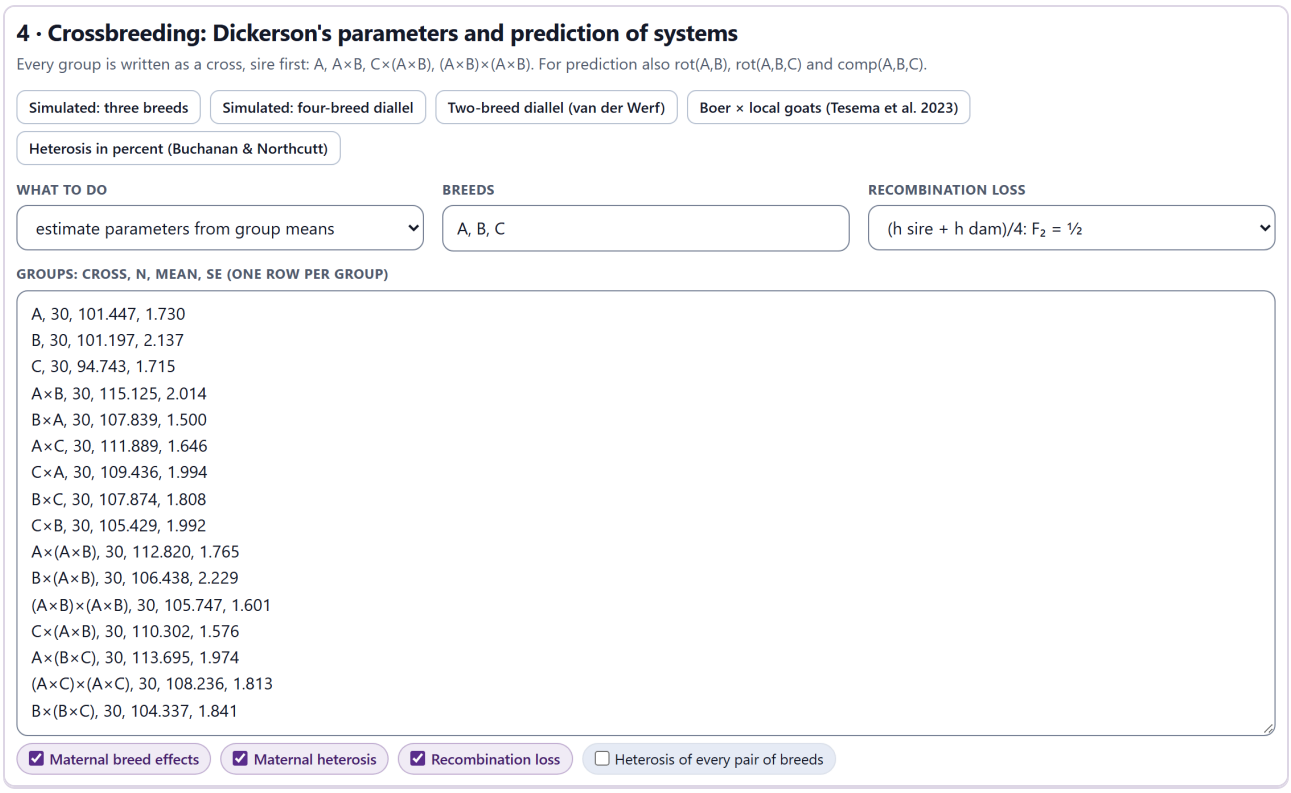


Figure 10.13. Card 4 with the three-breed experiment: one row per group with its cross, its number of animals, its mean and the standard error of the mean, and beneath them the checkboxes of the model parameters. Further down, outside the screenshot, come the list of crosses or systems to predict and the **Compute** button.

To estimate, each row of **Groups** carries the cross, n, the mean and its standard error. If all groups have a standard error, the weights are $1/SE^2$ and there is a lack-of-fit test; if not, the weights are n, or equal if there is no n either. Breed effects are estimated with $\Sigma g^I = \Sigma g^M = 0$, as deviations from the average of the breeds. The checkboxes remove or add the maternal breed effects, maternal heterosis, recombination loss and the heterosis of each pair of breeds; a parameter that the groups cannot separate is marked as not estimable, and so are the predictions that depend on it.

Simulated with a fixed seed: μ = 100; g^I = 6, −2, −4; g^M = −3, 2, 1; h^I = 8; h^M = 5; r^I = −4; 30 individuals per group, SD 10.

Crossbreeding parameters

Parameter	Estimate	SE	z	p	True
mean (μ)	**99.0068**	1.0193	—	—	100
direct additive A	**7.2011**	1.3546	5.32	< 0.0001 ***	6
direct additive B	**−2.5347**	1.3346	−1.90	0.0575 ns	−2
direct additive C	**−4.6664**	1.3915	−3.35	0.0008 ***	−4
maternal additive A	**−3.1845**	1.0184	−3.13	0.0018 **	−3
maternal additive B	**2.0007**	1.1116	1.80	0.0719 ns	2
maternal additive C	**1.1838**	1.0352	1.14	0.2528 ns	1
direct heterosis	**10.3996**	1.2011	8.66	< 0.0001 ***	8
maternal heterosis	**3.5919**	2.2317	1.61	0.1075 ns	5
recombination loss	**−4.1126**	6.0572	−0.68	0.4972 ns	−4

Weighted least squares on the group means with weights $1/SE^2$ and the constraints $\Sigma g^I = \Sigma g^M = 0$ (breed effects are deviations from the average of the breeds). Lack of fit: χ^2 = 10.8075 with 8 df, p = 0.2128. Recombination loss with the convention $(h_{sire} + h_{dam})/4$ (F_2 = ½).

Figure 10.14. The parameters of the three-breed experiment next to the true ones of the simulation. The lack of fit is not significant (χ^2 = 10.81 with 8 degrees of freedom, p = 0.21).

The direct and maternal effects come close to the true ones. Direct heterosis, 10.40 ± 1.20 against 8, falls just outside its 95% interval (8.05 to 12.75); in 1,000 simulated experiments those intervals cover between 93 and 95% of the time, as they should. The recombination loss, −4.11, has a standard error of 6.06: only the groups whose parents are already crosses separate it, and with 30 animals per group that is not enough. That is the usual situation in real experiments.

A×(B×C) combines the best direct effect, that of A, with a crossbred dam that brings maternal heterosis and the favorable maternal effects of B and C. The rotations retain ⅔ (two breeds) and 6/7 (three breeds) of the heterosis; the three-breed composite retains ⅔, but its recombination coefficient, ⅓, is larger than that of the rotations. The folded box **Coefficients of every group** gives the breed proportions of each group and of its dam with their coefficients, useful for checking that a cross was written as intended.

Predictions

Cross or system	Predicted	SE	h^I	h^M	r^I
A	**103.0234**	1.4164	0	0	0
B	**98.4728**	1.5488	0	0	0
C	**95.5242**	1.4567	0	0	0
A×B	**113.7403**	1.3824	1	0	0
C×(A×B)	**110.2116**	1.1790	1	1	0.25
A×(B×C)	**115.3627**	1.1487	1	1	0.25
rot(A,B)	**109.3903**	0.7144	0.6667	0.6667	0.1667
rot(A,B,C)	**110.1182**	0.7566	0.8571	0.8571	0.2143
comp(A,B,C)	**106.9636**	0.8563	0.6667	0.6667	0.3333

rot(A,B) and rot(A,B,C) are two- and three-breed rotations at equilibrium, averaged over the cycle ($h^I = h^M = ⅔$ and 6/7); comp(A,B,C) is a composite with equal shares after random mating ($h = 1 - \Sigma p^2$); comp(A:0.5,B:0.25,C:0.25) takes proportions. A "not estimable" prediction depends on a parameter these groups do not separate.

Figure 10.15. The predictions with their standard errors and their heterosis and recombination coefficients. The terminal cross A×(B×C) is the best (115.36).

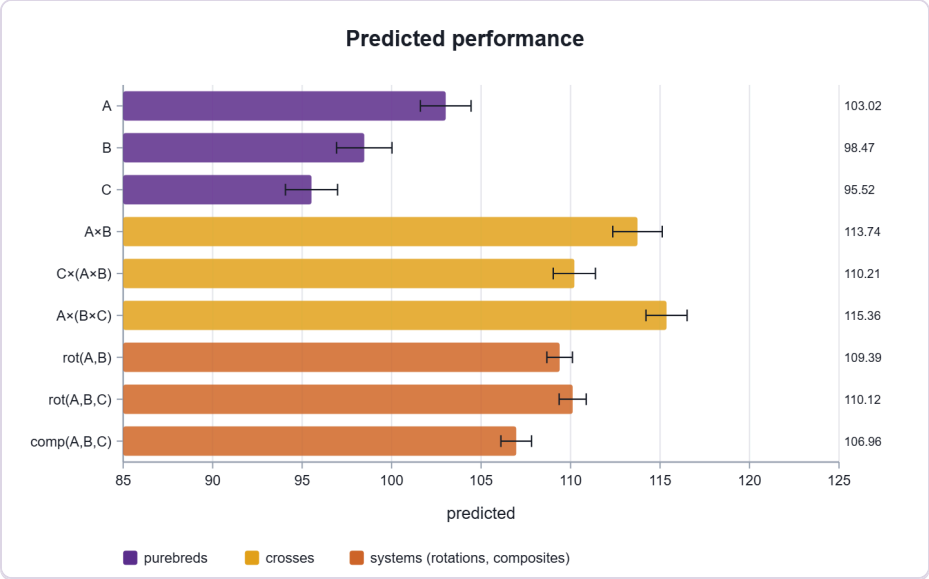


Figure 10.16. The predicted performance with ± SE, by color: purebreds, crosses and systems.

The breed diallel. With the checkbox Heterosis of every pair of breeds, each pair has its own direct heterosis, and if the groups form a diallel, the table Breed diallel: average, breed and specific heterosis splits it into average, breed and specific heterosis (Eisen and colleagues, 1983). In the simulated four-breed diallel, with true heterosis values of 4, 6, 2, 3, 5 and 4, the estimates are 4.15, 4.70, 2.83, 0.77, 5.37 and 2.85 (lack of fit p = 0.18).

Breed diallel: average, breed and specific heterosis

Component	Estimate	SE
average heterosis $\bar{h}$	3.4437	0.5693
breed heterosis h_A	1.3405	1.1898
breed heterosis h_B	−0.0411	1.1882
breed heterosis h_C	−2.0149	1.3049
breed heterosis h_D	0.7155	1.1991
specific heterosis $s_{A·B}$	0.0572	0.4025
specific heterosis $s_{A·C}$	1.5893	0.4316
specific heterosis $s_{A·D}$	−1.6465	0.4135
specific heterosis $s_{B·C}$	−1.6465	0.4135
specific heterosis $s_{B·D}$	1.5893	0.4316
specific heterosis $s_{C·D}$	0.0572	0.4025

Figure 10.17. Average heterosis is 3.44 ± 0.57; A is the breed that gives the most heterosis in its crosses and C the least. With four breeds, specific heterosis has only two degrees of freedom and its values come in equal pairs.

Known parameters. With predict with known parameters, the card asks for a list of parameters with their values (mu, gI:B, gM:B, hI, hM, rI) and predicts any cross; those not written are zero, so the breed with no effect written is the reference. The example of Tesema and colleagues (2023) uses their estimates for litter weaning weight in Boer × Central Highland goats: μ = 14.3 (the local breed), Boer g^{I} = −3.18, heterosis −0.04 and recombination loss −6.20. The predictions are 12.67 for the F_1, 10.35 for the backcross to Boer, 11.94 for the backcross to the local breed and 9.59 for the F_2.

Heterosis in percent. The third option starts from the mean of each breed and the percentages of individual and maternal heterosis and of recombination loss, and predicts multiplicatively: average of the breeds × $(1 + H_I h^I) \times (1 + H_M h^M) \times (1 - R \cdot r^I)$. This is how extension guides teach it.

Predictions

Cross or system	Predicted	h^I	h^M	r^I	Breed average	With individual heterosis
A×B	**492.0900**	1	0	0	470.0000	492.0900
C×(A×B)	**529.1224**	1	1	0.25	485.0000	507.7950
rot(A,B)	**498.2990**	0.6667	0.6667	0.1667	470.0000	484.7267
rot(A,B,C)	**517.3133**	0.8571	0.8571	0.2143	480.0000	499.3371

Multiplicative heterosis: breed average × $(1 + H_I h^I) \times (1 + H_M h^M) \times (1 - R \cdot r^I)$, with the percentages typed above.

Figure 10.18. The example of Buchanan and Northcutt: breeds of 460, 480 and 500 with individual heterosis of 4.7% and maternal heterosis of 4.2%. The F_1 A×B weighs 470 × 1.047 = 492.09.

THEORY IN BRIEF · TWO SCALES OF THE RECOMBINATION LOSS

The Recombination loss menu has two conventions: $(h_{sire} + h_{dam})/4$, in which the F_2 has $r^I = ½$ (Kinghorn, 1980; it is the one Tesema and colleagues use), and $(h_{sire} + h_{dam})/2$, in which it has 1 (Dickerson, 1973). The predictions are the same with both; only the scale of the parameter changes, which is half as large with the second. When you take a published value, use the convention of the paper.

VERIFIED · THE PUBLISHED VALUES

Van der Werf's two-breed diallel gives $g^I_1 - g^I_2 = 20$, $g^M = \mp 2.5$ and direct heterosis 20, exactly. The predictions of Tesema and colleagues (2023, tables 3 and 4) are 14.30, 12.67, 10.35, 11.94 and 9.59, as published; those of Buchanan and Northcutt, 492.09 for A × B and 507.80 for C × (A × B) with individual heterosis only; with maternal heterosis, the paper gives 529.13 because it rounds the intermediate step, and the app, 529.12. The parameters of the three-breed experiment and of the four-breed diallel agree with an independent fit to within 3.6×10^{-13}.

10.7 Downloads and next step

Card 5 · Assumptions, formulas and references summarizes the pedigree and the relationships, BLUP and REML, the models and crossbreeding, with their references. Results (.xlsx) downloads, each on its own sheet, the pedigree with F and *b*, the variance components, the breeding values, the means of each fixed factor, the BLUP of the trial entries, Dickerson's parameters, the breed diallel decomposition and the predictions. Each figure has its download panel.

Continue takes you to Block 11 (genomics), where the relationships come from the markers instead of the pedigree. The back button returns to Block 9 (← G×E) when the Block 3 data have several environments, and to Block 8 (← Selection) when they do not.

B11

CHAPTER ELEVEN · BLOCK 11 · GENOMICS

Genomic selection and hybrid prediction

Markers measure the actual relationship between two individuals, not the one expected from the pedigree, and with it you can predict the value of an individual that was never evaluated. Block 11 checks the markers, builds the genomic relationship matrix, fits GBLUP with the genomic heritability and the marker effects, measures by cross-validation how well it predicts and how much is gained per year, and predicts every possible single cross between two heterotic groups.

11.1 What the block does

11.2 Markers and quality control

11.3 Genomic relationships

11.4 GBLUP: genomic values

11.5 Cross-validation and response per year

11.6 Hybrid prediction

11.7 Downloads and next step

11.1 What the block does

Block 11 works with a marker file and, depending on the question, with phenotypes of individuals or with evaluated hybrids. The phenotypes can come with the example, from a file of your own or from the Block 3 trial, if its entries have markers. Everything is computed in the browser: with 600 lines and 1,200 markers, GBLUP takes a few seconds, and a cross-validation with 5 folds and 2 replicates, about fifteen.

Table 11.1. The cards of Block 11.

Card	What it gives
1 · Markers	Reading of any common coding, quality control, imputation, allele frequencies and population structure
2 · Genomic relationships	VanRaden's G matrix (methods 1 and 2), a ridge to invert it and an optional blend with the Block 10 pedigree
3 · GBLUP: genomic values	Components by REML, genomic heritability, genomic value of each individual and marker effects
4 · Cross-validation and response per year	Predictive ability for new individuals and genetic gain per year, phenotypic against genomic
5 · Hybrid prediction	GCA of two heterotic groups (and SCA), all single crosses, T2, T1 and T0 validation, three-way and double crosses
6 · Assumptions, formulas and references	The models, their formulas and the references

Table 11.2. The examples of Block 11.

Example	Data	Used for
CIMMYT wheat: 599 lines, 1279 DArT, 4 environments	Standardized yield of 599 lines in four environments (Crossa and colleagues, 2010)	Cards 1 to 4: the thread of the chapter
Dent × Flint maize: 1254 hybrids	123 dent and 86 flint lines with 2,000 SNPs; 1,254 of their 10,578 crosses evaluated (Technow and colleagues, 2014)	Card 5
Simulated: 300 lines, 1000 SNPs, h^2 = 0.4	Doubled-haploid lines with 60 QTL off the panel; the true value is known	Checking GBLUP and the validation
Simulated: two-group hybrids, 200 of 750	30 × 25 lines, 100 QTL with dominance; the true value of the 750 crosses is known	Checking hybrid prediction

11.2 Markers and quality control

Card 1 · Markers reads one row per individual, with its identifier in the first column and one column per marker; if the file comes the other way around, the checkbox Markers in rows transposes it. It accepts the codings of Table 11.3 and recognizes on its own which one it is; an empty cell, NA, "-" or "N" is a missing value.

Table 11.3. The codings card 1 reads. All are converted to dosages 0, 1 and 2 of the counted allele.

Coding	How it is read
Dosage 0/1/2	As is
−1/0/1	1 is added
Presence/absence 0/1	The menu A 0/1 file means decides: pure lines (0 → 0, 1 → 2) or dosages kept as they are
Letters (AA/AB/BB, A/C/G/T)	The second allele of each marker is counted; the two-base IUPAC codes (R, Y, S, W, K, M) are heterozygotes

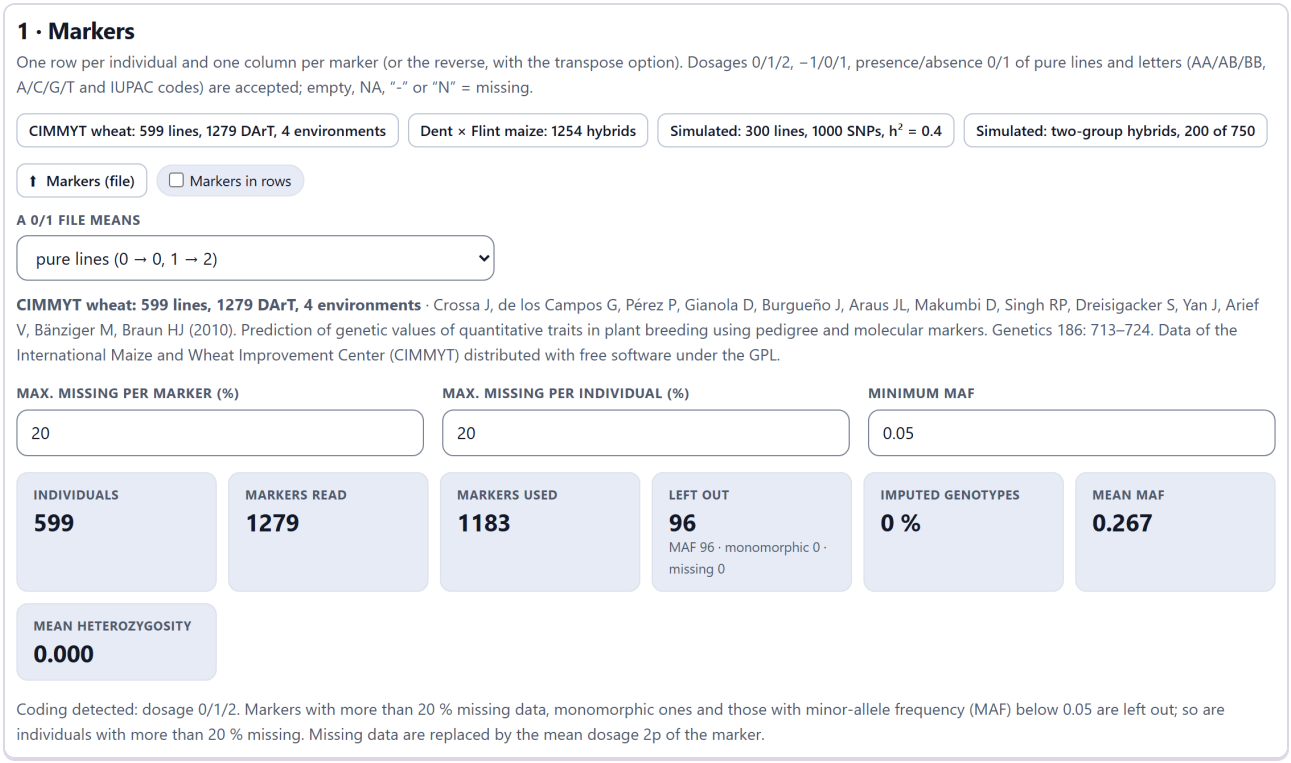


Figure 11.1. Card 1 with the wheat: of 1,279 DArT markers, 1,183 remain, because 96 have a minor allele frequency below 0.05. The lines are pure, so heterozygosity is 0.

Quality control has three limits: the maximum missing rate per marker and per individual (20% each) and the minimum minor allele frequency (MAF, 0.05). Monomorphic markers are also removed, and missing data are replaced by the mean dosage of the marker, $2p$. The tiles say how many individuals and markers remained, why each one was removed, what fraction was imputed, and the mean MAF and heterozygosity.

The card's second figure shows the principal components of the G matrix, which reveal the structure of the population. When hybrids are loaded, each line is colored by its heterotic group (Figure 11.3). In the wheat, the first three components explain 15.4, 6.2 and 4.5% of the variation.

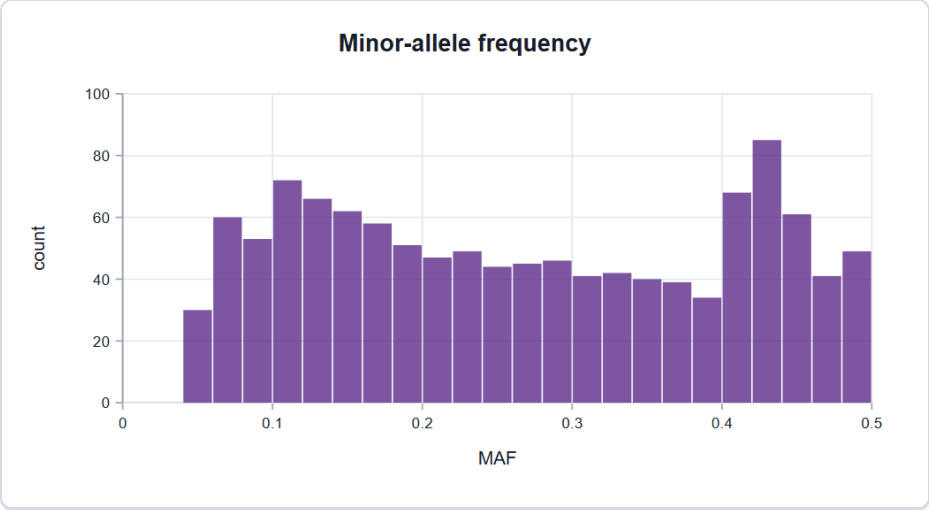


Figure 11.2. Minor allele frequency of the 1,183 wheat markers. A panel with many rare markers says little about relationships.

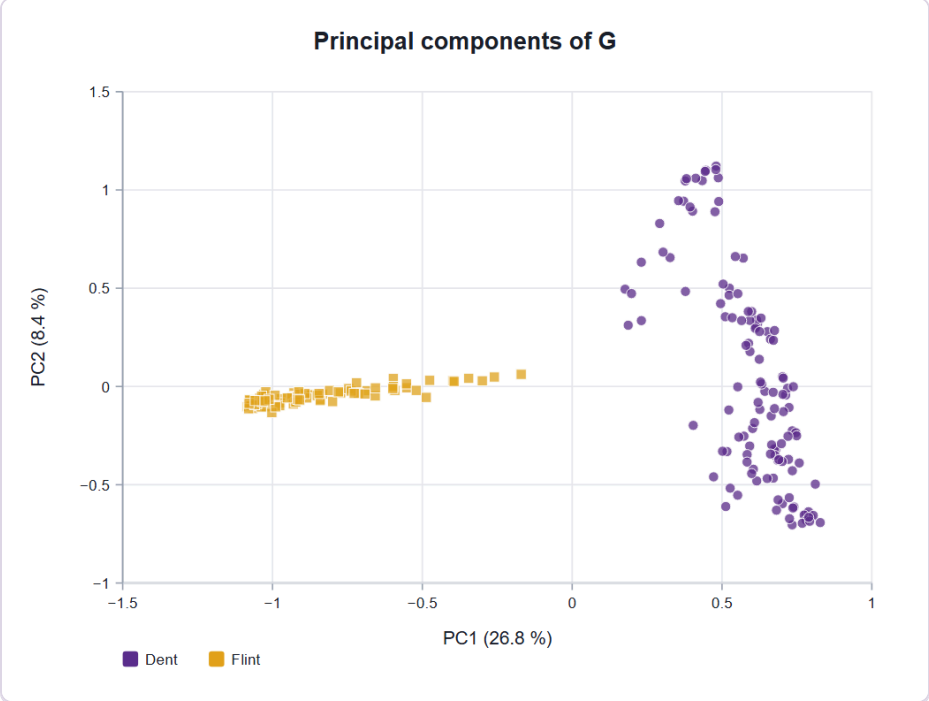


Figure 11.3. Principal components of G in Technow's maize: the first, with 26.8% of the variation, completely separates the dent lines from the flint lines.

11.3 Genomic relationships

Card `2 · Genomic relationships` builds VanRaden's (2008) G matrix. The `Matrix` menu offers method 1, with centered markers, and method 2, with each marker standardized, which gives more weight to rare ones. Because the allele frequencies come from the data themselves, each row of G sums to zero and G cannot be inverted; the `Ridge λ` adds a small amount to the diagonal (by default, 0.001 times the mean diagonal).

method 1: $G = ZZ'/(2\Sigma p_j q_j)$, with $Z = M - 2P$ · method 2: $G = Z^*Z^{*\prime}/m$, with each column divided by $\sqrt{(2p_j q_j)}$ · $K = G + \lambda I$

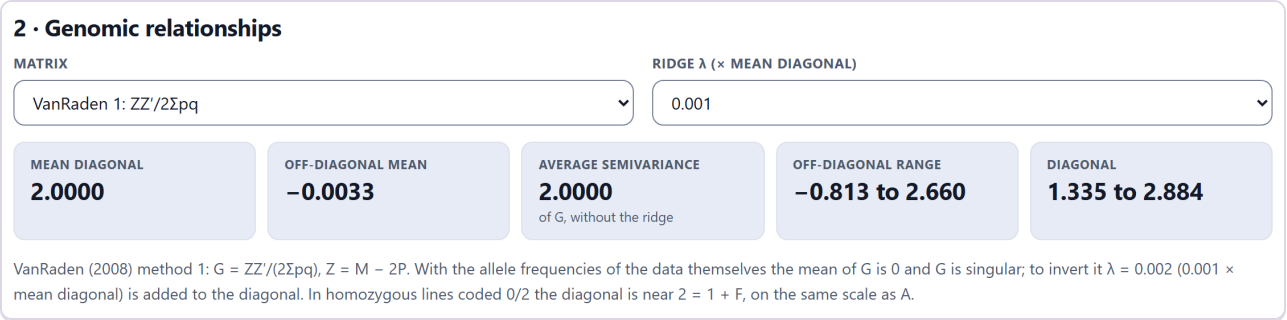


Figure 11.4. Card 2 with the wheat. In pure lines coded 0/2 the mean diagonal is 2 = 1 + F, the same scale as the pedigree A matrix.

If Block 10 has a pedigree that includes every individual with markers, the menu Weight of the pedigree (Block 10) appears, which blends $(1 - w)G + wA$, along with a figure that compares G against A pair by pair. The blend smooths G when there are few markers (VanRaden, 2008; Aguilar and colleagues, 2010); the comparison uncovers pedigree errors, such as an individual whose genomic relationship with its supposed sire is close to zero.

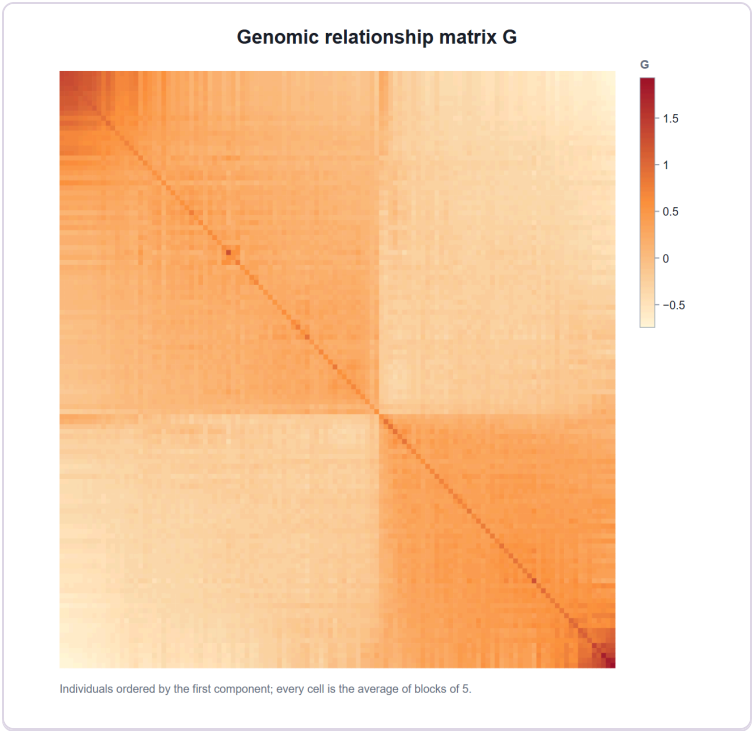


Figure 11.5. The G matrix of the wheat, with the lines sorted by the first component and averaged in blocks of 5 so that it fits. Two large groups of related lines are visible.

11.4 GBLUP: genomic values

Card **3 · GBLUP: genomic values** has two menus. **Phenotypes** says where they come from: those loaded with the example, those of a file (**Phenotypes (file: identifier and traits)**, with the identifier in the first column) or the Block 3 trial. With the trial, the model is fitted in one stage on the plots, with environments and replicates fixed and the interaction with the environment random if there are replicates within each environment. The card reports how many entries have markers; the rest are left out. **Trait** picks the trait, and **Fit GBLUP** solves.

$$\mathbf{y} = \mu + \mathbf{g} + \mathbf{e},\ \text{with}\ \mathbf{g} \sim N(0, K\sigma^2_g) \quad \cdot \quad h^2_{genomic} = \sigma^2_g \cdot ASV / (\sigma^2_g \cdot ASV + \sigma^2_e),\ \text{with}\ ASV = tr(K)/n - 1'K1/n^2$$

ASV is the average semivariance of K (Legarra, 2016; Feldmann and colleagues, 2022): it converts σ^2_g into the genetic variance among the individuals analyzed. In pure lines it is almost 2, because the variance among lines is twice that of a non-inbred population.

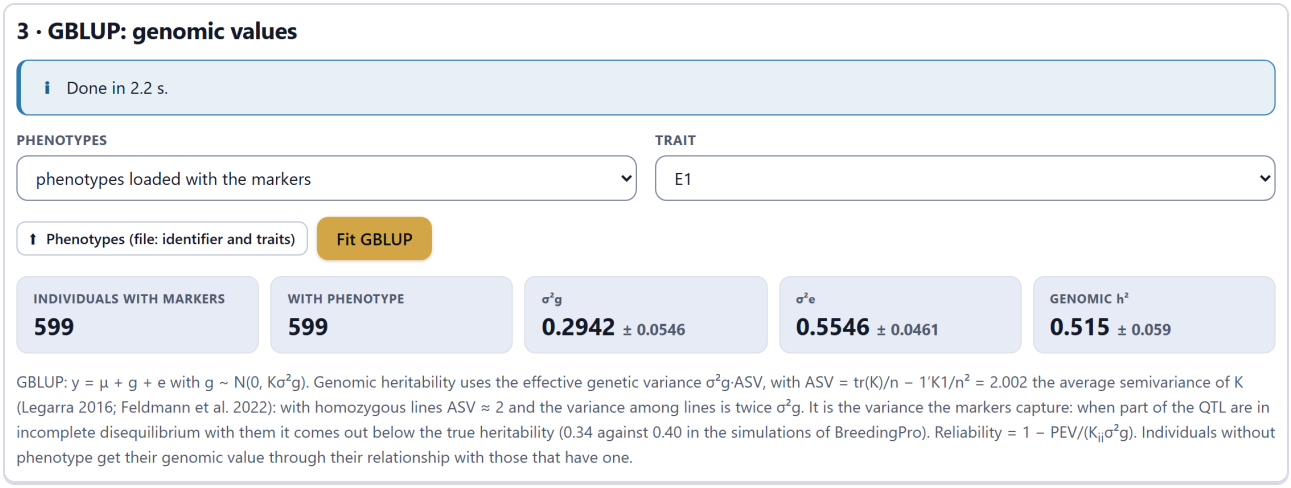


Figure 11.6. GBLUP of the wheat yield in environment E1: σ^2_g = 0.294 ± 0.055, σ^2_e = 0.555 ± 0.046 and a genomic heritability of 0.515 ± 0.059.

The genomic heritability is the one the markers capture. If some of the QTL are not in linkage disequilibrium with the panel, it comes out lower than the true one: in the simulated example, with 60 QTL off the panel and h^2 = 0.40, it gives 0.385 ± 0.088.

The table sorts the individuals by their genomic value and gives the phenotype, the predicted value (mean plus genomic value), the standard error of prediction and the reliability, 1 − PEV/($K_{ii}\sigma^2_g$); in the wheat it ranges from 0.60 to 0.96. An individual without a phenotype also receives its value, from its relationship with those that have one, and its row is highlighted: this is how lines not yet sown are predicted.

Rank	Individual	Phenotype	Genomic value (GEBV)	Predicted	SEP	Reliability
1	664062	2.2344	**1.3245**	1.3245	0.3250	0.808
2	77128	1.3097	**1.2863**	1.2863	0.3799	0.734
3	424702	0.9392	**1.1230**	1.1230	0.3635	0.800
4	12867	1.6858	**1.0772**	1.0772	0.3119	0.868
5	39359	1.4950	**1.0749**	1.0749	0.3121	0.867
6	458835	0.8811	**1.0375**	1.0375	0.3497	0.787
7	367751	1.8363	**1.0023**	1.0023	0.3565	0.777
8	305029	1.2708	**0.9965**	0.9965	0.3278	0.853
9	623960	0.7961	**0.9901**	0.9901	0.4056	0.741
10	41542	1.1422	**0.9826**	0.9826	0.3915	0.774

Figure 11.7. The 10 best genomic values of the wheat. Because yield comes standardized within each environment (mean 0, variance 1), the estimated mean is 0 and the predicted value equals the genomic one.

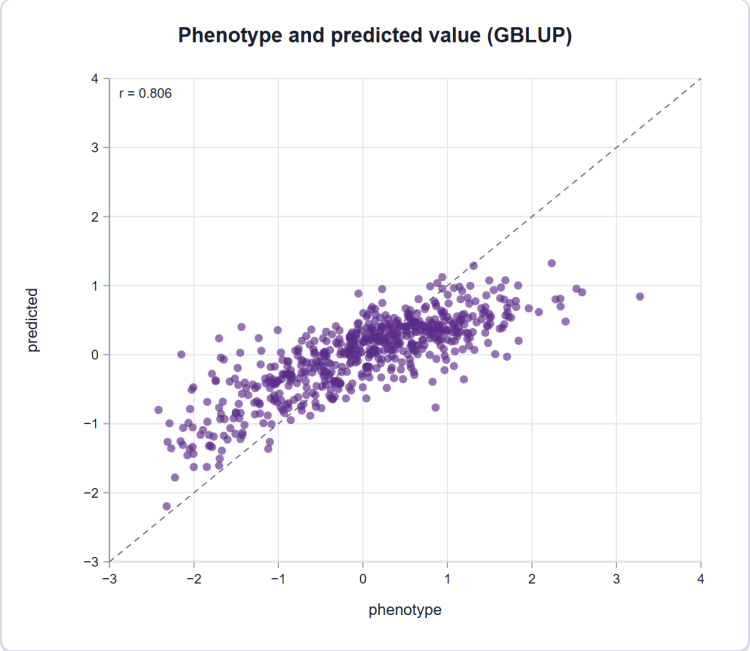


Figure 11.8. Phenotype against predicted value in the wheat. The correlation of 0.806 is optimistic: each line is predicted with its own record; cross-validation measures the one for new lines.

Marker effects. GBLUP is equivalent to ridge regression on all the markers (RR-BLUP; Meuwissen, Hayes and Goddard, 2001), and the effect of each marker is recovered from the genomic values: $\hat{\beta} = Z'K^{-1}\hat{g}/(2\Sigma pq)$. The figure plots them in absolute value in the order of the file, and the folded box **The 15 largest effects** lists them with their allele frequency.

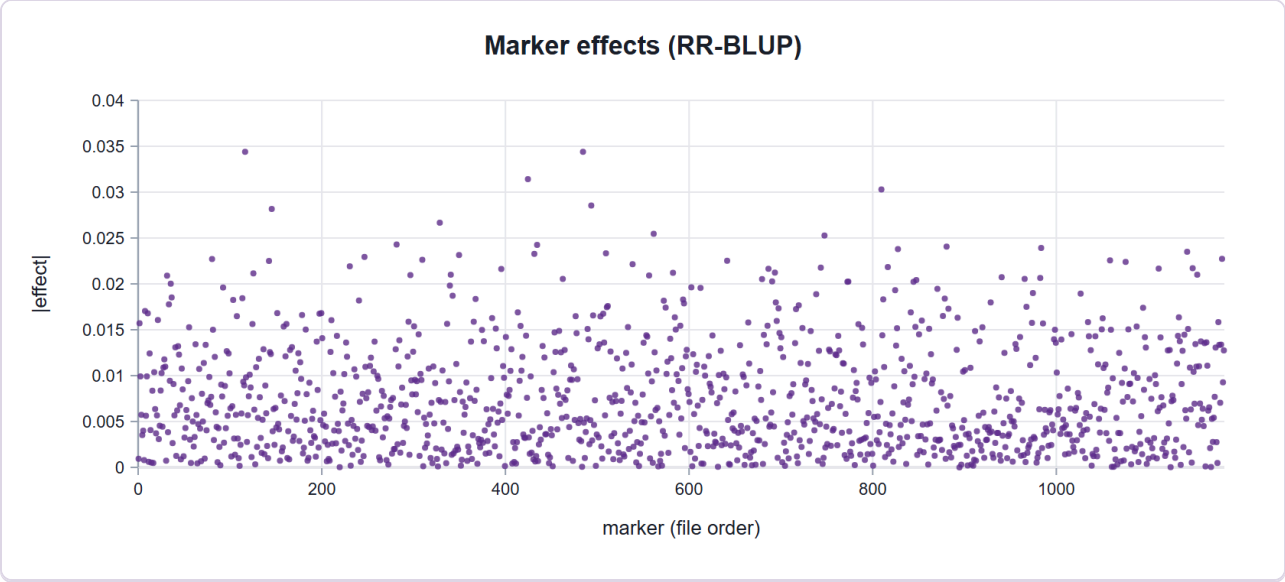


Figure 11.9. The marker effects of the wheat. None stands out: the largest is 0.034 standard deviations, which corresponds to a trait with many genes of small effect.

> **! IMPORTANT · A LARGE EFFECT IS NOT A GENE FOUND**
>
> GBLUP spreads the effect of a QTL among all the markers in disequilibrium with it, and shrinks them all equally. A marker with a large effect deserves a look, not a conclusion: association is tested with other methods. Because of the ridge λ, the effects reproduce the genomic values almost exactly: $Z\hat{\beta} = \hat{g} - \lambda K^{-1}\hat{g}$.

11.5 Cross-validation and response per year

What matters about a genomic model is how well it predicts individuals that were not evaluated. Card **4 · Cross-validation and response per year** splits the individuals with phenotypes into **Folds** (2, 5 or 10), predicts each fold with a model fitted without it and repeats the split as many times as **Replicates** says. It is the CV1 scheme of Burgueño and colleagues (2012): new individuals. The checkbox **Re-estimate the variances in every training set**, ticked by default, refits REML without the test fold; estimating them beforehand with all the data inflates the predictive ability. **Compare with the pedigree (A)** adds the same exercise with A, if Block 10 has the pedigree.

The table gives the predictive ability r(predicted, observed), with its deviation among replicates; the approximate accuracy, $r/\sqrt{h^2}$, which corrects for the noise of the phenotype it is compared with; the slope of the regression of observed on predicted, whose value close to 1 indicates that the predictions neither exaggerate nor shrink the differences; and the mean squared error. A figure plots each replicate.

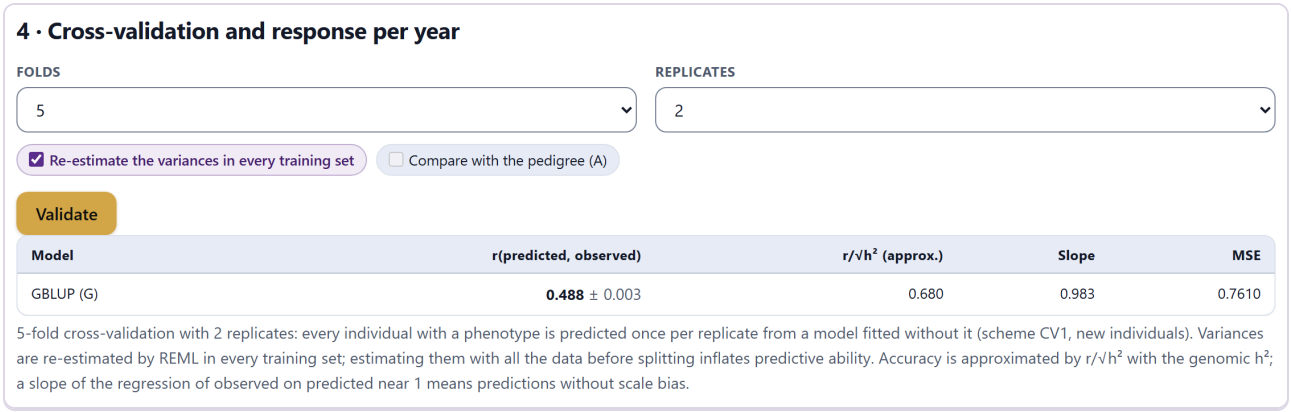


Figure 11.10. The validation of the wheat, with 5 folds and 2 replicates: predictive ability 0.488, an approximate accuracy of 0.680 and a slope of 0.983.

VERIFIED · WITH THE TRUE VALUE

In the simulated lines, the validation gives r = 0.320 and $r/\sqrt{h^2} = 0.516$; the correlation with the true breeding value, which is unknown in real data, is 0.510. In 15 simulated populations, $r/\sqrt{h^2}$ overstated the true accuracy by only 0.015 on average.

Response per year. Below, the card compares phenotypic with genomic selection in gain per year, which is where genomics wins: its cycle is shorter even if its accuracy is lower (Heffner and colleagues, 2010). The fields are filled from the GBLUP fit and can be changed: Proportion selected, σ_A (by default, $\sqrt{(\sigma^2_g \cdot ASV)}$), Phenotypic accuracy (h) (by default, $\sqrt{h^2}$) and the years per cycle of each scheme. The genomic accuracy comes from the validation.

ΔG per year = $i \cdot r \cdot \sigma_A / L$, with $r = h$ in phenotypic selection and r = accuracy from the validation in genomic selection

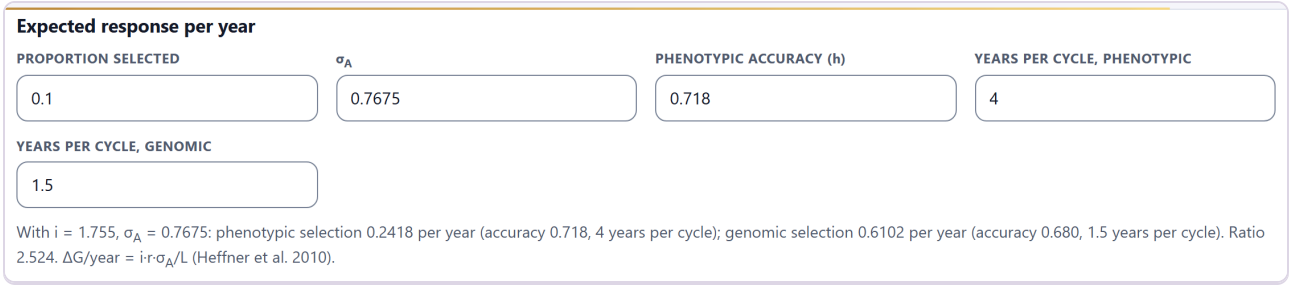


Figure 11.11. With 10% selected and cycles of 4 years for phenotypic and 1.5 for genomic selection, genomic selection gains 2.5 times more per year in the wheat.

The ratio depends almost entirely on the years per cycle: with equal cycles, genomic selection would gain less (0.680 against 0.718 accuracy). That is why the calculation makes sense only with the program's real cycles.

11.6 Hybrid prediction

In a hybrid program, a small fraction of the possible crosses between two heterotic groups is evaluated. Card **5 · Hybrid prediction** predicts all the others. It takes a table with one row per evaluated hybrid: the line of group 1, that of group 2 and the traits (adjusted means); the names of the first two columns are those of the groups. The markers of all the lines go in card 1.

$$y = \mu + GCA_1 + GCA_2 \; [+ SCA] + e, \text{ with } Var(GCA_1) = K_1\sigma^2_1,\; Var(GCA_2) = K_2\sigma^2_2 \text{ and } Var(SCA) = (K_1 \otimes K_2)\sigma^2_{SCA}$$

K_1 and K_2 are the genomic relationships within each group, computed with that group's frequencies (Bernardo, 1994; Technow and colleagues, 2014). An untested hybrid is predicted with the GCA of its two lines, and with the SCA if the checkbox **Include the SCA ($K_1 \otimes K_2$)** is ticked. With more than 700 evaluated hybrids, that checkbox is disabled: the SCA kernel would be too large to solve exactly in the browser.

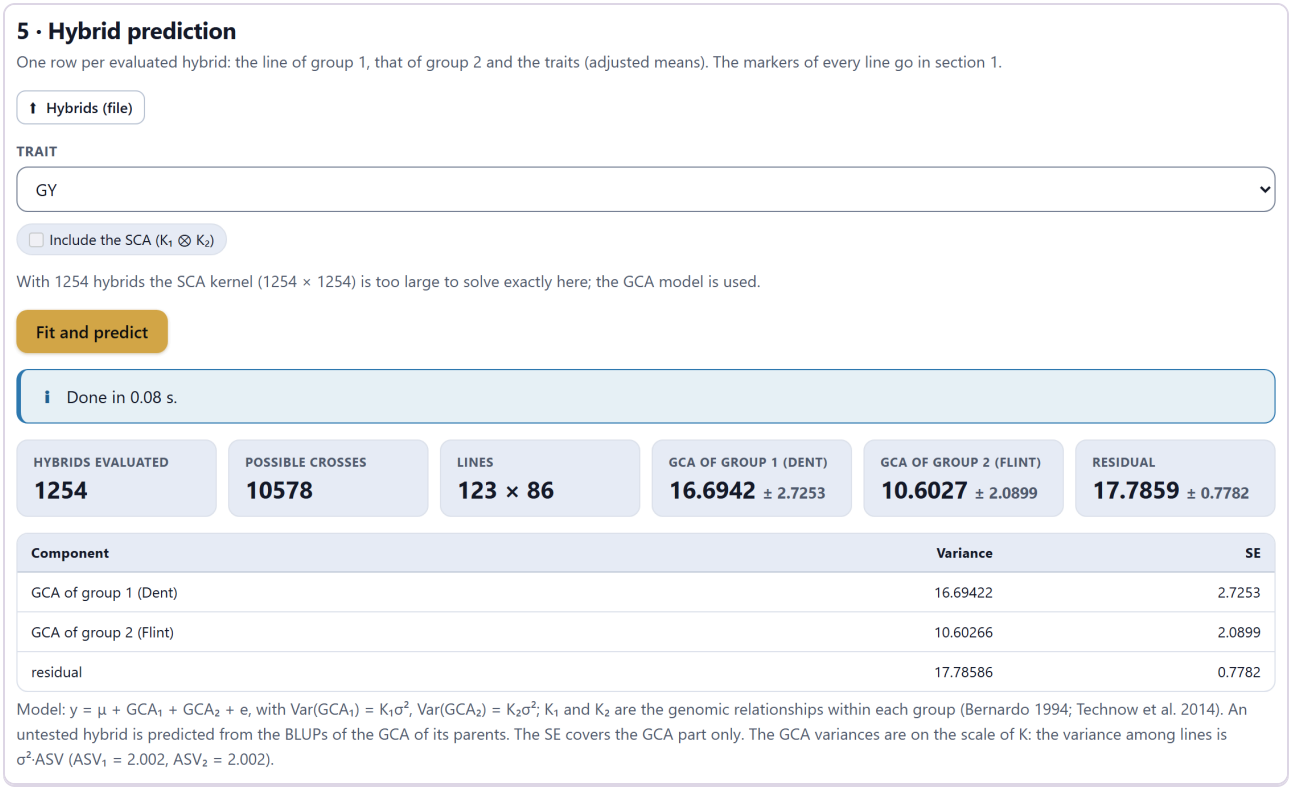


Figure 11.12. Technow's maize: 1,254 hybrids evaluated out of 10,578 possible. The GCA variance of the dent lines is 16.7 ± 2.7, that of the flint lines 10.6 ± 2.1, and the residual 17.8 ± 0.8.

The trait is GY, grain yield. The GCA variances are on the scale of K; the variance among lines is σ²·ASV, and the note gives the ASV of each group. The table **Best untested single crosses** ranks the 9,324 crosses that were not sown.

The standard error in the table includes only the GCA part. When two lines are almost identical, such as D3606 and D3607, keep only one: their crosses will give practically the same.

Best untested single crosses

Hybrid	GCA 1	GCA 2	Predicted	SE (GCA)
D3606 × F351	17.2748	8.1604	**25.6646**	2.0725
D3607 × F351	17.2740	8.1604	**25.6637**	2.0718
D3606 × F349	17.2748	6.7175	**24.2216**	2.0445
D3607 × F349	17.2740	6.7175	**24.2208**	2.0464
D3606 × F350	17.2748	6.5883	**24.0925**	2.0304
D3607 × F350	17.2740	6.5883	**24.0916**	2.0322
D3606 × F396	17.2748	6.0794	**23.5836**	1.4781
D3607 × F396	17.2740	6.0794	**23.5827**	1.4888
D3606 × F377	17.2748	5.9368	**23.4410**	1.4945
D3607 × F377	17.2740	5.9368	**23.4401**	1.4991

Figure 11.13. The 10 best untested crosses. D3606 and D3607 can hardly be told apart (GCA of 17.2748 and 17.2740): they are almost identical sister lines, and the list alternates them with each flint line.

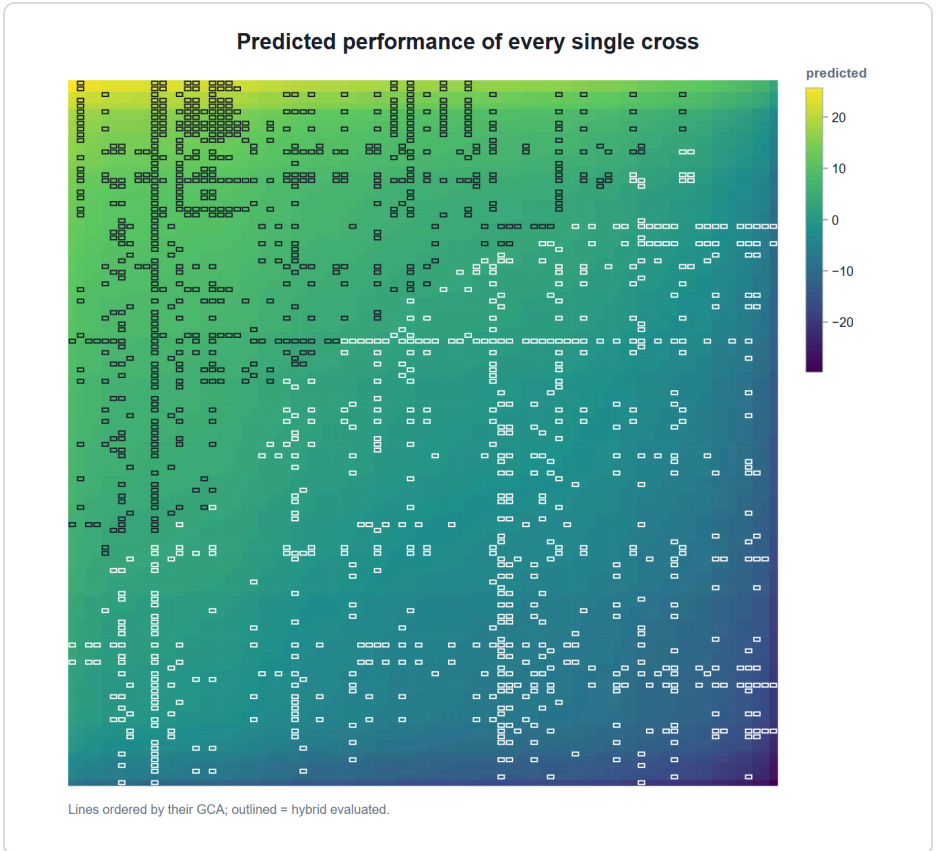
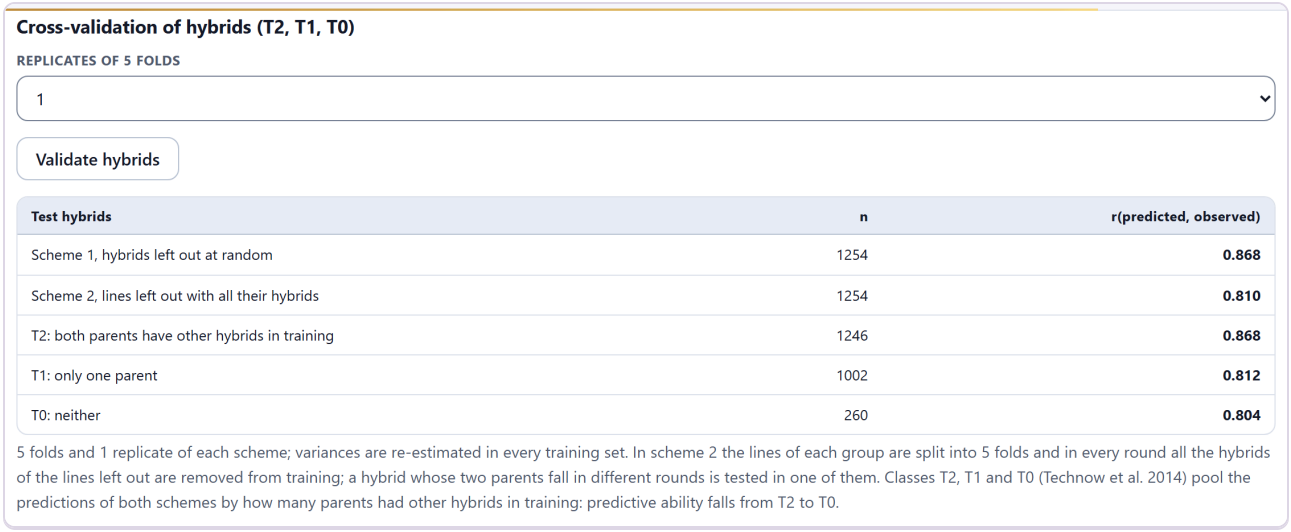


Figure 11.14. The predicted performance of the 10,578 crosses, with the lines sorted by their GCA. The outlined cells are the evaluated hybrids; the best new crosses are in the upper left corner.

A second figure, not shown here, compares observed with predicted values in the evaluated hybrids. In the simulated example, with SCA, its variance is estimated as 0 (on the boundary), and the correlation of the predictions with the true value of the 550 untested crosses is 0.780.

Cross-validation of hybrids

Validate hybrids runs two 5-fold schemes, repeated as many times as Replicates of 5 folds says (1, 2 or 5). In the first, hybrids are set aside at random; in the second, lines with all their hybrids (Technow and colleagues, 2014). The predictions are grouped by how many parents of the test hybrid had other hybrids in the training set: T2 (both), T1 (one) and T0 (none). T0 is the hardest and most useful case: predicting crosses of new lines, solely from their relationship with the evaluated ones.



Cross-validation of hybrids (T2, T1, T0)

REPLICATES OF 5 FOLDS

1

Validate hybrids

Test hybrids	n	r(predicted, observed)
Scheme 1, hybrids left out at random	1254	**0.868**
Scheme 2, lines left out with all their hybrids	1254	**0.810**
T2: both parents have other hybrids in training	1246	**0.868**
T1: only one parent	1002	**0.812**
T0: neither	260	**0.804**

5 folds and 1 replicate of each scheme; variances are re-estimated in every training set. In scheme 2 the lines of each group are split into 5 folds and in every round all the hybrids of the lines left out are removed from training; a hybrid whose two parents fall in different rounds is tested in one of them. Classes T2, T1 and T0 (Technow et al. 2014) pool the predictions of both schemes by how many parents had other hybrids in training: predictive ability falls from T2 to T0.

Figure 11.15. The validation of the maize hybrids: 0.868 in T2, 0.812 in T1 and 0.804 in T0. The drop is small, probably because many lines have almost identical sisters among the evaluated ones, such as D3606 and D3607.

In the simulated example the drop is sharp, as expected when new lines have no close relatives among the evaluated ones: 0.668 in T2, 0.328 in T1 and −0.074 in T0, with only 35 hybrids in that class. With the true values, 0.802, 0.454 and −0.094. GCA predicts well when the lines already have evaluated hybrids; for new lines, it depends on how much they resemble those that do.

Three-way and double crosses

The last box predicts crosses of three and four lines with Jenkins's (1934) rules: a three-way cross (A×B)×C with ½(AC + BC) and a double cross (A×B)×(C×D) with the average of the four nonparental single crosses AC, AD, BC and BD. You write one cross per line, with the lines on each side from different groups; when the model is fitted, the app proposes three with the best lines.

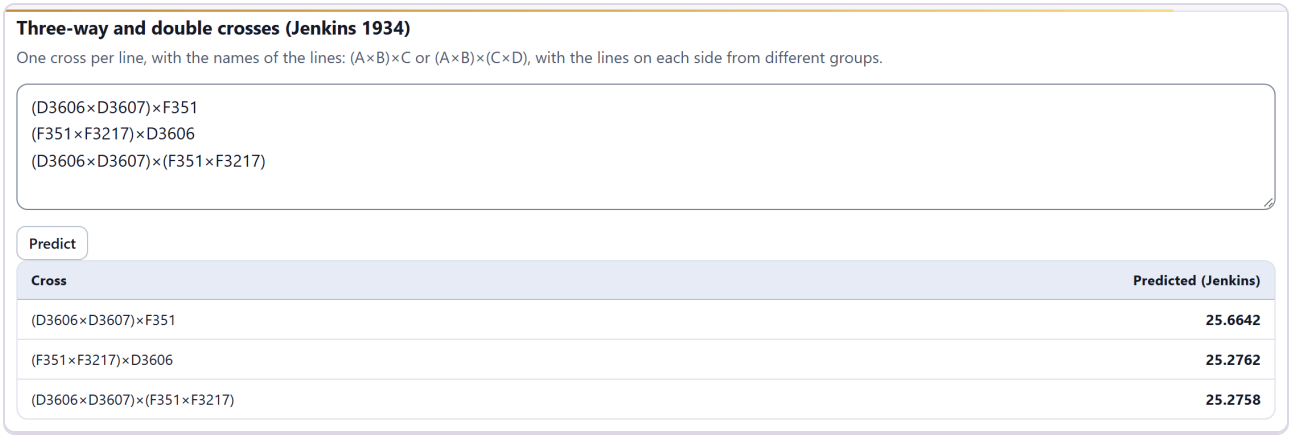


Figure 11.16. Two three-way crosses and one double cross with the best lines of each group. A cross that needs crosses within the same group is not predicted: the model knows only crosses between groups.

VERIFIED · INDEPENDENT RECONSTRUCTION

With the wheat, G (methods 1 and 2), its principal components, the REML components and the 599 genomic values agree with a reconstruction that maximizes the restricted likelihood with a general-purpose optimizer and solves Henderson's equations directly (differences of 10^{-8} or less). With the maize, the GCA variances 16.6942 and 10.6027 and the residual 17.7859, the 12 untested crosses that were checked and Jenkins's rules agree just as well. In 40 simulated populations with a true heritability of 0.40, the genomic heritability averaged 0.344: the markers captured 85% of the genetic variance, as the note of card 3 warns.

11.7 Downloads and next step

Card 6 · Assumptions, formulas and references summarizes genomic relationships, GBLUP and RR-BLUP, cross-validation and hybrid prediction, with their references. Results (.xlsx) downloads, each on its own sheet, the frequencies and quality of each marker, the G matrix (up to 1,500 individuals), the genomic values, the marker effects, the validation of each replicate and the prediction of every single cross. Each figure has its download panel.

Continue takes you to Block 12, the report, which gathers the analyses of all the blocks; ← BLUP returns to Block 10.

B12

CHAPTER TWELVE · BLOCK 12 · REPORT

Report, figures and results package

The last block gathers what the others did. It builds a report with the tables and figures of each block as they were left, a methods section written from the settings that were actually used and the list of references of each procedure. The report is printed as a PDF from the browser, and a ZIP package keeps everything separately: tables in CSV, figures in SVG and as high-resolution images, results workbooks and data.

12.1 What the block does

12.2 The content of the report

12.3 Preview, PDF and HTML

12.4 The written methods

12.5 Tables and figures

12.6 The ZIP package

12.7 How to cite and end of the tour

12.1 What the block does

Block 12 computes nothing new: it takes the results already on screen in the other blocks and arranges them into a document. The rule is simple and worth keeping in mind: **each block enters with what it shows right now**. If you change a trait, a method or a figure in a block, come back here and build the report again.

Table 12.1. The cards of Block 12.

Card	What it gives
1 · Content of the report	The blocks that go in, title, authors, notes and the optional parts
2 · Preview, PDF and HTML	The built report, to print it as a PDF, download it as HTML, open it in another tab or copy its methods
3 · ZIP package	The report and each of its pieces separately, ready for a paper or a thesis
4 · How to cite	The citation of the program, ready to copy

The example of the chapter is the 4 × 4 maize diallel of Pérez López and colleagues (2020), loaded in Block 3 and analyzed in Blocks 4 (Griffing), 5 (Hayman–Jinks and Gardner–Eberhart) and 8 (genetic parameters). With those four blocks, the report has 28 tables, 16 figures and 17 references, and it is built in a fraction of a second.

12.2 The content of the report

Card `1 · Content of the report` shows a box for each analysis block, from 2 to 11. Those that already have results say `with results` and come ticked; the rest say `no results: open the block and run it`, and cannot be chosen. Unticking a box leaves that block out.

A block counts as analyzed only if its results correspond to the data loaded now: if you change files in Block 3, the blocks you have not reopened appear without results, even if they keep the earlier ones. There are two special cases. Block 10 runs a crossbreeding example the first time it is opened, and Block 8 fills its recurrent selection table with the example of Fehr (1987); the report takes them only if you ran the crossbreeding yourself or if the table uses values from Block 6 or values you typed. Block 2 contributes the crossing and trial plan, if you built one.

Below come the `Title`, the `Author(s)` and some `Notes at the top (optional)`, which appear as they are at the beginning of the report. Five checkboxes decide the optional parts:

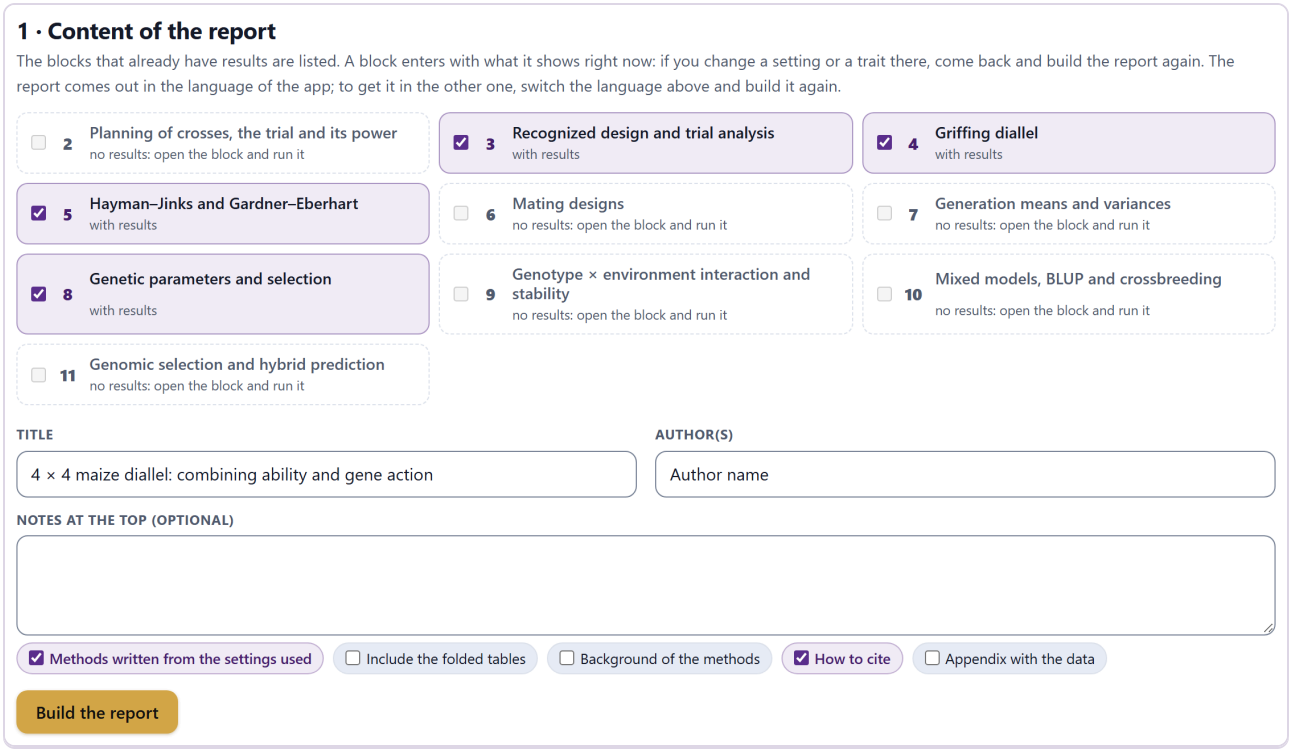


Figure 12.1. Card 1 after analyzing the maize diallel in Blocks 3, 4, 5 and 8.

- Methods written from the settings used (ticked): the methods section of Section 12.4.
- Include the folded tables: the tables that sit inside folded boxes in the app, such as those for each entry or those of coefficients.
- Background of the methods: an appendix with the assumptions, formulas and references cards of each block included.
- How to cite (ticked): the citation of the program at the end.
- Appendix with the data: the complete data table, as analyzed.

The report comes out in the language of the app. To have it in the other one, switch the language with the ES · EN buttons on the top bar and build it again: tables, figures, methods and references change together.

12.3 Preview, PDF and HTML

Build the report opens card 2 · Preview, PDF and HTML. Its tiles count the blocks, tables, figures and references, and give the size of the report; below, the whole report can be scrolled inside the card.

The four buttons do the following:

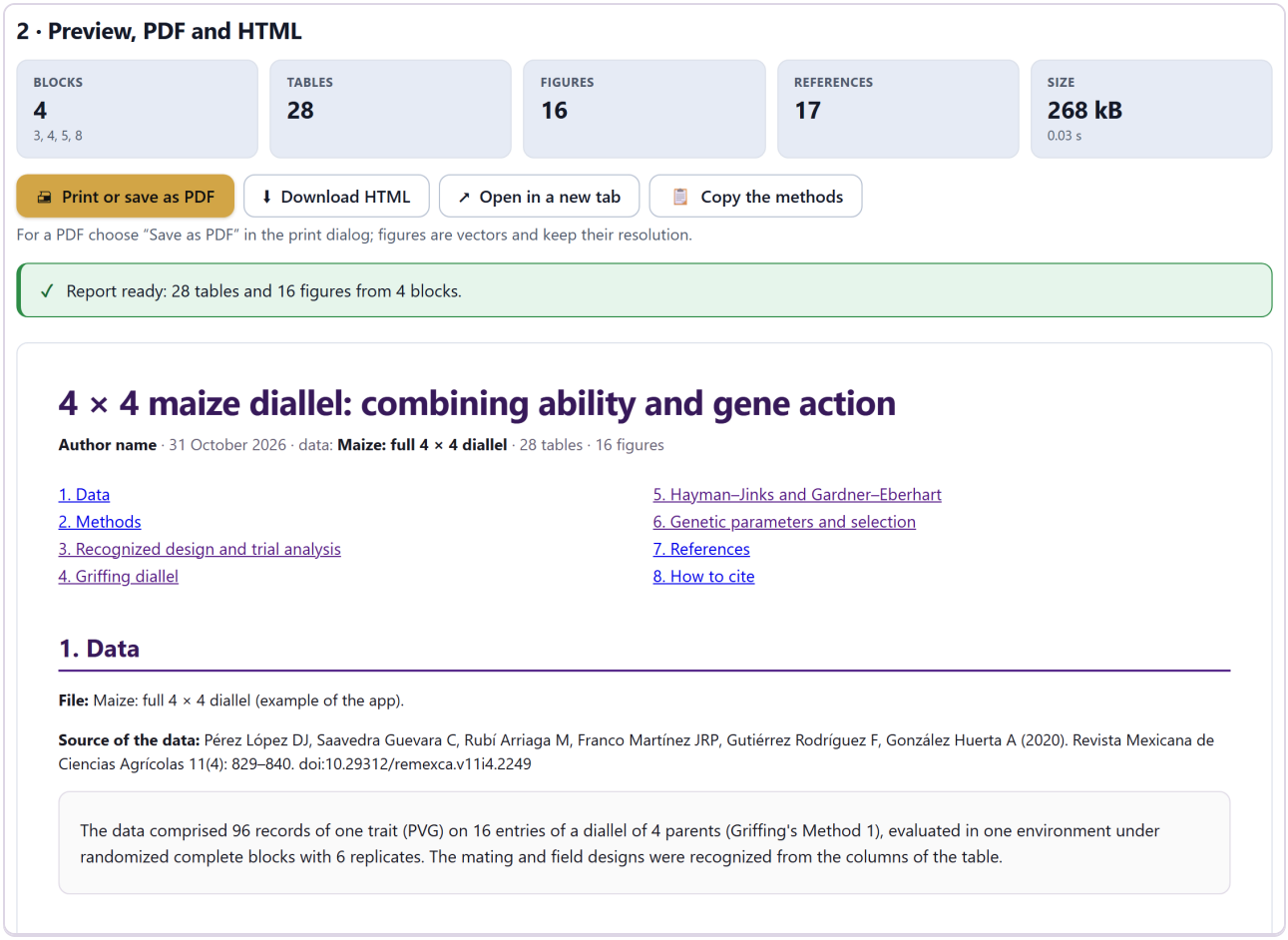


Figure 12.2. The report of the maize diallel: title, author, date, contents and the data section, with the source of the example and a paragraph describing the trial.

- Print or save as PDF opens the browser's print dialog; for the PDF, choose Save as PDF. Figures go as vectors and lose no resolution at any size.
- Download HTML saves the report as a single file that opens in any browser, offline and without the app.
- Open in a new tab shows it full screen, to read it or print it from there.
- Copy the methods copies the methods section with its references to the clipboard, as plain text, ready to paste into a manuscript.

Table 12.2. The sections of the report.

Section	What it contains
Header and contents	Title, authors, date, data file, number of tables and figures, your notes and the links to each section
1. Data	The file, the source of the data if it is an example and a paragraph describing the trial
2. Methods	One paragraph per block, with the settings used and the original citation of each procedure
One section per block	Its tables and figures, in the order of the app, numbered and with their titles

Table 12.2. The sections of the report. *(continued)*

Section	What it contains
References	Only the works cited in the methods, by author and year
Background of the methods	Optional: the assumptions, formulas and references cards of the blocks included
How to cite and data	The citation of the program and, if requested, the appendix with the analyzed data

12.4 The written methods

The methods section is not a template with blanks: each block writes its paragraph from what it actually did. It says which design it recognized, which method and which model it used, how it estimated each quantity, what significance level it applied and what checks it made, and it cites each author by name and year.

2. Methods

Recognized design and trial analysis

Every trait was analyzed per environment according to its field design: randomized complete blocks with the model $y = \mu + \text{replicate} + \text{entry} + e$ and Tukey's (1949) one-degree-of-freedom test for non-additivity. Outliers were searched with externally studentized residuals and a Bonferroni correction (Cook & Weisberg 1982). Trial heritability was estimated with random entries by REML, as $H^2 = \sigma^2 g/(\sigma^2 g + \sigma^2 e/r)$ and as the generalized H^2 of Cullis et al. (2006).

Griffing diallel

General (GCA) and specific combining ability (SCA; Sprague & Tatum 1942) of the 4 parents were estimated with Griffing (1956) Method 1 (parents, F_1 crosses and reciprocals), Model I (fixed parents), by least squares weighted by the number of plots of each entry under Griffing's restrictions; with complete balanced data the procedure agrees with his closed forms. Reciprocal effects were split into maternal and non-maternal parts (Cockerham 1963). Variance components were obtained from the expected mean squares and translated into $\sigma^2 A = 4\sigma^2 GCA/(1 + F)$ and $\sigma^2 D = 4\sigma^2 SCA/(1 + F)^2$ with $F = 1$ (Wright 1985); Baker's (1978) ratio was computed. Tests used $\alpha = 0.05$. As a check, the four Griffing methods were fitted to the same data.

Hayman–Jinks and Gardner–Eberhart

The full diallel table of 4 parents was analyzed with Hayman's (1954a) partition into items a, b_1, b_2, b_3, c and d, plus the genic item of Walters & Gale (1977), testing against the pooled experimental error. The genetic components D, H_1, H_2, F, h^2 and E and the regression of Wr on Vr followed Hayman (1954b) and Jinks (1954), with the corrections derived for single crosses; the mean degree of dominance $\sqrt{(H_1/D)}$, the allele ratio $(H_2/4H_1)$, the dominant-to-recessive ratio and the dominance order of the parents were derived (Mather & Jinks 1982). Variety effects, average heterosis, variety heterosis and specific heterosis were estimated with Gardner & Eberhart (1966) Analyses II and III by restricted least squares, and the heterosis of every cross relative to the mid-parent and the better parent with its exact standard error. The identities between the Griffing, Hayman and Gardner–Eberhart methodologies were verified on the data themselves.

Genetic parameters and selection

For the trait PVG, $\sigma^2 G$ and $\sigma^2 e$ were estimated from the mean squares, together with heritability on a plot and an entry-mean basis with the exact interval of Knapp et al. (1985), the genetic and phenotypic coefficients of variation (Burton & DeVane 1953) and the expected genetic advance (Johnson et al. 1955), with a selected proportion of 5 % ($i = 2.063$).

Figure 12.3. The methods of the maize diallel: one paragraph for each of the four blocks, with Griffing's method 1, model I, Hayman's partition with the Walters and Gale item, and the proportion selected in Block 8.

References are registered as the text cites them, so the final list has exactly the works named, not one more; the program's tests check that none is left uncited. When two works by the same author and year are cited, such as Hayman (1954a) and Hayman (1954b), they carry a letter; if only one is cited, the year goes alone. In Spanish, two authors are joined with "y" and three or more are abbreviated with "et al.", the usual form in papers; in English, with "&". If some block were left without a paragraph, the card warns about it.

> **IMPORTANT · REVIEW THE METHODS BEFORE USING THEM**
>
> The text describes what the app did with the data, but it cannot know what happened in the field: the location, the dates, crop management or how each trait was measured. Complete it with that information and read it carefully before pasting it into a manuscript; you are the one who signs the paper.

12.5 Tables and figures

Each block section gathers the tables and figures of its cards, in the order in which they appear in the app. Tables are numbered consecutively throughout the report and take as their title the heading they sit under; when a card has two tables under the same heading, they are told apart as "part 1 of 2" and "part 2 of 2". Buttons, menus, figure editors and the text of the other language are removed.

General combining ability

Table 7. General combining ability.

#	Parent	$\hat{g}_i$	SE	t	p
1	L3	20.0104	2.0311	9.85	< 0.0001 ***
2	L4	−0.1146	2.0311	−0.06	0.9552 ns
3	L1	−6.0729	2.0311	−2.99	0.0038 **
4	L2	−13.8229	2.0311	−6.81	< 0.0001 ***

Figure 6. General combining ability effects · PVG.

Figure 12.4. General combining ability in the report: Table 7 and Figure 6, just as they were in Block 4 (between them goes Table 8, of least significant differences, which is omitted here).

Figures go in as you left them in their editor: with your titles, colors, font sizes and proportions. If you want a different figure in the report, change it in its block before building it. They go as vector graphics (SVG), so they print sharp at any size.

12.6 The ZIP package

Card 3 · ZIP package saves the report and all its pieces separately, which is what a journal asks for or what is worth archiving with a thesis. The Figures besides SVG menu adds each figure as PNG, as TIFF (the format many journals ask for) or in no other format; Resolution goes from 150 to 900 dpi. Two checkboxes add the results workbook (.xlsx) of each block and the analyzed data.

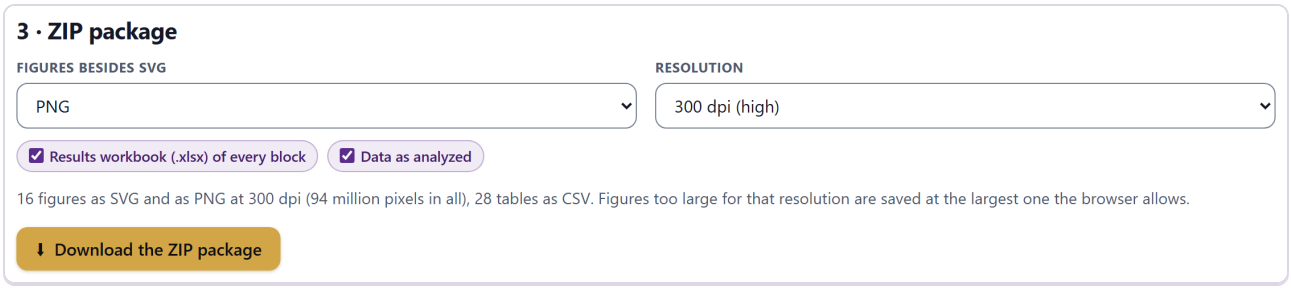


Figure 12.5. Card 3. The note works out how many pixels the images will have at the chosen resolution: here, 94 million for 16 figures at 300 dpi.

Table 12.3. What the package contains.

File or folder	Content
`report.html`	The same report as the preview
`tables/`	Each table in CSV, with its number and title in the file name
`figures/svg/` and `figures/png/` (or `tiff/`)	Each figure as a vector and as an image at the chosen resolution
`results/`	The results workbook of each block, the same as its download button
`data/`	The data table as analyzed
`methods.txt`	The methods section and the references as plain text
`README.txt`	What each file is, with the program version and the date

TIFF goes uncompressed and weighs about 3 bytes per pixel: the 16 figures of the example at 600 dpi would add up to more than 1 GB. For a journal it is usually enough to download as TIFF, from each figure's panel, only the ones you will submit. If a figure is too large for the requested resolution, it is saved at the highest the browser allows and the note warns about it. With many figures, the conversion takes a few seconds and a notice shows the progress.

12.7 How to cite and end of the tour

Card 4 · How to cite gives the citation of the program with its version and its DOI (10.5281/zenodo.23005938, which always leads to the most recent version), and Copy the citation puts it on the clipboard. The same citation appears at the end of the report and on the home page of the app.

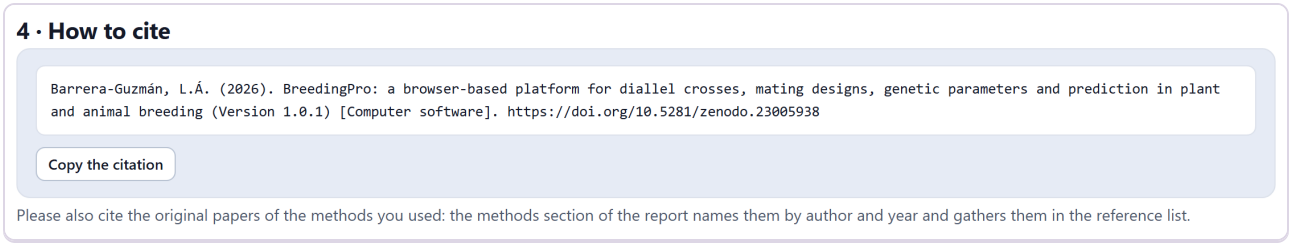


Figure 12.6. The citation of BreedingPro, version 1.0.1.

Besides the program, cite the original papers of the methods you used: the methods section names them and the reference list gathers them, so all you need to do is keep them when you edit the text.

> **VERIFIED · THE FIDELITY OF THE REPORT**
>
> In a session with six blocks, an independent routine went through the app page and compared each cell with the report: 40 tables and 4,052 cells, none different; the 26 figures, identical to those of the app; no leftover buttons, menus, editors, text of the other language or empty values. The package, opened with the operating system's ZIP reader, had its 103 files complete and error-free, and the report printed as a PDF kept all the figures as vectors.

This ends the tour of the blocks. ← Genomics returns to Block 11 and Back to home, to the home page. The appendices that follow gather the file formats, the decision rules, a glossary, troubleshooting and the references.

A–F

APPENDICES · REFERENCE MATERIAL

Appendices

The appendices are not read from start to finish: they are consulted. Here you will find the exact rules by which the app reads and writes each file, the decision rules of the whole manual gathered in one place, a glossary of terms, acronyms and symbols, the solutions to the most common problems, the full references of the methods and the data, and the third-party components with their licenses.

A File formats

B Summary of decision rules

C Glossary

D Troubleshooting

E References

F Third-party components

A File formats

Chapter 3 explains how to lay out a plot table and how the app recognizes the role of each column and the design (Tables 3.1 to 3.3); Chapters 10 and 11 describe the tables specific to their blocks. This appendix gathers the exact rules in one place: which files are read, what counts as missing, the line that declares the design, the generation codes, the tables of Blocks 10 and 11 and the format of each file the app downloads.

A.1 Files that are read

Table A.1. Ways of giving the data to the app and how they are read.

Source	How it is read	What you should know
Spreadsheet `.xlsx .xls .xlsm .ods`	In Block 3 the sheet with the most data is opened, and the Workbook sheet menu switches to another.	Numbers are stored as numbers, so the decimal separator does not matter.
Delimited text `.csv .tsv .txt`	The separator is inferred from the first 30 lines: tab if it appears at least half as often as the semicolon; otherwise semicolon if there are more semicolons than commas; otherwise comma.	UTF-8 encoding, with or without a byte-order mark. Double quotes are accepted. Spaces do not separate columns. Empty rows are ignored, and so are lines that start with `#` before the headers, except the declaration line (Section A.3).
The app's **data sheet**	A block copied from any spreadsheet, or text separated by commas, semicolons or tabs, is pasted with Ctrl+V.	Up to 20,000 rows and 5,000 columns. It is saved in the browser every time it changes (Section 3.2).
Block 2 field book	It passes to Block 3 without downloading anything, empty or with a simulated yield.	It brings the declaration line, so the design is not inferred: it is read (Section 2.8).

MISSING VALUES

The empty cell, `NA`, `N/A`, `NaN`, `null`, `.`, `..`, `-`, `--`, `?`, `*`, `missing`, `perdido`, `faltante`, `s/d`, `sd`, `nd`, `n.d.` and the errors `#N/A`, `#VALUE!` and `#DIV/0!` are missing, in upper or lower case. In a trait, any other text (`12.5a`) also becomes missing, and card 2 says how many there were and in which row the first one is.

DECIMAL SEPARATOR

On automatic, each column uses a decimal comma if it has more values like `12,5` or `1.234,5` than like `12.5`. The Decimal separator menu imposes point or comma on all of them. In a comma-separated CSV, an unquoted `12,5` splits the row into two columns: save that file with semicolons.

A.2 The plot table

One row per plot (or per plant) and one column per datum; the first row with content carries the column names, in Spanish or in English. Names are compared without accents, case, spaces or signs, and Table 3.1 gives the ones recognized for each of the 16 roles. Two identical column names are told apart by adding `(2)`, `(3)`… The rest is summarized in Table A.2.

Table A.2. How the pedigree, the type of each entry and the generations are written.

Datum	How it is written	See
Pedigree in two columns	Female (or line, F_2 plant, parent 1) and male (or tester, parent 2). If there is also an entries column with numbers that correspond one to one with the pedigree, the entry is named by its pedigree (`L1 × L2`) and the number is kept.	3.3
Pedigree in the name	`A × B`, `A x B`, `A/B`, `A*B` or compact codes such as `1x3`, provided at least half of the names are crosses and the parents recur among them. A name equal to a parent is that parent.	3.3

Table A.2. How the pedigree, the type of each entry and the generations are written. *(continued)*

Datum	How it is written	See
Entry type	`testigo`, `check`, `control` or `comercial`; `progenitor` or `parental`; `recíproca`; `línea`; `probador`; `F1`, `cruza` or `híbrido`. Without that column it is inferred: female equal to male is a parent, female different from male an F_1, and an entry without parents in data with crosses, a check.	3.3
Generations	`P1`, `P2`, `F1`, `F2`, `F3`; the reciprocal F_1 `RF1`, `F1R` or `F1rec`; the backcrosses `BC1`, `RC1`, `B1`, `retrocruza1` (and those to P_2); the selfed backcrosses `BC1S`, `BC11`, `B11`, `BS1` (and `BC2S`, `B22`, `BS2`). They go in the generation column, in the type column or in the entry name.	7.1
Check in a generation trial	One more entry that is not a generation: its name in the entries column, or written in the generation column itself (`Testigo`, `H-520`), which Block 3 reads as a check. It enters only standard heterosis.	7.7
Data per plant	Several rows with the same plot, or a `Plant` column. Each plot is summarized by the mean of its plants and the sampling error is kept apart.	3.8
Unreplicated means	One value per entry. The error of the original experiment (mean square, degrees of freedom, scale and replicates) is typed in card 3 of Block 3.	3.8

IMPORTANT · DETAILS THAT CONFUSE WHEN LOADING A TABLE

- **Identification numbers.** A numeric column with a name the app does not recognize (`Lot`, `Furrow`, `Code`) is taken as a response trait. Switch it to Ignore.
- **Females and males that share codes.** In the North Carolina designs and line × tester, `1 × 1` is a cross between two different plants, not a parent; the app marks them with ♀ and ♂. If the codes recur across sets, it prefixes the set (`S1·3`).
- **Entry names.** They are compared exactly between tables: to join the trial with the pedigree of Block 10 or with the markers of Block 11, each entry must carry the same name in both files.

A.3 The declaration line of the field book

Every Block 2 field book begins, if the checkbox Include the line that describes the design is left ticked, with a line like this one:

```
#BreedingPro fieldbook, mating=griffing, method=4, field=alpha, reps=3, k=5, locations=2, entries=3
0, seed=2026, F=1
```

These are `key=value` pairs separated by commas. If the spreadsheet spread the line over several cells, Block 3 joins it back together. Keys are case-sensitive; a numeric value is read as a number.

Table A.3. Keys of the declaration line and who uses them.

Key	Values	Use
`mating`	`griffing`, `partial` (partial diallel), `nc1`, `nc2`, `nc3` (North Carolina designs), `ttc` (triple test cross), `lxt` (line × tester), `generations`; the Block 3 menu adds `none` (genotype trial).	Block 3 takes the mating design without inferring it (confidence *declared*).
`method`	1 to 4; only with `griffing`.	Griffing's method.
`field`	`rcbd` (randomized complete blocks), `alpha` (α-lattice), `crd` (completely randomized), `augmented`; Block 3 also accepts `ibd` (incomplete blocks), `rowcol` (row–column) and `means`.	The field design.
`F`	Inbreeding of the parents: 0, 0.5, 0.75 or 1.	Block 6 takes it as the starting value of the control `Inbreeding of the parents (F)`.
`reps`, `k`, `blocks`, `locations`, `entries`, `seed`	Replicates, lattice block size, blocks of the augmented design, locations, entries and random seed.	Record of the plan. With the seed, Block 2 regenerates the same field map.

Below the line go the headers of eleven fixed columns—`Location`, `Plot`, `Replicate`, `Block`, `Row`, `Column`, `Entry`, `Genotype`, `Female`, `Male` and `Type`—and one empty column for each trait typed in card 3 of Block 2 (Table 2.8). With the app in Spanish, the headers come out in Spanish; Block 3 understands them in both languages. The data are recorded in the empty columns without changing the names of the fixed ones.

A.4 The tables of Block 10

Table A.4. What the cards of Block 10 read. The column roles are described in Table 10.3.

Table	How it is written	See
Records card 1	One row per record with the individual, its sire and its dam; an unknown parent is written `0` or `NA`, or left empty. Roles are proposed from the name: `Animal`, `ID`, `Individual` or `Genotype` is the individual; `Sire`, `Father` or `Bull`, the sire; `Dam` or `Mother`, the dam; `CG`, `Litter` or `Group`, a random effect; a numeric column with few integer values, a fixed effect, and one with many values, the response trait. In plants, the seed parent goes in as the dam.	10.2
Separate pedigree optional	Individual, sire and dam columns with ancestors that have no records, so that the relationships reach further back. Parents without a row of their own are added as founders.	10.2
Groups card 4, estimate	One row per group: `cross, n, mean, SE`. The number and the standard error can be left empty (`1×2, , 309`); with every SE given, the weights are $1/SE^2$ and there is a lack-of-fit test.	10.6
Crosses	Sire first: `A`, `A×B`, `C×(A×B)`, `(A×B)×(A×B)`. The sign can be `×`, `*`, `·` or an `x` between spaces. For prediction also `rot(A,B)` and `rot(A,B,C)` (rotations) and `comp(A,B,C)` (composite with equal parts) or `comp(A:0.5,B:0.25,C:0.25)`.	10.6

Table A.4. What the cards of Block 10 read. The column roles are described in Table 10.3. *(continued)*

Table	How it is written	See
Parameters predict with known parameters	One row per parameter with its value: `mu, 14.3`, `gI:B, −3.18`, `gM:B, …`, `hI, −0.04`, `hM, …`, `rI, −6.20`. Those not written are zero: the breed with no effect written is the reference.	10.6
Breed means heterosis in percent	One row per purebred (`A, 460`) and, separately, the percentages of individual and maternal heterosis and of recombination loss.	10.6

A.5 The tables of Block 11

Table A.5. What the cards of Block 11 read. The marker codings are in Table 11.3.

Table	How it is written	See
Markers card 1	One row per individual, with its identifier in the first column and one column per marker; if the file comes the other way around, the checkbox Markers in rows transposes it. Dosages 0/1/2, −1/0/1, presence or absence 0/1 (the menu A 0/1 file means decides whether they are pure lines or dosages) or letters (`AA/AB/BB`, `A/C/G/T`, and the two-base codes `R Y S W K M` as heterozygotes). Missing: empty, `NA`, `–` or `N`.	11.2
Phenotypes card 3	The identifier in the first column and one column per trait. Those of the example or the Block 3 trial also serve, if its entries have markers.	11.4
Hybrids card 5	One row per evaluated hybrid: the line of group 1, that of group 2 and the traits (adjusted means). The names of the first two columns are those of the groups. The markers of all the lines go in card 1.	11.6
Three-way and double crosses	One per line, `(A×B)×C` or `(A×B)×(C×D)`, with the lines on each side from different groups.	11.6

A.6 Files the app produces

All CSV tables are written comma-separated, with a decimal point and up to 12 significant digits, in UTF-8 with a byte-order mark (spreadsheets open them with the accents right) and with Windows line endings. A field that contains a comma, a semicolon, quotes or a line break goes in double quotes. In the `.xlsx` workbooks, each table takes its own sheet; sheet names follow the language of the app, and some column headers use fixed symbols or abbreviations (`sigma2A`, `SE`, `GEBV`). The file name is formed from the block and the trait, without accents or signs (`griffing_Yield.xlsx`). The names below are those of the app in English.

Table A.6. What each block downloads. The figures, the report and the package are described in Sections I.6, 12.3 and 12.6.

Block	File	Content
B2	`field_book.xlsx` · `field_book.csv` · `entries.csv`	Sheets `Field book`, `Entries`, `Crossing block` and `Design`; the CSV, only the field book, with the declaration line if requested; the list of crosses and entries.
B3	`data_and_design_….xlsx` · `adjusted_means.csv` · `cleaned_data.csv`	Sheets `Means`, `ANOVA`, `Quality`, `Data` and `Design`; the adjusted means by trait and environment; the original table with the `Excluded` column (Table 3.7).

Table A.6. What each block downloads. The figures, the report and the package are described in Sections I.6, 12.3 and 12.6. *(continued)*

Block	File	Content
B3	Data sheet: Save as	`.xlsx`, `.ods`, `.csv` with commas or with semicolons, or `.tsv`, with the values as they were typed.
B4	`griffing_….xlsx` · `griffing_effects.csv`	Sheets `ANOVA`, `Effects`, `Components`, `Parameters`, `Crosses` and, if the methods were compared, `Comparison`; the CSV, the effects with their standard errors.
B5	`hayman_gardner_….xlsx` · `diallel_table.csv`	Sheets `Hayman`, `Components`, `Arrays` (V_r, W_r), `Gardner–Eberhart`, `Heterosis` and `Diallel table`.
B6	`mating_….xlsx`	Sheets `ANOVA`, `Components`, `Parameters`, `Effects` and, depending on the design, `Sums and differences` or `Heterosis`.
B7	`generations_….xlsx`	Sheets `Generations`, `Scaling`, `Models`, `Parameters`, `Perfect fit`, `Variances`, `Factors` and `Heterosis`.
B8	`selection.xlsx`	Sheets `Parameters`, `rG`, `rP`, `rE`, `P`, `G`, `Indices`, `Selection` and `Methods`.
B9	`gxe_interaction_….xlsx`	Sheets `GxE table` (with the imputed cells), `ANOVA`, `Regression`, `Variances`, `Ranks`, `AMMI axes`, `AMMI genotypes`, `AMMI environments`, `GGE` and `Two stages`.
B10	`blup_and_crossbreeding.xlsx`	Sheets `Pedigree` (with F and b), `Components`, `Breeding values`, one for each fixed factor, `Trial BLUP`, `Dickerson`, `Breed diallel` and `Predictions`.
B11	`genomics.xlsx`	Sheets `Markers`, `G` (up to 1,500 individuals), `GEBV`, `Effects`, `Validation` and `Hybrids`.
B12	Report `.html` · package `.zip`	The name comes from the title of the report. The package brings `report.html`, `tables/`, `figures/`, `results/`, `data/`, `methods.txt` and `README.txt` (Table 12.3).
Figures	Panel of each figure	PNG, JPG, WEBP and TIFF (uncompressed) up to 900 dpi, or SVG, which is vector; the panel gives the size in pixels and in centimeters.

Only the sheets that have results go into a workbook: a Block 6 with a North Carolina II brings no `Sums and differences`, and a Block 10 without crossbreeding brings no `Dickerson`.

B Summary of decision rules

The decision rules of the manual, in the order of the app. Those of the introduction and of Chapters 1 to 5 are repeated just as they appear there; those of Blocks 6 to 12, whose chapters give the criteria within the text, are summarized here in the same format. Each box names the section that explains the theory it rests on. They are guides for interpretation, not laws: knowledge of your species and of your program always comes first.

In	**Introduction**	**I.8** Which block to use **I.9** Reading a P value
B1	**Home, labs and theory**	**1.2** How the app words the interpretation
B2	**Planning crosses, trial and power**	**2.4** Which field design to use
B3	**Data and field analysis**	**3.6** The quality of the trial **3.9** Before moving on to the genetic blocks
B4	**Griffing diallels**	**4.4** Which effect is useful for what
B5	**Hayman–Jinks and Gardner–Eberhart**	**5.5** The Wr–Vr graph
B6	**Mating designs**	**6.2** The inbreeding F of the parents **6.5** Components and tests
B7	**Generations and heterosis**	**7.3** The variance of each mean **7.4** Scaling, models and heterosis
B8	**Genetic parameters and selection**	**8.2** Reading a heritability **8.6** When to trust path analysis **8.7** Which index to use
B9	**Genotype × environment interaction**	**9.2** Interaction and stability **9.7** AMMI and the GGE biplot
B10	**Mixed models, BLUP and crossbreeding**	**10.3** The mixed model **10.5** Breeding values and crossbreeding
B11	**Genomics and hybrid prediction**	**11.2** Markers and GBLUP **11.5** Validation and hybrids
B12	**Report, figures and package**	**12.3** Which output to use

In Introduction to BreedingPro

DECISION RULE · WHICH BLOCK TO USE — SECTION I.8

If the table has the female and male of a group of parents crossed among themselves…	**then** Griffing (4), Hayman–Jinks (5) and parameters (8).
If males and females are different groups (NC, L × T) or the diallel is partial…	**then** mating designs (6) and parameters (8).
If there is a generation column (P_1, P_2, F_1, F_2…)…	**then** generations (7).
If they are genotypes in several locations or years…	**then** parameters (8), stability (9) and, with a pedigree, BLUP (10).
If you have a pedigree or markers…	**then** BLUP (10) or genomics (11).

Block 3 makes this choice for you when it recognizes the design, and states it on the 3 · Recognized design card.

READING RULE · A P VALUE	SECTION 1.9
If P < 0.001 (***)...	**then** very strong evidence.
If 0.001 ≤ P < 0.01 (**)...	**then** strong evidence.
If 0.01 ≤ P < 0.05 (*)...	**then** moderate evidence.
If P ≥ 0.05 (ns)...	**then** not significant at 5%.

A P value does not measure the importance of an effect: also look at its size and its standard error.

B1 Home, labs and theory

DECISION RULE · HOW THE APP WORDS THE INTERPRETATION	SECTION 1.2
If b does not differ from 1 (|t| smaller than the critical value at 5% with p – 2 degrees of freedom)...	**then** the additive–dominance model holds.
If b departs from 1...	**then** there is non-allelic interaction or another broken assumption, and the components are biased.
If $(H_1/D)^{½} < 0.25$...	**then** gene action is practically additive.
If the intercept of the line is positive and significant...	**then** partial dominance.
If it is negative and significant...	**then** overdominance.
If it does not differ from zero and $(H_1/D)^{½}$ is between 0.8 and 1.2...	**then** complete dominance. Outside that interval, the app warns that the intercept test has little power (Exercise 4).
If r(Yr, Wr + Vr) < –0.3 · > 0.3...	**then** the dominant · recessive alleles increase the trait; between both values the direction is unclear.

The app also names the two parents closest to the origin: they are the ones that carry the most dominant alleles.

B2 Planning crosses, trial and power

DECISION RULE · WHICH FIELD DESIGN TO USE	SECTION 2.4
If there are few entries (up to 15 or 20) and the land is uniform over the size of a replicate...	**then** randomized complete blocks.
If there are many entries and a full replicate no longer fits on homogeneous soil...	**then** α-lattice, with blocks of a size close to $\sqrt{v}$.
If there are many new entries and seed for a single plot of each...	**then** augmented, with at least two checks and 12 or more error degrees of freedom.
If the plots are pots or homogeneous units (greenhouse, laboratory)...	**then** completely randomized.

The app requires at least two replicates, except in the augmented design, and warns when the error is left with fewer than 12 degrees of freedom.

B3 Data, recognized design and field analysis

READING RULE · THE QUALITY OF THE TRIAL SECTION 3.6

Test	Fine	To check
Normality of residuals (Shapiro–Wilk)	p ≥ 0.05: holds	p < 0.05: check the Q–Q plot
Replicate × entry **additivity** (Tukey, 1 df)	p ≥ 0.05: additive	p < 0.05: non-additive: try a transformation
Outliers (Bonferroni)	none	some: review in card 4
Repeatability (H^2_c)	≥ 0.5: informative trial	0.3 to 0.5: moderate; < 0.3: low
Precision (CV)	≤ 12%: good	12 to 20%: acceptable; > 20%: low
Gain from incomplete blocks	≥ 5%: blocks helped	< 5%: little gain

The CV limits are indicative: they depend on the crop and the trait. If Bartlett's test of the combined analysis gives p < 0.05, use the two-stage analysis of Block 9.

RULE OF THUMB · BEFORE MOVING ON TO THE GENETIC BLOCKS SECTION 3.9

If a column was left with a wrong role or as Ignore…	**then** correct it in card 2: everything else depends on it.
If the mating design has medium or low confidence…	**then** read the reasons and check Recognized entries and the figure of crosses present.
If there are outliers…	**then** compare them with the field book and exclude only proven errors.
If normality or additivity fail…	**then** consider transforming the trait before loading it again.

The genetic blocks use exactly what remains in Block 3: its data, its design, the excluded observations and the external error.

B4 Griffing diallels

READING RULE · WHICH EFFECT IS USEFUL FOR WHAT SECTION 4.4

High, significant GCA ($\hat{g}_i$)	The parent passes its value on to all its crosses: useful for **forming populations and synthetics** and for choosing the parents of a program.
High, significant SCA ($\hat{s}_{ij}$)	The pair performs better than its GCA effects predict: useful for **hybrids** of that combination, not for choosing parents.
Maternal effect (m_i)	Which parent is the female matters: use the parent in the favorable direction (cytoplasm, endosperm, seed size).
Non-maternal reciprocal (n_{ij})	The direction matters only in that pair; confirm it before drawing conclusions.

Baker's ratio sums up the balance: close to 1, GCA predicts the crosses well and additive action dominates (Section 4.5).

B5 Hayman–Jinks and Gardner–Eberhart

READING RULE · THE WR–VR GRAPH	SECTION 5.5
Slope that does not depart from 1	The additive–dominance model is adequate; if it departs, there is epistasis or genes correlated between parents.
Intercept positive · zero · negative	Partial dominance · complete dominance · overdominance.
Distance from the origin of each point	Near: the parent carries mostly dominant alleles; far: recessive ones.
Correlation between the parent's value and Wr + Vr	Positive: dominant alleles decrease the trait; negative: they increase it.

B6 Mating designs

DECISION RULE · THE INBREEDING F OF THE PARENTS	SECTION 6.2
If the parents are inbred lines (line × tester, partial diallels of lines)…	**then** F = 1.
If they are plants from a population (North Carolina designs, triple test cross)…	**then** F = 0.
If they are lines with few generations of selfing…	**then** F = 0.5 or 0.75, depending on the generation.
If the Block 2 field book declared another value…	**then** the app uses that one: confirm it.

With the same mean squares, going from F = 0 to F = 1 divides σ^2_A by 2 and σ^2_D by 4.

READING RULE · COMPONENTS AND TESTS OF A DESIGN	SECTIONS 6.5 TO 6.8
If a component comes out negative…	**then** the true variance is close to zero or replicates are lacking. It is reported as it comes out; with a negative σ^2_D there is no degree of dominance, H^2 or Baker's ratio.
If the two σ^2_A of a North Carolina II, that of males and that of females, differ greatly…	**then** suspect maternal effects or that males and females do not come from the same population.
If in the triple test cross the contrast $L_1 + L_2 - 2L_3$ is significant…	**then** there is epistasis, provided L_3 really is the F_1: check F₁ tester (L₃).
If in line × tester the interaction dominates and σ^2_A is imprecise…	**then** almost all the variation among crosses is specific; a few purposely chosen testers do not estimate the σ^2_A of a population.

The test that a component is zero is the F of its source, not its standard error.

B7 Generations and heterosis

✓ DECISION RULE · THE VARIANCE OF EACH MEAN	SECTION 7.3
If the plants were randomized one by one (or there is one plant per plot)…	**then** the variance among individuals divided by their number, s^2/n.
If there are replicated plots of several plants…	**then** the plot error without blocks, which is what `automatic` does.
If there is a single value per plot…	**then** the residual mean square; there is no D, H, F, E nor effective factors.
If there are plots and `among individuals / n` is forced…	**then** the standard errors come out too small and any difference looks significant: do not do it.

The blocks do not enter the variance of the means: each generation is in all of them and their effect cancels out in the comparisons.

✓ READING RULE · SCALING, MODELS AND HETEROSIS	SECTIONS 7.4 TO 7.7
If a single scaling test is significant…	**then** the additive–dominance model is not enough; the joint test says which epistasis is needed.
If the scaling tests fail…	**then** before adding epistasis try a transformation of the trait: it may be a scale effect.
If [h] and [l] are significant with the same sign · with opposite signs…	**then** complementary · duplicate epistasis (in the chosen metric).
If D comes out negative…	**then** there are few plants, linkage or epistasis; read Warner's h^2 and H^2 from the F_2.
If the question is gene action · whether the F_1 is worthwhile…	**then** read heterosis over the parents · standard heterosis against the check.

The difference is what is tested, with its standard error and Satterthwaite degrees of freedom; the percentage only helps to read it.

B8 Genetic parameters and selection

✓ READING RULE · A HERITABILITY	SECTIONS 8.2 TO 8.4
If H^2 on a mean basis < 0.3 · 0.3 to 0.6 · > 0.6…	**then** low · medium · high (the gray band of Figure 8.2).
If Knapp's interval is wide (few genotypes)…	**then** the class is uncertain: report the interval, not just the value.
If there are several environments…	**then** use the combined H^2, with σ^2_{GE}; that of a single environment mixes the interaction with the genetic part.
If the trial has incomplete blocks or is unbalanced…	**then** Cullis's H^2.

With a negative σ^2_G the heritability is reported as it is, but genetic advance is taken as zero.

DECISION RULE · WHEN TO TRUST PATH ANALYSIS	SECTION 8.6
If the app does not show the table (the matrix is not positive definite or $R^2 > 1$)…	**then** use the phenotypic correlations, remove traits with small σ^2_G or increase the ridge constant.
If the condition number exceeds 100 or there are large VIFs…	**then** the direct effects are unstable even if the matrix is valid.
If a direct effect is small and the total correlation large…	**then** the trait acts through others: its indirect effects tell you so.

DECISION RULE · WHICH SELECTION INDEX TO USE	SECTIONS 8.7 AND 8.8
If P, G and the economic weights are well estimated…	**then** Smith–Hazel.
If G and P are unreliable…	**then** the base index (b = w).
If a trait must not change…	**then** the restricted index.
If the ratio between gains matters…	**then** the proportional gains index, with $\theta > 0$.
If there are no economic weights…	**then** the desired gains index, or the eigen index (ESIM) for the most heritable combination.
If there is neither P nor G…	**then** the rank-sum of Mulamba and Mock or Elston's index.

The coefficients are not interpreted one by one: what matters are the expected gains in each trait.

B9 Genotype × environment interaction

READING RULE · INTERACTION AND STABILITY	SECTIONS 9.2 TO 9.6
If the environments are a sample of the region · are the only ones of interest…	**then** test the genotypes against the interaction · against the error.
If Bartlett's test rejects equal errors…	**then** read the two-stage analysis.
If b does not differ from 1 and S^2_d is not significant…	**then** broad, predictable adaptation; b > 1, it takes advantage of good environments; b < 1, it holds up in poor ones.
If S^2_d is significant…	**then** the genotype's response is unpredictable, even if its mean is high.
If you sort by the mean rank of the summary…	**then** use it to order the discussion, not to decide: it adds up statistics that measure different things.

Static stability, in which the genotype yields the same in every environment, is rarely desirable for yield: a genotype that does not change does not take advantage of good environments.

✓ DECISION RULE · AMMI AND THE GGE BIPLOT	SECTIONS 9.7 AND 9.8
If you must decide how many AMMI axes to retain…	**then** Cornelius's residual test; Gollob's is liberal.
If a genotype has a high mean and a low WAAS…	**then** it is desirable: quadrant IV of WAAS against the mean.
If the first two GGE components explain less than 60%…	**then** read the biplot with caution.
If the which-won-where view forms sectors…	**then** they are mega-environments only if the pattern recurs in other years.
If you are looking for a test environment…	**then** one that is long and close to the average-environment axis; a long one with a large angle serves to discard unstable genotypes.

B10 Mixed models, BLUP and crossbreeding

✓ DECISION RULE · THE MIXED MODEL	SECTIONS 10.3 AND 10.4
If there are hundreds of records · only a few…	**then** estimate the variances by REML · give them as known, from the literature or from a larger population.
If you want to know whether a random effect is needed…	**then** read the likelihood ratio against the $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ mixture; compare only models with the same fixed effects.
If there are many small contemporary groups…	**then** the group as a random effect.
If there are individuals with several records…	**then** add the permanent environment.
If the model exceeds 2,600 unknowns…	**then** trim the pedigree to the ancestors of the individuals with records.

✓ READING RULE · BREEDING VALUES AND CROSSBREEDING	SECTIONS 10.3, 10.5 AND 10.6
If the standard error of prediction is as large as the differences…	**then** still choose by breeding value: it already discounts that uncertainty.
If plots are missing, replicates are unequal or the heritability is low…	**then** the trial BLUP beats the mean; with families of few plots, the relationships (K = A) help most.
If you split the additive and non-additive parts with few families…	**then** the split is not reliable: also look at the additive-only model.
If you take the recombination loss from a paper…	**then** use its same convention, /4 or /2.
If a parameter comes out not estimable…	**then** the groups do not separate it: add backcrosses, F_2 or three-breed crosses.

B11 Genomic selection and hybrid prediction

DECISION RULE · MARKERS AND GBLUP	SECTIONS 11.2 TO 11.4
If you have no other criterion for quality control…	**then** the default values: up to 20% missing per marker and per individual, and a MAF of 0.05 or more.
If there are few markers and Block 10 has the pedigree…	**then** blend G with A (Weight of the pedigree).
If an individual hardly resembles its supposed sire genomically…	**then** suspect a pedigree error.
If the genomic heritability comes out lower than that of the trial…	**then** some of the QTL are not in disequilibrium with the panel.
If a marker has a large effect…	**then** it deserves a look, not a conclusion: association is tested with other methods.

READING RULE · VALIDATION AND HYBRID PREDICTION	SECTIONS 11.5 AND 11.6
If you want to know how well the model predicts…	**then** read the cross-validation, not the correlation with the individual's own record, and leave Re-estimate the variances in every training set ticked.
If the slope of the validation is close to 1…	**then** the predictions neither exaggerate nor shrink the differences.
If you compare genomic with phenotypic selection per year…	**then** use the program's real years per cycle: the ratio depends almost entirely on them.
If T0 drops sharply compared with T2…	**then** the new lines have no close relatives among the evaluated ones.
If two lines are almost identical…	**then** keep one: their crosses will give practically the same.

B12 Report, figures and results package

DECISION RULE · WHICH OUTPUT TO USE	SECTIONS 12.3 AND 12.6
If you are going to read, review or print…	**then** the preview and Print or save as PDF.
If you share it with someone who does not have the app…	**then** Download HTML: it opens in any browser, offline.
If you are writing a manuscript…	**then** Copy the methods, and complete them with what happened in the field.
If you are submitting to a journal or archiving a thesis…	**then** the ZIP package; TIFF only for the figures you submit.
If you need it in the other language…	**then** switch ES · EN and build it again.

Each block enters with what it shows at that moment: if you change something in a block, build the report again.

C Glossary

The terms, acronyms and symbols of the manual with a brief definition and the section where they are explained at leisure. Below each term, in parentheses, goes its Spanish name when it differs, as it appears in the app with the ES button and in the literature in Spanish.

C.1 Acronyms

Table C.1. Acronyms of the manual and the app. In parentheses, the Spanish acronym when the app in Spanish or the Spanish literature uses a different one.

Acronym	Meaning	Acronym	Meaning
AMMI	Additive main effects and multiplicative interaction	**GCA**	General combining ability (ACG)
ANOVA	Analysis of variance	**GEBV**	Genomic estimated breeding value, in the results sheets
ASV	AMMI stability value (Block 9); in Block 11, the average semivariance of K (SM)	**GGE**	Genotype plus genotype × environment: what the GGE biplot draws
BC	Backcross (RC): BC_1 to P_1, BC_2 to P_2	**IPCA**	Interaction principal component axis, in AMMI
BLUP	Best linear unbiased predictor	**L × T**	Line × tester (L × P)
CI	Confidence interval (IC)	**LSD**	Least significant difference (DMS)
CRD	Completely randomized design (CA)	**MAF**	Minor allele frequency
CV	Coefficient of variation; CV_g, CV_p and CV_e: genetic, phenotypic and environmental	**MS · MSE**	Mean square · error mean square (CM · CME)
CV1	Cross-validation scheme that predicts new individuals	**NC I, II, III**	North Carolina designs (CN I, II, III)
DArT	Diversity Arrays Technology markers	**PEV**	Prediction error variance
df	Degrees of freedom (gl)	**QTL**	Quantitative trait locus
dpi	Dots per inch (ppp)	**RCBD**	Randomized complete block design (BCA)
EBV	Estimated breeding value (VG)	**REML**	Restricted maximum likelihood
EM	Expectation–maximization algorithm	**RR-BLUP**	Ridge-regression BLUP on the markers
ESIM	Eigen selection index	**SCA**	Specific combining ability (ACE)
F_1, F_2, F_3	First, second and third filial generation	**SD · SE**	Standard deviation · standard error (DE · EE)
G×E	Genotype × environment interaction (G×A)	**SEP**	Standard error of prediction: √PEV (EEP)
GA	Expected genetic advance (AG)	**SNP**	Single nucleotide polymorphism
GBLUP	Genomic BLUP: BLUP with the marker relationship matrix	**SS**	Sum of squares (SC)
T0, T1, T2	Classes of the hybrid validation: none, one or both parents with hybrids in the training set	**TA**	Transmitting ability: half the breeding value (HT)
VIF	Variance inflation factor	**WAAS · WAASY**	Weighted average of absolute AMMI scores · combined with the mean

C.2 Terms

Table C.2. Glossary in alphabetical order.

Term	Definition	Section
A		
Accuracy (exactitud)	Correlation between the true and the predicted value. In cross-validation it is approximated by $r/\sqrt{h^2}$.	10.3, 11.5
Additive and dominance variance (σ^2_A, σ^2_D) (varianza aditiva y de dominancia)	Variance of the breeding values and of the dominance deviations. Selection exploits σ^2_A; hybrids also exploit σ^2_D.	I.3, 4.5
Adjusted mean (media ajustada)	Mean of an entry corrected for the effects of the design (blocks, replicates), comparable across entries.	3.6
Alpha lattice (látice α)	Resolvable incomplete block design: each replicate is divided into small blocks, which absorb the heterogeneity of the land.	2.4
AMMI	Model that fits the effects of genotypes and environments and decomposes the interaction into a few multiplicative axes (IPCA); the rest is taken as noise.	9.7
Animal model (modelo animal)	Mixed model in which each individual in the pedigree has its breeding value, whether or not it has records.	10.3
Augmented design (bloques aumentados)	Design in which some checks are repeated in each block and the new entries appear only once; the checks estimate the block effects.	2.4, 3.8
Average degree of dominance (grado medio de dominancia)	$\sqrt{(H_1/D)}$ or $\sqrt{(2\sigma^2_D/\sigma^2_A)}$: 0 without dominance, between 0 and 1 partial, 1 complete and more than 1 apparent overdominance.	4.5, 5.4
Average semivariance (ASV) (semivarianza media, SM)	$\mathrm{tr}(K)/n - 1'K1/n^2$: converts σ^2_g into the genetic variance among the individuals analyzed.	11.4
B		
Baker's ratio (razón de Baker)	$2\sigma^2_{GCA}/(2\sigma^2_{GCA} + \sigma^2_{SCA})$: close to 1, GCA predicts the crosses well.	4.5
Bending (doblado de una matriz)	Correction of an impossible covariance matrix (negative eigenvalues) to make it possible, changing it as little as possible.	8.7
BLUP	Prediction of a random effect that shrinks extreme values toward the mean according to their precision and makes use of relationships.	10.3
Boundary (frontera)	A variance estimated by REML that ends up at zero. It is reported as such, without a standard error, and its test uses a mixture of χ^2.	I.9, 10.3
Breeding value (valor genético)	Value of an individual as a parent: twice the mean deviation of its progeny from the population.	10.3

Table C.2. Glossary in alphabetical order. *(continued)*

Term	Definition	Section
Broad-sense heritability (H^2) (heredabilidad en sentido amplio)	Genetic fraction of the phenotypic variance, σ^2_G/σ^2_P, on a plot or a genotype-mean basis.	I.3, 8.2
C		
Check (testigo)	Known variety or hybrid sown with the entries to compare them; it does not enter selection.	3.3, 7.7
Coancestry (coancestría)	Probability that two alleles taken at random, one from each individual, are identical by descent. The inbreeding of an offspring is the coancestry of its parents.	10.2
Combined analysis (análisis combinado)	The analysis of all environments together, with the entry × environment interaction. It is fitted in one stage or, if the model is very large, in two.	3.7
Condition number (número de condición)	Ratio between the largest and the smallest eigenvalue of the correlation matrix; more than 100 indicates strong collinearity.	8.6
Contemporary group (grupo contemporáneo)	Animals raised together, under the same management and season; it is fitted as a fixed or random effect.	10.4
Cross-validation (validación cruzada)	Predicting a group of individuals with a model fitted without them and comparing with what was observed; it measures the real predictive ability.	11.5
Cullis heritability (heredabilidad de Cullis)	Generalized heritability $1 - v_\Delta/(2\sigma^2_g)$, with the average variance of a difference between two means; valid with unbalanced data and incomplete blocks.	3.6
D		
Diallel cross (dialelo)	All possible crosses among a group of parents. Griffing distinguished four methods, depending on whether the parents and the reciprocals are included.	4.1
E		
Ecovalence (ecovalencia)	Sum of the squared residuals of a genotype in the additive model: its contribution to the interaction (Wricke).	9.5
Effective factors (factores efectivos)	Minimum number of genes that separate two parents, estimated with the Castle–Wright formula. It is a lower bound.	7.8
Efficiency (eficiencia)	How much a design with incomplete blocks improves on complete blocks in the precision of comparisons among entries.	2.4, 3.6
Environmental index (índice ambiental)	Mean of an environment minus the grand mean: it measures how good the environment was.	9.4
Epistasis	Interaction between genes at different loci. In generation means: additive × additive [i], additive × dominance [j] and dominance × dominance [l]; complementary or duplicate depending on the signs of [h] and [l].	7.1, 7.5
Expected mean square (cuadrado medio esperado)	Combination of variances that a mean square estimates. Block 6 computes it for the actual experiment, by traces, instead of copying it from a table.	6.3

Table C.2. Glossary in alphabetical order. *(continued)*

Term	Definition	Section
F		
F∞ and F_2 metrics (métrica F∞ y F_2)	Two origins for the parameters of generation means: the mean of the pure lines the cross can yield (Mather and Jinks) or the F_2 (Hayman).	7.5
Founder (fundador)	Individual in the pedigree with no known parents; the founders form the base, whose mean breeding value is 0.	10.2
G		
GBLUP	BLUP with the genomic relationship matrix G. It is equivalent to ridge regression on all the markers (RR-BLUP).	11.4
General combining ability (GCA) (aptitud combinatoria general, ACG)	Average performance of a parent in its crosses, as a deviation from the grand mean. It depends mainly on the additive variance.	I.3, 4.4
Genetic advance (GA) (avance genético, AG)	How much the mean would rise if the chosen proportion were selected on its phenotypic mean: $i \cdot \sigma_P \cdot H^2$.	8.2
Genomic heritability (heredabilidad genómica)	The part of the variance the markers capture. It is lower than the true one if some of the QTL are not in disequilibrium with the panel.	11.4
Genotype × environment interaction (interacción genotipo × ambiente)	Change in the ranking of or the differences among genotypes from one environment to another.	9.2
H		
Heterosis	Superiority of the F_1 over the mean of its parents or over the better of them. In a variety diallel it is split into average, variety and specific heterosis.	5.6, 7.7
Heterotic group (grupo heterótico)	Set of lines that are crossed with those of another group and not among themselves, such as the dent and flint lines of maize.	11.6
I		
Ideal genotype (genotipo ideal)	Point of the GGE biplot with the highest mean and the greatest stability; it serves as a reference for comparing the others.	9.8
Inbreeding coefficient (F) (consanguinidad)	Probability that the two alleles at a locus of an individual are identical by descent. It is 1 in a pure line.	10.2
Inbreeding depression (depresión endogámica)	What the F_2 loses relative to the F_1: $F_1 - F_2$. Without epistasis, the F_2 keeps half the heterosis.	7.7
Individual and maternal heterosis (heterosis individual y materna)	In crossbreeding, the advantage of the crossbred individual and that of having a crossbred dam.	10.6

Table C.2. Glossary in alphabetical order. *(continued)*

Term	Definition	Section
J		
Joint scaling test (prueba conjunta de escala)	Weighted least-squares fit of the means of all generations; its χ^2 says whether the model is enough (Cavalli).	7.5
L		
Least significant difference (LSD) (diferencia mínima significativa, DMS)	The smallest difference between two means that is significant at the chosen level.	I.9, 3.6
Line × tester (línea × probador)	Each line is crossed with a few testers; it gives the GCA of lines and testers and the SCA of their crosses.	6.8
M		
Maternal effect (efecto materno)	Part of an individual's value that depends on its mother and not on its own genes. In a diallel, the systematic difference between reciprocal crosses; in the animal model, the maternal breeding value of the dam.	4.4, 10.4
Mega-environment (megaambiente)	Group of environments with the same winning genotype, which recurs over the years.	9.8
Mendelian sampling (muestreo mendeliano)	The part of an offspring's breeding value that is not the average of its parents; its relative variance, b, enters the inverse of A.	10.2
Model I and model II (modelo I y modelo II)	Fixed parents or environments (only those are of interest) or random ones (a sample from a population). It changes the denominator of some tests.	I.9, 4.1
N		
Narrow-sense heritability (h^2) (heredabilidad en sentido estrecho)	Additive fraction of the phenotypic variance, σ^2_A/σ^2_P: the one that determines the response to selection.	4.5, 7.6
North Carolina designs I, II and III (Carolina del Norte I, II y III)	Mating designs of Comstock and Robinson: hierarchical (each male with its own females), factorial (males × females) and each F_2 plant crossed to both parents of the F_1.	6.1
P		
Partial diallel (dialelo parcial)	Each parent is crossed with only s of the others, almost always in the circulant order of Kempthorne and Curnow.	6.9
Path analysis (análisis de sendas)	Decomposes the correlation of each trait with a dependent one into a direct effect and indirect effects through the others.	8.6
Permanent environment (ambiente permanente)	An individual's own effect, with no relationships, that is repeated in all its records: what it has of good or bad from its history and not from its genes.	10.4

Table C.2. Glossary in alphabetical order. *(continued)*

Term	Definition	Section
Plot error (error de parcela)	Variation among plots of the same entry not explained by their plants; in Block 7, σ^2_p.	7.3
Potence ratio (relación de potencia)	$(F_1 - MP)/(½\|P_1 - P_2\|)$: 0 intermediate inheritance, ±1 complete dominance, more than 1 apparent overdominance.	7.7
Power (potencia)	Probability that a test detects a difference that exists. Block 2 computes it before sowing.	2.6
Q		
Quasi-F (cuasi-F)	F test whose denominator is a combination of mean squares, with Satterthwaite degrees of freedom; it is used when no single source serves as denominator.	6.3
R		
Reciprocal effects (efectos recíprocos)	Differences between A × B and B × A. They are split into maternal (of the female) and non-maternal (specific to the pair).	4.4
Recombination loss (pérdida por recombinación)	What is lost when the favorable gene combinations of each breed are broken up; it appears in crosses whose parents are already crosses.	10.6
Relationship matrix (A, G) (parentesco)	Twice the coancestry between two individuals: A from the pedigree, G from the markers.	10.2, 11.3
Reliability (confiabilidad)	Square of the correlation between the true and the predicted breeding value: $1 - PEV/[(1 + F)\sigma^2_a]$. Its square root is the accuracy.	10.3
Reliability index (Annicchiarico) (confiabilidad de Annicchiarico)	Relative performance, as a percentage of the mean of each environment, that a genotype exceeds with probability $1 - \alpha$.	9.5
Repeatability (repetibilidad)	Correlation between repeated records of an individual, $(\sigma^2_a + \sigma^2_{pe})/\sigma^2_p$. The trial heritability of Block 3 is also a repeatability.	3.6, 10.4
Response to selection (respuesta a la selección)	Advance of the mean in the next generation: $R = h^2S = i \cdot h \cdot \sigma_A$.	I.3, 8.9
Restricted maximum likelihood (REML) (máxima verosimilitud restringida)	Method for estimating variance components that discounts the degrees of freedom of the fixed effects; with balanced data it matches the moment estimators when these are not negative.	3.6, 10.3
Ridge (cresta)	Small amount added to the diagonal of a matrix so that it can be inverted or stabilized: the constant k of path analysis and the λ of the G matrix.	8.6, 11.3
Rotation and composite (rotación y compuesta)	Crossbreeding systems: in rotation, sires of two or three breeds are used in turn; the composite is a population formed from several breeds that mates at random.	10.6

Table C.2. Glossary in alphabetical order. *(continued)*

Term	Definition	Section
S		
Sampling error (error de muestreo)	Variation among the plants of the same plot; it is separated from the plot error when there are data per plant.	3.8
Scaling tests (pruebas de escala)	Contrasts A, B, C and D among generation means that are zero if the additive–dominance model is enough (Mather).	7.4
Selection index (índice de selección)	Linear combination of traits, I = b′y, for choosing the genotypes that most improve an aggregate breeding value.	8.7
Selection intensity (i) (intensidad de selección)	Superiority of the selected individuals in standard deviations. Some books call it k.	8.3
Sire model (modelo de padres)	Model in which the random effect is the transmitting ability of each sire; smaller, but it ignores the dams.	10.4
Specific combining ability (SCA) (aptitud combinatoria específica, ACE)	How much a cross yields above or below what the GCA of its parents predicts. It reflects dominance and epistasis.	I.3, 4.4
Stability variance (varianza de estabilidad)	The contribution of a genotype to the interaction in the form of a variance, which can be tested against the error (Shukla).	9.5
Standard heterosis (heterosis estándar)	Difference between the F_1 and a check sown in the same trial, as a percentage of the check mean.	7.7
Static and dynamic stability (estabilidad estática y dinámica)	Yielding the same in every environment, or following the average pattern of the environments. Stability statistics measure one or the other.	9.9
Studentized residual (residual estudentizado)	Residual of a plot divided by its standard deviation estimated without it; it is used to flag outliers.	3.5
Superiority measure (Lin and Binns) (superioridad de Lin y Binns)	Mean distance of a genotype from the best one in each environment; the smaller, the better. It combines mean and stability.	9.5
T		
Three-way and double crosses (cruzas triples y dobles)	Crosses of three or four lines; they are predicted with Jenkins's rules from the nonparental single crosses.	11.6
Triple test cross (cruza triple de prueba)	Each F_2 plant is crossed with both parents and with their F_1; besides σ^2_A and σ^2_D, it tests epistasis.	6.7
Two-stage analysis (análisis en dos etapas)	First the adjusted means of each environment with their error; then a model of the genotype × environment table in which each mean is weighted by the precision of its environment.	9.3

Table C.2. Glossary in alphabetical order. *(continued)*

Term	Definition	Section
W		
WAAS	Average of the absolute values of a genotype's scores on the retained AMMI axes, weighted by the interaction each axis explains; the smaller, the more stable.	9.7
Wr–Vr graph (gráfico Wr–Vr)	Covariance of each array with the parents against its variance; its line tells the degree of dominance, and the position of each parent, how many dominant alleles it carries.	5.5

C.3 Symbols with more than one meaning

Table C.3. The literature uses the same letters for different quantities; the block and the section make clear which one is meant.

Symbol	Meanings (section)
F	Inbreeding of the individual or of the parents (6.2, 10.2) · component F of the variances, Σdh (5.4, 7.6) · F statistic (I.9) · generations F_1, F_2, F_3 (7.1).
D, H	Hayman's additive and dominance components, D, H_1 and H_2 (5.4), and those of the generation variances (7.6) · scaling test D (7.4) · aggregate breeding value H = w'g of the index (8.7).
E	Environmental component (5.4, 7.6) · efficiency factor of a lattice (2.4).
h, H^2	Narrow-sense heritability h^2 and broad-sense H^2 (4.5, 8.2) · dominance effect [h] of the means (7.1) · Hayman's h^2, the net dominance (5.4) · heterosis coefficients h^l and h^M (10.6) · phenotypic accuracy h (11.5).
i	Selection intensity (8.3) · additive × additive epistasis [i] (7.1).
b	Slope on the environmental index (9.4) · slope of the Wr–Vr line (5.5) · Mendelian sampling variance (10.2) · coefficients of a selection index (8.7).
r	Number of replicates (3.6) · correlation (8.5) · reliability r^2 (10.3) · predictive ability (11.5) · recombination loss r^l (10.6).
A	Pedigree relationship matrix (10.2) · scaling test A (7.4) · additive variance σ^2_A (4.5).
G	Genomic relationship matrix (11.3) · genotypic covariance matrix of an index (8.7) · effect of the genotypes in the GGE biplot (9.8).
P	Phenotypic covariance matrix (8.7) · parents P_1 and P_2 (7.1) · P value (I.9).
W	Wricke's ecovalence W_i (9.5) · Annicchiarico's reliability W_i (9.5) · Hayman's covariance W_r (5.5).
S	Selection differential (I.3, 8.8) · deviations from regression S^2_d (9.4) · Huehn's rank statistics $S^{(1)}$ to $S^{(6)}$ (9.6) · in Spanish, the average semivariance SM, which the app in English calls ASV (11.4).
d	Additive effects [d] of the means (7.1) · desired gains of an index (8.7) · dominance deviation at a locus, d/a (1.2).
k	Plants per plot (7.3) · ridge constant of path analysis (8.6) · block size of a lattice (A.3) · selection intensity in some books (8.3).
λ	Ridge of the G matrix (11.3) · eigenvalue λ^2 of the eigen index (8.7).
s	Crosses per parent in a partial diallel (6.9) · SCA effect $\hat{s}_{ij}$ (4.4) · variance among plants s^2 (7.3).

Table C.3. The literature uses the same letters for different quantities; the block and the section make clear which one is meant. *(continued)*

Symbol	Meanings (section)
m	Mean of the generation means (7.1) · maternal effect m_i of a diallel (4.4) · fraction m^2 of the maternal genetic effect (10.4).
σ^2_p, σ^2_P	Plot error of Block 7 (7.3) · phenotypic variance of the animal model (10.3).

D Troubleshooting

The most common problems, grouped by the moment they appear. Messages go in quotation marks, as they are seen on screen with the app in English; `{n}` marks a part that changes. Messages in red stop the analysis; amber ones only warn.

D.1 Opening the app and keeping your work

Table D.1. Problems when opening the app and when coming back to it.

Symptom or message	Cause	What to do	See
The page shows no colors or buttons, or does not respond.	Folders next to `index.html` are missing (`js/`, `css/`, `vendor/`) or the browser is old.	Copy the whole `BreedingPro` folder and use an up-to-date browser.	I.5
The institution does not allow pages to be opened as local files.	A browser or network policy.	Use `Open BreedingPro.bat`: it opens the same app with a server on your own computer (port 9200). Nothing leaves the machine.	I.5
When the page is reloaded, the data and results disappear.	The app remembers only the language, the theme, the figure style and the Block 3 data sheet.	Download Results (.xlsx) from each block or the Block 12 package before closing, and load the file again.	I.5
The data sheet does not appear on another computer, in another browser or in a private window.	It is saved in the browser where it was typed, and separately for each way of opening the app (double click or local server).	Always open the app the same way; to take the sheet elsewhere, Save as.	3.2
The Block 2 field map came out different when built again.	Another random seed.	Type the seed saved in the `Design` sheet of the workbook or in the declaration line.	2.5

D.2 Loading the data

Table D.2. Problems when reading a table in Block 3.

Symptom or message	Cause	What to do	See
"The entries column (genotype, cross or parents) is missing." Cards 3 to 5 do not appear.	No column has the entry role, nor those of female and male.	In card 2, give the role Entry / genotype to the names column, or Female / line / parent 1 and Male / tester / parent 2 to the pedigree columns.	3.3
"There is no response trait: mark at least one numeric column as "Response trait"."	The data columns were read as text or given another role.	Check the decimal separator and any text inside the numbers; mark the traits.	3.3
A column of lot, furrow or code numbers appears as a trait.	Its name is not recognized and 90% or more of its cells are numbers.	Switch it to Ignore.	3.3
Card 2 warns of non-numeric values in a trait.	Cells such as `12.5a` or notes written next to the datum: they are taken as missing.	Correct in the data sheet, fix the value and Use these data →.	3.2
Decimals are misread: `12,5` as text or as 125.	The column mixes decimal comma and point, or a comma-separated CSV carries decimal commas.	Choose the Decimal separator, or save the CSV with semicolons.	3.3
The wrong sheet of the workbook was opened.	The one with the most data is opened.	Choose another in Workbook sheet.	3.2
"The workbook has no sheets with data."	The workbook is empty or its sheets have no values.	Check the file in the spreadsheet and save it with the values.	3.2
The data come in wide form: one column per replicate or per environment.	The app needs one row per plot.	Columns → rows...; a diallel in matrix form, with Diallel table → rows.	3.3
Pasting does not work in the data sheet grid.	The browser does not allow the clipboard to be read.	Open Pasting does not work in this browser? Paste into this box instead and paste into the text box.	3.2

D.3 Recognized design and field analysis

Table D.3. Problems with the design and the analysis of each environment.

Symptom or message	Cause	What to do	See
The mating design comes out with medium or low confidence.	Entries of the complete design are missing, the partial diallel is not circulant, crosses of the factorial are missing, the females of a North Carolina II repeat codes across males or there are fewer than 20 F_2 plants.	Read the reasons, check Recognized entries and, if needed, choose the design in the Mating design menu.	3.4
A North Carolina I is recognized as II.	The females of each male repeat codes with those of other males.	Choose North Carolina I in the menu, or give the females unique codes.	3.4
"{n} plots hold more than one entry (e.g., plot {k})."	A data-entry error in the plot or entry column.	Correct the data: a plot cannot have two entries.	3.4
The row–column design is not used even though the data have row and column.	It is never chosen on its own.	Choose Rows and columns in the Field design menu.	3.4
A lattice or a row–column design was analyzed as complete blocks.	The data have no real incomplete blocks, or the row and column columns are missing.	That is correct; if it is not, check the roles of the block, row and column columns.	3.8
The environment is left without an error, without tests or LSD.	A single value per entry: they are means.	Type the Error of the original experiment: mean square, degrees of freedom, scale and replicates.	3.8
Two plots of the same entry come out as outliers with the same |t|.	With two replicates, the two residuals of an entry are equal and of opposite sign.	Decide with the field book which one is wrong; exclude only proven errors.	3.5
An adjusted mean carries a dagger (†).	The entry is not connected with the others in the design.	Check that it is in blocks that share entries with others.	3.6
The combined analysis says it was done in two stages.	The one-stage model would have more than 1,400 parameters.	That is normal; entries that are not in every environment are left out of the second stage.	3.7
The combined analysis warns that the F tests are approximate.	Bartlett's test rejects that the errors of the environments are equal.	Use the two-stage analysis of Block 9.	3.7

D.4 Genetic and selection blocks

Table D.4. Messages and missing results in Blocks 4 to 9.

Symptom or message	Cause	What to do	See
A block says the data are not of its design.	Block 3 recognized another mating design.	Go to the block that card 1 indicates, or correct the design in Block 3.	3.9
A variance component comes out negative and carries a dagger.	The true variance is close to zero, or there are few replicates or parents.	It is reported as it comes out; the parameters that use it are not computed. More parents or replicates estimate it better.	4.5, 6.5
σ^2_A and σ^2_D change a lot from one analysis to another with the same data.	The inbreeding F of the parents changed.	Check the control Inbreeding of the parents (F).	6.2
The triple test cross shows highly significant epistasis for no apparent reason.	The F_1 tester is not the one the app assumed.	Choose the right one in F_1 tester (L_3).	6.7
Block 7 gives no D, H, F, E or effective factors.	There is one value per plot: without plants there is no variance among individuals.	Load the data per plant.	7.3
In Block 7 everything is significant and the standard errors are tiny.	The variance of each mean was forced to among individuals / n with data in plots.	Go back to automatic.	7.3
In Block 8, a genotypic correlation comes out outside [–1, 1] or is left empty.	Small σ^2_G (the moment estimator is not bounded) or negative σ^2_G.	It is not truncated: read it with caution or remove that trait.	8.5
In Block 8, the path table does not appear.	The correlation matrix is not positive definite, or R^2 would come out greater than 1.	Use the phenotypic correlations, remove traits with small σ^2_G or increase the ridge constant.	8.6
"Some index has an accuracy or heritability above 1..."	The estimated G matrix does not fit inside P in some direction.	Remove the traits with very small σ^2_G or use the base index.	8.7
In Block 8, the genetic advance of a trait is zero.	With several environments, its σ^2_G came out negative.	That is as it should be: the trait has almost no genetic variance across environments.	8.4
"At least 3 genotypes and 2 environments with data are needed."	Block 3 has no location or year column.	Give the role Location / environment or Year / season to the column that tells them apart.	9.1
In Block 9, some cells carry an asterisk.	They were missing and were estimated by EM-AMMI; the interaction loses one degree of freedom for each.	Check whether those combinations are really missing.	9.2
In Block 9, Annicchiarico's reliability does not appear.	It is computed only when the larger value is the better one.	That is expected for days to flowering or disease incidence.	9.5
The GGE biplot carries a caution notice.	The first two components explain less than 60% of G + G×E.	Read it with caution and rely on AMMI and the stability statistics.	9.8

D.5 BLUP and genomics

Table D.5. Problems in Blocks 10 and 11.

Symptom or message	Cause	What to do	See
"The individual column is missing."	No column has the individual role.	Assign it in Role of every column.	10.2
"The pedigree file needs individual, sire and dam columns."	The separate pedigree does not have three recognizable columns.	Name them as in Table A.4 (individual, sire and dam).	10.2
Pedigree warnings: an individual repeated with different parents, an individual that is its own parent, a loop, or the same individual as sire and dam.	Data-entry errors in the pedigree, or selfs in plants.	The app keeps the first row and deletes the impossible parent; correct the file. In plants, tick Plants: selfing allowed.	10.2
"The system of equations has {n} unknowns; this block solves exactly (dense matrices) up to 2600..."	The pedigree is too large for the exact calculation in the browser.	Trim the pedigree to the ancestors of the individuals with records or analyze it in parts.	10.3
"Type every variance (greater than zero) or choose to estimate them by REML."	A given variance is missing.	Complete them or switch to REML.	10.4
REML gives meaningless values or estimates nothing useful.	Too few records.	Give the variances as known.	10.4
The permanent environment is left out with a notice.	No individual has more than one record.	That is correct: the effect is not estimable.	10.4
Card 3 does not find the pedigree of the entries.	The names in the trial and in the pedigree do not match.	Use the same name in both files.	10.5
"Type at least two breeds." · "unknown breed in "{cross}" (position {k})" · "missing ")" in "{cross}""	The list of breeds or the way a cross is written.	Write the breeds separated by commas and the crosses as in Table A.4.	10.6
A Dickerson parameter comes out not estimable.	The groups do not separate it.	Add groups that contain it (backcrosses, F_2, three-breed crosses) or remove the parameter with its checkbox.	10.6
The markers are read the wrong way around: one individual per column.	The file has the markers in rows.	Tick Markers in rows.	11.2
"Too few individuals have both phenotype and markers."	The identifiers of the phenotype file, or the entries of Block 3, do not match those of the markers.	Use the same identifiers in both files.	11.4
Weight of the pedigree (Block 10) does not appear.	Block 10 has no pedigree that includes every individual with markers.	Load that pedigree in card 1 of Block 10.	11.3

Table D.5. Problems in Blocks 10 and 11. *(continued)*

Symptom or message	Cause	What to do	See
The checkbox Include the SCA (K_1 ⊗ K_2) is disabled.	There are more than 700 evaluated hybrids: the SCA kernel would be too large.	Predict with the GCA.	11.6
"Too few hybrids have a phenotype and genotyped parents."	The lines in the hybrid table are not in the marker file.	Check the names of the lines in both files.	11.6
"Write a three-way or double cross, for example (A×B)×C." or a cross that is not predicted.	The way it is written, or a cross that needs crosses within the same group.	Write one cross per line, with the lines on each side from different groups.	11.6

D.6 Report and figures

Table D.6. Problems when building the report and exporting.

Symptom or message	Cause	What to do	See
A block appears with *no results* and cannot be chosen.	It has not been opened with the data loaded now.	Open the block, run it and come back to Block 12.	12.2
The report does not include the crossbreeding of Block 10 or the recurrent selection of Block 8.	The examples those blocks load on their own do not count as your analyses.	Run the crossbreeding, or type your values in the recurrent selection table.	12.2
The report shows values from before a change.	Each block enters with what it showed when the report was built.	Build the report again.	12.1
A figure in the report is not as you wanted it.	It goes in as it was left in its editor.	Change it in its block and build the report again.	12.5
The ZIP package weighs more than a gigabyte.	TIFF figures go uncompressed: about 3 bytes per pixel.	Leave the package figures in SVG or PNG and download as TIFF, from their panel, only the ones you submit.	12.6
A figure was saved at fewer dpi than requested, with a notice.	The browser limits the image size.	Lower the resolution or the size of the figure, or use SVG.	12.6

E References

The works cited in the manual and in the texts of the app (the home page, the assumptions, formulas and references cards of each block, and the examples), with their full details.

The symbol ◆ marks those that the methods section of Block 12 can name: with them you complete the reference list of a paper (Section 12.4). Five original titles include the name of a program; in its place goes **[...]**. Chapters 3 to 5 cite "Hayman (1954)" and "Griffing (1956)" without a letter: they are Hayman (1954a) and Griffing (1956b). DOIs are given for the open-access papers from which example data come. The titles of works published in Spanish are kept in Spanish.

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NOTE · CORRECTIONS MADE IN THE APP WHILE PREPARING THE APPENDICES

Comparing the references and notices of the app with the original sources made it possible to correct:

- The eigen index (ESIM) was attributed to Cerón-Rojas and colleagues (2008); the paper that proposes it is from 2006. It was corrected in the app and in Table 8.3.
- The reference list of Block 8 omitted eleven works that its notes cite: Williams (1962), Elston (1963), Burrows (1972), Mallard (1972), Bulmer (1980), Hayes and Hill (1981), Tallis (1985), Itoh and Yamada (1987), Carvalho and Cruz (1996), Cerón-Rojas and colleagues (2006) and Schaeffer (2014).
- The entry for Saavedra Guevara and colleagues (2021) on the home page had the wrong title and last page, and that for Jasso-Bobadilla and colleagues (2022), the wrong last page.
- The third-party notices file did not list the examples of Blocks 8 to 11 or their terms of use.

F Third-party components

BreedingPro is an original work: its code, its figure engine, its own palettes and scales, its illustrations and its simulated data were made for this program. Only one software component is third-party; the palettes and published data it uses are declared here with their terms.

Table F.1. Third-party software component included in the app.

Component	Version	License	Use and notice
Spreadsheet reading and writing library `vendor/xlsx.full.min.js`	0.18.5	Apache 2.0	Reads `.xlsx`, `.xls`, `.xlsm` and `.ods` workbooks and writes the `.xlsx` files of every block and the `.ods` of the data sheet. It is an unmodified copy that is not part of the original work. Its copyright notice, its source and the full text of the license are in `vendor/THIRD-PARTY-NOTICES.txt`, which must always accompany the app.

! IMPORTANT · WHEN SHARING OR REDISTRIBUTING THE APP

The code of BreedingPro is distributed under the GNU General Public License, version 3 or later (file `LICENSE`). If you share the folder, keep it complete: `LICENSE`, `vendor/THIRD-PARTY-NOTICES.txt` and the library in `vendor/` travel together. If you publish a figure with the Okabe and Ito palette, Tol's, or the Cividis scale, or a result obtained with an example, cite their authors. The program is cited with its DOI, 10.5281/zenodo.23005938 (see below and the manual's credits).

Table F.2. Published color palettes and scales offered by the figure editor.

Resource	Authorship	Terms	In the editor
Color-blind-safe palette	Okabe and Ito (2008): eight colors; the app adds a gray and a dark brown.	Freely accessible guide; the app and the manual cite its authors.	Okabe & Ito 2008 (color-blind safe)
Qualitative color schemes	Tol (2021): *bright* scheme plus black and two colors of the *vibrant* scheme.	Freely accessible technical note; the author is cited.	Tol 2021 (color-blind safe)
Viridis, Magma, Inferno and Plasma scales	Nathaniel J. Smith and Stéfan van der Walt; Viridis also with Eric Firing (2015).	Dedicated to the public domain (CC0 1.0); their authors appreciate credit.	Continuous scales of the heat maps, with five colors of each.
Cividis scale	Núñez, Anderton and Renslow (2018).	Open-access paper (CC BY 4.0); it is cited.	Cividis (color-blind safe)

The other thirteen qualitative palettes of the editor, such as Purple maize (BreedingPro) or Harvest, and the eight scales of Figure F.1 are the app's own.

THE APP'S OWN HEAT-MAP SCALES

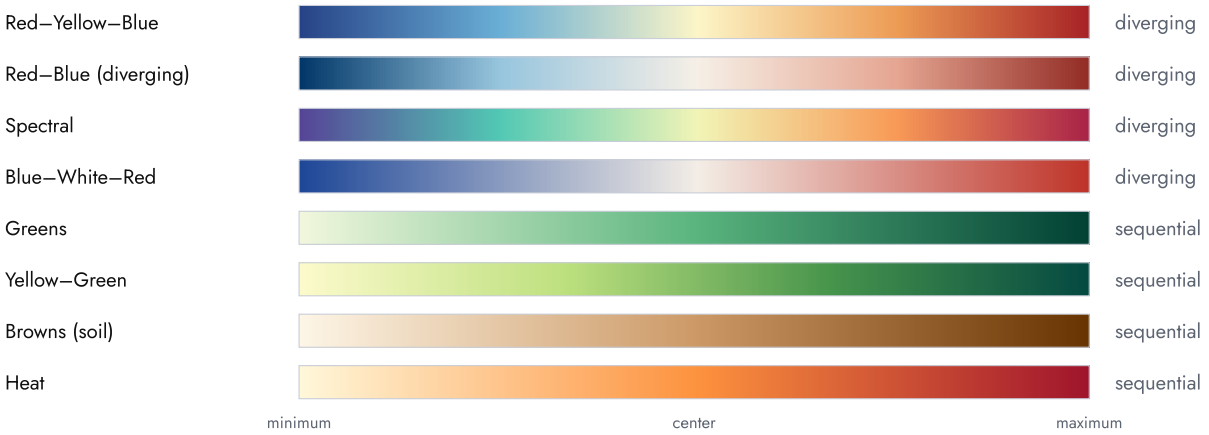

Figure F.1. The eight heat-map scales designed for the app, with their names in the menu. They start from colors anchored in a perceptual color space: in the diverging ones, lightness rises toward the center and falls equally toward the ends; in the sequential ones, it falls continuously. The ends of the diverging scales remain distinguishable with red–green color blindness.

Table F.3. Published data used by the examples. If you publish something made with one of them, cite the original work.

Example	Source	Terms
Maize: full 4 × 4 diallel	Pérez López and colleagues (2020)	Open-access paper, CC BY-NC 4.0: non-commercial use with attribution.
Maize: partial diallel of 8 lines	Jasso-Bobadilla and colleagues (2022); design of Kempthorne and Curnow (1961)	CC BY-NC 4.0.
Nicotiana rustica: 8 × 8 diallel	Hayman (1954a), table 5, reproduced by Onofri and colleagues (2021)	Open copy under a CC BY 4.0 license.
Diallel with parents in two environments	Zhang, Kang and Lamkey (2005), reproduced by Onofri and colleagues (2021)	CC BY 4.0.
Crosses between maize varieties	Lonnquist and Gardner (1961); Gardner and Eberhart (1966); Onofri and colleagues (2021)	CC BY 4.0.
Textbook 8 × 8 diallel; North Carolina I and III; line × tester	Singh and Chaudhary (1979)	Transcriptions distributed under the GPL in free software.
Maize: means of 36 single crosses	Griffing (1956b)	Transcription distributed under the GPL.
Potato: North Carolina II; potato: 28 clones in 6 locations	International Potato Center (CIP), Peru: Majes; Ayacucho, Huancayo, La Molina and San Ramón (2002–2003)	Distributed under the GPL in free software.
Maize: 13 hybrids (one and four environments); oat: 10 genotypes in 14 environments	T. Olivoto, author of the data	Distributed by their author under GPL v3.
Means of 17 genotypes in 12 environments	Kang (1993); Kang and Magari (1995)	Distributed under the GPL.

Table F.3. Published data used by the examples. If you publish something made with one of them, cite the original work. *(continued)*

Example	Source	Terms
N. rustica: six generations	Gale, Mather and Jinks (1977), table 1, first set	Only the published statistics are used (number, mean and variance of each generation); the plants were reconstructed with those exact values.
Block 10: animal and sire models	Mrode (2014), examples 3.1, 3.2 and the litter example; Schaeffer (2019), tables 3.1, 6.1 and 6.2	Short textbook examples, reproduced with their source to check the calculations.
Block 10: crossbreeding	van der Werf (n.d.); Tesema and colleagues (2023); Buchanan and Northcutt (n.d.)	A few published means and parameters, with their source; the paper by Tesema and colleagues is CC BY 4.0.
Block 11: wheat with DArT markers; dent × flint maize hybrids	Crossa and colleagues (2010), data of the International Maize and Wheat Improvement Center (CIMMYT); Technow and colleagues (2014), with 2,000 of the 35,478 SNPs	Distributed under the GPL in free software.
All the simulated ones	BreedingPro, with fixed seeds	Free of rights; their true values come with the example.

Manual resources. The typefaces Cormorant, Crimson Pro and Jost (Christian Thalmann, Jacques Le Bailly and Owen Earl) are under the SIL Open Font License (OFL) 1.1; the app uses no external typefaces. The illustrations and diagrams are original, and the screenshots come from the app itself.

How to cite

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The citation of the program is the one given by the How to cite section of the app's home page and card 4 · How to cite of Block 12, and the one that closes every Block 12 report in English. The DOI is the concept DOI: it always leads to the most recent version. Each archived version also has its own (version 1.0.0, 10.5281/zenodo.23005939).